

REPRODUCTION AND RECOMBINATION

Prokaryotes usually reproduce by binary fission, which is a form of asexual reproduction. During binary fission, shown in Figure 23-11, the DNA replicates and the two daughter DNA molecules are pulled to opposite ends of the cell. Finally, the cell membrane pinches in two to form two identical daughter cells.

Prokaryotes can exchange pieces of DNA that can be added to the cell's DNA without reproduction. This process is called *recombination*. Prokaryotes can get DNA from other prokaryotes in three ways: transformation, bacterial conjugation, and transduction. **Transformation** occurs when a prokaryote takes in DNA from its outside environment. **Conjugation** is the process by which two prokaryotes bind together and one cell transfers DNA to the other cell through a structure called a *sex pilus*. In **transduction**, a virus obtains a small part of DNA from a host prokaryote. The virus then copies itself inside the host, and new copies of the prokaryotic DNA are made with the viral DNA. After the new viruses have been released, they carry the prokaryotic gene to the next prokaryote.

Understanding these methods of bacterial genetic exchange is of great importance to our understanding of genetics. Transformation provided the evidence that DNA is the genetic material of cells. Investigations on the process of conjugation gave scientists key information about the mechanisms of genetic recombination common to all organisms. Transformation is used every day in thousands of labs to transfer bits of human and other DNAs into bacterial cells to identify and isolate certain genes. The sequencing of the human genome, one of the greatest feats in history, relied on recombining human genes into artificial plasmids and artificial chromosomes called *bacterial artificial chromosomes* (BACs). The BACs and plasmids were then inserted into bacterial cells by transformation. The results of such studies are being used to develop a better understanding of human health and to find improved treatment for disease.

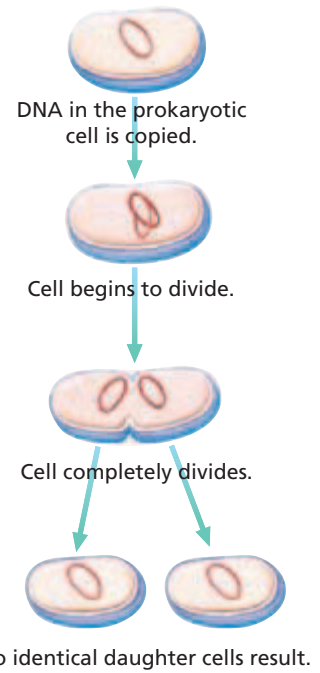


FIGURE 23-11

Most prokaryotes reproduce by binary fission, in which two identical cells are produced from one cell.

SECTION 2 REVIEW

1. List the functions of six structures of a prokaryotic cell.
2. Describe three ways in which prokaryotes move.
3. Identify how the oxygen requirements of a prokaryote can affect the types of environments in which it can live.
4. How do autotrophs and heterotrophs differ?
5. List and summarize the three methods of prokaryotic recombination.

CRITICAL THINKING

6. **Relating Concepts** What eukaryotic organelle is functionally similar to the prokaryotic cell membrane and its infoldings?
7. **Applying Information** Why is improperly sterilized canned food an ideal environment for the obligate anaerobe *Clostridium botulinum*?
8. **Analyzing Information** Identify why recombination is not a form of reproduction.

SECTION 3

OBJECTIVES

- **Describe** the ways in which bacteria can cause disease in humans.
- **Explain** how a bacterial population can develop resistance to antibiotics.
- **Identify** reasons for recent increases in the numbers of certain bacterial infectious diseases.
- **Identify** ways of preventing a foodborne illness at home.
- **List** four industrial uses of bacteria.

VOCABULARY

pathology
exotoxin
endotoxin
antibiotic resistance
zoonosis
bioremediation

BACTERIA AND HUMANS

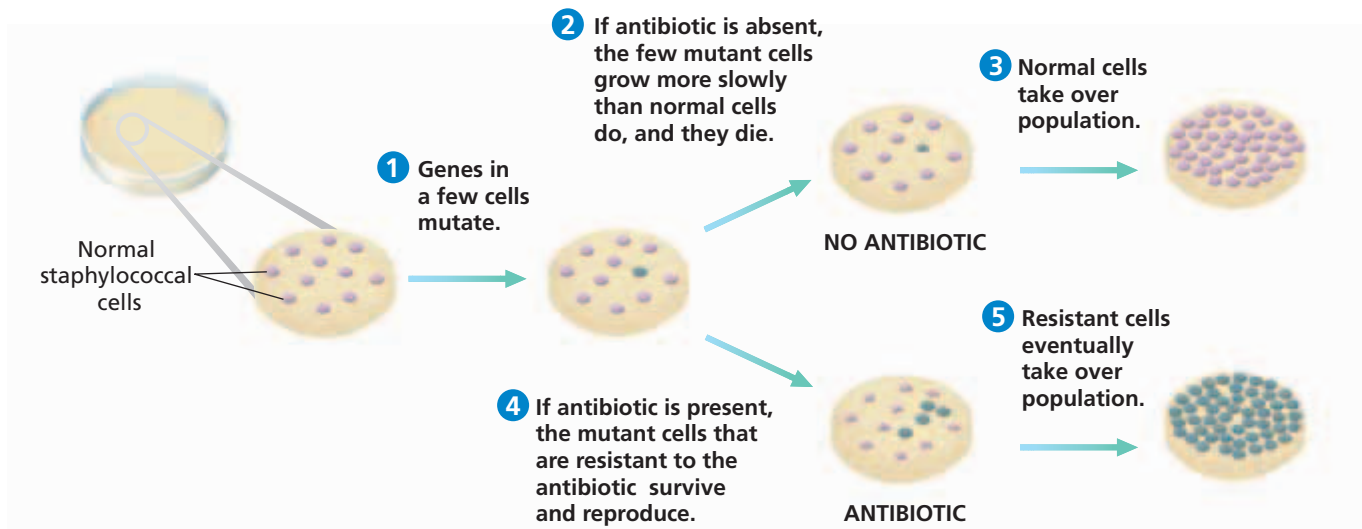
Bacteria are probably best known for the diseases that they cause in humans. However, most bacteria that live on human skin are harmless, and other bacteria are used biotechnologically, as in the production of yogurt. Still other bacteria that live in soil change atmospheric nitrogen into ammonia and nitrites, which plants use for growth.

BACTERIA AND HEALTH

The scientific study of disease is called **pathology** (puh-THAHL-uh-jee), and bacteria and other organisms that cause disease are called **pathogens**. Table 23-3 summarizes several bacterial diseases. Some bacteria cause disease by making certain poisons called **toxins** (TAHKS-ins). **Exotoxins** (EKS-oh-TAHK-sins) are toxic substances that bacteria secrete into their environment. Some Gram-positive bacteria secrete protein exotoxins. For example, *Clostridium tetani* secretes an exotoxin that causes tetanus in humans. **Endotoxins** are toxic substances made of lipids and carbohydrates associated with the outer membrane of Gram-negative bacteria, such as *E. coli*. Endotoxins are not released until the cell dies. After they are released, endotoxins can cause fever, body aches, diarrhea, hemorrhage, and weakness.

TABLE 23-3 Bacterial Diseases

Disease	Bacterium	Areas affected by illness	Mode of transmission
Anthrax	<i>Bacillus anthracis</i>	lungs, skin, or intestines	airborne spores; may be picked up from environmental surfaces
Botulism	<i>Clostridium botulinum</i>	nerves	improperly preserved foods
Cholera	<i>Vibrio cholerae</i>	intestine	feces-contaminated water
Dental caries (tooth decay)	<i>Streptococcus mutans</i> , <i>S. sanguis</i> , and <i>S. salivarius</i>	teeth and gums	dense collections of bacteria normally found in the mouth
Gonorrhea	<i>Neisseria gonorrhoeae</i>	urethra, fallopian tubes, and epididymis	person-to-person transmission by sexual contact
Lyme disease	<i>Borrelia burgdorferi</i>	skin, joints, and heart	tick bite
<i>Salmonella</i> food poisoning	species of the genus <i>Salmonella</i> , such as <i>S. typhimurium</i>	intestine	feces-contaminated water and food
Tetanus	<i>Clostridium tetani</i>	nerve cells at synapses	soil-contaminated skin wounds
Staph infection	<i>Staphylococcus aureus</i>	skin, soft tissue, lungs, and blood	direct contact of skin wounds with <i>S. aureus</i>



Bacteria also cause disease by destroying body tissues. As bacteria stick to body cells, they secrete digestive enzymes that break down tissue for its nutritional value, which allows further bacterial invasion. For example, some species of the genus *Streptococcus* make a blood clot-dissolving enzyme that allows bacteria to spread easily from tissue to tissue.

Antibiotics and Antibiotic Resistance

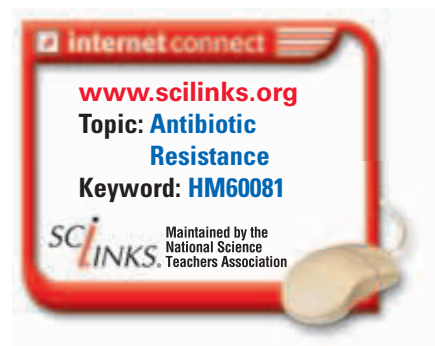
Antibiotics affect bacteria by interfering with certain cellular activities. Penicillin (PEN-i-SIL-in), for example, blocks the ability to build new cell wall material, while tetracycline (TE-truh-SIE-klin) blocks protein synthesis. Antibiotics are made naturally by some fungi and bacteria, and they kill neighboring bacteria or fungi that compete for resources. Scientists have worked to improve the effectiveness of these and other chemicals. Antibiotics that can kill more than one kind of organism are called *broad-spectrum antibiotics*.

A big worry for modern medicine is **antibiotic resistance**, the evolution of populations of pathogenic bacteria that antibiotics are unable to kill. Antibiotic resistance may develop in a population of bacteria in several ways. In one case, mutations in bacterial DNA give a bacterium resistance to antibiotics. A cell with such a mutation has a selective advantage when antibiotics are present. Mutant bacteria multiply and take over the population and thus stop the antibiotic's curing power. Figure 23-12 shows how resistance can evolve in a population of bacteria.

Unfortunately, antibiotics have been misused by doctors who overprescribe them and patients who do not finish their course of prescribed antibiotics. Many resistance genes are now present on plasmids called *R-plasmids*, which can pass easily between many bacteria by transformation. R-plasmids can carry resistance genes to many antibiotics. In this way, they give the bacterium *multiple resistances* to antibiotics. Many diseases that were once easy to treat with antibiotics, such as TB and staphylococcal infections, are now becoming more difficult to treat because of multiple resistances.

FIGURE 23-12

- 1 Individual bacteria in a large population may have mutated genes that give the bacteria resistance to an antibiotic.
- 2 If the antibiotic is absent, these resistant cells usually grow very slowly.
- 3 The nonmutated cells eventually take over the population.
- 4 If the antibiotic is present, however, it kills off the normal cells and leaves the mutant, antibiotic-resistant cells.
- 5 The mutant cells continue to grow and to take over the population. Antibiotics provide a selective advantage to antibiotic-resistant bacteria in this way.



Emerging Infectious Diseases Caused by Bacteria

Physicians and scientists have become worried by emerging infectious diseases. Diseases such as TB, which is caused by species of the genus *Mycobacteria*, and staphylococcal post-surgical infections, which are caused by *Staphylococcus aureus*, are becoming more difficult to treat because of the increased numbers of antibiotic-resistant bacteria. Other diseases, such as Lyme disease, are becoming more common than they were in the past. Lyme disease is caused by the bacterium *Borrelia burgdorferi*. If a tick infected with the bacterium bites a person, the bacterium can enter the person's bloodstream and cause flulike symptoms, arthritis, and sometimes heart inflammation.

Most emerging diseases develop when infectious agents, such as bacteria, pass from wild animals to humans. A disease that can pass from animals to humans is called a **zoonosis** (ZO-uh-NOH-sis). One reason for the emergence of Lyme disease is the flow of human populations into previously untouched natural areas. As people move into these areas, the chance of contact with infected ticks increases. The increase in global travel has also added to the spread of new and difficult-to-treat infectious diseases around the world.



Quick Lab

Modeling the Spread of Disease

Materials disposable gloves, lab apron, and safety goggles; clear plastic cup with liquid

Procedure



1. Put on your disposable gloves, lab apron, and safety goggles.
2. Obtain a cup of liquid from your teacher.
3. Pour the contents of your cup into a classmate's cup. Have the classmate pour the liquid back into your cup. Repeat this step with two other classmates. Do not touch anyone else's hands.
4. Predict the number of classmates that have been "exposed" to the "disease." Your teacher will test the liquid in each student's cup to see who has caught the "disease." Count the number of "infected" students.

Analysis How did the actual results compare with your prediction? Was direct contact needed?

Food Hygiene and Bacteria

Foodborne illnesses can be a major public health hazard, especially if an outbreak happens in a place such as a restaurant, where food is served to large numbers of people. However, the improper preparation, handling, or storage of food at home may cause most cases of foodborne illness in the United States. Many illnesses, such as campylobacteriosis (caused by *Campylobacter jejuni*), are relatively mild. Usual symptoms are nausea, vomiting, and intestinal cramps that last less than a week. However, certain foodborne pathogens, such as the *E. coli* O157:H7 strain, can cause serious illness and complications such as kidney damage in children and older adults. *E. coli* O157:H7-related illness is usually due to eating undercooked, contaminated hamburger meat.

Many foodborne illnesses can be avoided by selecting, storing, cooking, and handling food properly. For example, choose only fresh, clean foods, and wash raw fruit and vegetables well in clean water before eating them. Wash hands, kitchen towels, and kitchen utensils often in hot, soapy water to avoid cross-contamination during food preparation. Also, avoid cross-contamination of raw and cooked foods by keeping these foods separated during preparation and storage. Always refrigerate raw foods (including eggs) and deli meats at 4°C (39°F). Cook foods, especially meats, thoroughly and heat them to recommended temperatures to kill possible pathogens. Also, refrigerate leftovers promptly, and then reheat them thoroughly before eating them. Wash your hands well with soap and water for 10 to 15 seconds. Always wash your hands after using the bathroom, changing a diaper, handling animals, working outdoors, and coughing or sneezing into your hands.

MILESTONES

IN

Treating Bacterial Diseases

Timeline

1677 Anton van Leeuwenhoek first observes living bacteria.



1861 Louis Pasteur invents sterilization.

1876 Joseph Lister uses sterile surgical techniques.

1876 Robert Koch proves *Bacillus anthracis* causes anthrax.



1910 Paul Ehrlich invents Salvarsan, a drug for syphilis.

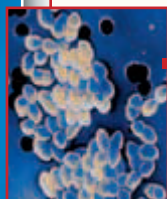
1928 Alexander Fleming discovers antibiotic effects of penicillium mold.

1939 Howard Florey and Ernst Chain isolate penicillin and create first antibiotic drug.



1950s Faster antibiotic development and successful prevention and treatment of disease occur.

1980 Molecular techniques aid the study of pathogens.



Present Development of antibiotic resistance spurs more research into antibiotic treatment.

Most pharmacies today carry many antibiotics for treating bacterial diseases, yet until World War II, no such medicines existed. Not until the 19th century was it discovered that bacteria could cause disease. Much of modern medicine rests on the pioneering work of the researchers discussed here.

In 1677, Dutch tradesman Anton van Leeuwenhoek first observed bacteria. His hand-held microscopes magnified objects much better than other microscopes of his day did. In the sticky plaque of his own teeth, he observed “many little living animalcules, very prettily a moving.”

In 1850, Hungarian physician Ignaz Semmelweis instructed his medical students to wash their hands between seeing patients. This first simple antiseptic procedure spared thousands of new mothers from deadly puerperal (pyoo-UHR-puhr-uhl), or child-bed, fever.

In 1861, French chemist Louis Pasteur proved that germs did not grow by spontaneous generation. He also showed that heating broth can kill bacteria and sterilize the broth. By sterilizing his hands, surgical instruments, and his patient’s incisions with carbolic acid, British surgeon Joseph Lister applied Pasteur’s ideas. This action greatly reduced the loss of life from infection.

In 1876, Robert Koch, a German physician, was the first to prove the link between a bacterium and a disease. He isolated *Bacillus anthracis* from a sick animal and established the bacterium’s role in the disease anthrax. Within 20 years, Koch and colleagues across Europe and the United States had discovered the causal agents for dozens of infectious diseases.

The growing knowledge of bacteria and disease in the late 1800s encouraged researchers to look for safer antibacterial drugs, because heavy metals such as mercury often harmed the patient.

In 1910, German physician Paul Ehrlich developed the compound Salvarsan, which successfully treated syphilis. In 1928, British microscopist Alexander Fleming accidentally discovered the antibacterial effect of penicillium mold, which was to become the source of penicillin, the first mass-produced antibiotic.

During the last half of the 20th century, many new antibiotics were created. More recently, molecular techniques have opened a new era of medical research for the treatment of bacterial diseases. Computer programs are now commonly used to design more-effective antibiotics and vaccines. This advancement is especially important because of the increasing numbers of bacteria that are not affected by traditional antibiotics, such as penicillin and streptomycin.

Review

1. Identify two conclusions drawn from Louis Pasteur’s 1861 experiment.
2. **Critical Thinking** Is hand washing still an important way to prevent the spread of disease? Explain.
3. **Critical Thinking** It was once believed that poisonous vapors called *miasmas* caused disease. Why do you think this belief existed?

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(a)



(b)

FIGURE 23-13

Bacteria are important in many industries. *Bacillus thuringiensis*, or Bt, is commonly sold as an organic pesticide to kill pests, such as the corn earworm caterpillar in (a). After an oil spill, cleanup crews spray fertilizer, which encourages the growth of naturally occurring prokaryotes to break down the oily sludge (b).

BACTERIA IN INDUSTRY

Bacteria are used to make and process many common foods and important chemicals. By fermenting the lactose in milk, bacteria make sour-milk products such as buttermilk, sour cream, and yogurt. Bacteria are also used to make cheeses such as mozzarella, ricotta, cheddar, and Roquefort. Fermented foods, such as sauerkraut, pickles, kimchi, coffee, and soy sauce, are products of bacterial fermentation. Bacteria are also grown in huge fermenters to make certain chemicals, such as acetone, acetic acid, enzymes, antibiotics, and insulin. They help break down organic waste in sewage and recycle carbon and nitrogen. Mining engineers also use bacteria in mining minerals and recovering petroleum.

Heat-loving prokaryotes are a major source of heat-resistant enzymes for biomedicine and industry. The heat-tolerant enzyme *Taq* polymerase allows biotechnologists to copy pieces of DNA rapidly, while other enzymes from thermophilic archaea, when added to laundry detergent, help dissolve stains.

Bacteria can cause diseases in crops and farm animals, but certain species are also useful for agriculture. *Bacillus thuringiensis*, or Bt, makes tiny crystals of protein that act as an endotoxin for certain plant pests, such as the corn earworm caterpillar shown in Figure 23-13a. After the insect feeds on a plant with Bt, the Bt crystals break down the insect's gut and kill the insect. Geneticists have inserted the gene that encodes the Bt toxic protein into the genome of some corn strains, so the corn now kills insects directly.

Biologists have also learned to harness bacteria to recycle compounds in a process called **bioremediation**, which uses bacteria to break down pollutants. After an oil spill, as shown in Figure 23-13b, bioremediation workers can spray contaminated beaches with nitrogen- and phosphorus-containing fertilizers. These nutrients, along with the oil along the shoreline, aid the growth of naturally occurring prokaryotes that can break down the petroleum and use it as an energy source.

SECTION 3 REVIEW

1. Describe two ways that bacteria can cause disease.
2. Describe the function of antibiotics in nature.
3. Explain how antibiotic resistance can evolve in bacteria.
4. Identify the main problem associated with antibiotic resistance.
5. List ways to avoid a foodborne illness.
6. Explain how bioremediation works to clean up environmental pollution.

CRITICAL THINKING

7. **Analyzing Concepts** How does the development of antibiotic resistance in a population of bacteria support the theory of evolution through natural selection?
8. **Applying Information** Why would a pickle processor carry out the preparation of pickles in anaerobic conditions?
9. **Drawing Conclusions** A student said that one should never allow bacteria near any foods. Explain whether this conclusion is appropriate.

CHAPTER HIGHLIGHTS

SECTION 1

Prokaryotes

- Prokaryotes are single-celled organisms that do not have a membrane surrounding their DNA. Prokaryotes occupy two domains of life: Archaea and Bacteria.
- Members of the domain Archaea contain certain genes more similar to eukarya, which include plants, animals, and fungi, than to the genes of bacteria.
- The domain Archaea includes methanogens, which produce methane gas and live in anaerobic environments; extreme halophiles, which live in very salty environments; and thermoacidophiles, which live in extremely acidic, hot environments.
- Gram staining distinguishes between Gram-positive and Gram-negative bacteria. Bacteria also occur in several variations of three shapes: rods, spheres, and spirals.
- The domain Bacteria is classified into five groups: Proteobacteria, Gram-positive bacteria, Cyanobacteria, Spirochetes, and Chlamydia.
- Nitrogen-fixing bacteria fix nitrogen gas from the air into a form that both prokaryotes and plants can use.

Vocabulary

prokaryote (p. 461)
peptidoglycan (p. 462)
methanogen (p. 462)
halophile (p. 462)

thermoacidophile (p. 462)
bacillus (p. 463)
coccus (p. 463)
spirillum (p. 463)

streptococcus (p. 463)
staphylococcus (p. 463)
Gram-negative bacterium (p. 463)

Gram-positive bacterium (p. 463)
antibiotic (p. 465)

SECTION 2

Biology of Prokaryotes

- The major structures of a prokaryotic cell include a cell wall, a cell membrane, cytoplasm, a capsule, pili, endospores, ribosomes, and sometimes structures for movement, called *flagella*.
- Endospores are thick-coated, resistant cells that some Gram-positive bacteria form when conditions become unfavorable. Endospores can survive high temperatures, strong chemicals, drying, and radiation and can survive for long periods of time.
- Prokaryotes obtain nutrients either from the nonliving environment or by utilizing the products or bodies of living organisms. Heterotrophs obtain carbon from other organisms. Autotrophs obtain their carbon from CO₂.
- Different prokaryotic species live in different environments. Temperature requirements range from 0°C to 110°C. Most prokaryotic species grow best at a neutral pH.
- Genetic recombination in bacteria can occur through transformation, conjugation, and transduction.

Vocabulary

plasmid (p. 468)
capsule (p. 468)
glycocalyx (p. 468)
pilus (p. 468)

endospore (p. 469)
heterotroph (p. 469)
autotroph (p. 469)
phototroph (p. 469)

chemotroph (p. 469)
obligate anaerobe (p. 470)
facultative anaerobe (p. 470)
obligate aerobe (p. 470)

transformation (p. 471)
conjugation (p. 471)
transduction (p. 471)

SECTION 3

Bacteria and Humans

- Human diseases may result from toxins produced by bacteria or from the destruction of body tissues.
- A mutation in the DNA of a single bacterium can confer resistance to an antibiotic. Cells with the mutant gene have a selective advantage when the antibiotic is present. Mutant cells take over the population when the normal cells die.
- The number of certain bacterial diseases has increased because of the increase in the number of antibiotic-resistant bacteria, the movement of people into previously untouched areas, and global travel.
- Foodborne illnesses can be avoided by selecting, storing, cooking, and handling food properly. Frequent hand washing in hot, soapy water is also very important.
- Many species of bacteria are used to produce and process different foods, to produce industrial chemicals, to mine for minerals, to produce insecticides, and to clean up chemical and oil spills.

Vocabulary

pathology (p. 472)
exotoxin (p. 472)

endotoxin (p. 472)
antibiotic resistance (p. 473)

zoonosis (p. 474)
bioremediation (p. 476)

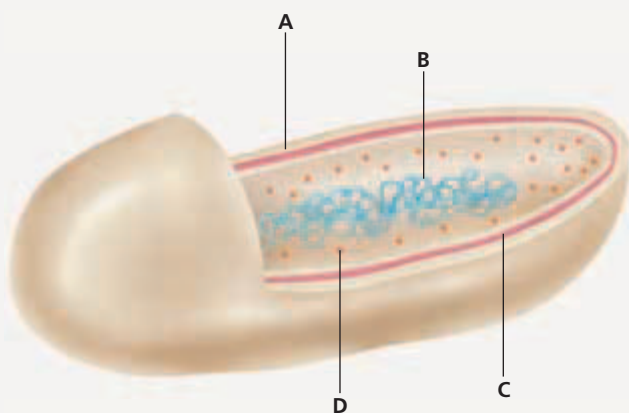
CHAPTER REVIEW

USING KEY TERMS

- For each pair of terms, explain how the meanings of the terms differ.
 - halophile* and *thermoacidophile*
 - phototroph* and *chemotroph*
 - transduction* and *conjugation*
- Explain the relationship between antibiotics and antibiotic resistance.
- Word Roots and Origins** The term *autotroph* comes from the Greek *autos*, meaning “self,” and *trophikos*, meaning “food.” Using this information, explain why the term *autotroph* is a good name for cyanobacteria.

UNDERSTANDING CONCEPTS

- Identify** the method that scientists have used to investigate how eukaryotes, bacteria, and archaea are related.
- List** one distinguishing characteristic of each of the three main groups of archaea.
- List** three common methods that are used to identify bacteria.
- State** reasons that cyanobacteria are considered the most self-sufficient organisms in nature.
- Identify** a nitrogen-fixing bacterium commonly found in the root nodules of legumes.
- Name** one bacterium that produces an antibiotic that is used in medicine.
- Label** the parts of the bacterium below.

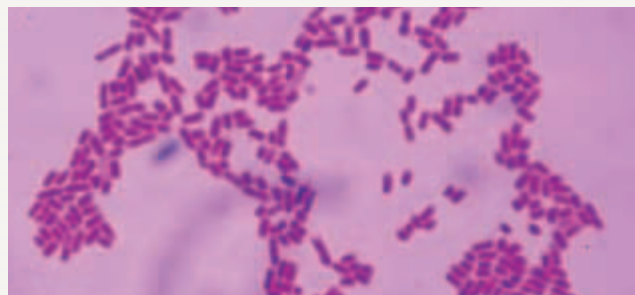


- Summarize** how chemoautotrophs get carbon and energy from their environment.
- Name** the term used to describe bacteria that can live in the presence or absence of oxygen.
- Describe** one way that bacteria can exchange genetic information.

- List** four diseases caused by bacteria and the organs that the diseases affect.
- Summarize** how antibiotic resistance can develop in a bacterial population.
- Relate** three things that you can do to prevent foodborne illness in your home.
- Identify** the metabolic process that bacteria use to make food products such as pickles, coffee, and sauerkraut.
- CONCEPT MAPPING** Create a concept map that identifies the three main cell shapes of bacteria and describes the nutritional modes of bacteria. Include the following terms in your concept map: *prokaryotes*, *bacilli*, *cocci*, *spirilla*, *chemotrophic*, *heterotrophic*, *endotoxin*, *exotoxin*, *Gram-negative*, and *Gram-positive*.

CRITICAL THINKING

- Applying Information** Scientists have only recently discovered fossilized prokaryotes. Explain why these fossils may have taken so long to discover.
- Predicting Results** *Clostridium perfringens*, the soil-dwelling bacterium that causes gas gangrene, is an obligate anaerobe. Using this information, predict which would be more likely to become infected with *C. perfringens*: a deep puncture wound or a surface cut. Explain the reason for your inference.
- Recognizing Relationships** Penicillin works by interfering with the ability of bacteria to polymerize the peptidoglycan cell wall. Given this fact, explain why Gram-positive bacteria are more susceptible to the effects of penicillin than Gram-negative bacteria are.
- Analyzing Graphics** Examine the photograph below, which shows bacteria that have been treated with a Gram stain. Would you hypothesize that these bacteria produce endotoxins? Explain your answer.





Standardized Test Preparation

DIRECTIONS: Choose the letter of the answer choice that best answers the question.

- How do bacteria produce yogurt from milk?
A. by conjugation
B. by fermentation
C. by nitrogen fixation
D. by aerobic respiration
- What are rod-shaped bacteria called?
F. cocci
G. bacilli
H. spirilla
J. halophiles
- What are thermoacidophiles?
A. bacteria
B. archaea
C. spirochetes
D. cyanobacteria
- Genetic recombination in bacteria can occur during which process?
F. conjugation
G. binary fission
H. heterocyst formation
J. endospore production

INTERPRETING GRAPHICS: The table below lists the response of a bacterium to several antibiotics. A score of 0 means not sensitive, and sensitivity increases as the score increases. Use the table to answer the questions that follow.

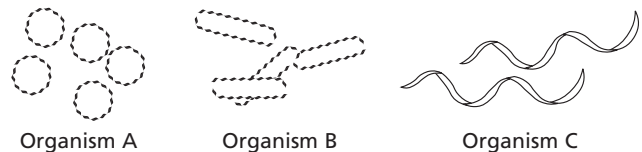
Antibiotic	Sensitivity
Ampicillin	3
Bacitracin	0
Cephalosporin	0
Penicillin	0
Rifampin	0
Streptomycin	3
Tetracycline	2

- Which of the following antibiotics killed the bacteria most effectively?
A. penicillin
B. bacitracin
C. tetracycline
D. streptomycin
- Which of the following antibiotics had no effect on the bacterium?
F. ampicillin
G. tetracycline
H. streptomycin
J. cephalosporin

DIRECTIONS: Complete the following analogy.

- obligate aerobe : oxygen :: obligate anaerobe :
A. low pH
B. absence of oxygen
C. presence of methane
D. high salt concentration

INTERPRETING GRAPHICS: The image below shows different shapes in bacteria. Use the image to answer the questions that follow.



- What shape is represented by organism C?
F. coccus
G. bacillus
H. spirillum
J. filamentous
- You would expect the members of the bacterial genus *Bacillus* to be what type of shape?
A. rod shaped
B. spiral shaped
C. square shaped
D. sphere shaped

SHORT RESPONSE

The Gram stain is used to distinguish bacteria based on a certain physiological difference between the bacteria.

Describe the color of Gram-positive and Gram-negative bacteria when they are treated with the Gram stain.

EXTENDED RESPONSE

Species of the symbiotic bacterial genus *Rhizobium* live within the root nodules of legumes, such as soybeans and peanuts. Some farmers take advantage of the nitrogen-fixing abilities of these species when they rotate their crops every few years and grow legumes instead of their normal food crops.

Part A Define *nitrogen fixation*.

Part B Explain the benefits of crop rotation on the soil.

Test TIP

When asked to complete an analogy, focus on the completed pair of terms provided. Identify the relationship that exists between the two terms; then, find the answer choice that has the same relationship with the third term provided.

Culturing Bacteria

OBJECTIVES

- Test common surfaces for the presence of bacteria.
- Learn a simple procedure for culturing bacteria.

PROCESS SKILLS

- observing
- hypothesizing
- experimenting
- collecting data
- analyzing data

MATERIALS

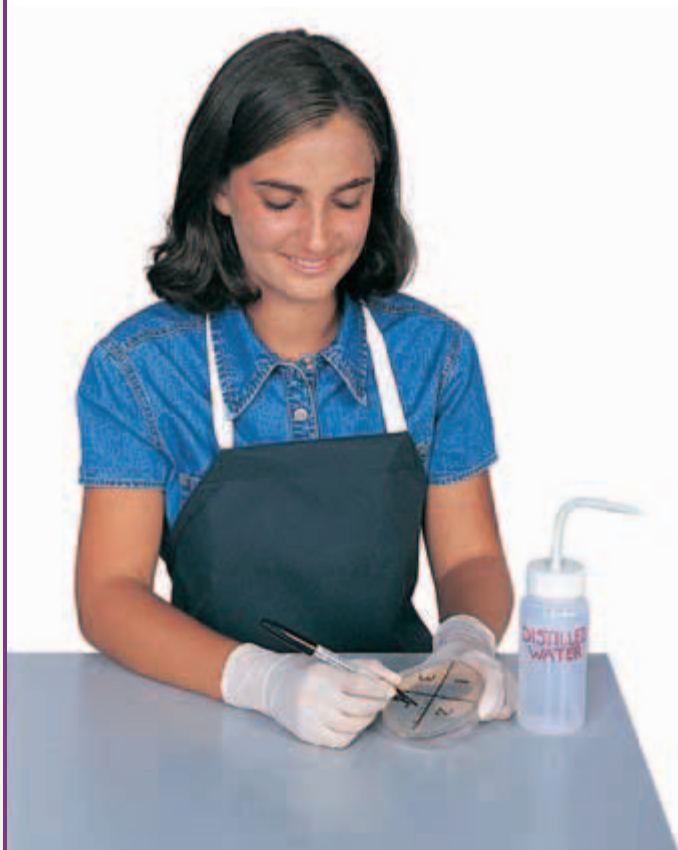
- Petri dish with nutrient agar
- glass-marking pen
- 4 sterile cotton swabs
- distilled water
- tape
- protective gloves

Background

1. Culturing bacteria involves growing microorganisms in a nutrient medium that is favorable for growth.
2. What are favorable conditions for the growth of most bacteria?
3. List some places in your school where bacteria are likely growing.

PART A Setting Up the Experiment

1. Make a hypothesis about places in your school that would be likely to contain bacteria or that have conditions under which bacteria grow best. Make a list of three places where you will test for the presence of bacteria. Do not choose places outside your laboratory without your teacher's permission.
2.  **CAUTION** Put on a lab apron and disposable gloves. Obtain a Petri dish with nutrient agar from your teacher. Divide your Petri dish into four equal quadrants by writing on the outer bottom surface of the dish with the marking pen. Label the quadrants 1, 2, 3, and 4, as shown in the photograph at left. Also label the bottom of the dish with your group name or number. Do not mark on the top of the Petri dish because it will rotate with respect to the agar.
3. Use quadrant 1 as a control. Take a sterile cotton swab, and moisten it with distilled water. Be careful not to touch the cotton swab to any other surfaces, including your fingers. Remove the cover to the Petri dish, and rub the cotton swab across the nutrient agar in quadrant 1. Be careful not to tear the surface of the agar. Replace the cover immediately.
4. Take another sterile cotton swab, and moisten it with distilled water. Swipe the moistened swab across a surface that you have decided to test for the presence of bacteria. Be careful not to touch the cotton swab to any other surface. Touching the swab to other surfaces could contaminate the swab with bacteria other than the bacteria from your selected surface.



5. Remove the cover to the Petri dish, and rub the cotton swab across the nutrient agar in quadrant 2. This process transfers bacteria from the surface you have sampled to the nutrient agar in the Petri dish. Replace the cover immediately.
6. Using a clean cotton swab each time, repeat steps 4–5 for quadrants 3 and 4. Be sure to note in your lab report the areas that you swipe for each quadrant.
7. After you have swiped a sample in each quadrant of the Petri dish, seal the Petri dish with tape.
CAUTION Do not open the Petri dish again. Treat the contents of the Petri dish as you would any other pathogens.
8. Place the Petri dish in a warm place with the cover side down for 24 hours or in an incubator at 37°C overnight.
9.  Dispose of the cotton swabs and gloves according to your teacher's directions.
10.   Clean up your lab materials and wash your hands before leaving the lab.

PART B Collecting Data

11. In your lab report, create a data table similar to the model shown below. Allow plenty of space to record your observations for each quadrant swabbed.
12. Check the Petri dish daily for bacterial growth until you no longer find new bacterial colonies (about seven days). Bacterial colonies tend to look shiny and wet, whereas fungal colonies tend to look dry and fuzzy. Record your observations in your data table. What is enabling you to see bacteria? What does each colony represent?

13.  Discard your Petri dish as directed by your teacher.

Analysis and Conclusions

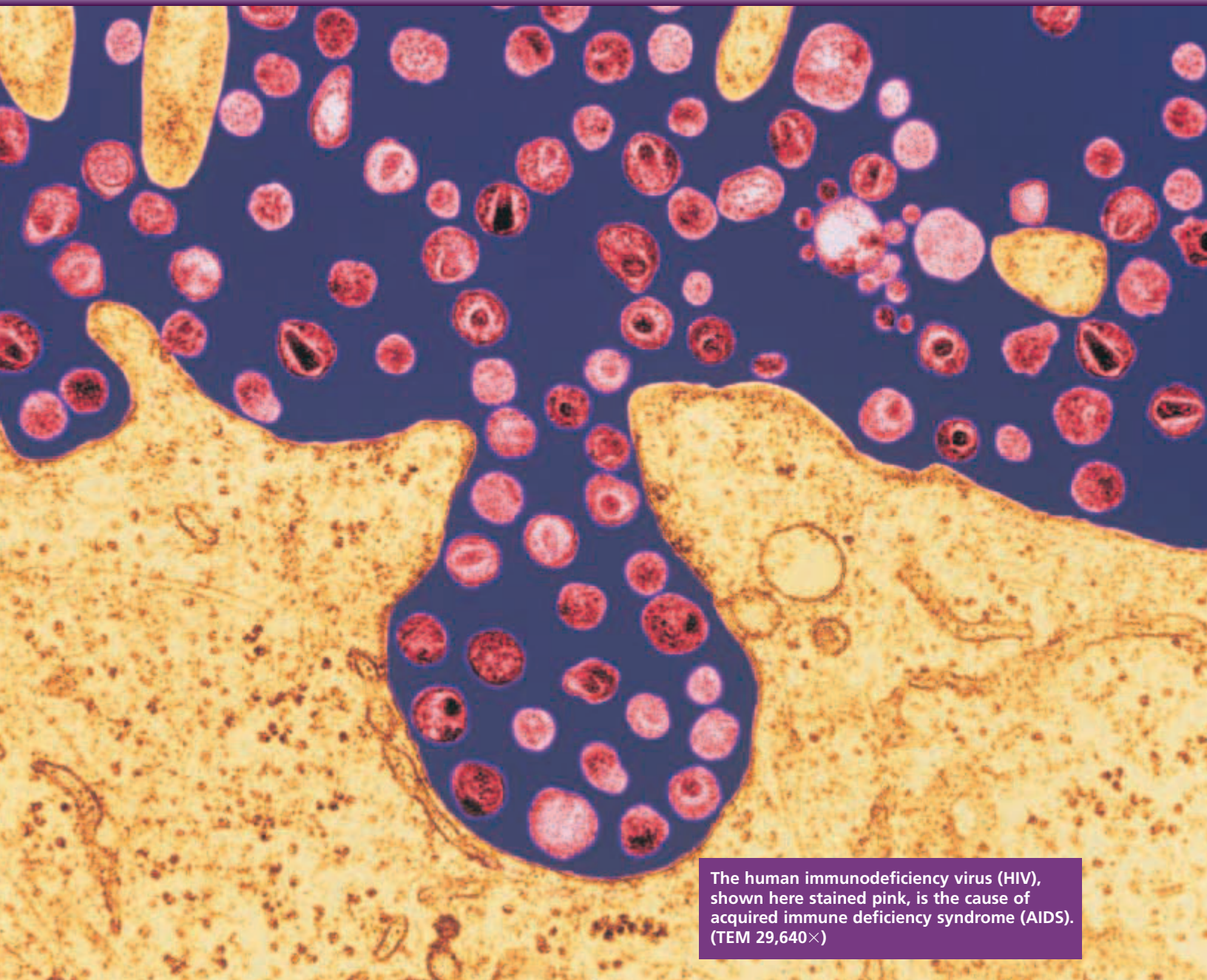
1. On which surfaces did you find the most bacteria? the fewest bacteria? Did your results conform to your expectations? Explain.
2. Compare the colonies of bacteria that grew in each quadrant. What can you tell about the bacteria from the kinds of colonies the bacteria produced?
3. What are some possible sources of error in the procedure you followed?
4. Combine the data obtained by the entire class. Which surfaces yielded the most bacteria?
5. Which of the surfaces that you sampled would you prefer to use as a food-preparation area? Explain your choice.
6. Would the amount of surface area that you sampled with your swab affect the number of colonies that grew on your dish? Explain your answer.
7. Can you tell by looking at the colonies on the dish if they would cause disease? Why or why not?
8. What test described in this chapter could you use to partially identify your bacteria? What characteristics of the bacteria could you learn from this test?

Further Inquiry

Design an experiment to test the effects of one variable, such as temperature or the presence of antibacterial soap, on the growth of bacteria. What would you use as a control in your experiment?

PRESENCE OF BACTERIA ON COMMON SURFACES

Surface swabbed	Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7



The human immunodeficiency virus (HIV), shown here stained pink, is the cause of acquired immune deficiency syndrome (AIDS). (TEM 29,640 $\times$)

SECTION 1 *Viral Structure and Replication*

SECTION 2 *Viral Diseases*

VIRAL STRUCTURE AND REPLICATION

In 2003, some people in China started showing symptoms of a new illness. These symptoms were similar to those of pneumonia. The condition was highly infectious. Soon, scientists found that the disease was caused by a virus. They called the disease severe acute respiratory syndrome, or SARS.

DISCOVERY OF VIRUSES

A **virus** is a nonliving particle made up of nucleic acid and a protein coat or nucleic acid and a lipid-protein (lipoprotein) coat. Even though viruses are not living organisms, they are of interest to biologists because they cause many diseases in living organisms and they are useful tools for genetic research.

Scientists began studying viruses in the late 1800s after they found that a factor smaller than bacteria could cause disease. At that time, scientists did not have the technology to see viruses. But they wanted to know if viruses were very small cells or simply non-living groups of molecules.

In 1935, Wendell Stanley crystallized the tobacco mosaic virus (TMV). TMV is a virus that infects plants, such as tobacco and tomato plants. The disease causes plants to wither and develop mosaic-like spots on their leaves, as shown in Figure 24-1. Scientists concluded that an infective agent that could be crystallized was unlikely to be made up of cells.

CHARACTERISTICS OF VIRUSES

Viruses are not alive because they lack some of the key characteristics of living organisms. For example, viruses do not have cytoplasm or organelles. They cannot carry out cellular functions such as metabolism and homeostasis. They do not grow as cells do by dividing in two. Even though viruses do have genetic material, or a genome—either DNA or RNA—they cannot reproduce outside their host cell. They must enter a living cell and use the host cell's ribosomes, ATP, enzymes, and other molecules to reproduce.

OBJECTIVES

- **Summarize** the discovery of viruses.
- **Describe** why viruses are not considered living organisms.
- **Describe** the basic structure of viruses.
- **Compare** the lytic and lysogenic cycles of virus replication.
- **Summarize** the origin of viruses.

VOCABULARY

virus
capsid
envelope
provirus
retrovirus
reverse transcriptase
bacteriophage
lytic cycle
virulent
lysis
lysogenic cycle
temperate virus
prophage

FIGURE 24-1

Tobacco mosaic virus (TMV) was the first virus that was crystallized. When TMV infects a plant, it causes small mosaic-like (patchy) blotches on the leaves, as shown in this infected tobacco plant.



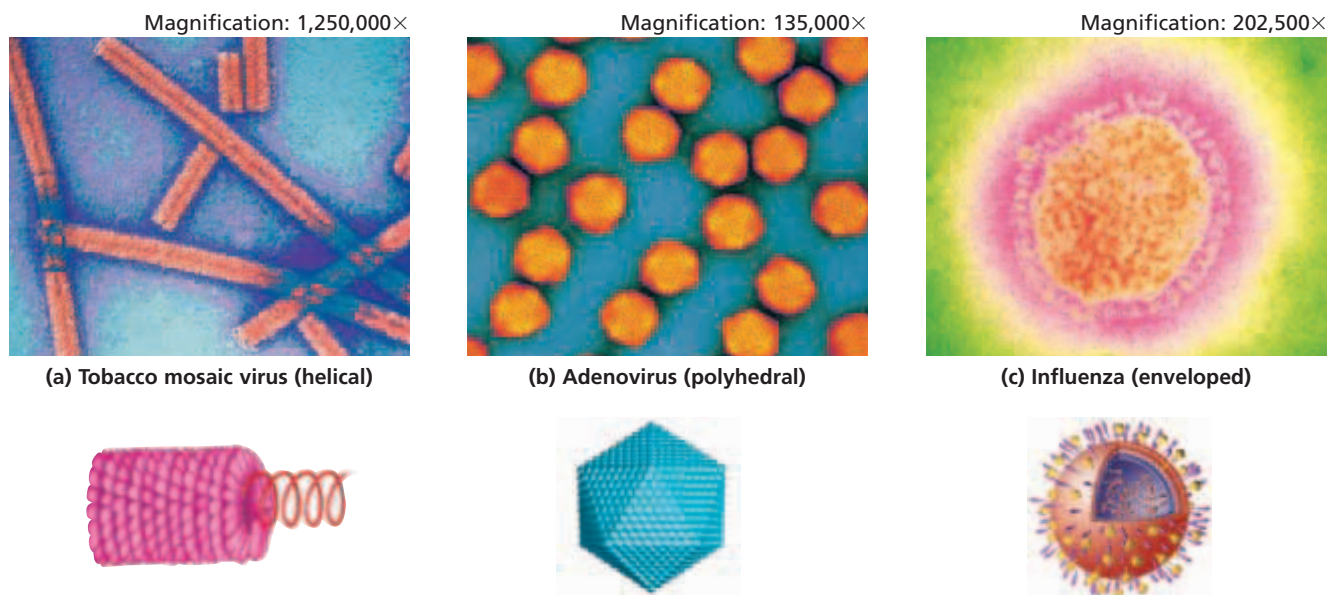


FIGURE 24-2

Viruses have a variety of sizes and shapes. (a) The tobacco mosaic virus is about 18 nm in diameter and has a helical shape. (b) The adenoviruses are about 80–110 nm in diameter and have the shape of an icosahedron. (c) The spherical influenza viruses are between 50–120 nm in diameter.

Viral Size and Structure

Viruses are some of the smallest particles that are able to cause disease. But they vary in size and shape, as shown in Figure 24-2. The shape of a virus is the result of its genome and the protein coat that covers the genome. A protein coat, or **capsid** (KAP-sid), is the only layer surrounding some viruses. The capsid of some viruses, such as TMV, forms a *helix*, shown in Figure 24-2a. The rabies and measles viruses are also helical viruses. As shown in Figure 24-2b, the adenovirus capsid has the shape of an *icosahedron* (IE-koh-suh-HEE-druhn), a shape with 20 triangular faces and 12 corners. Other viruses with this shape include those that cause herpes simplex, chickenpox, and polio. The influenza virus, shown in Figure 24-2c, is spherical in shape.

Some viruses have a bilipid membrane called an **envelope** that surrounds the capsid. The envelope is formed from either the nuclear membrane or the cell membrane of the host cell as the viral capsid buds from the host cell. Proteins in the envelope, such as those of the influenza virus shown in Figure 24-2c, help new viruses recognize host cells. Enveloped viruses include the chickenpox virus (varicella virus) and human immunodeficiency virus (HIV), which causes acquired immunodeficiency syndrome (AIDS).

Classification of Viruses

Viruses can be classified by whether they have RNA or DNA as their genome and whether their genome is single stranded or double stranded and linear or circular. Viruses are also classified based on the nature of their capsid and on the presence or absence of an envelope. Table 24-1 describes some viruses that affect human health. For example, the virus that causes severe acute respiratory syndrome (SARS) is a *coronavirus*. *Corona* is the Latin word for “crown.” The SARS virus has single-stranded, linear RNA and an envelope with lollipop-shaped proteins that make the envelope look like a crown.



Quick Lab

Calculating Nanometers

Materials meterstick with millimeter marks, paper, scissors, tape, pencil

Procedure Cut the paper into strips. Tape the strips together to form one strip that is 2 m long, and label 1 m, 20 cm, 2 cm, and 2 mm.

Analysis

1. Write an equation at the 1 m mark and at the end of the strip that shows the relationship between the length of the paper in meters and nanometers.
2. Write equations beside the 2 cm and the 20 cm marks to show the relationship of centimeters to nanometers.
3. Write an equation by the 2 mm mark to show its relationship to nanometers.

TABLE 24-1 Some Important Viruses That Infect Humans

Viral group	Genetic material	Envelope	Shape and structure	Examples of diseases
Papovaviruses	DNA, circular, ds	no	icosahedral	warts, cervical cancer
Adenoviruses	DNA, linear, ds	no	icosahedral	respiratory infections
Herpesviruses	DNA, linear, ds	yes	icosahedral	cold sores, genital sores
Poxviruses	DNA, linear, ds	yes	brick-shaped, enveloped	smallpox, cowpox
Parvoviruses	DNA, linear, ss	no	icosahedral	roseola, fifth disease
Picornaviruses	RNA, linear, ss	no	icosahedral	polio, hepatitis, colds
Orthomyxoviruses	RNA, linear, ss	yes	oval or filamentous	influenza A, B, and C
Rhabdoviruses	RNA, linear, ss	yes	helical	rabies
Retroviruses	RNA, linear, ss	yes	spherical	AIDS, leukemia
Coronaviruses	RNA, linear, ss	yes	helical, surrounded by lollipop-shaped proteins	upper respiratory infections, SARS

VIRAL REPLICATION

Outside the host cell, a virus is a lifeless particle with no control over its movements. It is spread by air, in water, in food, or by body fluids. Viruses infect both prokaryotes and eukaryotes.

Viruses first need to recognize a host cell before they can infect it. An enveloped virus can do so by a “lock-and-key” fit between certain envelope proteins and specific receptor molecules on the host cell. A viral infection begins when a virus enters the host cell. The viral genome takes over the metabolic machinery of the cell and makes new viruses. Viruses are *obligate intracellular parasites*—they replicate only by using host cell enzymes and organelles to make more viruses. DNA and RNA viruses differ in the way they replicate.

Replication in DNA Viruses

When the DNA of some DNA viruses enters a host cell, it makes mRNA, which is the template for making proteins during protein synthesis. The DNA of other DNA viruses inserts into the host cell’s chromosome. This inserted viral DNA is known as a **provirus**. The host cell’s enzymes then transcribe the provirus into mRNA and translate this RNA into viral proteins. DNA viruses also use the host cell’s enzymes to make new viral DNA. The replicated viral DNA and proteins assemble to make new viral particles.

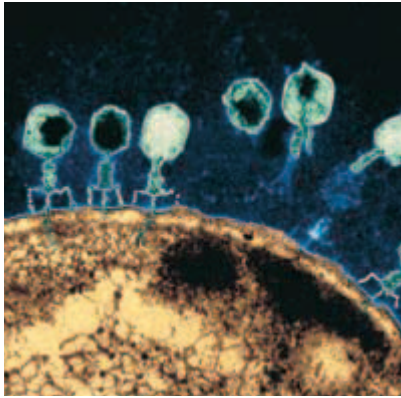
Replication in RNA Viruses

The genome of some RNA viruses enters host cells and serves directly as mRNA, which is translated into new viral proteins immediately after infection. The genome of other RNA viruses is first transcribed and thus serves as both a template for the synthesis of mRNA and as a template for the synthesis of more copies of the viral genome.

Word Roots and Origins

retrovirus

from the first two letters of the two words *reverse transcriptase* combined with the word *virus*



Some RNA viruses, called **retroviruses** (RE-troh-VIE-ruhs-uhz), contain the enzyme *reverse transcriptase* (tran-SKRIP-tays) in addition to RNA. **Reverse transcriptase** uses RNA as a template to make DNA, which then inserts into the host cell's genome. Reverse transcriptase reverses the normal process of transcription, in which DNA serves as a template for producing RNA. The host cell's enzymes transcribe the virus DNA, and cell ribosomes translate the RNA into proteins that become part of the new viruses. Human immunodeficiency virus (HIV) is a retrovirus.

Replication in Viruses That Infect Prokaryotes

Scientists have gained a better understanding of virus replication by studying **bacteriophages** (bak-TIR-ee-uh-FAYJ-uhz), viruses that infect bacteria. Bacteriophages, or phages, have complex capsids, shown in Figure 24-3. Phage capsids are made up of a hexagonal head filled with DNA. Attached to the head is a protein tail with one or more tail fibers. The tail fibers attach the virus to a cell. The tail helps the virus inject its genome into the host cell. The most commonly studied bacteriophages, T phages, infect *Escherichia coli*, a bacterium found in the digestive tract of many animals and humans. The TEM in Figure 24-3 shows an *E. coli* cell infected with many T phages. Research led to the discovery that many phages and other viruses can reproduce by one or both of two different processes: the lytic cycle or the lysogenic cycle.

Lytic Cycle

During the **lytic cycle**, a virus, such as a T4 phage, invades a host cell, produces new viruses, and ruptures (lyses) the host cell when releasing newly formed viruses. Viruses that reproduce only by the lytic cycle are called **virulent**. T phages are virulent viruses. Virulent viruses destroy the cells that they infect.

During the lytic cycle, a phage first attaches its tail fibers to specific receptor molecules on the cell surface of a susceptible bacterium, as shown in step ❶ of Figure 24-4. Recall that viruses cannot efficiently infect cells that do not have the specific protein receptors for that virus. The phage then injects its DNA into the cell but leaves its protein-containing head and tail outside the host cell. In step ❷, the ends of the viral DNA attach to each other, forming a circle. The viral DNA remains separate from the host cell's DNA. In step ❸, virulent viruses continue the lytic cycle. The viral DNA takes control of the host's protein-synthesis pathway, and the viral genome is copied. Enzymes transcribe mRNA from the viral DNA. Host ribosomes translate the mRNA into viral proteins, and enzymes replicate the viral DNA. In step ❹, head proteins bind to the newly made phage genomes. Heads containing DNA bind to tails, and tails assemble into tail fibers. Finally, the phage enzyme called *lysozyme* digests the cell wall, and up to 200 new phage particles burst from the bacterial cell in a process called **lysis** (LIE-sis), shown in step ❺.

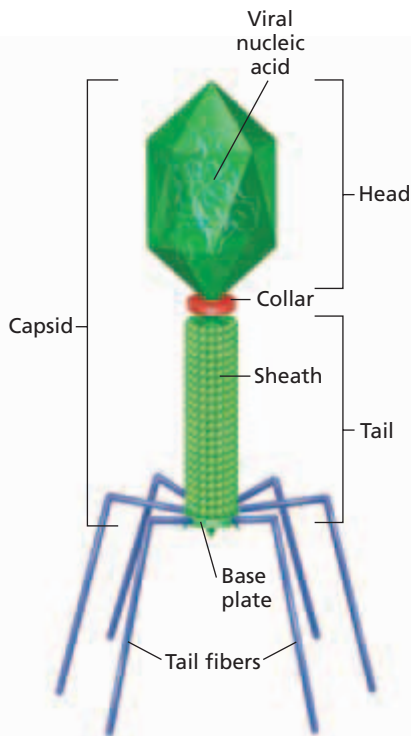


FIGURE 24-3

This TEM and diagram show the structural complexity of a T phage. Scientists named the seven T phages—T1, T2, T3, T4, T5, T6, and T7—to match the order in which they were discovered. (TEM 138,600×)

Word Roots and Origins

lysis

from the Greek *lysis*,
meaning "loosening" or "dissolving"

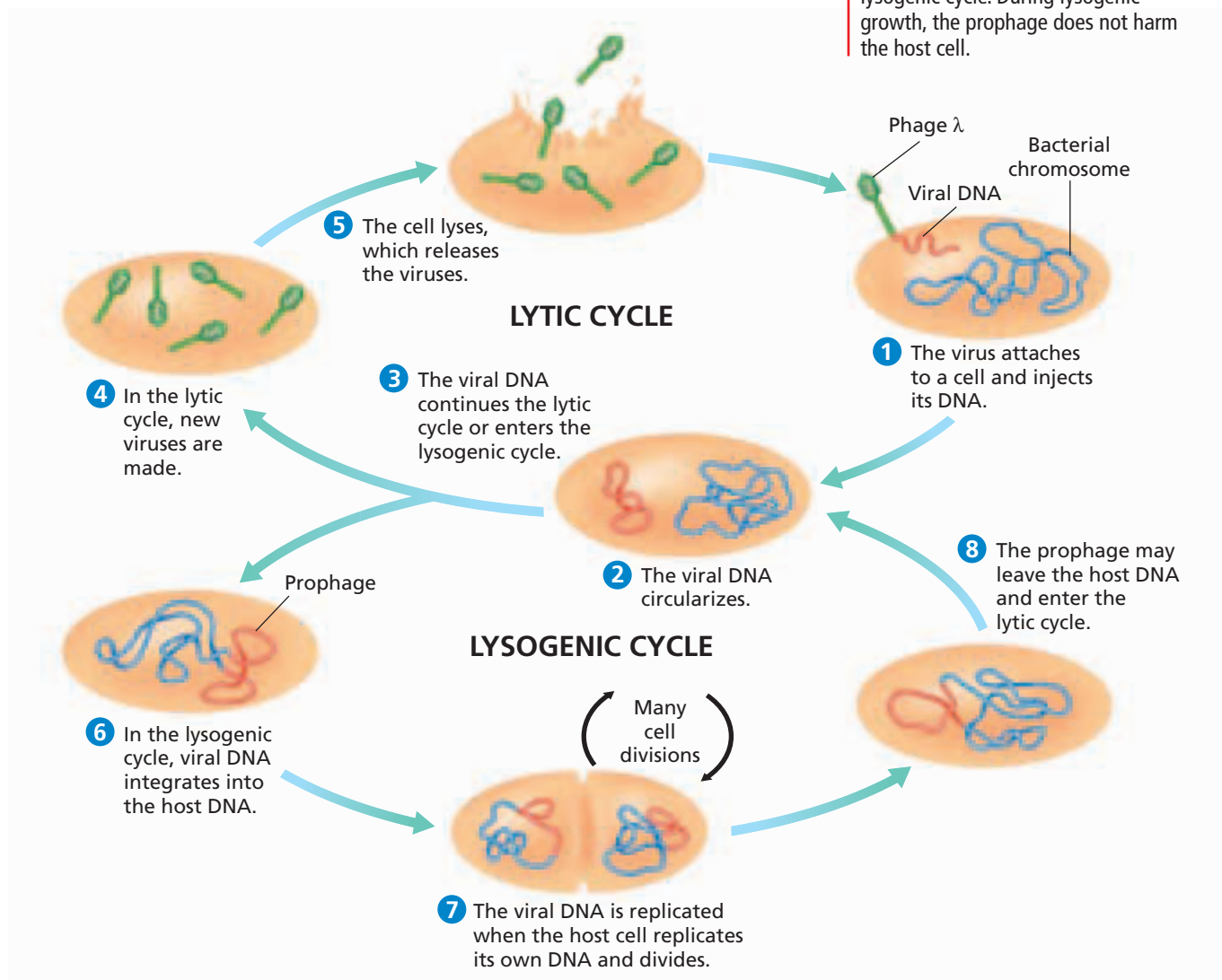
Lysogenic Cycle

A lytic cycle directly bursts an infected cell, but an infection cycle called a **lysogenic** (LIE-soh-JEN-ik) **cycle** allows viruses to hide in their host cell for days, months, or years. A virus whose replication includes the lysogenic cycle is called a **temperate virus**.

As shown in step ① of Figure 24-4, temperate phages, such as phage lambda (λ), enter bacteria in the same way that virulent phages do. In a lysogenic cycle, however, the phage DNA that enters the bacterial cell integrates into the host cell's chromosome, shown in step ⑥. Phage DNA that is integrated into a specific site of the host cell's chromosome is called a **prophage** (PROH-fayj). In step ⑦, the prophage is replicated when the host bacterium replicates its own DNA. Each daughter cell is therefore infected with a prophage. In this way, a single infected cell can give rise to a large population of infected cells. In step ⑧, the prophage can exit the bacterial chromosome and enter the lytic cycle. Radiation or certain chemicals can cause the phage DNA to leave the bacterial chromosome. Phage particles are replicated and assembled and are released as the host cell lyses.

FIGURE 24-4

After entering the host cell, the DNA of a temperate virus can immediately start the production of new viruses in the lytic cycle. Alternatively, it can insert itself into the bacterial DNA in the lysogenic cycle. During lysogenic growth, the prophage does not harm the host cell.



Viruses: Tools for Biotechnology

The virus life cycle makes viruses important research tools. A researcher can replace large pieces of the DNA of a phage with DNA of particular interest, such as the human gene for cystic fibrosis. The researcher can then insert this recombinant DNA into empty phage heads and allow these viruses to infect bacteria. As the virus grows over and over through the lytic cycle, the number of copies of the foreign gene increases until millions of copies are available for study. Bacteriophages have also been invaluable tools for medical research.

THE ORIGIN OF VIRUSES

Because viruses need host cells for replication, most scientists reason that viruses evolved from early cells. One hypothesis is that the first viruses were probably naked pieces of nucleic acid that could travel from one cell to another. The viruses entered cells through damaged cell membranes. Over time, genes evolved that coded for protective protein coats as well as special proteins that bind to target cells, allowing viruses to invade healthy cells.

An example of evolution by natural selection can be seen in the influenza virus. The immune system usually destroys most influenza viruses, but a few viruses may have genetic mutations that change the proteins on their surfaces. These genetic changes make the viral particles unrecognizable to the immune system, and the mutated viruses then take over the viral population.

Viruses that mutate quickly, such as influenza virus and HIV, make it difficult for the immune system to recognize and destroy them. Rapid mutation also makes it difficult to develop vaccines that prevent these viral infections over long periods of time. Therefore, each year, makers of influenza vaccine try to produce a specific “flu shot” that targets the strain of influenza virus that is most likely to infect the greatest number of people during that influenza season.

SECTION 1 REVIEW

1. Describe how the tobacco mosaic virus was discovered.
2. Why are viruses not considered living organisms?
3. Describe the structure of a bacteriophage.
4. Compare the lytic and lysogenic cycles.
5. Discuss how the earliest viruses may have originated.

CRITICAL THINKING

6. **Applying Information** Viruses have genetic material, but they are not alive. Explain.
7. **Drawing Conclusions** The assembly of new viral particles can sometimes take place in the host cell’s nucleus. However, such assembly does not occur with phage particles. Why not?
8. **Evaluating Information** Evaluate the following statement: Antibiotics will cure a cold.

VIRAL DISEASES

In recent years, several viral diseases, such as SARS, have appeared and spread quickly. Where do viral diseases such as these come from? How do these viral diseases spread, and how are they prevented and treated?

VECTORS OF VIRAL DISEASE

Because viruses are lifeless particles, their spread depends on other agents. A **vector** is an intermediate host that transfers a pathogen or a parasite to another organism. Vectors of viral diseases include humans, animals, mosquitoes, ticks, and fleas. The West Nile virus, a virus that causes fever and headache and, in very rare cases, coma and paralysis, infects mainly birds, such as crows and jays. If a mosquito bites a bird infected with West Nile virus and then bites a human, the virus can be spread. Mosquitoes can transmit several other viruses, such as the yellow fever virus.

HUMAN VIRAL DISEASES

Viruses cause many diseases in humans, such as flu, chickenpox, measles, polio, and viral hepatitis. Viral infections can affect various human organs, including the brain, liver, heart, lungs, and skin.

Chickenpox and Shingles

Chickenpox and shingles are caused by the same varicella-zoster herpesvirus. The virus multiplies in the lungs and travels to blood vessels in the skin. The symptoms of chickenpox include fever and skin rash. The virus is spread through direct contact with the skin rash and through the air. After recovery, a person has lifelong resistance to reinfection. The virus, however, can sometimes stay in nerve cells as a provirus. The virus can later cause a disease called *shingles*. The shingles rash, shown in Figure 24-5, can shed new chickenpox viruses and infect susceptible children and adults.

Viral Hepatitis

Hepatitis (HEP-uh-TIET-is), or inflammation of the liver, can be caused by at least five viruses. Hepatitis A and hepatitis E can be spread by fecally contaminated food and water. Hepatitis B, C, and D are spread by sexual contact, by contact with infected blood and serum, and by the use of contaminated needles. Symptoms of hepatitis can include fever, nausea, jaundice, and liver failure.

OBJECTIVES

- **Name** several vectors of viral diseases.
- **Identify** four viral diseases that result in serious human illnesses.
- **Discuss** the relationship between viruses and cancer.
- **Name** three examples of emerging viral diseases.
- **Compare** the effectiveness of vaccination, vector control, and drug therapy in fighting viruses.
- **Contrast** viroids, prions, and viruses.

VOCABULARY

vector
protease inhibitor
oncogene
proto-oncogene
emerging disease
inactivated virus
attenuated virus
viroid
prion

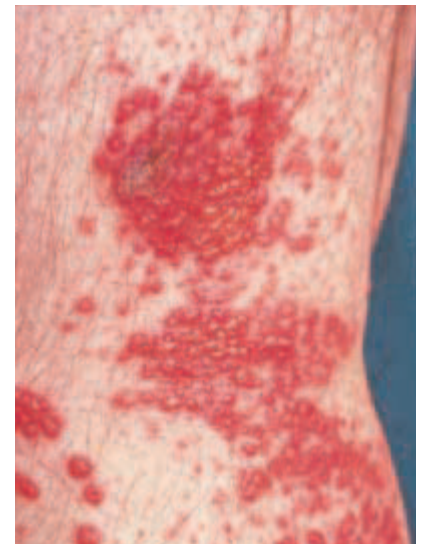


FIGURE 24-5

The painful shingles rash, caused by a herpes virus, is limited to an area of the skin innervated by a particular nerve branch, for example, on the side of the chest.

Acquired Immune Deficiency Syndrome (AIDS)

The human immunodeficiency virus (HIV) causes AIDS. HIV gradually destroys an infected person's immune system. HIV is spread by sexual contact; by contact with infected body fluids, such as blood, semen, or vaginal fluids; and from mother to fetus. Glycoproteins on the surface of HIV, shown in Figure 24-6, bind to specific receptor proteins found on *macrophages*, which are immune system cells.

Figure 24-7 summarizes how HIV infects macrophages. First, the virus attaches to the CD4 and CCR5 receptors on the cell surface. To enter the cell, the HIV glycoprotein must bind to both CD4 and the CCR5 coreceptor. Then, the viral envelope fuses with the cell membrane and releases the capsid into the host cell. The enclosed capsid enters the cell cytoplasm, where viral RNA and the enzyme reverse transcriptase are released. Reverse transcriptase uses the viral RNA as a template for making a double-stranded DNA version of the viral genome. The HIV DNA enters the cell's nucleus and inserts itself into the DNA of the host's chromosome, becoming a provirus. Cellular enzymes transcribe and then translate the HIV genes into viral proteins. Then, some of the HIV particles assemble. Finally, the cell membrane pinches off and forms a viral envelope as the newly made HIV particles separate from the cell. Viral replication of HIV results in many mutations. Eventually, the virus recognizes other coreceptors, such as those found on helper T cells. In helper T cells, the HIV particles are released by lysis.

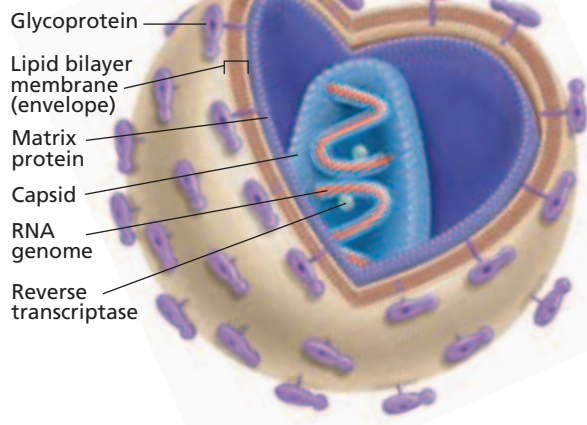


FIGURE 24-6

Human immunodeficiency virus (HIV) contains two identical RNA molecules and two molecules of reverse transcriptase, which makes double-stranded DNA from the RNA.

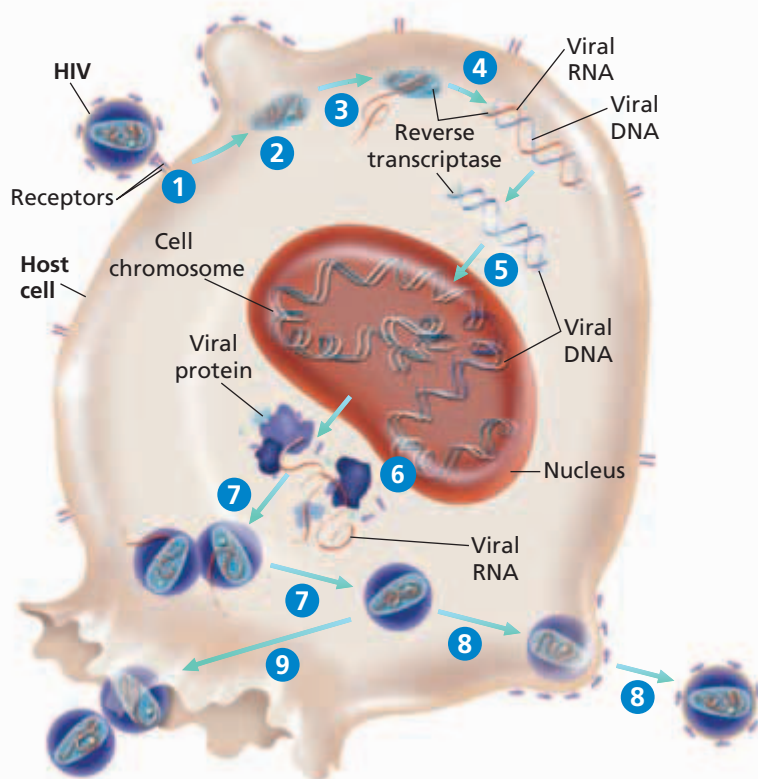


FIGURE 24-7

The human immunodeficiency virus (HIV) infects certain cells of the immune system. In step 1, HIV attaches to receptors on the cell surface. In step 2, the capsid is released into the host cell. In steps 3 and 4, the viral RNA is copied into double-stranded DNA. As shown in step 5, the viral DNA inserts into the host cell's DNA. In step 6, viral proteins are made. Then, in step 7, HIV particles assemble. In step 8, newly made HIV particles pinch off from the cell membrane. In some immune cells, HIV particles are released by lysis, shown in step 9.

Researchers working on a cure for AIDS are studying the HIV life cycle for weak points to target with new therapies. Because normal human cells do not make reverse transcriptase, this enzyme provided the first target for a chemical therapy. For example, the drug *azidothymidine* (uh-ZIE-doh-THIE-muh-DEEN) (AZT) inhibits the ability of reverse transcriptase to make a DNA copy of the viral RNA. Another class of drugs, called **protease inhibitors**, blocks the synthesis of new viral capsids. Combinations of protease inhibitors and AZT have been shown to be helpful in slowing the progression of HIV infection to full-blown AIDS.

Viruses and Cancer

Cancer results when cells divide at an uncontrolled rate and form a tumor that invades surrounding tissue. Some viruses contain viral **oncogenes**, genes that cause cancer by blocking the normal controls on cell reproduction. Other viruses cause cancer because the viral DNA inserts itself into a host's chromosome near a **proto-oncogene**, which usually controls cell growth. The proto-oncogene is converted to an oncogene. Human papillomavirus (HPV) can cause cervical cancer, and hepatitis B virus can cause liver cancer. Other viruses, such as human T-lymphotrophic virus (HTLV), can cause leukemia, and Epstein-Barr virus (EBV) can cause Burkitt's lymphoma, a malignant tumor of the jaw.

EMERGING VIRAL DISEASES

Emerging diseases are illnesses caused by new or reappearing infectious agents that typically exist in animal populations—often in isolated habitats—and can infect humans who interact with these animals. For example, some animals living in tropical forests of central Africa most likely harbor Ebola virus. Researchers think that as people clear these forests for agriculture or housing, the people become exposed to infected animals. The Ebola virus is one of many viruses known to cause *hemorrhagic fever*, an often fatal syndrome characterized by fever, vomiting, internal bleeding, and circulatory system collapse. Hantavirus, which is harbored by deer mice and other rodents, caused a fatal outbreak of pneumonia in the southwestern United States in 1993. The SARS virus may have transferred to humans from civet cats.

Both wild and domestic animals can harbor viruses that can be transmitted to people. Close interaction of animals with humans, as shown in Figure 24-8, can lead to the transfer of many diseases. Such diseases are of great concern to public health agencies worldwide. Vaccines are expensive to develop and often difficult to distribute, especially in underdeveloped countries. The cost of public health education to help people understand how they can prevent outbreaks of disease is also high. In addition, laws banning the sale of certain animals as pets or for food can be difficult to enforce.

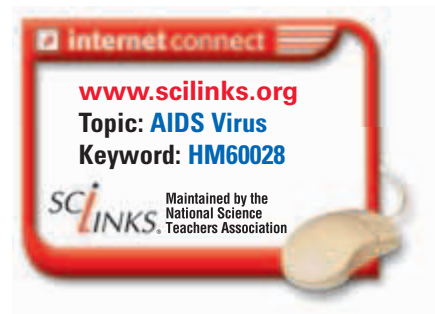
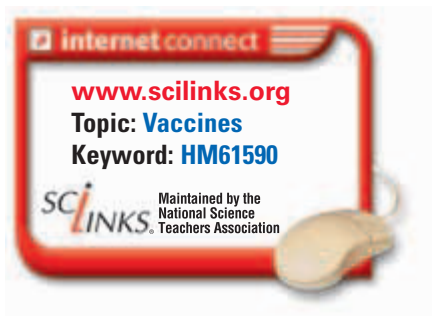


FIGURE 24-8

Avian, or bird, flu is an emerging viral disease. The close interaction of people and poultry on farms and in markets, such as this one in Hong Kong, contributes to the transfer of viruses from birds to humans.



PREVENTION AND TREATMENT

Healthcare providers and public health officials prevent and treat viral diseases through vaccination, vector control, and drug therapy. So far, the first two measures have been the most successful.

Vaccinations

A vaccine is a solution that contains a harmless version of a virus, bacterium, or a toxin that causes an immune response when introduced to the body. Vaccination is a highly effective way to prevent viral infection. Viral vaccines can be made from inactivated viruses, attenuated viruses, or parts of the viral coat. An **inactivated virus** is not able to replicate in a host. An **attenuated virus** is a weakened form of the virus that cannot cause disease. In general, attenuated viruses provide greater protection from disease. Vaccines against measles, mumps, rubella, polio, hepatitis A and B, and chickenpox have greatly reduced the incidence of these diseases. The genetic diversity of HIV makes the development of an AIDS vaccine a difficult task. Educating people about HIV transmission is currently the best approach to slowing the spread of AIDS.

Smallpox once killed 40 percent of the people it infected, leaving survivors scarred and often blind. The smallpox virus is a DNA virus that is spread by nasal droplets from sneezing or coughing. Symptoms include fever, headache, backache, and development of a lumpy skin rash, shown in Figure 24-9. The World Health Organization (WHO) began a smallpox eradication program in 1967 through vaccination and the quarantine of sick people. The last naturally acquired smallpox case occurred in Somalia in 1977. In 1980, WHO declared that smallpox had been eradicated in nature.



FIGURE 24-9

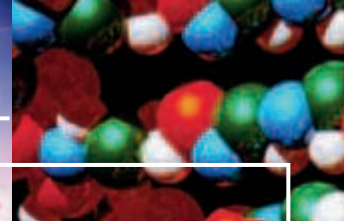
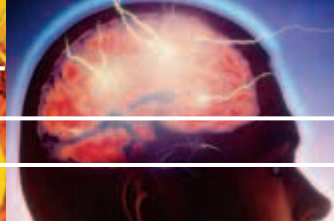
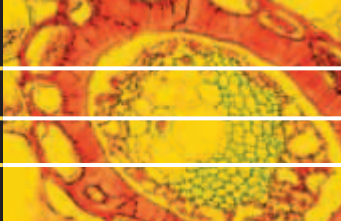
The smallpox virus causes painful lesions that cover the face, shoulders, and chest and, in late stages, the arms and legs. These lesions can leave disfiguring scars, can cause blindness, and may result in death due to hemorrhaging.

Vector Control

An important part of preventing viral disease is the control of animal vectors. Mosquito-control programs eradicated yellow fever in the United States. Rabies vaccinations keep pets free of infection and also protect humans. Wildlife officials set out meat that contains rabies vaccine to control rabies in coyotes and wolves.

Drug Therapy

Several kinds of antiviral drugs interfere with viral nucleic acid synthesis. Unfortunately, the number of antiviral drugs is small compared to that of drugs that treat bacterial, fungal, and parasitic infections. Because viruses use host cells in their life cycles, it is difficult to design a drug that blocks the virus but doesn't harm cells. The drug *acyclovir* (ay-SIE-kloh-VIR) blocks the DNA polymerase of herpes viruses and chickenpox virus. Such drugs do not destroy a given virus but allow time for the body to build up an immune response to the virus.



MARINE VIRUSES: What Is Their Role?

In the early 1990s, marine biologists discovered something startling: every milliliter of sea water contains millions of virus particles. Prokaryotic marine organisms, such as cyanobacteria, and eukaryotic protist producers form the base of nearly all marine food chains. These tiny organisms, or *phytoplankton*, fix most of Earth's carbon and release most of its oxygen annually. The discovery of so many virus particles in sea water raised immediate questions: Do the viruses infect the phytoplankton and keep their numbers in check? How do viruses affect food chains and carbon and oxygen cycles? How must biologists factor marine viruses into their models for ocean ecosystems? Do any of these viruses pose a health risk to humans?

The Role of Marine Viruses

Since the 1990s, scientists have learned more about marine viruses. For example, Lita Proctor of the University of California at Los Angeles carried out pioneering studies on marine viruses.

Using electron microscopy, Proctor discovered that many marine viruses live on or in host cells and float freely only after their cellular hosts die.

Marine ecologist Curtis Suttle and his colleagues at the University of British Columbia identified specific types of marine viruses and their activities. Suttle used antibodies labeled with a fluorescent dye to identify particular viruses and their host cells. Suttle discovered that many types of marine viruses infect cells in phytoplankton and stop photosynthesis. He tried removing specific types of viruses to protect their hosts. However, Suttle found that the rest of the phytoplankton in the water sample stopped growing! Apparently, the other organisms depended on the nutrients released when the viruses killed their hosts.

Suttle also found that viral infection seems to be a constant occurrence in phytoplankton—about 20 percent of the organisms are killed by viruses at any given time.

Suttle's studies helped show that marine viruses are an important part of the carbon cycle and other ecological interactions.

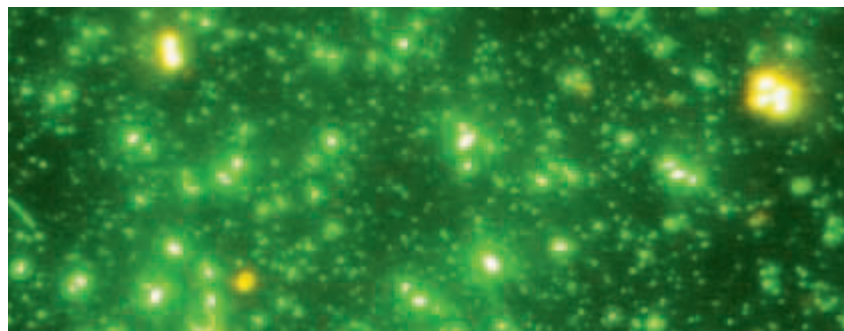
The Search for Marine Viruses That Infect Humans

Researchers have used the tools of molecular genetics to search the waters around many coastal cities for viruses known to cause human disease. Sewage dumped into the oceans contains many kinds of pathogenic viruses. So far, poliovirus, hepatitis A, and possibly HIV have been found in sea water. However, scientists are unsure whether these submerged viruses can infect swimmers.

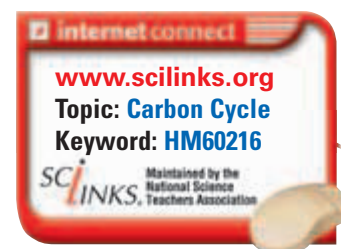
Marine viruses are clearly important, but the study of them is in its infancy. At this point, researchers can culture only about 2 percent of phytoplankton, so the study of marine viruses is hampered in this very basic way.

REVIEW

1. What is phytoplankton?
2. Identify two effects of viral infection on phytoplankton.
3. **Critical Thinking** Why were marine viruses not discovered until the early 1990s?



Fluorescent dyes allow scientists to see viruses in marine waters. The small dots in the photo are viruses; the larger, brighter dots are bacteria.



VIROIDS AND PRIONS

Even simpler than viruses is a group of disease-causing agents called viroids. **Viroids** (VIE-roydz) are the smallest known particles that are able to replicate. A viroid is made up of a short, circular, single strand of RNA that does not have a capsid. Viroids infect plants. These naked RNA molecules can disrupt plant-cell metabolism and damage entire crops. Economically important plants affected by viroids include coconuts, potatoes, and oranges.

In late December 2003, the U.S. Department of Agriculture diagnosed a single cow in Washington State with bovine spongiform encephalopathy (BSE), commonly known as *mad cow disease*. BSE is a degenerative brain disease that results in muscle paralysis, wasting, and death. A cow with BSE is shown in Figure 24-10. Research suggests that mad cow disease is caused by prions (PRIE-AHNZ). **Prions** are infectious protein particles that do not have a genome. They are abnormal forms of a natural brain protein that appear to convert normal brain proteins into prion particles. Scientists hypothesize that the prions then clump together inside cells and cause cell death.

Prions appear to cause a number of other degenerative brain diseases, including scrapie in sheep and Creutzfeldt-Jakob (KROYTZ-felt-YAH-kohb) disease (CJD) and kuru in humans. The BSE epidemic in the United Kingdom in the early 1990s was most likely caused by the feeding of scrapie-contaminated meat and bone meal to cattle. Scientific evidence indicates that BSE has been transmitted to humans, causing a variant form of CJD (vCJD). This evidence comes from studies that followed the unusually high numbers of young people in the United Kingdom that developed this rare disease in the late 1990s. Eating BSE-contaminated beef and beef products was the probable cause of vCJD in these cases. Measures to protect the food supply are the best safeguards. However, the overall risk to human health from BSE is very low.

FIGURE 24-10

This cow, which is unable to stand and walk, is showing signs of mad cow disease, a disease caused by prions.



SECTION 2 REVIEW

1. Give an example of a vector of a viral disease and the disease it transmits.
2. Identify four viral diseases of humans.
3. What is the relationship between viruses and cancer?
4. Explain how human actions have contributed to the increase of emerging viral diseases.
5. What are three ways to fight viruses?
6. Compare viroids, prions, and viruses.

CRITICAL THINKING

7. **Applying Information** Consider how emerging viruses develop. Would you consider emerging viruses to be new viruses? Explain.
8. **Evaluating Conclusions** Explain why microbiologists oppose the use of antibiotics in patients with viral infections.
9. **Applying Information** Once inside the body, HIV's surface glycoproteins can mutate to recognize a new type of cell receptor. How does this mutation aid the virus in its life cycle?

CHAPTER HIGHLIGHTS

SECTION 1

Viral Structure and Replication

- Researchers in the late 1800s discovered that something smaller than bacteria could cause disease.
- Wendell Stanley demonstrated that viruses were not cells when he crystallized TMV, the virus that causes tobacco mosaic disease in tobacco and tomato plants.
- Viruses are nonliving particles containing DNA or RNA and are surrounded by a protein coat called a capsid. Some viruses also have an envelope that is derived from a host cell's nuclear membrane or cell membrane.
- Viruses do not have all of the characteristics of life and are therefore not considered to be living.
- Viruses can be classified based on whether they have RNA or DNA, whether the RNA or DNA is single or double stranded and circular or linear, by capsid shape, and whether or not they have an envelope.
- DNA viruses can enter host cells and directly produce RNA, or they can insert into a host's chromosome, where they are transcribed to RNA along with the host's DNA.
- The RNA genome of some RNA viruses can be directly translated to make viral proteins. Others use reverse transcriptase and RNA as a template to make DNA, which is then used to produce viral RNA and proteins.
- Bacteriophages can follow a lytic cycle (making new viral particles immediately) or a lysogenic cycle (becoming part of the host genome and making new particles later).
- Viruses are important tools for biotechnology.
- Most scientists think viruses originated from fragments of host-cell nucleic-acid material.

Vocabulary

virus (p. 483)
capsid (p. 484)
envelope (p. 484)
provirus (p. 485)

retrovirus (p. 486)
reverse transcriptase
(p. 486)
bacteriophage (p. 486)

lytic cycle (p. 486)
virulent (p. 486)
lysis (p. 486)
lysogenic cycle (p. 487)

temperate virus (p. 487)
prophage (p. 487)

SECTION 2

Viral Diseases

- Vectors of viral diseases include humans, animals, and insects.
- Viruses cause many human diseases, including the common cold, flu, hepatitis, rabies, chickenpox, certain types of cancer, and AIDS.
- Some viruses contain oncogenes that can cause cancer, while other viruses convert proto-oncogenes to oncogenes.
- The human immunodeficiency virus (HIV) is an RNA virus spread by sexual contact, by contact with infected body fluids, and from mother to fetus. HIV targets macrophages and thus damages the body's immune system in the disease called acquired immunodeficiency syndrome (AIDS).
- Emerging viruses usually infect animals isolated in nature but can jump to humans when contact occurs in the environment.
- Vaccines have helped to greatly reduce certain viral diseases. Control efforts, including killing mosquitoes and other vectors and quarantining ill patients, have helped reduce the spread of certain viral diseases.
- Antibiotics are ineffective against viral diseases. Viral drugs, such as acyclovir, block specific steps in viral replication.
- Viroids are short, circular, single strands of RNA lacking a capsid that infect plant cells.
- Prions are infectious particles containing protein but no nucleic acids. Prions cause mad cow disease and similar degenerative brain diseases.

Vocabulary

vector (p. 489)
protease inhibitor (p. 491)
oncogene (p. 491)

proto-oncogene (p. 491)
emerging disease (p. 491)

inactivated virus (p. 492)
attenuated virus (p. 492)

viroid (p. 494)
prion (p. 494)

CHAPTER REVIEW

USING VOCABULARY

- For each pair of terms, explain how the meanings of the terms differ.
 - virus* and *viroid*
 - oncogene* and *proto-oncogene*
 - capsid* and *envelope*
 - provirus* and *vector*
- Use the following terms in the same sentence: *virus*, *lytic cycle*, *lysogenic cycle*, and *bacteriophage*.
- Use each of the following terms in a separate sentence: *prion*, *prophage*, *temperate virus*, and *bacteriophage*.
- Word Roots and Origins** The word *virus* is derived from the Greek *ios*, which means “poison.” Using this information, explain why the term *virus* is a good name for these particles.

UNDERSTANDING KEY CONCEPTS

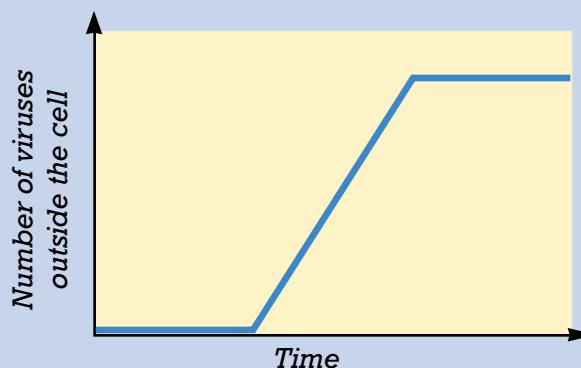
- Summarize** how the structure of viruses was discovered by Wendell Stanley.
- Discuss** why viruses are not considered living organisms.
- Describe** three different shapes viruses can have.
- Compare** replication in DNA viruses to replication in RNA viruses.
- Differentiate** the lytic cycle of viral replication from the lysogenic cycle of viral replication in bacteriophages.
- Summarize** how viruses are thought to have originated.
- Describe** four diseases caused by viruses that occur in humans.
- Name** three vectors of viral diseases that can spread viruses to humans.
- Discuss** the role of viruses and oncogenes in the onset of cancer.
- Describe** the structure of HIV.
- Explain** the activity of reverse transcriptase in the replication cycle of the human immunodeficiency virus (HIV).
- Summarize** how emerging diseases can occur.
- Discuss** three methods humans use to control the spread of viral diseases.
- Describe** how viruses, viroids, and prions differ from one another.

- CONCEPT MAPPING** Use the following terms to create a concept map that describes the lytic cycle: *viruses*, *virulent*, *phage*, *injects*, *DNA*, *replicates*, *assemble*, *protein*, and *lyse*.

CRITICAL THINKING

- Evaluating Information** The drug azidothymidine (AZT) works by blocking the enzyme reverse transcriptase. Explain how AZT can help patients infected with HIV.
- Applying Information** Shingles is a disease caused by the same herpesvirus that causes chickenpox. How do you account for the fact that shingles often appears years after the initial chickenpox attack?
- Applying Current Research** Based on your knowledge of HIV structure and replication, describe one way to interrupt the replication of HIV.
- Applying Information** How does the increase of resistance to antiviral drugs in HIV relate to the theory of evolution by natural selection?
- Applying Information** Tobacco mosaic virus does not infect humans, but humans can transmit TMV from infected plants to healthy plants. What role do humans play in this mode of transmission?
- Making Real-Life Connections** For viral diseases without known cures, such as AIDS, certain types of hepatitis, and Ebola, identify ways in which the incidence of such diseases can be reduced.
- Interpreting Graphics** Look at the graph below. Discuss how the sharp jump in the number of viruses outside the cell corresponds to the phases of the lytic cycle.

Growth Curve of Viruses



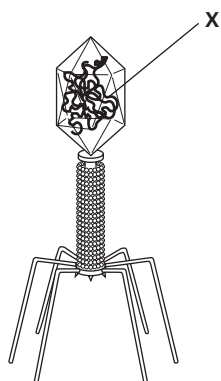


Standardized Test Preparation

DIRECTIONS: Choose the letter of the answer choice that best answers the question.

- What are viruses made of?
 - enzymes and fats
 - carbohydrates and ATP
 - protein and nucleic acids
 - mitochondria and lysosomes
- How do viroids differ from viruses?
 - Viroids are larger in size.
 - Viroids do not have a capsid.
 - Viroids do not have nucleic acids.
 - Viroids can cause disease in plants.
- During which of the following processes does a phage kill its host?
 - conjugation
 - transcription
 - the lytic cycle
 - the lysogenic cycle
- Which of the following is one reason why viruses are not considered living organisms?
 - Viruses are able to grow.
 - Viruses do not metabolize.
 - Viruses can reproduce by splitting.
 - Viruses are too small to be easily observed.

INTERPRETING GRAPHICS: Study the figure below to answer the following questions.

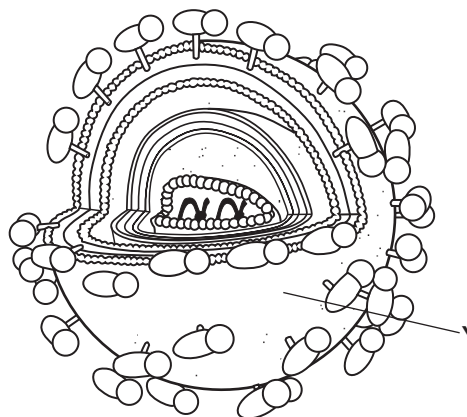


- Which of the following does the diagram represent?
 - a virus
 - a prion
 - a viroid
 - a bacterium
- To which of the following is label X pointing?
 - envelope
 - nucleic acid
 - protein coat
 - cell membrane

DIRECTIONS: Complete the following analogy.

7. skin : person :: capsid :
- virus
 - insect
 - fungus
 - bacterium

INTERPRETING GRAPHICS: The figure below represents the human immunodeficiency virus. Use the figure to answer the question that follows.



8. The structure labeled Y represents which of the following?
- capsid
 - envelope
 - RNA genome
 - reverse transcriptase

SHORT RESPONSE

Reverse transcriptase is an enzyme that catalyzes the synthesis of DNA from RNA.

Explain why retroviruses must have reverse transcriptase to replicate.

EXTENDED RESPONSE

Viruses share several characteristics of living organisms. However, viruses are not considered to be living.

Part A Compare the characteristics viruses share with living organisms to the characteristics they do not share with living organisms.

Part B Would you anticipate more or fewer emerging viral diseases to appear in the future? Explain.

Test TIP

When using a diagram to answer questions, carefully study each part of the figure as well as any lines or labels used to indicate parts of the figure.

Infecting Plants with Tobacco Mosaic Virus

OBJECTIVES

- Study the effect of cigarette tobacco on leaves of tobacco plants.

PROCESS SKILLS

- observing
- comparing and contrasting
- experimenting

MATERIALS

- lab apron
- protective gloves
- safety goggles
- 2 tobacco or tomato plants
- glass-marking pencil
- tobacco from several brands of cigarettes
- mortar and pestle
- 10 mL 0.1 M dibasic potassium phosphate solution
- 100 mL beaker
- cotton swabs
- 400 grit carborundum powder

Background

1. The tobacco mosaic virus, TMV, infects tobacco as well as other plants.
2. Plants that are infected with TMV have lesions and yellow patches on their leaves.

3. In what form do viruses exist outside host cells?
4. The tobacco mosaic virus is an RNA virus with rod-shaped capsids and proteins arranged in a spiral.
5. Plants damaged by wind, low temperatures, injury, or insects are more susceptible to plant viruses than healthy plants are.
6. Plant viruses are transmitted by insects, gardening tools, inheritance from the parent, and sexual reproduction.
7. In this investigation, you will test whether tobacco from cigarettes can infect tobacco plants with TMV.

PART A Setting Up the Experiment

1.   Put on a lab apron, gloves, and goggles before beginning this investigation.
2. Obtain two tobacco plants that have not been infected with TMV. Label one of the plants "control plant." Label the other plant "experimental plant."
3.   **CAUTION** Use poisonous chemicals with extreme caution. Keep your hands away from your face when handling plants or chemical mixtures. Place pinches of tobacco from different brands of cigarettes into a mortar. Add 5 mL of dibasic potassium phosphate solution, and grind the mixture with a pestle as shown in the figure at left.
4. Pour the mixture into a labeled beaker. This mixture can be used to test whether cigarette tobacco can infect plants with TMV.
5.  Wash your hands and all laboratory equipment used in the previous steps with soap and water to avoid the accidental spread of the virus.
6. Moisten a sterile cotton swab with the mixture, and sprinkle a small amount of carborundum powder onto the moistened swab. Apply the mixture to two leaves on the "experimental" plant by swabbing the surface of the leaves several times. Why do you think swabbing the leaves with carborundum powder might facilitate infection?



7. Moisten a clean swab with dibasic potassium phosphate solution, and sprinkle a small amount of carborundum powder onto the moistened swab. This swab should *not* come into contact with the mixture of cigarette tobacco. Swab over the surface of two leaves on the control plant several times.
8. Do not allow the control plants to touch the experimental plants. Keep both plants away from other plants that may be in your investigation area, such as houseplants or garden plants. Wash your hands after handling each plant to avoid the accidental spread of TMV.
9. Treat both plants in precisely the same manner. The only difference between the two plants should be the experimental factor—exposure to cigarette tobacco. Both plants should receive the same amount of light and water.
10.   Clean up your materials according to your teacher's instructions and wash your hands before leaving the lab.



PART B Collecting Data

11. In your lab report, create a data table similar to the model shown below. Allow plenty of space to record your observations of each plant.
12. Check the control and experimental plants each day for one week. Record your observations of each plant in your lab report. Wash your hands after handling each plant to prevent contaminating your results.

Analysis and Conclusions

1. What differences, if any, did you detect in the two plants after one week?
2. Did the plants exposed to cigarette tobacco become infected with the tobacco mosaic virus?
3. Why do you think it was necessary to use tobacco from different brands of cigarettes?
4. What are some of the possible sources of error in the experiment?
5. Greenhouse operators generally do not allow smoking in their greenhouses. Aside from health and safety issues, how might your results support this practice?

Further Inquiry

The tobacco mosaic virus is capable of infecting different species of plants. Design an experiment to determine which of several types of plants are susceptible to the virus.

OBSERVATIONS OF TOBACCO PLANTS

Day	Control plant	Experimental plant
1		
2		
3		
4		
5		
6		
7		

PROTISTS



The single-celled organisms shown here are protists from the genus *Stentor*, named after a Greek announcer who had a very loud voice. These trumpet-shaped cells are highly complex and adapted for capturing and consuming prey.

SECTION 1 *Characteristics of Protists***SECTION 2** *Animal-like Protists***SECTION 3** *Plantlike and Funguslike Protists***SECTION 4** *Protists and Humans*

CHARACTERISTICS OF PROTISTS

Protists are a diverse collection of eukaryotic organisms, such as protozoa, algae, slime molds, and water molds. Protists are sometimes described as animal-like, plantlike, or funguslike. However, these organisms lack the cellular differentiation found in animals, plants, and fungi.

A DIVERSE GROUP OF EUKARYOTES

Single-celled or simple multicellular eukaryotic organisms that generally do not fit in any other kingdom are called **protists**. Most protists are microscopic, but a few protists, such as some algae, are several meters in length. Protists are defined by exclusion—most protists are eukaryotic organisms that cannot be classified as fungi, plants, or animals. As a result, protists are the most diverse group of eukaryotes. Protists have varying body plans, types of movement, and means of obtaining food. Protists are made up of eukaryotic cells, each containing a nucleus and other organelles. Most living protists contain mitochondria. Some protists, such as *Euglena* shown in Figure 25-1, also contain chloroplasts.

The First Eukaryotes

Protists emerged early in the history of domain Eukarya. Some of the oldest eukaryotic cells were protists. Protists emerged as much as 2 billion years ago.

By analyzing genetic information in the nucleus, mitochondria, and chloroplasts of protists, scientists have found strong evidence that the first protists arose from prokaryotic cells. Nuclear genes in protists and in other eukaryotes resemble archaeal and bacterial genes. Additionally, the genetic information found in the mitochondria and chloroplasts of protists and other eukaryotes is similar to genetic information found in bacteria and cyanobacteria. Because of these genetic similarities, scientists hypothesize that protists and other eukaryotes arose from ancient prokaryotes that lived in *endosymbiosis*. In endosymbiosis, an organism lives inside a larger organism. Scientists think that mitochondria and chloroplasts arose from ancient prokaryotes that lived inside larger prokaryotes. Over time, the endosymbiotic prokaryotes became organelles within eukaryotic protists.

OBJECTIVES

- Define *protist*.
- Describe a hypothesis for the origin of eukaryotic cells.
- Explain how protists are classified.
- Describe the two major ways by which protists obtain energy.
- List three structures protists use for movement.
- Describe how protists reproduce.

VOCABULARY

protist
binary fission
multiple fission
conjugation

FIGURE 25-1

Most *Euglena* are unicellular, contain chloroplasts, move about, and sometimes consume other organisms. Because *Euglena* have both plantlike and animal-like characteristics, they cannot be classified in the plant or animal kingdoms. So, they are classified as protists.



CLASSIFICATION

Often, protists are classified by the characteristics that resemble those of fungi, plants, and animals. For example, the reproduction of some protists resembles the reproduction of fungi. Other protists capture light energy for photosynthesis, as plants do. Some protists move and consume other organisms, as animals do.

The great diversity among protist species makes them difficult to classify. Protists have traditionally been classified in kingdom Protista. Because of ongoing molecular analyses, many scientists now hypothesize that kingdom Protista should be divided into several kingdoms. Scientists are still revising the classification of protists.

CHARACTERISTICS

Table 25-1 summarizes the characteristics of several protist phyla. There are few general characteristics of protists, but they can be characterized by body plan, means of obtaining food, and motility.

TABLE 25-1 Characteristics of Various Protist Phyla

Phylum	Common name	Body plan	Motility	Nutrition type	Representative genera
Protozoa	protozoa, sarcodines, mycetozoans	unicellular	pseudopodia	heterotrophic, some parasitic	<i>Amoeba</i> , <i>Entamoeba</i> , <i>Radiolaria</i>
Ciliophora	ciliates	unicellular	cilia	heterotrophic, some parasitic	<i>Paramecium</i> , <i>Balantidium</i> , <i>Stentor</i>
Sarcomastigophora	mastigophorans	unicellular	flagella	heterotrophic, some parasitic	<i>Trypanosoma</i> , <i>Leishmania</i>
Apicomplexa	apicomplexans	unicellular	none in adult	heterotrophic, parasitic	<i>Plasmodium</i> , <i>Toxoplasma</i>
Chlorophyta	green algae	unicellular, colonial, multicellular	some with flagella	autotrophic	<i>Spirogyra</i> , <i>Volvox</i> , <i>Chlamydomonas</i> , <i>Protococcus</i>
Bacillariophyta	diatoms	unicellular, colonial	mostly no flagella	autotrophic	<i>Cocconeis</i> , <i>Bacillaria</i>
Dinoflagellata	dinoflagellates	unicellular	flagella	autotrophic, some heterotrophic	<i>Noctiluca</i> , <i>Ceratium</i>
Euglenophyta	euglenoids	unicellular	flagella	autotrophic, some heterotrophic	<i>Euglena</i>
Dictyostelida	cellular slime molds	unicellular, multicellular	pseudopodia	heterotrophic	<i>Dictyostelium</i>
Oomycota	water molds	unicellular, multicellular	some with flagella	heterotrophic, some parasitic	<i>Phytophthora</i>

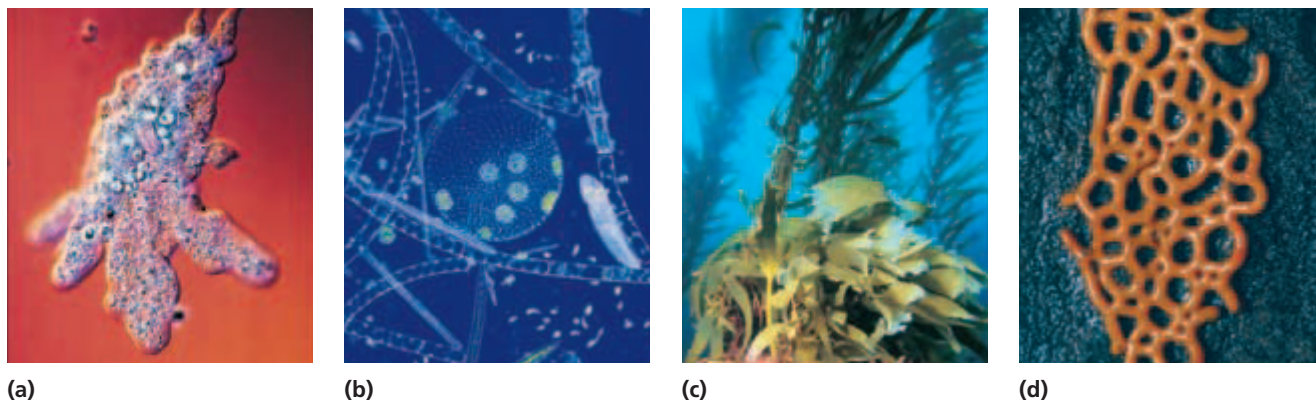


FIGURE 25-2

Protists have a variety of body plans. (a) The single-celled *Amoeba proteus* constantly changes its body shape. (b) Cells from *Volvox*, a colonial protist, have coordinated activity. (c) Brown algae, such as this kelp (*Macrocystis pyrifera*), are multicellular giants among protists. (d) This pretzel slime mold (*Hemitrichia serpulata*) reproduces similarly to fungi.

Unicellular and Multicellular

As shown in Figure 25-2, protists come in a wide variety of body plans. Most protists are unicellular, such as the amoeba shown in Figure 25-2a. Some protists, such as the *Volvox* in Figure 25-2b, form colonies in which several cells are joined into a larger body. Some of these colonies have a division of labor; certain cells specialize in reproduction, and other cells specialize in obtaining energy.

A few protists, such as the brown algae in Figure 25-2c and the pretzel slime mold in Figure 25-2d, form large multicellular bodies. Some brown algae may grow to more than 60 m in length. These marine giants have specialized regions for reproduction, photosynthesis, and attachment to the ocean floor. However, these regions lack the cellular differentiation found in true tissues and organs.

Nutrition

Protists obtain energy in a number of ways. Many protists are *autotrophs*, organisms that can make their own food molecules. These protists make food in much the same way that plants do. The protists absorb energy from sunlight with the aid of specialized light-absorbing pigments. Protists often utilize chlorophyll, as plants do, but they may use additional pigments. The protists use the captured light energy, water molecules, and carbon dioxide molecules to make carbohydrates.

Some protists are *heterotrophs*, organisms that must get their food by eating other organisms or their byproducts and remains. Some heterotrophic protists engulf smaller protists and digest them. Other heterotrophic protists obtain energy in the same way that fungi do. These protists secrete digestive enzymes into the environment. The enzymes break down cells or bits of food into small molecules that the protists can absorb and use.

Motility

Most protists are able to move at some time during their life cycles. Some protists move with the aid of long, whiplike structures called *flagella* (singular, *flagellum*). Other protists move with the aid of *cilia* (singular, *cilium*), which are shorter than flagella and often form rows. Finally, some protists, such as amoebas, move by temporarily extending structures called *pseudopodia* (singular, *pseudopodium*).

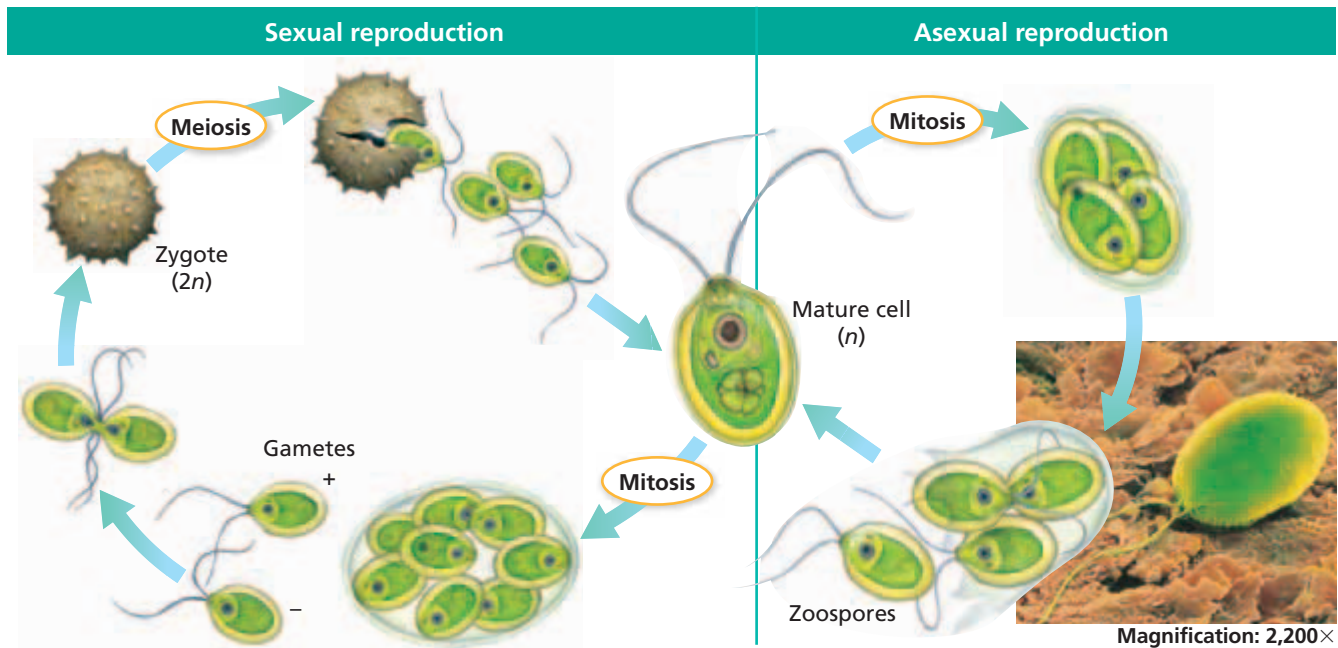


FIGURE 25-3

Algae of the genus *Chlamydomonas* are unicellular green algae that undergo both asexual and sexual reproduction. For many protist producers, the type of reproduction alternates by generation. For example, a parent may reproduce asexually, but its offspring may reproduce sexually. For other protists, sexual reproduction occurs only when environmental conditions are stressful.

REPRODUCTION

Many protists reproduce asexually. During **binary fission**, a single protist cell divides into two cells. Some protists reproduce by **multiple fission**, a form of cell division that produces more than two offspring. Both types of fission produce offspring that are genetically identical to the parent cell.

One way by which many protists reproduce sexually is conjugation. During **conjugation**, two individuals join and exchange genetic material stored in a small second nucleus. Then, the cells divide to produce four offspring. The offspring are genetically different from the parent cells. As shown in Figure 25-3, many protists can reproduce both asexually and sexually.

SECTION 1 REVIEW

1. What is the definition of *protist*?
2. What is a hypothesis for how eukaryotic cells arose?
3. How are protists often classified?
4. Describe unicellular and multicellular protists.
5. What are two ways by which protists obtain energy?
6. How do protists move?
7. Describe how some protists reproduce both asexually and sexually.

CRITICAL THINKING

8. **Applying Information** Why might scientists argue that protists should be classified in several kingdoms rather than in one kingdom?
9. **Relating Concepts** Compare binary fission and conjugation.
10. **Calculating Information** Make a bar graph comparing the volumes of the following: a protist (cone shaped, 500 μm long, 200 μm in diameter); a human white blood cell (sphere shaped, 10 μm in diameter); and a bacterial cell (cylinder shaped, 2 μm long, 0.8 μm in diameter).

Science in Action

How Did Eukaryotic Cells Evolve?

The idea that cellular organelles were originally separate organisms was first proposed in the early 1900s. The few scientists who later became aware of this hypothesis thought that it was preposterous. In the 1960s, Lynn Margulis thought that the idea should be reexamined. Her research led her to form a theory that shook the foundations of the biological community.



Lynn Margulis

HYPOTHESIS: Cellular Organelles Were the Result of Endosymbiosis

Endosymbiosis is a relationship in which one organism lives inside another. Lynn Margulis, an American biologist, thought that endosymbiosis led to the evolution of eukaryotic cells. Margulis proposed that mitochondria and chloroplasts, the cell's energy-producing organelles, were initially separate organisms that had become integral parts of larger cells.

Margulis hypothesized that the first cells with chloroplasts probably came about when a host organism ingested photosynthetic bacteria. A few of these bacteria were not digested. These unlikely partnerships provided something for everyone: The guest gave the host a source of food, while the host protected the guest. Margulis also surmised that mitochondria arose in a similar way.

METHODS: Use Electron Microscopy to Observe Organelles and Cells

Margulis took advantage of advances in electron microscopy and molecular biology to obtain evidence in support of her hypothesis. Working with her graduate research advisor, Hans Ris, Margulis used high-powered electron microscopy to carefully observe and compare mitochondria, chloroplasts, eukaryotic cells, and prokaryotic cells. In addition, she looked for the presence of DNA in the chloroplasts of *Euglena gracilis* by using radioactively labeled nucleotides.

RESULTS: Organelles Are Similar to Bacteria

Margulis found that bacteria and organelles show many similarities. Like prokaryotes, mitochondria and chloroplasts have circular DNA. These organelles also have their own ribosomes, which are the same size as those found in prokaryotes but smaller than those found in the cytoplasm of eukaryotes. Both mitochondria and chloroplasts are found enclosed in membranes as though they were captured in a vacuole of a larger cell. Also, mitochondria and chloroplasts divide by fission, as bacteria do.

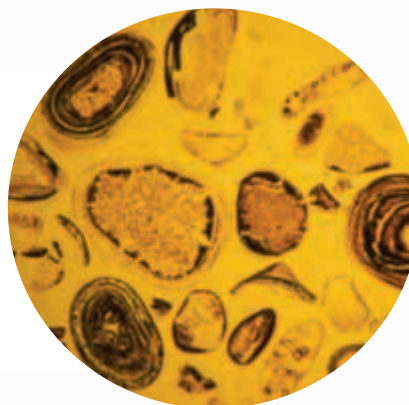
An endosymbiotic relationship was witnessed by Kwang Jeon at the University of Tennessee.

One day, Jeon noticed that bacteria had infected his amoebas and that a few of the amoebas did not die. Their new partners replicated along with the hosts. Generations later, these amoebas are thriving; yet the formerly separate organisms are totally dependent upon one another. If the invader is removed from the host, both die.

CONCLUSION: Organelles Arose Through Endosymbiosis

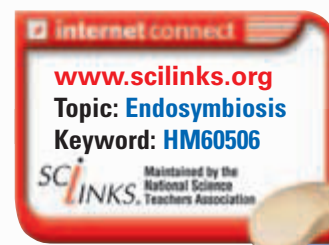
Today, endosymbiosis is the accepted explanation for how eukaryotes arose. Additional support for Margulis's theory comes from examining modern organisms that contain intracellular symbiotic bacteria and photosynthetic protists. For example, sea slugs incorporate chloroplasts into some of their cells, which provides a source of energy for the sea slugs when other food is scarce.

Endosymbiosis led to the formation of eukaryotic cells. Multicellular eukaryotes, such as these fossil algae, arose around 2 billion years ago.



REVIEW

1. Restate the endosymbiosis hypothesis in your own words.
2. In what ways do prokaryotes resemble chloroplasts and mitochondria?
3. **Critical Thinking** Why must modern plants be the result of at least two separate endosymbiotic events?



SECTION 2

OBJECTIVES

- **Discuss** the key characteristics of Protozoa, Ciliophora, Sarcomastigophora, and Apicomplexa.
- **Describe** how protozoa use pseudopodia to move and to capture food.
- **Explain** how ciliates move and reproduce.
- **Describe** how mastigophorans move and capture food.
- **Describe** the role of apicomplexans in disease.

VOCABULARY

pseudopodium
amoeboid movement
test
cilium
pellicle
oral groove
mouth pore
gullet
anal pore
contractile vacuole
macronucleus
micronucleus
flagellum

ANIMAL-LIKE PROTISTS

Protists aren't animals. However, four major groups of protists move and obtain food in animal-like ways. Having animal-like characteristics would seem to suggest that animal-like protists are closely related to animals in an evolutionary sense, but animal-like protists and animals are not closely related.

PHYLUM PROTOZOA

Most animal-like protists are heterotrophs that move about capturing and consuming prey. Animal-like protists are sometimes called *protozoa*. Protozoa are single-celled protists that can move independently without cilia or flagella. Biologists group more than 40,000 species of protozoa in the phylum Protozoa, which includes the subphyla Sarcodina (SAHR-kuh-DIE-nuh) and Mycetozoa (mie-SEET-uh-ZOH-uh). Of these species, about three-quarters are identified only by fossil remains. Figure 25-4 shows two examples of protozoans.

A key characteristic of most protozoa is the formation of pseudopodia (SOO-doh-POH-dee-uh). **Pseudopodia** are large, rounded cytoplasmic extensions that function both in movement and feeding. A pseudopodium forms when the cytoplasm flows forward to create a blunt, armlike extension. Simultaneously, other pseudopodia retract, and the cytoplasm flows in the direction of the new pseudopodium, causing the cell to move. This type of locomotion is called **amoeboid movement**. Amoeboid movement is a form of *cytoplasmic streaming*, the internal flowing of a cell's cytoplasm.

Protozoa actively prey on smaller cells, such as bacteria and smaller protists, and food particles. A sarcodine feeds by surrounding the food with pseudopodia and trapping the food in a vesicle. The sarcodine releases enzymes to digest the food trapped inside the vesicle. Figure 25-4a shows a sarcodine, an amoeba, using pseudopodia to capture food.

FIGURE 25-4

(a) Some sarcodines, such as amoebas, send out pseudopodia that engulf smaller organisms (indicated by the yellow circle). (b) Some mycetozoans, such as radiolarians, have distinctive glassy shells.



(a)

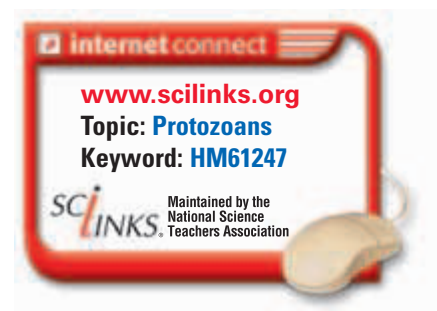


(b)

Protozoan Diversity

Sarcodines include hundreds of species that inhabit freshwater environments, marine environments, and soil. The cell membranes of some sarcodines, such as amoebas, are exposed directly to the environment. Other sarcodines are covered with a protective **test**, or shell. For example, *foraminifera* (fuh-RAM-uh-NIF-uh-uh) are found primarily in oceans and are covered with intricate tests of calcium carbonate. The shells of ancient foraminifera accumulated at the ocean bottom where, after millions of years, they became limestone. The tests of a group of mycetozoans, the *radiolarians* (RAY-dee-oh-LER-ee-uhnz), contain silicon dioxide. Radiolarians often have radially arranged spines. Both foraminiferans and radiolarians have slender pseudopodia that extend through tiny openings in the test.

Although most amoebas live freely, some species live in human intestines and may cause disease. One such amoeba, *Entamoeba histolytica*, enters the body through contaminated food and water. It releases enzymes that attack the lining of the large intestine and cause ulcers. A sometimes fatal disease called *amebiasis* may result, causing acute abdominal pain.

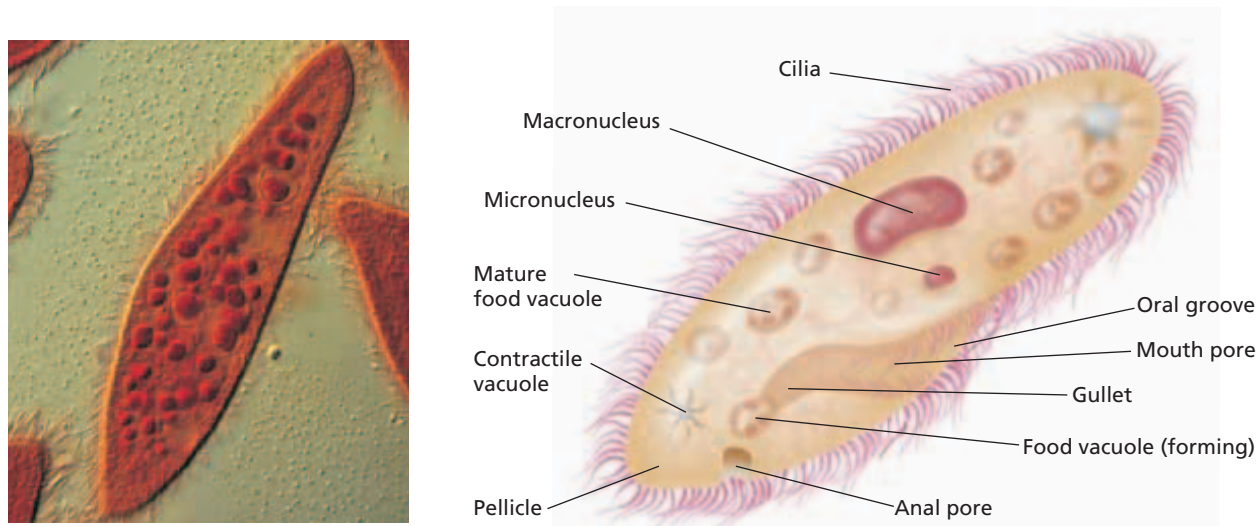


PHYLUM CILIOPHORA

The nearly 8,000 species that make up the phylum Ciliophora (SIL-ee-AHF-uh-ruh) share one key feature: they have cilia. **Cilia** are short, hairlike cytoplasmic projections that line the cell membrane. Cilia make it possible for these protists to move. Members of the genus *Paramecium*, shown in Figure 25-5, are among the most thoroughly studied ciliates. Paramecia are found in ponds and slow-moving streams that contain plants and decaying organic matter. A paramecium has cilia arranged in rows across its cell membrane. The cilia beat in waves, moving the cell through the water. Ciliates often feed on bacteria, algae, and other small organisms in their marine and freshwater habitats.

FIGURE 25-5

Like other ciliates, paramecia move by using hundreds of short projections called *cilia*. Paramecia have a large macronucleus, which controls many cell functions, and one or more micronuclei, which are involved in reproduction. The oral groove, mouth pore, and gullet collect food into vacuoles.



Word Roots and Origins

macronucleus

from the Greek *makros*, meaning "long," and the Latin *nucleus*, meaning "nut" or "kernel"

Characteristics

Ciliates have the most elaborate organelles of any protist. A clear, elastic layer of protein, called a **pellicle**, surrounds the cell membrane. The pellicle has a funnel-like depression called an **oral groove**. Cilia lining the oral groove create currents that sweep food down the groove to the **mouth pore**. The mouth pore opens into a **gullet**, which forms food vacuoles that move throughout the cytoplasm. Enzymes in the vacuoles digest food into small organic molecules that enter the cytoplasm. Undigested materials move to the **anal pore**, which contracts and expels them. Ciliates also have **contractile vacuoles**, saclike organelles that expand to collect excess water and contract to squeeze the water out of the cell.

Ciliates have two types of nuclei. The large **macronucleus** contains multiple copies of DNA that direct the cell's metabolism and development. The smaller **micronucleus** participates in the exchange of genetic material during conjugation.

Reproduction

Asexual reproduction in ciliates occurs by binary fission. In this process, the micronucleus divides by mitosis. The macronucleus, which has up to 500 times more DNA than the micronucleus, simply elongates and splits in half. One half goes to each new cell.

Sexual reproduction in ciliates involves conjugation. During conjugation, two cells join, and their macronuclei disintegrate. Each diploid micronucleus then undergoes meiosis, producing four haploid micronuclei. In each cell, all but one micronucleus disintegrates. The remaining micronucleus divides by mitosis, producing two identical haploid micronuclei. The two cells then exchange one micronucleus. The two micronuclei in each cell then fuse to form one diploid micronucleus. The two cells separate, and a macronucleus forms in each cell from products of mitotic divisions of the micronucleus. Although the cells exchange genetic material during conjugation, they produce no new cells. After conjugation, the two cells divide, forming four offspring.

FIGURE 25-6

Flagella are a key characteristic of mastigophorans, such as these trypanosomes (*Trypanosoma* sp.), shown with red blood cells. Trypanosome flagella are structurally similar to the tails of human sperm.



PHYLUM SARCOMASTIGOPHORA

The phylum Sarcomastigophora (SAR-koh-mas-ti-GAHF-uh-ruh) includes 1,500 species and the subphylum Mastigophora. Members of this phylum are characterized by one or more flagella. **Flagella** are long, whiplike structures that are made up of microtubules and used for movement. Flagella can be seen in Figure 25-6. The rapid motion of flagella moves the protist through water. Many mastigophorans are free-living species that inhabit lakes and ponds, where they feed on smaller organisms.

Some mastigophorans are parasites, including many in the genus *Trypanosoma*. Trypanosomes live in the blood of fish, amphibians, reptiles, birds, and mammals and are carried from host to host by bloodsucking insects. Two species of *Trypanosoma* cause sleeping sickness. The tsetse fly, which lives only in Africa, transmits these parasites. The disease is characterized by increasing fever, lethargy, mental deterioration, and coma.

PHYLUM APICOMPLEXA

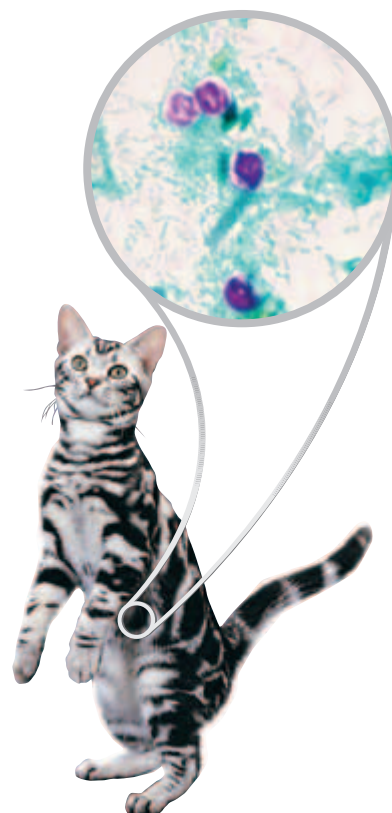
The phylum Apicomplexa (AY-PI-kuhm-PLEKS-uh) includes at least 4,000 species that have adult forms with no means of locomotion. Many apicomplexans were once known as *sporozoans* (SPOH-ruh-ZOH-uhnz). All apicomplexan species are animal parasites.

Apicomplexans have complex life cycles involving both asexual and sexual reproduction. The reproduction of apicomplexans often results in *sporozoites*, infectious cells that are protected by resistant coats. Apicomplexan sporozoites are characterized by an *apical complex*, or a group of organelles specialized for entering host cells and tissues. Because of their complex life cycles, many apicomplexans require two or more different hosts to complete their life cycle. For example, *Plasmodium*, the parasite that causes malaria, needs both mosquitoes and humans to complete its life cycle.

As apicomplexans inhabit a host's cells and tissues, the apicomplexans absorb nutrients and destroy host cells. Apicomplexans may have killed more people than any other group of pathogens on Earth. Each year, millions of people die as a result of *Plasmodium* infection. Another apicomplexan, *Toxoplasma gondii*, causes few symptoms in healthy adult humans but can be dangerous to developing fetuses and newborn babies. *Cryptosporidium*, which is shown in Figure 25-7, is often transmitted by contaminated drinking water or contact with the feces of infected animals.

FIGURE 25-7

Cryptosporidium (stained purple) is a genus of apicomplexans that cause diarrhea, abdominal pain, and vomiting. They sometimes spread to humans through the feces of infected animals, such as cats.



SECTION 2 REVIEW

1. Describe the key characteristics of each phylum of animal-like protists.
2. How are pseudopodia used for movement and the capture of food?
3. Describe reproduction in Ciliophora.
4. How do protozoa, ciliates, and mastigophorans move?
5. What is the role of apicomplexans in disease?
6. List five protists that cause disease, and briefly describe each disease.

CRITICAL THINKING

7. **Applying Information** How might controlling mosquito populations prevent the spread of malaria? Explain your answer.
8. **Making Inferences** Apicomplexans lack a means of locomotion. How does this relate to the fact that apicomplexans are parasites?
9. **Analyzing Information** A scientist found an animal-like protist that lacks cilia or flagella and does not form spores. To which phylum does the protist most likely belong? Explain.

SECTION 3

OBJECTIVES

- **Describe** four main body forms of algae.
- **List** the common name for each of the seven phyla of plantlike protists.
- **Explain** how green algae and plants are similar.
- **Describe** four phyla of funguslike protists.
- **Compare** plasmodial slime molds, cellular slime molds, and water molds.

VOCABULARY

alga
gametangium
phytoplankton
thallus
accessory pigment
diatom
shell
bioluminescence
red tide
euglenoid
fruiting body
water mold
plasmodial slime mold
cellular slime mold

FIGURE 25-8

Multicellular algae, such as kelp, often have air bladders, which help the algae float upright underwater or in mats on the ocean surface.



PLANTLIKE AND FUNGUSLIKE PROTISTS

Algae are plantlike protists that vary in size from tiny unicellular organisms to multicellular giants that reach 60 m in length. Although many funguslike protists are called molds, they do not fit in kingdom Fungi. Funguslike protists include slime molds and water molds.

CHARACTERISTICS OF ALGAE

Many plantlike protists are known as algae. Unlike animal-like protists, which are heterotrophic, **algae** (singular, *alga*) are autotrophic protists. Algae have chloroplasts and produce their own carbohydrates by photosynthesis, as plants do. Older classification systems placed algae in kingdom Plantae. However, algae lack tissue differentiation and thus have no true roots, stems, or leaves. The reproductive structures of algae also differ from those of plants. Algae form gametes in single-celled gamete chambers called **gametangia** (GAM-uh-TAN-jee-uh). By contrast, plants form gametes in multicellular gametangia.

Despite their diversity, algae share several features. Algae have the photosynthetic pigment chlorophyll *a*. Most algae are aquatic and have flagella at some point in their life cycle. Algal cells often contain *pyrenoids* (pie-REE-NOYDZ), structures associated with algal chloroplasts that synthesize and store starch.

Algae have four basic body forms. *Unicellular algae* consist of a single cell. Most unicellular algae are free-living aquatic organisms and together are known as **phytoplankton**. Phytoplankton form the base of nearly all marine and freshwater food chains. *Colonial algae*, such as *Volvox*, consist of groups of individual cells acting in a coordinated manner. Some of the cells may become specialized, allowing for a division of labor. *Filamentous algae*, such as *Spirogyra*, are multicellular algae that are slender, rod shaped, and composed of cells joined end to end. Some filamentous algae have structures that anchor the organism to the ocean bottom. *Multicellular algae*, such as the kelp shown in Figure 25-8, are usually large and complex and often appear plantlike. Many large multicellular algae are also known as *seaweeds*. The plantlike body portion of a seaweed is called a **thallus** (THAL-uhs), and its cells are usually haploid.

TABLE 25-2 *Phyla of Plantlike Protists*

Phylum	Body form	Photosynthetic pigments	Form of food storage	Cell-wall composition
Chlorophyta (green algae)	unicellular, colonial, filamentous, multicellular	chlorophylls <i>a</i> and <i>b</i> , carotenoids	starch	polysaccharides, cellulose
Phaeophyta (brown algae)	multicellular	chlorophylls <i>a</i> and <i>c</i> , carotenoids, fucoxanthin	laminarin (a polysaccharide)	cellulose with alginic acid
Rhodophyta (red algae)	multicellular	chlorophyll <i>a</i> , phycobilins, carotenoids	starch	cellulose or pectin, many with calcium carbonate
Bacillariophyta (diatoms)	unicellular, colonial	chlorophylls <i>a</i> and <i>c</i> , carotenoids, fucoxanthin, some with none	laminarin, leucosin (an oil)	pectin, many with silicon dioxide
Dinoflagellata (dinoflagellates)	unicellular	chlorophylls <i>a</i> and <i>c</i> , carotenoids, xanthophyll, some with none	starch	cellulose
Chrysophyta (golden algae)	unicellular, colonial	chlorophylls <i>a</i> and <i>c</i> , carotenoids, fucoxanthin, xanthophyll, some with none	laminarin, leucosin	cellulose
Euglenophyta (euglenoids)	unicellular	chlorophylls <i>a</i> and <i>b</i> , carotenoids, many with none	paramylon (a starch)	no cell wall, protein-rich pellicle

PLANTLIKE PROTISTS

Plantlike protists are classified into seven phyla based on type of pigments, form of food storage, and cell-wall composition. The characteristics of the seven phyla are summarized in Table 25-2.

Phylum Chlorophyta (Green Algae)

The phylum Chlorophyta (klaw-RAHF-uh-tuh) contains more than 17,000 identified species of protists called *green algae*. Green algae have an amazing number of body forms, ranging from single cells and colonies to filamentous and multicellular forms.

Green algae share several characteristics with plants. Both green algae and plants have chlorophylls *a* and *b*. They both have carotenoids, which are **accessory pigments** that capture light energy and transfer it to chlorophyll *a*. Green algae and plants store food as starch and have cell walls made up of cellulose. The similarities between green algae and plants suggest that they may have had a common ancestor or that ancient green algae gave rise to land plants.

Most species of green algae are aquatic. Some species, such as the *Protococcus* shown in Figure 25-9, inhabit moist terrestrial environments, such as soil and tree trunks. Some green algae live as symbiotic partners with invertebrates, such as corals. Other green algae live with fungi as a part of organisms called *lichens*.

FIGURE 25-9

Many green algae, such as *Protococcus*, inhabit moist environments on land, such as the shady, often north-facing sides of tree trunks.





FIGURE 25-10

Giant kelp (*Macrocystis pyrifera*) forms kelp gardens along the Pacific coast and provides habitat for a rich variety of life, including crustaceans, fishes, sharks, and sea otters.

Word Roots and Origins

stipe

from the Latin *stipes*, meaning
“log” or “trunk of a tree”

FIGURE 25-11

Red algae, such as this *Corallina* species, are smaller than most brown algae, but members of both phyla are often referred to as *seaweeds*.



Phylum Phaeophyta (Brown Algae)

The phylum Phaeophyta (fee-AHF-uh-tuh) contains approximately 1,500 species of multicellular organisms called *brown algae*. Brown algae contain chlorophylls *a* and *c* and fucoxanthin (FYOO-koh-ZAN-thin), an accessory pigment that gives the algae their characteristic brown color. Brown algae store food as laminarin, a carbohydrate whose glucose units are linked differently than those in starch.

Brown algae are mostly marine organisms, and they include plantlike seaweeds and kelps. They are most common along rocky coasts where ocean water is cool. A few species of brown algae, such as *Sargassum*, can be found far offshore, where they form dense, floating mats.

Some of the largest algae known are classified in the phylum Phaeophyta. The large brown alga shown in Figure 25-10 is *Macrocystis pyrifera*, a giant kelp that thrives in intertidal zones and reaches 60 m in length. The thallus is anchored to the ocean bottom by a rootlike *holdfast*. The stemlike portion of the alga is called the *stipe*. And the leaflike region, modified to capture sunlight for photosynthesis, is the *blade*.

Phylum Rhodophyta (Red Algae)

The 4,000 species in the phylum Rhodophyta (roh-DAHf-uh-tuh) are known as *red algae*. Red algae contain chlorophyll *a* and accessory pigments called *phycobilins*. Phycobilins play an important role in absorbing light for photosynthesis. These pigments can absorb the wavelengths of light that penetrate deep into the water. As a result, phycobilins allow red algae to live at depths where algae lacking these pigments cannot survive. Some species of red algae have been found at depths of nearly 270 m, which is about three times deeper than organisms from any other algal phyla have been found.

A few species of red algae live in fresh water or on land, but most red algae are marine seaweeds. Despite their common name, not all red algae are reddish in appearance. The depth at which red algae live determines the amount of phycobilins they have. The *Corallina* alga shown in Figure 25-11 displays typical red algae pigmentation and body shape.

Phylum Bacillariophyta (Diatoms)

The phylum Bacillariophyta (BAS-uh-ler-ee-AHF-uh-tuh) contains as many as 100,000 species of unicellular protists called **diatoms**. Diatoms have cell walls, or **shells**, consisting of two pieces that fit together like a box and lid. Each half is called a *valve*, and the shells contain silicon dioxide. There are two basic types of diatoms. *Centric diatoms* have circular or triangular shells and are most abundant in marine environments. *Pennate diatoms* have rectangular shells and are most abundant in freshwater environments. Diatoms are an abundant component of phytoplankton and important producers in freshwater and marine food webs. In addition, diatoms release atmospheric oxygen.

Phylum Dinoflagellata (Dinoflagellates)

More than 2,000 species of organisms called *dinoflagellates* make up the phylum Dinoflagellata (DIE-noh-FLAJ-uh-LAH-tuh). Dinoflagellates, such as those shown in Figure 25-12, are small, usually unicellular organisms bearing two flagella of unequal length. The flagella are oriented perpendicular to each other. Dinoflagellates have cell walls made of cellulose plates that look like armor when seen under a microscope.

Most dinoflagellates are photosynthetic autotrophs, but a few species are colorless and heterotrophic. Photosynthetic dinoflagellates usually have a yellowish green to brown color due to large amounts of carotenoids, as well as chlorophylls *a* and *c*.

Some species of dinoflagellates, such as those in genus *Noctiluca*, can produce **bioluminescence**, the production of light by means of a chemical reaction in an organism. Other species produce toxins and red pigments. When the populations of these species explode, they turn the water brownish red, resulting in a phenomenon known as **red tide**. Red tides are fairly common in the Gulf of Mexico. Red tide toxins can kill large numbers of fish. When shellfish, such as oysters, feed on red tide dinoflagellates, they also consume the toxins, which are dangerous to humans who eat the shellfish.

Phylum Chrysophyta (Golden Algae)

The phylum Chrysophyta (kruh-SAHF-uh-tuh) contains about 1,000 species of *golden algae*. Most golden algae live in fresh water, but a few species are found in marine environments. Golden algae cells form highly resistant cysts that enable them to survive beneath the frozen surfaces of lakes during winter and dry lake beds during summer. Two flagella of unequal length are located at one end of each cell.

Most golden algae appear yellow or brown because of the presence of carotenoids. Golden algae also have chlorophylls *a* and *c* and store much of their surplus energy as oil. Golden algae likely played a role in the formation of petroleum deposits.

Phylum Euglenophyta (Euglenoids)

The phylum Euglenophyta (YOO-glueh-NAHF-uh-tuh) contains about 1,000 species of flagellated unicellular algae called **euglenoids**. Euglenoids are both plantlike and animal-like. Many are autotrophic, like plants, but they lack a cell wall and are highly motile, like animals. Euglenoids contain chlorophylls *a* and *b* and carotenoids. Most euglenoids live in fresh water, but a few occupy moist environments, such as soil or the digestive tracts of certain animals.

Euglena, shown in Figure 25-13, are abundant in fresh water, especially water polluted by excess nutrients. *Euglena* lack a cell wall, so they are flexible and can change their shape. *Euglena* have an elastic, transparent pellicle made of protein just beneath their cell membrane. Eyespots help *Euglena* sense their environment. *Euglena* also have a contractile vacuole that expels excess water. Although usually photosynthetic, *Euglena* raised in the dark do not form chloroplasts and become heterotrophs.

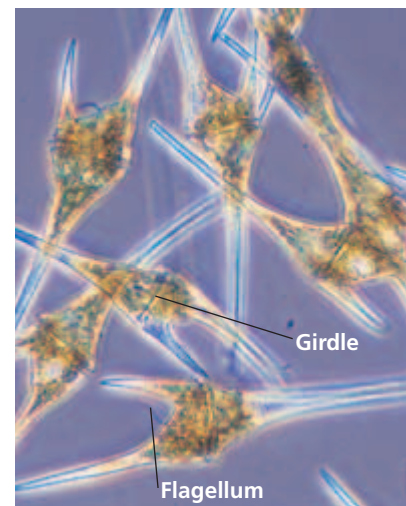
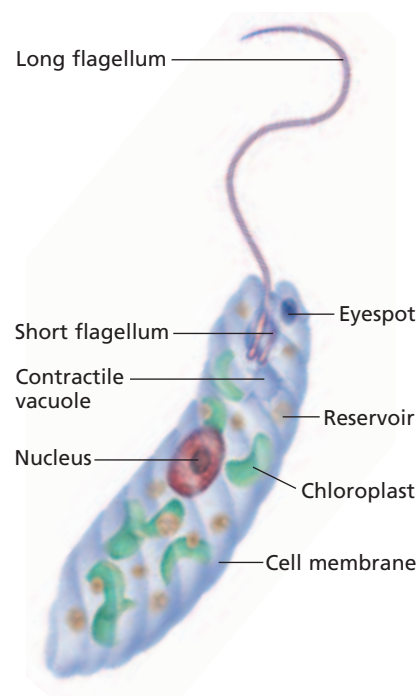


FIGURE 25-12

Flagella spin dinoflagellates, such as these members of genus *Ceratum*, through the water like tops. Only one of the two flagella is clearly visible in this picture. The second flagellum lies in the girdle at the middle of the organism. (450 $\times$)

FIGURE 25-13

Members of the genus *Euglena* contain chloroplasts like a plant but move around like an animal. *Euglena* are common in ponds. They are propelled by a long flagellum. An eyespot guides them toward light.



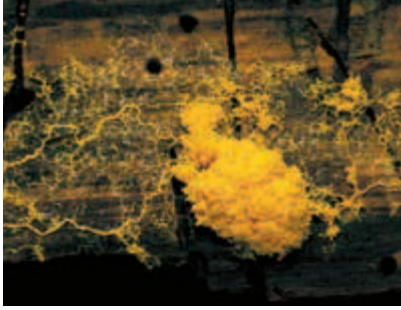


FIGURE 25-14

Slime molds often grow on decomposing organic matter, such as rotting logs. Slime molds consume small organisms and organic matter.

FUNGUSLIKE PROTISTS

Biologists recognize two groups of funguslike protists: slime molds and water molds. Slime molds are typically found on moist, decaying matter. They appear as glistening white, yellow, or red masses of slime, as shown in Figure 25-14. Slime molds have life cycles in which they spend part of their lives in a mobile, amoeba-like feeding form, consuming organic matter and bacteria much as protozoa do. However, these protists have a stationary reproductive stage in which they produce a funguslike, spore-bearing structure called a **fruiting body**, shown in Figure 25-15. A **water mold** is a funguslike protist composed of branching filaments of cells. Water molds are found mainly in bodies of fresh water but sometimes live in soil or as parasites.

Phylum Myxomycota (Plasmodial Slime Molds)

The phylum Myxomycota (MIKS-oh-mie-KOH-tuh) includes about 700 species of **plasmodial slime molds**. During the feeding stage of its life cycle, a plasmodial slime mold is a mass of cytoplasm, called a *plasmodium*, that may be as large as several square meters. Each plasmodium is multinucleate, containing many diploid nuclei that are not separated by cell walls. As the plasmodium creeps along the forest floor by cytoplasmic streaming, it consumes decaying leaves and other debris by phagocytosis.

When food or water is scarce, the plasmodium begins to reproduce. It forms stalked fruiting bodies in which haploid spores are produced by meiosis. Spores are resistant to harsh conditions. Under favorable conditions, spores crack open and give rise to haploid reproductive cells. Two such cells fuse, and their nuclei combine to form a diploid nucleus. Repeated mitotic divisions follow, but the cells do not undergo cytokinesis. The lack of cytokinesis results in the multinucleated plasmodium.

Phylum Dictyostelida (Cellular Slime Mold)

The phylum Dictyostelida (dik-TEE-oh-STEL-uh-duh) includes about 65 species of cellular slime molds. **Cellular slime molds** live as individual haploid cells that move about like amoebas. Each cell moves as an independent organism, creeping over the ground or swimming in fresh water and ingesting food.

When food or water becomes scarce, the cells release a chemical that attracts nearby cells, causing them to gather by the thousands into a dense structure called a *pseudoplasmodium*. A pseudoplasmodium is a coordinated colony of individual cells that resembles a slug and leaves a slimy trail as it crawls. Although the cells move as one unit, each cell retains its membrane and identity. Eventually, the pseudoplasmodium settles and forms fruiting bodies in which haploid spores develop. When a fruiting body breaks open, the spores are dispersed. Each spore may grow into an individual amoeboid cell, thus completing the life cycle.

FIGURE 25-15

Both plasmodial slime molds and cellular slime molds produce fruiting bodies that resemble those of fungi.





FIGURE 25-16

Some parasitic water molds attack aquatic organisms, such as fish, forming long filaments that eventually harm the fish.

Phylum Oomycota (Water Molds)

The phylum Oomycota (OH-mie-KOH-tuh) includes a number of organisms that are parasitic. For example, some oomycotes affect fish, as shown in Figure 25-16. Oomycotes reproduce asexually and sexually. During asexual reproduction, they produce flagellated zoospores. Zoospores germinate into threadlike cells. Some zoospores form a *zoosporangium*, which produces new zoospores. During sexual reproduction, water mold cells develop egg-containing and sperm-containing structures. Fertilization tubes grow between the two types of structures, enabling sperm to fertilize eggs, which forms diploid zygotes. A zygote grows into a new mass of filaments, which can again reproduce asexually or sexually.

Phylum Chytridiomycota (Water Molds)

Members of phylum Chytridiomycota (kie-TRID-ee-oh-mie-KOH-tuh), or the *chytrids* (KIE-tridz), are primarily aquatic protists characterized by gametes and zoospores with a single, posterior flagellum. Most chytrids are unicellular and parasitic.

Chytrids share many characteristics with fungi. They have similar means of obtaining nutrients, cell walls made of the same type of material, filamentous bodies, and similar enzymes and biochemical pathways. Because of these similarities, many biologists classify chytrids as fungi. Other biologists hypothesize that chytrids are a link between protists and fungi.



Quick Lab

Comparing Funguslike Protists

Materials paper and pencil

Procedure Create a chart that compares the phyla of funguslike protists. Include descriptions of cellular structure, locomotion, and means of reproduction.

Analysis Which funguslike protists are haploid? Which are diploid? Which funguslike protists reproduce sexually? Which reproduce asexually?

SECTION 3 REVIEW

1. What are four body forms of algae?
2. List seven phyla of plantlike protists.
3. Why do biologists theorize that ancient green algae gave rise to plants?
4. How are brown algae different from green and red algae?
5. What are the main characteristics of slime molds and water molds?
6. Compare plasmodial slime molds and cellular slime molds.

CRITICAL THINKING

7. **Applying Information** *Pseudo* means "false." Why do biologists call the migrating structure of a cellular slime mold a *pseudoplasmodium*?
8. **Identifying Relationships** Assume that chytrids are a link between protists and fungi, and draw a phylogenetic tree reflecting this information.
9. **Applying Information** An organism contains chlorophylls *a* and *c* and carotenoids, and it is unicellular. To which phylum does it belong? Explain your answer.

SECTION 4

OBJECTIVES

- **State** four environmental roles of protists.
- **Describe** algal blooms and red tides and their impact.
- **State** an important role for protists in research.
- **List** a use of protists as food and three uses of protist byproducts.
- **Describe** four protist-caused diseases.

VOCABULARY

algal bloom
chemotaxis
alginate
carrageenan
agar
diatomaceous earth
malaria
sporozoite
merozoite
gametocyte
giardiasis
cryptosporidiosis
trichomoniasis

PROTISTS AND HUMANS

A casual glance out the window usually will not include a sighting of protists, except perhaps for some green algae growing on a tree trunk or some slime collected at a pond's edge. Protists have a profound impact on human lives through their modification of the environment, their use in research and industry, and the harm they may cause to human health.

PROTISTS IN THE ENVIRONMENT

Protists are seldom apparent to the naked eye, but they play important roles in the environment. They produce large amounts of oxygen, form the foundation of food webs, recycle materials, and play a role in several symbiotic relationships.

Along with plants and photosynthetic bacteria, protists produce large quantities of atmospheric oxygen. Nearly every organism on Earth relies on this oxygen for *cellular respiration*, the process in which oxygen is used to obtain energy from organic molecules.

Photosynthetic protists are also critical components of marine and freshwater food webs. Photosynthetic protists produce carbohydrates. Larger protists and other aquatic organisms feed on these carbon-rich protists. For example, diatoms support large schools of anchovy, which sea lions eat before they, themselves, become the prey of orcas.

Protists play an important role in the carbon cycle. Photosynthetic protists use carbon dioxide and water from the environment to make carbohydrates, which are taken up by other living things or used by the protists for cellular processes. When protists use the carbohydrates, they release carbon dioxide and water back into the environment. Some protists, such as slime molds, are decomposers that aid in the cycling of other nutrients as well.

Protists form several important symbiotic relationships. For example, protists living in symbiosis with corals give the corals both their color and much of their carbon supply. Symbiosis between algae and fungi forms lichens. When lichens grow on newly exposed rocks, their secretions break down the rock. This process creates new soil that can support the growth of plants. Finally, protists found in the guts of some animals help the animals digest cellulose. For example, *Trichonympha* is one of the protists that help termites digest cellulose.



Ecology of Protists

Temperature and the availability of light and nutrients influence the growth of algal populations. Algae grow in the *photic zone*, the portion of a body of water through which light penetrates. Nitrate and phosphate availability usually limits algal growth rates. In the open ocean, the availability of iron ions may limit algal growth.

High water temperatures and nutrient concentrations can cause **algal blooms**, a vast increase in the concentration of diatoms and other photosynthetic protists. Algal blooms often occur during spring in bays and estuaries and frequently harm these environments. During an algal bloom, large numbers of dying protists sink to the bottom, where they decompose. Bacterial decomposers use up oxygen, which fish, crustaceans, and other sea life need to survive. Dropping oxygen levels can cause these larger species to die off.

Dinoflagellate blooms cause red tides, shown in Figure 25-17. Toxins produced by dinoflagellate blooms concentrate as they move up the food chain. These toxins can sicken or kill humans and other organisms that consume clams, oysters, or krill that have fed on the red tide dinoflagellates.



FIGURE 25-17

Because red tides produce dangerous toxins, people in some areas are often advised not to eat shellfish during the summer months, the time when red tides are most likely to happen.

PROTISTS IN RESEARCH

Research on protists has helped biologists understand a number of fundamental cellular functions, such as cell movement. For example, individual cells of the cellular slime mold *Dictyostelium discoideum* move toward a source of AMP, a precursor to the energy-storage molecule ATP. These cells crawl using amoeboid movement, shown in Figure 25-18.

Researchers are interested in learning how *Dictyostelium* cells can recognize AMP and crawl toward it, a process called **chemotaxis**. The researchers think that human leukocytes, or white blood cells, also perform chemotaxis as they crawl toward sites of infection. Biologists studying chemotaxis in human leukocytes have found the same proteins and cell movement seen in *Dictyostelium* cells. This discovery has improved the understanding of how leukocytes protect against disease. Some scientists believe this knowledge may help improve the treatment of diseases, such as cancers.



FIGURE 25-18

Dictyostelium uses amoeboid movement to reach a source of AMP. Understanding how *Dictyostelium* cells move improves the understanding of how human leukocytes move.

Eco Connection

Protist Biocontrol

Scientists from the United States Department of Agriculture are conducting field studies with the protist *Edhazardia aedis* to test its effectiveness as a control agent for disease-carrying mosquitoes. The chemical DDT was used previously, but it was banned in the 1970s after it was discovered to be harmful to bird populations. *E. aedis*, which was discovered in Argentina, infects mosquito larvae in the water and kills them. Scientists think that this protist will prevent disease while causing no environmental damage.

PROTISTS IN INDUSTRY

Humans make daily use of protists and their byproducts. Humans eat protists directly and use them as food additives, as fertilizers, and in cosmetics and medicines.

Protists as Food

For thousands of years, humans have been collecting seaweeds for food. People in Japan began cultivating seaweeds for food nearly 300 years ago. Today, about 10 percent of the Japanese diet is seaweeds. Among the most popular of these seaweeds is *nori*, a red alga from the genus *Porphyra*. Nori is used in soups, salads, and sushi, shown in Figure 25-19. More than half a million tons of nori are harvested each year. *Porphyra* has also been used for hundreds of years by people in Great Britain and is known there as *laverbread*. *Kombu*, made from kelps of the genus *Laminaria*, is another staple in Asian diets. More than 10 million tons of these kelps are harvested annually.

Protist Byproducts

The cell walls of most brown algae, especially large kelps, contain alginic acid. Alginic acid is a source of a commercially important polysaccharide called **alginate**. Alginate is used in cosmetics and various drugs and as a stabilizer in ice cream and salad dressings. Alginate is also used in textiles, water-soluble medical dressings for burns, and inks.

Certain species of red algae have cell walls that are coated with a sticky substance called **carrageenan** (KAR-uh-GEEN-uhn). Carrageenan is a polysaccharide that is used to control the texture of many commercial and food products. It is used in the production of cosmetics, gelatin capsules, and some types of cheese.

Agar is another polysaccharide that comes from the cell walls of red algae. Agar is commonly used in scientific research. It is the gel-forming base used for culturing microbes, such as bacteria. Agar is also widely used by the food industry in canned foods and bakery items.

Diatoms are quite abundant in aquatic ecosystems. As diatoms die, their shells sink and accumulate in large numbers at the bottom of lakes and oceans, forming a layer of material called **diatomaceous earth**. This slightly abrasive material is a component of many commercial products, such as detergents, paint removers, and toothpaste. Diatomaceous earth is also used in filters, and some people use it as a natural insecticide.

FIGURE 25-19

Sushi, a traditional food in Asian countries, has become more popular in the United States. The seaweeds used in sushi are high in protein, vitamins, and minerals.



PROTISTS AND HEALTH

Many protists are parasites. Table 25-3 summarizes the effects of three protist parasites.

Malaria

Apicomplexans from the genus *Plasmodium* cause **malaria**, which is characterized by severe chills, headache, fever, and fatigue. Each year, nearly 3 million people die from malaria. The four species of *Plasmodium* that infect humans all have life cycles involving an *Anopheles* mosquito. When an infected *Anopheles* mosquito bites a person, *Plasmodium* cells called **sporozoites** enter the bloodstream and infect the liver. New cells called **merozoites** emerge from the liver and infect red blood cells. At regular intervals, merozoites burst from red blood cells and release toxins, causing malarial symptoms. Without treatment, merozoites may remain in the liver for years, causing recurrent disease.

Some merozoites in the blood develop into the sexual forms of the parasite, **gametocytes**. When an *Anopheles* mosquito bites an infected person, the mosquito ingests these gametocytes. The gametocytes form gametes, which combine to form zygotes that develop into more sporozoites. When the insect bites another person, the life cycle begins again. Figure 25-20 shows the life cycle of *Plasmodium*.

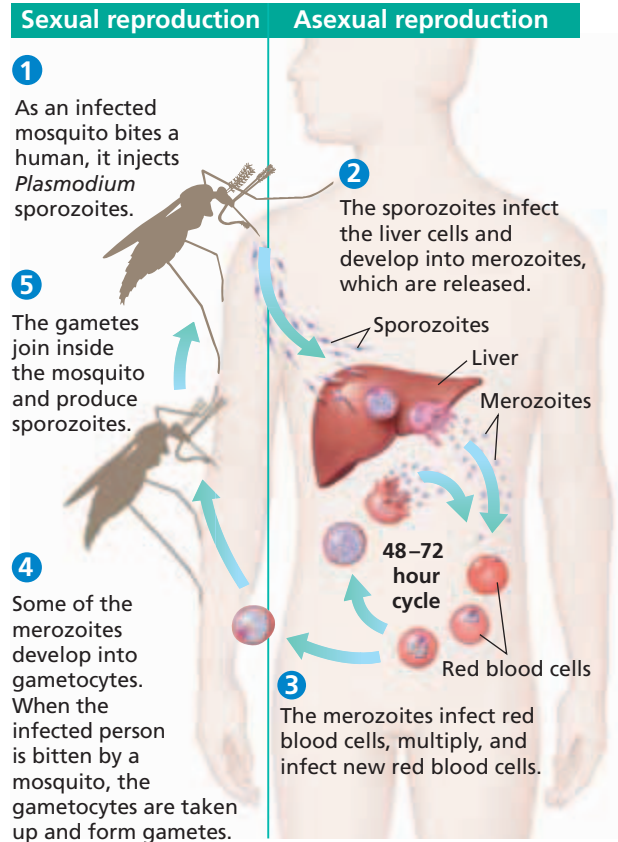


FIGURE 25-20

The life cycle of the malaria-causing *Plasmodium* requires two hosts.

TABLE 25-3 Examples of Disease-Causing Protists

Disease	Pathogen (Phylum)	Symptoms	Spread by	Precautions
Amebiasis	<i>Entamoeba histolytica</i> (Protozoa)	severe diarrhea, fever, and gastrointestinal tract hemorrhage	contaminated water or food	<i>E. histolytica</i> is common in developing countries. Drink bottled, boiled, or filter-purified water. Peel fresh fruit, and avoid unpasteurized dairy products.
Sleeping sickness	<i>Trypanosoma</i> sp. (Sarcomastigophora)	swollen lymph nodes, severe headaches, fever, fatigue, and coma	tsetse flies	Tsetse flies are found only in Africa. Wear protective clothing, such as long pants and long-sleeved shirts. Use insect repellent, and avoid open vehicles and bushes.
Leishmaniasis	<i>Leishmania donovani</i> (Sarcomastigophora)	skin sores, swollen glands, fever, and swollen spleen and liver	sand flies	Sand flies are found mostly in tropical and subtropical regions. Stay indoors during the day, when sand flies are most active. Wear protective clothing, and use insect repellents.



FIGURE 25-21

Giardia lamblia is a mastigophoran. The *Giardia* shown in this picture are shown in their natural habitat—the human intestine.

Giardiasis

Giardia lamblia, shown in Figure 25-21, causes an illness called **giardiasis** (JEE-ahr-DIE-uh-sis). Giardiasis is characterized by severe diarrhea and intestinal cramps. Cattle, beavers, and several other animals carry the parasite and contaminate water with their feces. Hikers and other people who are likely to drink this contaminated water are susceptible to giardiasis. Thousands of cases occur annually in the United States. The disease usually is not fatal, and drugs aid recovery. Drinking bottled, boiled, or filtered water prevents giardiasis.

Cryptosporidiosis

Cryptosporidiosis (KRIP-toh-spawr-I-dee-OH-sis) is characterized by diarrhea and is most often caused by *Cryptosporidium parvum*, an apicomplexan that lives on the surface of cells lining the small intestine. A stage called the *oocyst* (OH-sist) can pass in feces from infected animals to humans. An outer shell protects the parasite outside the body, allowing it to survive for long periods and making it resistant to chlorine disinfectants. Cryptosporidiosis is usually not fatal, but it can be dangerous for people with immune disorders. Washing hands thoroughly after coming in contact with feces and avoiding untreated water can prevent cryptosporidiosis.

Trichomoniasis

The mastigophoran *Trichomonas vaginalis* is responsible for **trichomoniasis** (TRIK-oh-moh-NIE-uh-sis), a sexually transmitted disease. The parasite does not survive well outside the body, so it is mainly spread by sexual contact. Trichomoniasis is one of the most common sexually transmitted diseases.

Many people do not have symptoms when infected with trichomoniasis. However, women are more likely to have symptoms of trichomoniasis than men are. Symptoms include discolored discharge, genital itching, and the urge to urinate. Abstinence from sexual intercourse prevents trichomoniasis.

SECTION 4 REVIEW

1. Outline three environmental roles of protist producers.
2. What are three examples of protist symbioses?
3. Why do researchers use slime molds to study chemotaxis?
4. How are protists used for food?
5. Identify three protist byproducts and how people use them.
6. Describe four protist-caused diseases, and name the organisms that cause the diseases.

CRITICAL THINKING

7. **Applying Information** What basic precautions can decrease exposure to protist diseases?
8. **Making Predictions** Some scientists hypothesize that global warming will increase mosquito populations. Predict what might happen to rates of malaria as a result.
9. **Calculating Information** If 500 million people contract malaria annually and 6.5 billion people live on Earth, what fraction of the human population becomes infected with malaria annually?

CHAPTER HIGHLIGHTS

SECTION 1

Characteristics of Protists

- Protists are unicellular or simple multicellular eukaryotic organisms that are not plants, fungi, or animals. Protists are classified by the characteristics that make them funguslike, plantlike, or animal-like.
- Evidence suggests that the first protists arose from endosymbiotic prokaryotes.
- Many protists are autotrophs. Other protists are heterotrophs. Protists use flagella, cilia, or pseudopodia for locomotion.
- Protists reproduce either asexually, sexually, or both. They reproduce asexually by binary fission or multiple fission. They often reproduce sexually by conjugation.

Vocabulary

protist (p. 501)

binary fission (p. 504)

multiple fission (p. 504)

conjugation (p. 504)

SECTION 2

Animal-like Protists

- Animal-like protists include the phyla Protozoa, Ciliophora, Sarcomastigophora, and Apicomplexa. For locomotion, protozoa use pseudopodia, ciliates use cilia, and sarcomastigophorans use flagella.
- Ciliates reproduce asexually by binary fission and sexually by conjugation.
- Apicomplexans lack locomotion, so they are parasites.

Vocabulary

pseudopodium (p. 506)

amoeboid movement (p. 506)

test (p. 507)

cilium (p. 507)

pellicle (p. 508)

oral groove (p. 508)

mouth pore (p. 508)

gullet (p. 508)

anal pore (p. 508)

contractile vacuole (p. 508)

macronucleus (p. 508)

miconucleus (p. 508)

flagellum (p. 508)

SECTION 3

Plantlike and Funguslike Protists

- Algae can be unicellular, colonial, filamentous, or multicellular. Seven phyla of plantlike protists are Chlorophyta, Phaeophyta, Rhodophyta, Bacillariophyta, Dinoflagellata, Chrysophyta, and Euglenophyta.
- Green algae and plants have the same pigments, store food as starch, and have cell walls made of cellulose.
- Slime molds are found on decaying matter and spend part of their lives as mobile feeding forms and the other part as stationary reproductive forms.
- Plasmodial slime molds are multinucleate, and cellular slime molds live as individual haploid cells. Water molds are composed of branching filaments.

Vocabulary

alga (p. 510)

gametangium (p. 510)

phytoplankton (p. 510)

thallus (p. 510)

accessory pigment (p. 511)

diatom (p. 512)

shell (p. 512)

bioluminescence (p. 513)

red tide (p. 513)

euglenoid (p. 513)

fruiting body (p. 514)

water mold (p. 514)

plasmodial slime mold

(p. 514)

cellular slime mold (p. 514)

SECTION 4

Protists and Humans

- Protists produce large amounts of oxygen, form the foundation of food webs, recycle materials, and play a role in several symbiotic relationships.
- Algal blooms can lead to the depletion of oxygen in water. Red tides produce harmful toxins.
- Protists can help scientists understand the movement of leukocytes, provide food, and provide important byproducts, such as algininate, carrageenan, and agar.
- Parasitic protists cause malaria, giardiasis, cryptosporidiosis, and trichomoniasis in humans.

Vocabulary

algal bloom (p. 517)

chemotaxis (p. 517)

algininate (p. 518)

carrageenan (p. 518)

agar (p. 518)

diatomaceous earth (p. 518)

malaria (p. 519)

sporozoite (p. 519)

merozoite (p. 519)

gametocyte (p. 519)

giardiasis (p. 520)

cryptosporidiosis (p. 520)

trichomoniasis (p. 520)

CHAPTER REVIEW

USING VOCABULARY

1. Choose the term that does not belong in the following group, and explain why it does not belong: *binary fission*, *macronucleus*, *multiple fission*, and *conjugation*.
2. For each pair of terms, explain the relationship between the terms.
 - a. *pseudopodia* and *amoeboid movement*
 - b. *macronucleus* and *micronucleus*
 - c. *algae* and *thallus*
 - d. *sporozoites* and *merozoites*
3. Use each of the following terms in a separate sentence: *algal bloom*, *red tide*, *fruiting body*, and *pellicle*.
4. **Word Roots and Origins** The word *diatom* comes from the Greek word *diatomos*, which means “cut in half.” Using this information, explain how *diatom* is an appropriate term for the organisms it describes.

UNDERSTANDING KEY CONCEPTS

5. **Define** *protist*.
6. **Describe** a hypothesis for the origin of eukaryotic cells.
7. **State** the way in which protists are generally classified.
8. **Compare** two ways by which protists obtain food and three ways by which they move.
9. **List** three ways by which protists reproduce.
10. **Differentiate** between Protozoa, Ciliophora, Sarcomastigophora, and Apicomplexa.
11. **Explain** how protozoa use pseudopodia to move and capture food.
12. **Relate** how ciliates reproduce.
13. **Compare** movement in sarcodines, ciliates, and mastigophorans.
14. **Identify** the relationship between apicomplexans and disease.
15. **List** four body forms of algae.
16. **State** the common name for each of the seven phyla of plantlike protists.
17. **Compare** green algae and plants.
18. **Summarize** the main characteristics of the four phyla of funguslike protists.
19. **Describe** plasmodial slime molds and cellular slime molds.

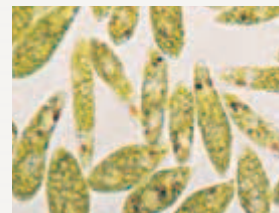
20. **Describe** four roles of protists in the environment.
21. **Describe** algal blooms and red tides.
22. **Name** an important role for protists in research.
23. **List** four uses of protists and protist byproducts.
24. **Summarize** the characteristics of four protist-caused diseases.
25. **CONCEPT MAPPING** Use the following terms to create a concept map that describes the characteristics of animal-like protists: *Protozoa*, *amoeboid movement*, *pseudopodia*, *Ciliophora*, *flagella*, *Sarcomastigophora*, *cilia*, *cytoplasmic streaming*, *test*, *pellicle*, *parasites*, and *Apicomplexa*.

CRITICAL THINKING

26. **Relating Concepts** Health officials advise that people should not eat shellfish from certain areas during the summer months. Explain the reason for this warning.
27. **Analyzing Data** Scientists are trying to develop a vaccine against malaria. Because malaria has several life stages, scientists must decide which life stage should be the target of the vaccine. Some scientists are trying to develop a vaccine against gametocytes as a way of controlling malaria. If they are successful in developing this vaccine, how will it help people living in areas where malaria regularly occurs? Explain your answer.
28. **Making Comparisons** How is the movement of *Dictyostelium* toward AMP similar to the movement of leukocytes? How is it different? Based on this information, is *Dictyostelium* a good model for leukocytes? Justify your answer.
29. **Interpreting Graphics** A scientist found two different euglenoids. The scientist concluded that specimen a is exclusively heterotrophic but specimen b is not. Examine the images of the specimens below, and determine if the scientist is correct in his or her conclusions.



(a)



(b)



Standardized Test Preparation

DIRECTIONS: Choose the letter of the answer choice that best answers the question.

- Most scientists believe that protists evolved from which of the following?
 - fungi
 - plants
 - euglenoids
 - prokaryotes
- Protist habitats are often characterized by the presence of which of the following?
 - soil
 - algae
 - blood
 - moisture
- Flagella are characteristic of members of which phylum?
 - Protozoa
 - Ciliophora
 - Apicomplexa
 - Sarcomastigophora

INTERPRETING GRAPHICS: The table below shows cases of amebiasis and malaria in the United States between 1986 and 1994. Use the table to answer the questions that follow.

Protist-Caused Diseases in the United States, 1986–1994

Year	Number of cases	
	Amebiasis	Malaria
1986	3,532	1,123
1988	2,860	1,099
1990	3,328	1,292
1992	2,942	1,087
1994	2,983	1,229

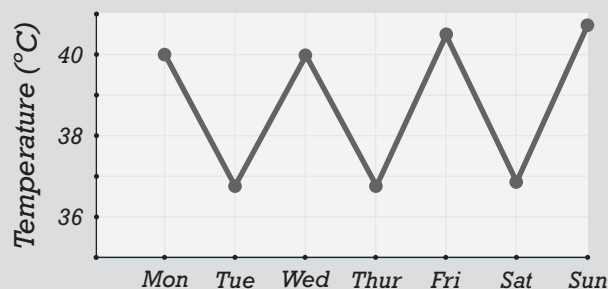
- How many people had malaria in 1992?
 - 1,087
 - 1,229
 - 1,292
 - 2,942
- How did the number of cases of amebiasis change between 1986 and 1994?
 - The number of cases increased.
 - The number of cases decreased.
 - The number of cases stayed the same.
 - The number of cases increased, then decreased.

DIRECTIONS: Complete the following analogy.

- Bacillariophyta : autotrophs :: Apicomplexa :
 - cilia
 - flagella
 - parasites
 - plasmodium

INTERPRETING GRAPHICS: The graph below shows the cycle of fever in a malaria patient. Use the graph to answer the question that follows.

Cycle of Fever in a Malaria Patient



- In this patient, how often does the cycle of fever repeat?
 - every 12 h
 - every 24 h
 - every 48 h
 - every 96 h

SHORT RESPONSE

Anopheles mosquitoes require water to breed.

What would happen to malaria cases during a dry season and during a wet season?

EXTENDED RESPONSE

A scientist wants to examine the effect of fertilizer on algal blooms. In the laboratory, the scientist adds increasing amounts of fertilizer to three samples of pond water and adds no fertilizer to a fourth sample of pond water.

Part A Which samples will show increased algal growth? Explain your answer.

Part B How can the scientist apply his or her laboratory results to a natural ecosystem? Compare the scientist's experiment to a natural ecosystem, such as a pond.

Test TIP

For a question involving experimental data, determine the constants, variables, and control **before** answering the question.

Classifying Green Algae

OBJECTIVES

- Observe live specimens of green algae.
- Compare unicellular, colonial, filamentous, and multicellular green algae.
- Classify genera of colonial green algae.

PROCESS SKILLS

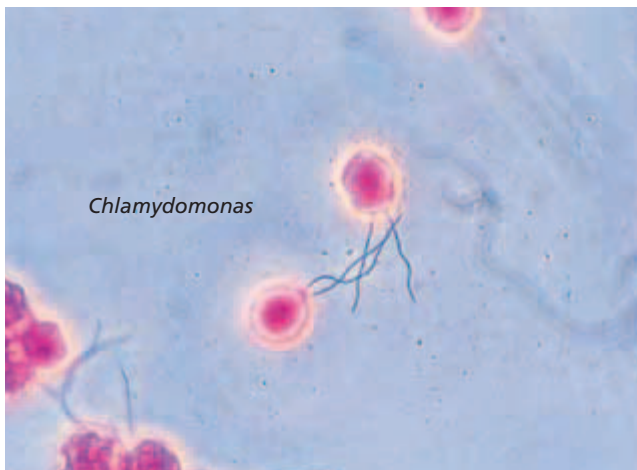
- observing
- comparing and contrasting
- classifying
- inferring

MATERIALS

- culture of *Chlamydomonas*
- colonial green algae culture (*Chlamydomonas*, *Eudorina*, *Gonium*, *Pandorina*, *Volvox*, and *Hydrodictyon*)
- culture of *Spirogyra*
- 3 depression slides
- 3 coverslips
- 3 medicine droppers
- compound light microscope

Background

1. Distinguish between the terms *protozoa* and *algae*.
2. Green algae are either unicellular, colonial, filamentous, or multicellular.
3. Explain why green plants are thought to have evolved from green algae.



4. List characteristics of green algae. Include ways that algae differ from plants.
5. How do green algae differ from other algae?

PART A Observing Unicellular Green Algae

1. Make a table similar to the one below in your lab report. Allow substantial space in your data table for labeled sketches of the different kinds of green algae you will view in this investigation. Use your data table to record your observations of each kind of green algae that you view.
2.  **CAUTION** Slides break easily. Use caution when handling them. Prepare a wet mount of the *Chlamydomonas* culture by placing a drop of the culture on a microscope slide with a medicine dropper and placing a coverslip on top of the specimen.

OBSERVATIONS OF GREEN ALGAE

Genus	Sketch of organism	Type of green algae (unicellular, colonial, filamentous, or multicellular)

- Examine the slide of *Chlamydomonas*, first under low power and then under high power.
- In your lab report, make a sketch of *Chlamydomonas*. Label the cell wall, flagella, nucleus, and chloroplasts if they are visible.



PART B Observing and Identifying Colonial Green Algae

- CAUTION** Slides break easily. Use caution when handling them. Prepare a wet mount of the colonial green algae culture using a clean medicine dropper, a clean microscope slide, and a clean coverslip.
- Examine the slide of mixed colonial green algae under low power, switching to the high-power setting as needed for clarity. Some of the organisms on your slide should resemble the photograph of *Volvox* above. How many different kinds of colonial green algae can you find?
- In your data table, draw each type of colonial green algae you observe. How are these algae different in appearance from *Chlamydomonas*?
- Use the dichotomous key, above right, to identify each type of colonial green algae. Choose an alga, and start at 1. Decide which choice best describes the alga. If the alga is not identified, go to the next number and make the next decision. Continue until you identify the alga. Then, label your drawing of the alga.

PART C Observing Filamentous Green Algae

- CAUTION** Slides break easily. Use caution when handling them. Prepare a wet mount of

DICHOTOMOUS KEY FOR COLONIAL GREEN ALGAE

1a	single cells	<i>Chlamydomonas</i>
1b	colony of cells	go to 2
2a	flattened or netlike colony	go to 3
2b	round colony	go to 4
3a	netlike colony	<i>Hydrodictyon</i>
3b	flattened colony	<i>Gonium</i>
4a	more than 100 cells in colony	<i>Volvox</i>
4b	fewer than 100 cells in colony	go to 5
5a	cells close together	<i>Pandorina</i>
5b	cells apart from each other	<i>Eudorina</i>

the *Spirogyra* culture using a clean medicine dropper, a clean microscope slide, and a clean coverslip.

- Examine *Spirogyra* under low power. Switch to the high-power setting if you need more magnification to see the specimen clearly. How does *Spirogyra* differ in appearance from the other kinds of algae you have observed in this investigation?
- In your data table, make a sketch of *Spirogyra*. Label the filaments and the individual cells in your drawing. Also, label the spiral-shaped chloroplasts if they are visible.
- Clean up your lab materials, and wash your hands before leaving the lab.

Analysis and Conclusions

- What characteristics did all of the algae you viewed have in common?
- Describe examples of specialization in the different kinds of algae you viewed. In which types of algae do some cells depend on others?
- What differences did you observe between small and large colonies of algae?
- Which specializations in algae are characteristic of green plants?

Further Inquiry

How might you search for evidence of evolutionary relationships among algae and between algae and green plants?



Some fungi—such as the fungus *Amanita muscaria*, commonly known as *fly agaric*—are important mutualists in nature. Other fungi are important decomposers of dead organic matter. Although beautiful, *A. muscaria* is poisonous if eaten.

SECTION 1 *Overview of Fungi*

SECTION 2 *Classification of Fungi*

SECTION 3 *Fungi and Humans*

OVERVIEW OF FUNGI

Fungi are in their own kingdom—kingdom Fungi. They differ from other organisms in many ways, including structure, method of reproduction, and methods of obtaining nutrients.

CHARACTERISTICS

Fungi are eukaryotic, nonphotosynthetic organisms, and most are multicellular heterotrophs. Most fungi are microscopic molds or yeasts. **Molds**, such as the fungi that grow on bread and oranges, are tangled masses of filaments of cells. **Yeasts** are unicellular fungi whose colonies resemble those of bacteria. Yeasts are best known as the microorganisms that make bread rise. The study of fungi is called **mycology** (mie-KAHL-uh-jee).

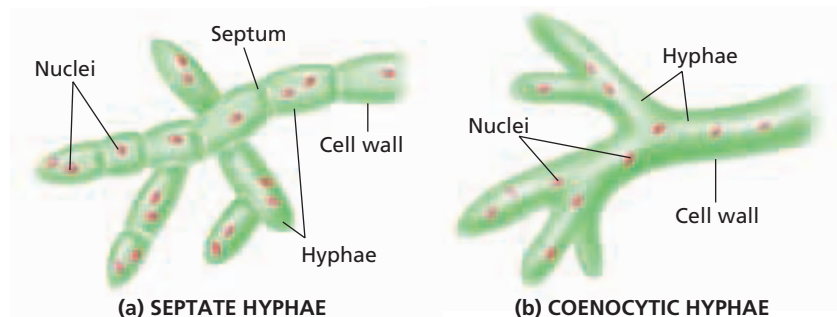
Obtaining Nutrients

Fungi get their nutrients by absorbing organic molecules from their environment. Fungi also secrete enzymes into their food and then absorb the digested nutrients through their cell walls. Like animals, fungi store energy in the form of glycogen. Most fungi are *saprophytic*—that is, they live on organic compounds that they absorb from dead organisms in the environment. This characteristic makes fungi a very important recycler of organic material in nature.

Structure of Fungi

Filaments of fungi are called **hyphae** (HIE-fee) (singular, *hypha*). The cell walls of hyphae contain **chitin** (KIE-tin), a polysaccharide that also makes up the exoskeleton of insects, crustaceans, and other arthropods. The presence of chitin distinguishes cell walls of fungi from those of plants, which have cellulose but no chitin.

A mat of hyphae that forms the body of a fungus is a **mycelium** (mie-SEE-lee-uhm). In some species, the cells that make up hyphae are divided by cross sections called **septa** (SEP-tuh) (singular, *septum*). Hyphae whose cells are divided by septa are called *septate hyphae*.



OBJECTIVES

- List the characteristics of fungi.
- Describe how fungi obtain nutrients.
- Distinguish between hyphae and a mycelium.
- Compare the ways fungi reproduce.
- Describe one hypothesis about the origin of fungi.

VOCABULARY

mold
yeast
mycology
hypha
chitin
mycelium
septum
coenocyte
dimorphism
sporangiophore
sporangium
sporangiospore
conidium
conidiophore
fragmentation
budding

FIGURE 26-1

The hyphae of some fungi have separating walls called *septa* (a). The hyphae of other fungi do not have septa and are called *coenocytic hyphae* (b).



FIGURE 26-2

This light micrograph shows the dimorphic fungus *Paracoccidioides brasiliensis* as it changes from its unicellular yeast form at 37°C (98.6°F) to a mycelial form at 25°C (77°F).

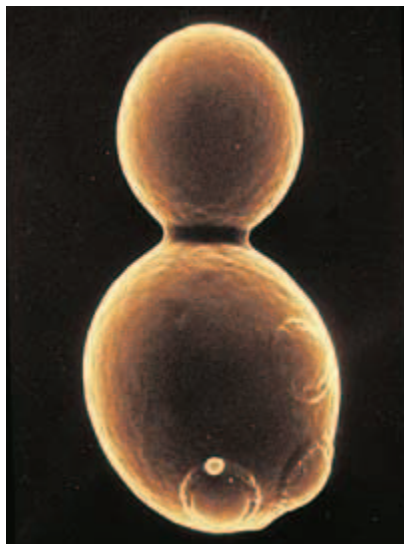
Word Roots and Origins

dimorphic

from the Greek *dis*, meaning "twice," and *morphe*, meaning "shape"

FIGURE 26-3

Fungi reproduce asexually in many ways. Yeasts, such as *Saccharomyces cerevisiae*, form buds that eventually pinch off to produce new cells.



Fungal species that have hyphae lacking septa are called **coenocytes** (SEE-noh-SIETS). The general structures of septate and nonseptate hyphae are shown in Figure 26-1. Hyphae increase in length by cellular growth and division at the tip. As the hyphae grow, the size of the mycelium increases. When hyphae encounter organic matter, such as a tree trunk or dead animal, they secrete digestive enzymes and then absorb the digested nutrients. Fungi range in size from the microscopic yeast to the largest single organism in the world. Fungi of the genus *Armillaria*, or honey mushrooms, can occupy an area of up to 800 hectares (3.1 mi²).

Several species of fungi can change form in response to changes in their environment. For example, *Histoplasma capsulatum*, which causes a severe disease in humans that can resemble tuberculosis, normally grows as a mycelium in soil. However, when it invades a human, the increased temperature and available nutrients cause the fungus to grow unicellularly as a yeast. This ability to exist in two different forms, demonstrated in Figure 26-2, is called **dimorphism** (die-MAWR-FIZ-uhm).

REPRODUCTION

Most fungi reproduce both sexually and asexually. Unlike most eukaryotes, most fungi are haploid throughout most of their life cycle.

Asexual Reproduction

Asexually, fungi produce thousands of genetically identical haploid spores, usually on modified cells of the hyphae. When these spores are placed in favorable environmental conditions, they germinate and grow new hyphae, each of which can form a mycelium and produce thousands of new asexual spores.

A variety of asexual spores are formed by different fungi. For example, **sporangiophores** (spoh-RAN-jee-oh-FAWHRZ) are specialized hyphae that look like upright stalks. On top of a sporangiophore is a sac called a **sporangium** (spoh-RAN-jee-UHM). Inside each sporangium, spores called **sporangiospores** (spoh-RAN-jee-oh-SPOHRZ) are made. *Rhizopus*, the black mycelial fungus that is commonly found growing on bread, is a sporangiospore-forming fungus.

Other fungi form spores called **conidia** (koh-NID-ee-uh) (singular, *conidium*), which are formed without the protection of a sac. Conidia are formed on top of a stalklike structure called a **conidiophore** (koh-NID-ee-uh-FAWHR). *Penicillium*, which is used to produce penicillin and certain types of cheeses, is a fungus that reproduces asexually by means of conidia.

Asexual reproduction may also occur by **fragmentation**. In this process, a septate hypha dries and shatters, releasing individual cells that act as spores. The fungus that causes athlete's foot reproduces this way. Yeasts reproduce by a process called budding, shown in Figure 26-3. **Budding** is an asexual process in which part of a yeast cell pinches itself off to produce a small offspring cell.

TABLE 26-1 Three Phyla of Fungi

Phylum and number of species	Structure	Asexual reproduction	Sexual reproduction (where identified)	Examples
Zygomycota 600 species	coenocytic hyphae	spores from sporangia	conjugation results in zygospores	<i>Mucor</i> , <i>Rhizopus</i>
Basidiomycota 25,000 species	septate hyphae	rare (conidia in a few species)	basidia produce basidiospores	<i>Puccinia</i> , <i>Ustilago</i> (mushrooms)
Ascomycota 60,000 species	septate or unicellular hyphae	conidia, budding	asci produce ascospores	yeasts, morels, <i>Penicillium</i> species

Sexual Reproduction

Many, but not all, species of fungi are also able to reproduce sexually. Fungi are neither male nor female. Instead, they occur in mating types that are sometimes called *minus* and *plus*. When two different mating types of the same species encounter one another, the hyphae of one mating type fuse with the hyphae of the opposite mating type. These fused hyphae give rise to a specialized structure, which produces and scatters genetically diverse spores.

The ability of some fungi to reproduce both sexually and asexually provides an adaptive advantage. When the environment is favorable, rapid asexual reproduction can ensure an increased spread of the species. During environmental stress, sexual reproduction can ensure genetic diversity, increasing the likelihood that offspring will be better adapted to the new environmental conditions.

Eco Connection

Fungi in the Food Chain

The northern spotted owl depends indirectly on a forest fungus for its survival. The owls prey on northern flying squirrels, and the squirrels depend on truffles for the bulk of their diet. Truffles are the sexual reproductive structure of some fungal species.

EVOLUTION

The first fungi were probably unicellular organisms that might have clung together after mitosis. Biologists hypothesize that fungi colonized land at about the same time that early plants did. They reason that fungi, like other eukaryotes, arose from endosymbiotic prokaryotes. According to the fossil record, fungi first appeared about 460 million years ago. Three phyla of modern fungi are listed in Table 26-1.



SECTION 1 REVIEW

1. Name three characteristics found in all fungi.
2. Describe how fungi obtain nutrients.
3. What is the difference between hyphae and a mycelium?
4. Identify three forms of asexual reproduction in fungi.
5. Describe one hypothesis about the origin of fungi.

CRITICAL THINKING

6. **Analyzing Concepts** How does being saprophytic make fungi important recyclers in nature?
7. **Making Inferences** Explain how fungi could have adapted to living on land *before* plants did.
8. **Evaluating Information** How can both sexual and asexual reproduction have an adaptive advantage over just asexual reproduction?

SECTION 2

OBJECTIVES

- **List** characteristics that distinguish three phyla of fungi.
- **Compare** the life cycles of zygomycetes, basidiomycetes, and ascomycetes.
- **Distinguish** between mycorrhizae and lichens.
- **Explain** the importance of mycorrhizae and lichens to the environment.

VOCABULARY

rhizoid
stolon
gametangium
zygosporangium
basidium
basidiocarp
basidiospore
ascogonium
antheridium
ascocarp
ascus
ascospore
mycorrhiza
lichen

CLASSIFICATION OF FUNGI

Fungi are classified in three phyla. Traditionally, fungi have been classified according to their structure and form of sexual reproduction. Though no longer the sole bases of classification, these characteristics are still useful in identifying fungi.

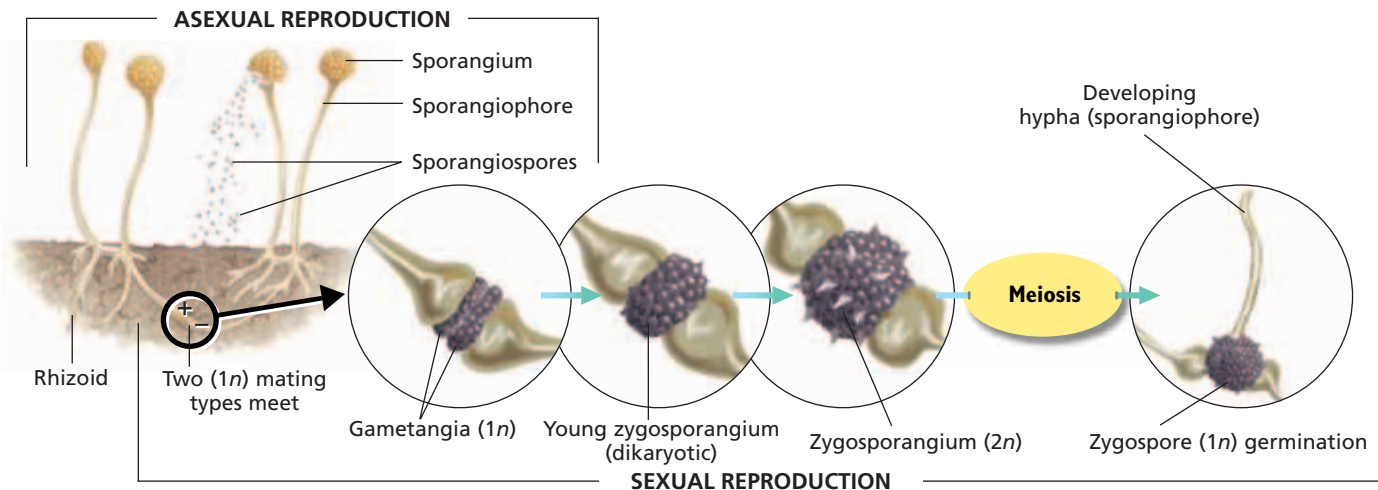
PHYLUM ZYGOMYCOTA

Most species in the phylum Zygomycota (ZIE-goh-mie-KOH-tuh) are terrestrial organisms found primarily in soil that is rich in organic matter. The hyphae of zygomycetes are coenocytic. *Rhizopus stolonifer*, the common bread fungus shown in Figure 26-4, is a zygomycete. The hyphae that anchor the mold to the surface of the bread and that penetrate the bread's surface are called **rhizoids** (RIE-ZOYDZ). Digestive enzymes produced by rhizoids break down the bread to release nutrients. Other hyphae, called **stolons** (STOH-LAHNZ), grow across the surface of the bread.

Sexual reproduction in zygomycetes is called *conjugation*. Conjugation in fungi occurs when the hyphae of two compatible mating types (parents) meet. Short branches form on the hyphae of both types and grow outward until they touch each other. A septum forms near the tip of each branch, and a cell called a gametangium develops. A **gametangium** (GAM-uh-TAN-jee-uhm) is a sexual reproductive structure that contains a nucleus of a mating type. The nucleus within each parent's gametangium divides several times. When the gametangia fuse, the nuclei mix and form into pairs, each pair containing a nucleus from each mating type.

FIGURE 26-4

During sexual reproduction zygomycetes produce gametangia. The nuclei from both mating types fuse and eventually produce genetically diverse spores.



Fused gametangia form a structure called a **zygosporangium** (ZIE-goh-spoh-RAN-jee-uhm), which contains many diploid nuclei. The zygosporangium then forms a thick wall and becomes dormant. Germination depends on environmental conditions. A sporangio-phore grows from the diploid zygosporangium and produces a sporangium, which releases numerous haploid spores.

PHYLUM BASIDIOMYCOTA

Basidiomycetes (buh-SID-ee-oh-MIE-SEETS) are often called *club fungi* because during sexual reproduction, they produce small, clublike reproductive structures called **basidia** (buh-SID-ee-uh) (singular, *basidium*). Sexual reproduction in basidiomycetes is shown in Figure 26-5. The spore-bearing structure of basidiomycetes is an aboveground structure called the **basidiocarp** (buh-SID-ee-oh-KAHRP). Mushrooms are basidiocarps. The basidiocarp consists of a stem called a *stalk* and a flattened structure known as a *cap*. On the underside of the cap are rows of *gills* that radiate out from the center. As shown in step 1, each gill is lined with thousands of dikaryotic (die-KAR-ee-AHT-ik) basidia. Dikaryotic cells contain two nuclei. Step 2 shows that in each basidium, two nuclei fuse to form a zygote ($2n$). In step 3, the zygote undergoes meiosis to form four haploid nuclei. These develop into four **basidiospores** (buh-SID-ee-oh-SPAWRZ), which are released into the air. As shown in step 4, under favorable environmental conditions, basidiospores germinate to produce haploid mycelia that grow underground. In steps 5 and 6, when compatible mating types encounter one another, their hyphae fuse and form a basidiocarp, which emerges aboveground.



Quick Lab

Dissecting a Mushroom

Materials mushroom, scalpel, hand lens, slides, coverslips, water, microscope



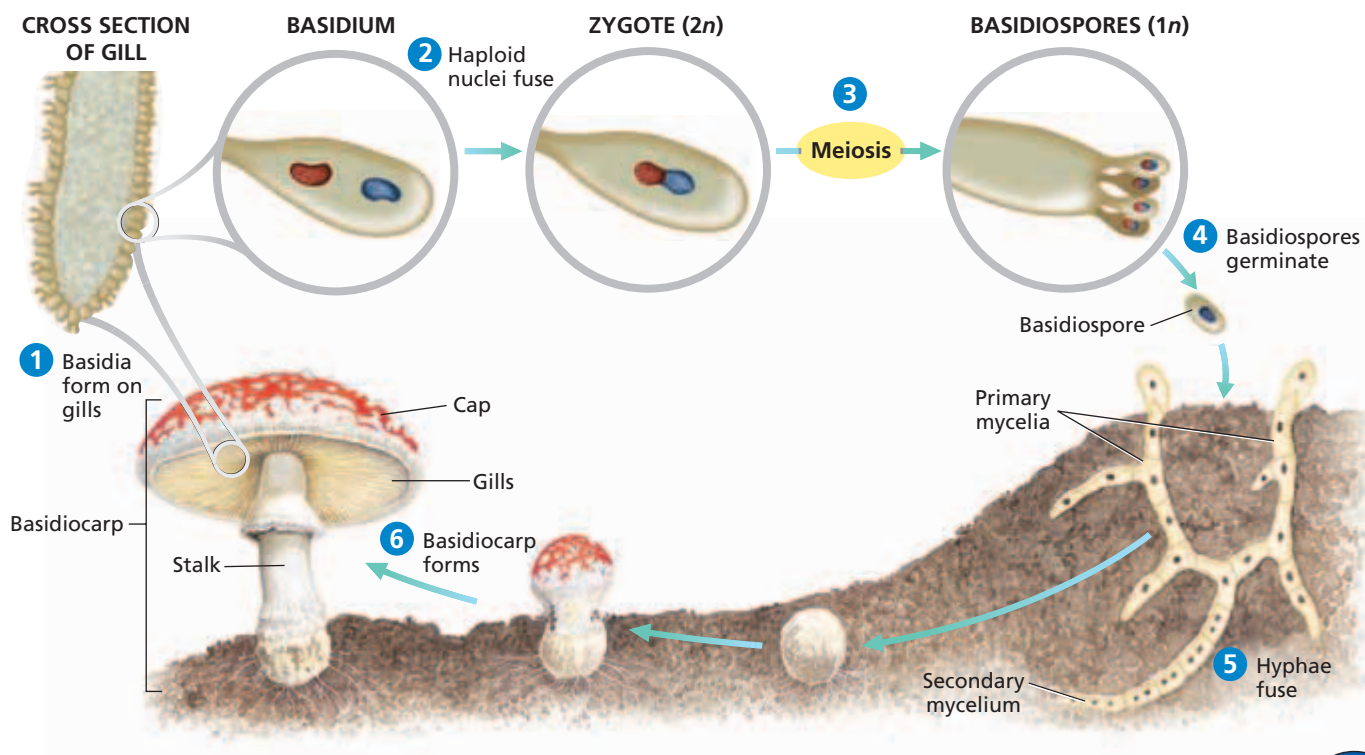
Procedure

1. Before beginning, tell your teacher if you are allergic to fungi. Cut the cap of a mushroom lengthwise.
2. Examine the mushroom with a hand lens. Draw and write down your observations.
3. Make a wet mount of the hyphae. Use a microscope to observe the cell structure of the hyphae.

Analysis What structures did you observe with the hand lens? with the microscope? What is the function of each structure?

FIGURE 26-5

Basidiomycetes usually reproduce sexually by means of a fruiting body, also called a *mushroom*.



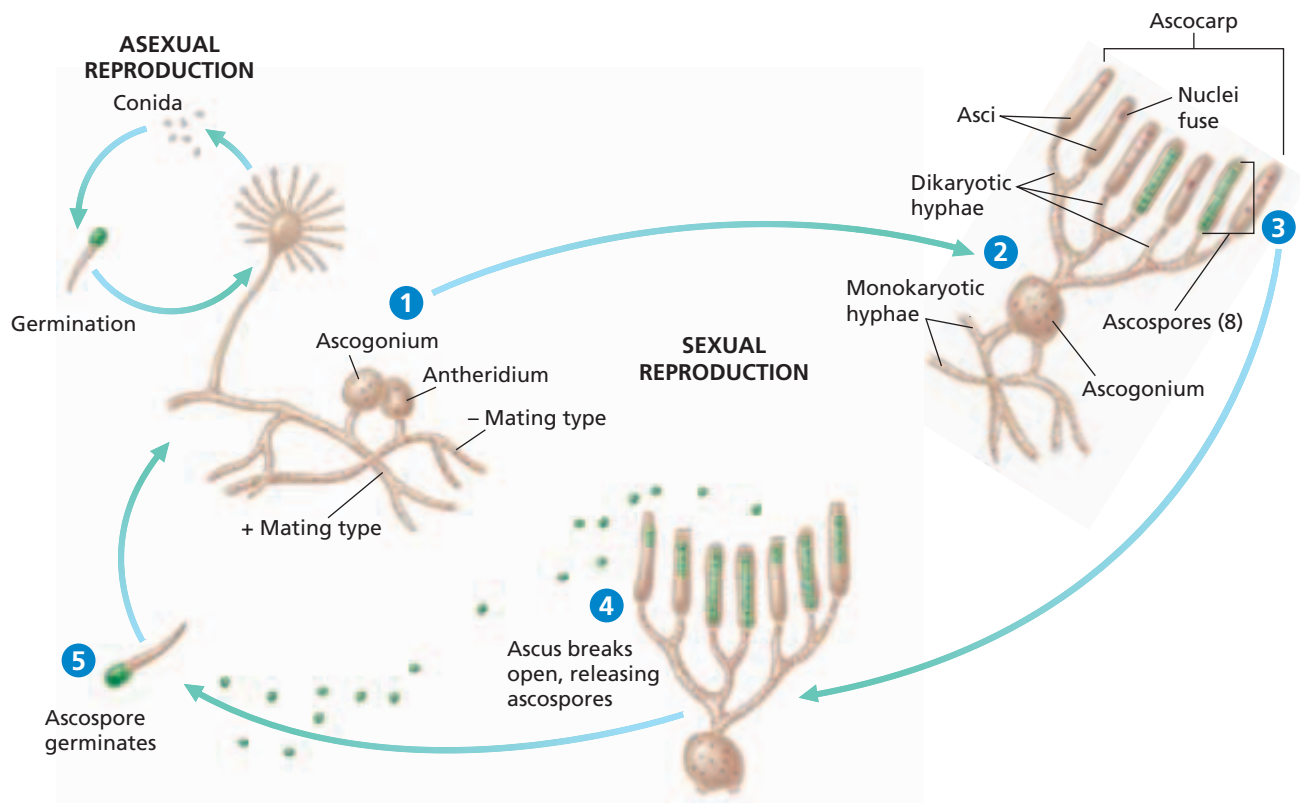


FIGURE 26-6

1 Compatible mating types form special structures that fuse to form an ascogonium. 2 From the ascogonium, dikaryotic hyphae grow and intertwine with monokaryotic hyphae to form an ascocarp. 3 The tips of the dikaryotic hyphae form asci inside, and ascospores form. 4 When the ascus ruptures, it releases the ascospores, which then germinate, 5, to form new monokaryotic hyphae.

PHYLUM ASCOMYCOTA

Ascomycetes (AS-koh-MIE-seets) are distinguished by the presence of saclike compartments where sexually produced spores form. Ascomycetes, also called *sac fungi*, live parasitically and in various habitats, including salt water, fresh water, and land.

Sexual reproduction in the ascomycetes begins when the hyphae of two different haploid mating types form male and female haploid gametangia. The female gametangium is called an **ascogonium** (AS-koh-GOH-nee-uhm), and the male gametangium is called an **antheridium** (AN-thuhr-ID-ee-uhm), as shown in Figure 26-6. As the ascogonium and antheridium approach one another, a tube forms between them, and the nuclei from the antheridium cross the tube and enter the ascogonium. Dikaryotic hyphae grow out of the ascogonium and intertwine with the monokaryotic hyphae of the original fungi (parents) to form a visible cuplike structure called the **ascocarp** (AS-koh-KAHRP). Cells that contain one nucleus are called *monokaryotic* (mahn-oh-KAR-ee-AHT-ik). An example of an ascocarp is shown in Figure 26-7.

Within the ascocarp, sacs called **asci** (AS-KIE) (singular, *ascus*) develop at the tips of the dikaryotic hyphae. Within the asci, the haploid nuclei fuse. The resulting zygotes undergo meiosis once and divide again by mitosis to form eight haploid nuclei. The nuclei form cell walls and become **ascospores** (AS-koh-SPAWRZ). When an ascospore is released and germinates, a new haploid hypha emerges.

Many yeasts, such as *Saccharomyces cerevisiae*, are ascomycetes. *S. cerevisiae* makes bread rise, ferments grapes to make wine and grain to make beer, and is an important research organism.



FIGURE 26-7

These ascocarps of the ascomycete *Sarcoscypha coccinea*, commonly known as the *scarlet cup fungus*, are the structures from which ascospores are released during sexual reproduction.

Deuteromycota

Fungi that do not have a sexual stage are classified in a group called *fungi imperfecti*, or deuteromycota. Continuing phylogenetic analyses have led some mycologists to place these fungi in the three established phyla. Most species of fungi that were formerly classified as *fungi imperfecti* can now be classified in the phylum Ascomycota. However, some biologists disagree with this reclassification.

Word Roots and Origins

mycorrhiza

from the Greek *mykes*, meaning "fungus," and *rhiza*, meaning "root"

MYCORRHIZAE AND LICHENS

A **mycorrhiza** (MIE-koh-RIE-zuh) is a symbiotic structure formed by a fungus and plant roots. More than 80 percent of vascular plants contain such fungi on their roots. The fungus absorbs and concentrates nitrogen, phosphate, and other ions for delivery to the plant root and increases surface area of the plant's root system. In turn, the fungus receives sugars that the plant makes during photosynthesis. All three fungal phyla form mycorrhizae. These mycorrhizal relationships appear to have coevolved with plants.

Lichens (LIE-kuhnz) represent symbiotic relationships between a fungus and a photosynthetic partner (usually a cyanobacterium or green alga). Most fungi in lichens are ascomycetes. The photosynthesizer makes sugars for the fungus, and the fungus provides moisture, shelter, and anchorage for the photosynthesizer. The fungus produces acids that decompose rocks, making minerals available to the lichen. The chemical decomposition of rocks by lichens contributes to the production of soil.

Lichens are identified according to their distribution and structure. *Crustose lichens* grow as a layer on the surface of rocks and trees. *Fruticose lichens* are shrublike, and some grow up to 1.5 m (5 ft) in length. *Foliose lichens* live on flat surfaces, where they form matlike growths with tangled bodies. One example of a lichen is shown in Figure 26-8.

FIGURE 26-8

Lichens are grouped according to their type of body. For example, this red lichen is a crustose lichen that grows on rocks.



SECTION 2 REVIEW

1. Compare and contrast the basidiocarp with the ascocarp.
2. What is the difference between a rhizoid and a stolon?
3. Describe the Ascomycota life cycle.
4. What are mycorrhizae and lichens?
5. How are mycorrhizae and lichens important to the environment?

CRITICAL THINKING

6. **Recognizing Differences** What is the difference between a dikaryotic fungal cell and a diploid fungal zygote?
7. **Predicting Results** If the mycorrhizae in a forest were to suddenly die off, what would probably happen to plants in the forest?
8. **Evaluating Information** What is the importance of conjugation for fungi?

SECTION 3

OBJECTIVES

- **Describe** three ways that fungi cause disease in humans.
- **List** three ways that fungi contribute to good health.
- **Provide** examples of fungi's industrial importance.
- **List** three types of food that fungi provide.

VOCABULARY

aflatoxin

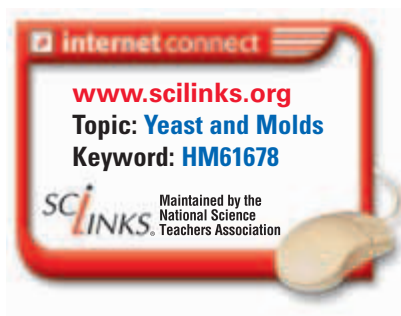


FIGURE 26-9

Ringworm is a fungal infection of the skin. The dried skin that falls off a lesion is contaminated with fungal spores. These spores can infect other people and spread the infection.



FUNGI AND HUMANS

Fungi are important to humans. Some fungi cause devastating human and plant diseases, while others serve as important food sources for humans. Fungi are also used to produce chemicals, fuels, and pharmaceutical compounds.

HUMAN FUNGAL DISEASES

Certain fungi can attack the tissues of living plants and animals and cause disease. Fungal disease is a major concern for humans because fungi attack not only us but also our food sources, making fungi competitors with humans for nutrients.

Mold spores can cause mild to serious allergies in some people. Mold spores can become airborne and may then be inhaled, triggering an allergic reaction. Sniffling, sneezing, and respiratory distress are symptoms of an allergic reaction. Some fungi can also infect and poison humans. Table 26-2 lists some human fungal diseases.

Common Fungal Infections

Fungi may infect the skin, hair, nails, and tissues of the body. For example, fungi on the skin can cause ringworm. Ringworm, as shown in Figure 26-9, can occur almost anywhere on the skin. Athlete's foot, another form of ringworm, occurs on the foot and between the toes.

Another fungal pathogen is *Candida albicans*. This yeast is commonly found in the mouth, intestine, and, in women, in the vaginal tract. Generally, *C. albicans* exists in balance with other microorganisms, such as bacteria that live in and on the body. However, if the normal balance of microorganisms changes, such as when some antibiotics are used or when pregnancy or illness occurs, *C. albicans* can flourish and cause a disease called *candidiasis*, commonly known as a *yeast infection*.

Other Fungal Illnesses

Serious fungal diseases that involve the internal organs are often caused by dimorphic fungi such as *Histoplasma capsulatum*, *Paracoccidioides brasiliensis*, *Coccidioides immitis*, and *Blastomyces dermatitidis*. If the spores of these fungi are inhaled, they can cause severe respiratory illness and spread to other organs, sometimes resulting in death. They grow as a mold in soil, but at human body temperature (37°C or 98.6°F), they become unicellular. *H. capsulatum* often grows as a mold in the feces of birds and can become airborne and inhaled when feces dry out. In some people, the mold causes *histoplasmosis*, which is characterized by fever and respiratory problems that resemble those of tuberculosis.

TABLE 26-2 Some Common Human Fungal Diseases

Disease	Symptoms	Fungus/Phylum	Route of transmission
Athlete's foot	fluid-filled blisters, scaly skin, itching	species of the genus <i>Trichophyton</i> or species of the genus <i>Epidermophyton</i> (Ascomycota)	contact with skin lesions or contaminated floors
Ringworm	ring-shaped skin lesions	<i>Microsporum</i> , <i>Trichophyton</i> (Ascomycota)	contact with skin lesions, contaminated floors, or contaminated objects
Candidiasis (yeast infection)	burning sensation, itching, thick discharge	<i>Candida</i> (Ascomycota)	poorly controlled diabetes; antibiotic treatments increase susceptibility
Tinea cruris (jock itch)	intense itching, ring-shaped lesions	<i>Microsporum</i> , <i>Trichophyton</i> (Ascomycota)	contact with skin lesions, contaminated floors, or contaminated objects
Histoplasmosis	fever, chills, headache, body aches, chest pains, nonproductive cough	<i>Histoplasma capsulatum</i> (Ascomycota)	inhalation of airborne conidia

Sometimes, humans accidentally eat poisonous mushrooms. For example, many species from genus *Amanita* resemble edible mushrooms. However, they contain extremely dangerous toxins that can quickly destroy a person's liver. Figure 26-10 shows an *Amanita* mushroom. Poisonous mushrooms cause varying symptoms when eaten, including upset stomach, vomiting, or diarrhea. Some species can cause death.

Other fungal toxins include the **aflatoxins** (AF-luh-tahks-ins), poisons produced by some species of *Aspergillus*. Aflatoxins can cause liver cancer. Fungi that make aflatoxin can contaminate corn, peanuts, cottonseed, and tree nuts.

FIGURE 26-10

Poisonous mushrooms, such as this *Amanita virosa* (death angel) harm people when they are mistaken for edible mushrooms.



FUNGI IN INDUSTRY

Fungi produce many products used in nonfood industries. Several fungal species are used in the production of important medicines. For example, species of the genus *Penicillium* produce the antibiotic penicillin, and species of the genus *Cephalosporium* produce cephalosporin antibiotics. Species of the genus *Rhizopus* cause chemical transformations of specific chemicals to make cortisone, an anti-inflammatory drug used to treat skin rashes and reduce joint swelling.

The yeast *Saccharomyces cerevisiae* is an important tool in genetic engineering. For example, the vaccine for hepatitis B was developed by inserting hepatitis B genes into yeast plasmids. The yeast uses the inserted viral genes to produce viral proteins that are used as vaccines. Yeast is also used to produce ethanol, a main ingredient in the automobile fuel gasohol.



FIGURE 26-11

Morels and truffles are prized for their delicate flavor. Truffles (top) grow in association with oak trees. Morels (bottom) grow wild in the Americas and are usually found in the spring.

TABLE 26-3 Food Products and Fungi

Type of food	Fungus
Cheeses: blue, brie, Camembert, Gorgonzola, Roquefort	species of the genus <i>Penicillium</i>
Beer, wine	<i>Saccharomyces carlsbergensis</i> , <i>Saccharomyces cerevisiae</i>
Soy products: miso (Japanese), soy sauce, tempeh (Indonesian), tofu (Japanese)	<i>Aspergillus oryzae</i> , species of the genus <i>Rhizopus</i> , species of the genus <i>Mucor</i>
Nutritional yeast	species of the genus <i>Saccharomyces</i>
Breads	<i>Saccharomyces cerevisiae</i>

Fungi and Food Industries

Many fungi are valuable food sources for humans. Yeast, such as species of the genus *Saccharomyces*, is an important nutritional supplement because it contains vitamins, minerals, and other nutrients. Species of the genus *Agaricus* (white button), shiitake, and portabella mushrooms are often found in grocery stores in the United States. People also prize the taste of other fungi, such as truffles and morels, which are pictured in Figure 26-11. Truffles and morels are ascomycetes found near the roots of trees. Table 26-3 summarizes some of the uses of fungi in food.

Fungi can also cause damage to foods. Many fungi are important plant pathogens that attack grain or fruit. For example, wheat rust is a basidiomycete that attacks wheat grains. Other fungi can attack crops such as corn, beans, onions, squashes, and tomatoes.

Fungi also produce several chemical compounds that are important to the food-processing industry, such as citric and gluconic acids. Citric acid is used in soft drinks and candies. *Ashbya gossypii* is a producer of vitamin B₂, an important nutritional supplement.

SECTION 3 REVIEW

1. Identify three ways in which fungi can cause disease in humans.
2. How are fungi used to promote health?
3. Name three nonfood substances that are produced by fungi.
4. List three types of foods that are derived from fungi.

CRITICAL THINKING

5. **Relating Concepts** How might upsetting the balance of microorganisms in the body cause a yeast infection?
6. **Applying Information** Why might it be better to rely on fungi to produce citric acid instead of the older method of extracting the acid from lemons?
7. **Comparing Concepts** How do lichens help the process of biological succession?

CHAPTER HIGHLIGHTS

SECTION 1

Overview of Fungi

- Fungi are eukaryotic, nonphotosynthetic organisms that can be unicellular or multicellular in form.
- Fungi are among the most important decomposers of organic matter. Fungi obtain nutrients by secreting enzymes and absorbing simple organic molecules from their environment.
- Fungi are made up of short filaments called hyphae. Mats of hyphae are called mycelium. Some species have partitions called septa in their hyphae, making individual cells. Fungal cell walls contain chitin rather than cellulose, which is found in plant cell walls.
- Most fungi reproduce both asexually and sexually.
- Fungi evolved about 460 million years ago. Fungi probably evolved from prokaryotes and then adapted to various terrestrial environments.

Vocabulary

mold (p. 527)
yeast (p. 527)
mycology (p. 527)
hypha (p. 527)

chitin (p. 527)
mycelium (p. 527)
septum (p. 527)
coenocyte (p. 528)

dimorphism (p. 528)
sporangiophore (p. 528)
sporangium (p. 528)
sporangiospore (p. 528)

conidium (p. 528)
conidiophore (p. 528)
fragmentation (p. 528)
budding (p. 528)

SECTION 2

Classification of Fungi

- The phylum Zygomycota is coenocytic (their hyphae lack septa). Asexual sporangiospores form within sacs called sporangia. Sexual reproduction results in zygospores.
- The phylum Basidiomycota includes mushrooms. Mushrooms are sexual reproductive structures called *basidiocarps*. Basidiocarps produce basidia, on which basidiospores form.
- Most fungi are in the phylum Ascomycota, or sac fungi. Hyphae form a cup-shaped ascocarp, in which ascospores form.
- Yeast are unicellular Ascomycota. They reproduce asexually by budding. Yeast are used in brewing, baking, and genetic engineering.
- Mycorrhizae are symbiotic structures that form between plant roots and a fungus. The fungus provides certain ions and other nutrients to the plant and, in turn, the fungus gets sugars from the plant.
- Lichens represent symbiotic relationships between fungi and photosynthetic organisms, such as cyanobacteria or green algae.

Vocabulary

rhizoid (p. 530)
stolon (p. 530)
gametangium (p. 530)
zygosporangium (p. 531)

basidium (p. 531)
basidiocarp (p. 531)
basidiospore (p. 531)
ascogonium (p. 532)

antheridium (p. 532)
ascocarp (p. 532)
ascus (p. 532)
ascospore (p. 532)

mycorrhiza (p. 533)
lichen (p. 533)

SECTION 3

Fungi and Humans

- Fungi can cause disease in humans when humans inhale airborne spores, when they eat food contaminated by toxic fungi, when toxic fungi come in contact with skin, or when they accidentally eat poisonous mushrooms.
- Pathogenic fungi that cause serious disease include *Histoplasma capsulatum*, *Blastomyces dermatitidis*, and *Coccidioides immitis*. *H. capsulatum* is associated with bird feces.
- Various fungi are used in the production of vitamin B₂, cortisone, penicillin and other antibiotics, and some genetically engineered drugs.
- Fungi are used in the production of familiar foods such as cheeses, bread, beer, wines, and soy products.

Vocabulary

aflatoxin (p. 535)

CHAPTER REVIEW

USING VOCABULARY

- For each pair of terms, explain how the meanings of the terms differ.
 - hypha* and *mycelium*
 - septate hyphae* and *coenocytic hyphae*
 - basidiocarp* and *ascocarp*
 - mycorrhiza* and *mycelium*
- Choose the term that does not belong in the following group, and explain why it does not belong: *hypha*, *mycelium*, and *lichen*.
- Use each of the following terms in a separate sentence: *mold*, *yeast*, *chitin*, and *rhizoid*.
- Word Roots and Origins** The word *gametangium* comes from two Greek words: *gamete*, meaning “spouse,” and *angeion*, meaning “case” or “capsule.” Using this information, explain why *gametangium* is a suitable term for what happens when two different mating types of fungal filaments meet.

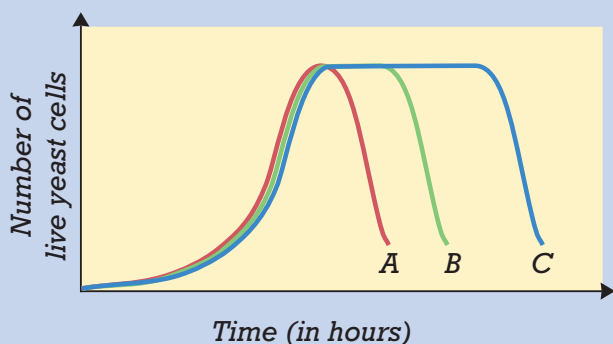
UNDERSTANDING KEY CONCEPTS

- State** how fungi and plants differ from each other.
- Explain** how hyphae provide nutrients for a fungus.
- Name** the structure that makes up the body of a fungus.
- List** three methods of asexual reproduction in fungi.
- Summarize** one hypothesis about the origin of fungi.
- Define** *dimorphism* and give an example of a fungus that is dimorphic.
- Name** an example of each of the three phyla of modern fungi.
- Describe** why some fungi are called *club fungi*.
- Differentiate** between a root and a rhizoid.
- Differentiate** between a gametangium and a zygosporangium.
- Summarize** the similarities between a mycorrhiza and a lichen.
- Explain** how mycorrhizae are of importance to both plants and fungi.
- Relate** how a fungus in bird feces can be potentially dangerous to humans.
- List** three ways in which fungi are beneficial to the health of humans.
- Identify** a chemical compound used in candies and soft drinks that is produced by certain fungi.
- Give** two examples of foods that are produced with the use of fungi.
- CONCEPT MAPPING** Use the following terms to create a concept map that shows the sequence of events in the formation of an ascocarp and that begins with a germinating ascospore: *ascocarp*, *ascogonium*, *asci*, *tube*, *ascospores*, *meiosis*, *antheridium*, and *mitosis*.

CRITICAL THINKING

- Interpreting Graphics** A winemaker experimenting with three different strains of yeast charts the number of viable yeast remaining in the wine as the grapes ferment. The three strains were grown in separate containers, at the same temperature, and with the same type of grape. Give reasons why strain C survived longer than strains A and B.

Growth of Three Strains of Yeast



- Critiquing Relationships** Comment on the validity of this statement: The relationship mycorrhizae have with trees is an example of mutualism.
- Relating Concepts** What adaptive advantage does dimorphism give a fungus?
- Analyzing Concepts** What advantages might a fungal spore have over a seed?
- Applying Information** A folk remedy for the treatment of an infected wound calls for a piece of moistened, moldy bread to be placed on the wound. How might this practice help a wound to heal?

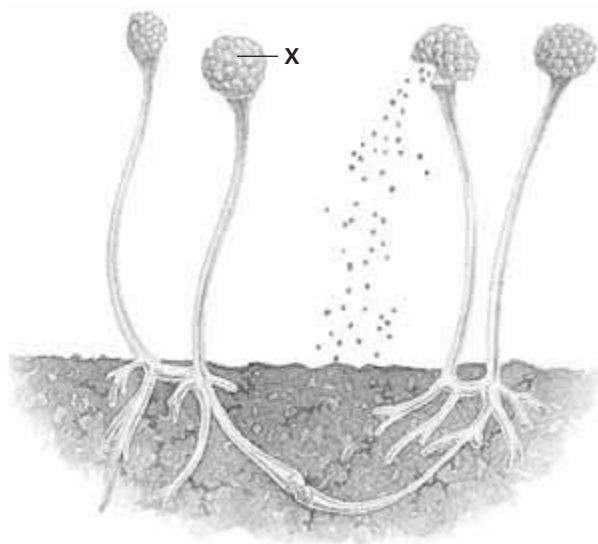


Standardized Test Preparation

DIRECTIONS: Choose the letter of the answer choice that best answers the question.

- What are fungi that feed on decaying organic matter called?
 - parasites
 - mutualists
 - symbionts
 - saprophytes
- Lichens represent a symbiotic association between a fungus and which other type of organism?
 - an alga
 - a plant
 - a mold
 - a rhizoid
- In a mycorrhiza, a fungus lives in a symbiotic relationship with which of the following?
 - a virus
 - a plant
 - a bacterium
 - a slime mold

INTERPRETING GRAPHICS: The diagram below shows the fruiting bodies of a type of fungus. Use the diagram to answer the questions that follow.



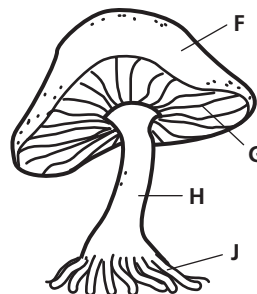
- What is the structure labeled X called?
 - a hypha
 - a zygote
 - a sporangium
 - a sporangiospore
- To what phylum does the fungus in the diagram above belong?
 - Ascomycota
 - Zygomycota
 - Basidiomycota
 - fungi imperfecti

DIRECTIONS: Complete the following analogy.

6. septa : septum :: asci :

- ascus
- ascocarp
- ascospore
- ascogonium

INTERPRETING GRAPHICS: The diagram below shows the typical structure of a member of the phylum Basidiomycota. Use the diagram below to answer the question that follows.



- Which structure is responsible for meeting the food requirements of the organism shown?
 - F
 - G
 - H
 - J

SHORT RESPONSE

Fungi are important decomposers that break down organic matter.

Explain how fungi contribute to nutrient recycling in the environment.

EXTENDED RESPONSE

Fungi are major competitors with humans for food.

Part A Explain what needs humans and fungi have in common.

Part B Explain why fungi could easily win the battle for nutrients.

Test TIP

When studying for a test on fungi life cycles, organize the information into a table. It will be easier to recall information from isolated parts of the table when you have to answer a question.

Exploring the Growth of Fungi on Food

OBJECTIVES

- Recognize fungal growth on food.
- Identify environmental conditions that favor the growth of fungi on food.

PROCESS SKILLS

- designing an experiment
- collecting data
- analyzing results
- hypothesizing
- organizing data

MATERIALS

- safety goggles
- lab apron
- disposable gloves
- 2 sterile Petri dishes with nutrient medium (such as potato dextrose agar)
- 2 sterile Petri dishes with nutrient medium and propionic acid
- fungal samples
- stereomicroscope
- toothpicks
- wax pencil
- masking tape



Background

1. How do multicellular fungi, such as molds, obtain nutrients?
2. How do multicellular fungi reproduce and grow?
3. What ecological role do fungi fulfill?
4. How might a fungus growing on food harm someone who eats it?

PART A Experimental Setup

1.  **CAUTION** Tell your teacher if you have any allergies to fungi. Put on safety goggles, a lab apron, and disposable gloves. Obtain four sterile Petri dishes, two with nutrient medium and two with nutrient medium plus propionic acid. Label the dishes for the presence or absence of propionic acid.
2. Examine the fungal samples through a dissecting microscope. Select a dense growth of a fungus from which you will take samples.
3. Use a toothpick to scoop up a small sample of the fungus you have selected. In the two Petri dishes without propionic acid, gently touch the sample to the medium in four places. Raise the lids of the dishes as little as possible while doing this. Do the same for the two Petri dishes with propionic acid, using a clean toothpick and another small sample of the same fungus. Dispose of the toothpicks according to the teacher's instructions.
4. Place a piece of masking tape on opposite sides of each dish to hold the lid and bottom together. Label each Petri dish with your name.
5. Design an experiment to determine which of two opposite environmental conditions is better for fungal growth. Some possible combinations are warm/cold, light/dark, and moist/dry. Label one of the dishes that contain propionic acid with one of the two environmental conditions you selected. Label the other dish with the other condition. Do the same for the two dishes that do not contain propionic acid. Then, incubate all four dishes under the appropriate conditions.

FUNGAL GROWTH IN DIFFERENT ENVIRONMENTS

Dish	Environmental condition	Propionic acid?	Source of fungus	Growth
1				
2				
3				
4				

6. Create a data table similar to the model above to record your experimental observations for your lab report. For example, the table above is designed to record any growth that may occur under two different environmental conditions (warm/cold, sun-light/dark, moist/dry), the source of the fungus, and if propionic acid is present. Design your data table to fit your own experiment. Remember to allow plenty of space to record your observations. Be sure to label your Petri dishes with the appropriate environmental conditions and with the date on which the experiment began.



PART B Comparing Amounts of Growth

- After one week, examine each dish under the stereo-microscope without opening the dish.
- Record your observations in your data table in your lab report.
- Examine the fungi in other groups' dishes, especially fungi grown under different environmental conditions.
- Add your observations of those dishes to your lab report.
-   Clean up your materials, and wash your hands before leaving the lab. Dispose of all materials according to instructions provided by your teacher.

Analysis and Conclusions

- Besides the environmental conditions you chose, what additional factor was tested in your experiment?
- What steps were taken in your experiment to avoid contamination of the plates?
- How would contamination of the plates affect the results of your experiment? What are the possible sources of contamination of your experiment?
- What does extensive fungal growth on a plate indicate? What does lack of fungal growth on a plate indicate?
- What effect does propionic acid have on fungal growth? How do you know?
- Why do you think propionic acid is added to foods?
- Which environmental conditions favor fungal growth? Which inhibit it?
- Compare the results for the different kinds of fungi grown. Did fungi from certain food sources grow more rapidly than others?
- Based on your conclusions, under what conditions would you keep a nonsterile food product if you wanted to prevent it from becoming moldy?
- Has this lab changed your attitude toward food storage? If so, how?

Further Inquiry

Design an experiment that determines which kinds of foods best support fungal growth and whether the presence of chemicals inhibits fungal growth.

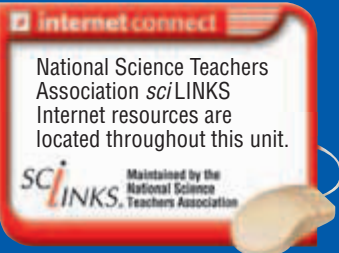
UNIT 7

CHAPTERS

- 27 *The Importance of Plants*
- 28 *Plant Evolution and Classification*
- 29 *Plant Structure and Function*
- 30 *Plant Reproduction*
- 31 *Plant Responses*



References to *Scientific American* project ideas are located throughout this unit.



PLANTS

“The day passed delightfully. Delight itself, however, is a weak term to express the feelings of a naturalist who, for the first time, has wandered by himself in a Brazilian forest. The elegance of the grasses, the novelty of the parasitical plants, the beauty of the flowers, the glossy green of the foliage, but above all the general luxuriance of the vegetation, filled me with admiration.”

Charles Darwin, *Voyage of the Beagle*, 1839



These autumn leaves show bright colors because they have lost most of their chlorophyll.



Rafflesia arnoldii has the largest known flowers. Although they are large and beautiful, these flowers emit the smell of rotting meat, which attracts flies as pollinators.

Strawberry plants have above-ground stems that are able to form new plants.



This cotton boll developed from the ovary of a flower of a cotton plant.

Orchid cactus

THE IMPORTANCE OF PLANTS

Rice fields, often called paddies, thrive in regions with abundant rainfall. Rice is a prime food source for more than 60 percent of the world's population. About 90 percent of the world's rice is grown in Asia, primarily in China and India.



SECTION 1 *Plants and People*

SECTION 2 *Plants and the Environment*

PLANTS AND PEOPLE

*Plants are essential to our survival because they produce virtually all our food. We eat plants either directly, in the form of fruits, vegetables, and grains, or indirectly, by eating animals that consume plants. Plants also provide medicines, clothing, paper, cosmetics, and many other products. Plants play a major role in the continuous cycling of the Earth's water, oxygen, carbon dioxide, and mineral nutrients. The study of plants is called **botany** (BAHT-nee).*

PLANT CULTIVATION

Of the more than 350,000 plant species, people use at least 10,000 species for food. Incredibly, fewer than 20 plant species provide more than 90 percent of our food supply. The cultivation of plants for food probably began about 11,000 years ago in the Middle East. Wheat, barley, lentils, and peas were the first domesticated food crops. Growing plants and raising animals for human use is called **agriculture** (AG-ri-KUHL-chuhr). People propagated, or reproduced, individual plants that had valuable characteristics, such as plants that produced the largest or tastiest fruits.

In the 11,000 years that humans have been cultivating plants, we have changed many of the plants so much that they could not grow and survive without us. For example, the wild wheat stalk, as shown in Figure 27-1, breaks easily in the wind, an adaptation that increases the dispersal of its seeds. But early farmers used seeds from plants with stalks that did not break easily for replanting. When these plants were grown, the seeds could be harvested before they fell from the plant. This form of selection—with people acting as selecting agents—has resulted in high-quality food plants.

You have probably eaten Thompson Seedless grapes, McIntosh apples, or Valencia oranges. They are just three examples of several hundred thousand different cultivars. The word *cultivar* is a contraction of the two terms *cultivated* and *variety*. **Cultivars** (KUHL-ti-VAHRZ) are selected by people, and they have at least one distinguishing characteristic that sets them apart from other members of their species. The famous Japanese flowering cherry trees in Washington, D.C., Yoshino cherries, are another example of a cultivar.

OBJECTIVES

- **Summarize** the history of plant cultivation.
- **Identify** the categories of food crops.
- **Explain** how humans have increased food production in the world.
- **Describe** non-food uses of plants.

VOCABULARY

botany
agriculture
cultivar
cereal
root crop
legume
fruit
vegetable
nut
spice
herb
quinine
fertilizer
pesticide
aspirin
gasohol

FIGURE 27-1

Wheat is one of the world's most important food crops. It is used to make breads, crackers, macaroni, and spaghetti.





Quick Lab

Making a Plant-Based Menu

Materials paper, pencil

Procedures

1. Prepare a written lunch menu that consists only of plant-derived foods. Be careful to design a fully nutritional meal. Then, write a description of a lunch setting that is also completely derived from plants, including utensils and furniture.
2. Share your menu and lunch-setting description with your classmates.

Analysis Was it difficult or easy to devise a lunch that includes only plant-based items? Write down any interesting or unusual plant choices or purposes for them that you and your classmates included.

FOOD CROPS

Food crops are usually classified partly by use and partly by family. The classification system in Table 27-1 is not like the taxonomic classification used by scientists because most categories contain species that are not closely related. Also, many crops fit into more than one category. For example, corn is a cereal, but it can also be classified as an oil crop, a sweetener, a vegetable, and a beverage.

Cereals

Cereals are grasses that contain grains. Grains are the edible, dry fruits of a cereal. Over half of the world's cultivated land is devoted to cereal crops, such as rice, wheat, corn, oats, sorghum, rye, and millet. Worldwide, cereals provide about 50 percent of the calories in the average human diet. In addition, much of the harvested grain is used for animal feed, so it is indirectly consumed by people as meat, poultry, eggs, and dairy products.

Wheat and corn are produced in the largest amounts. Wheat grows well in moderate to cold climates, including parts of the United States, Russia, and Canada. The United States is the leading producer of corn, also called maize. Rice is different from other cereals because it grows best in shallow water. Rice thrives in areas with warm temperatures.

TABLE 27-1 Food Crops

Category	Example plants
Cereals	rice, wheat, corn, oats, sorghum, rye, barley, millet
Root crops	potato, cassava, sweet potato, yam, taro
Legumes	soybean, peanut, bean, pea, alfalfa, lentils
Fruits	apple, peach, banana, grape, orange, blueberry, pineapple, cherry, mango, pear
Nuts	walnut, cashew, pecan, coconut, almond, macadamia, filbert, pistachio
Vegetables	spinach, cabbage, sweet corn, pea, turnip, asparagus, tomato, artichoke, zucchini
Forages	cereals, legumes, grasses
Oils	cottonseed, rapeseed, palm, sesame, soybean, corn, safflower, sunflower
Beverages	coffee, tea, cola, cacao, fruit juice, grape (wine), corn (whiskey), barley and hops (beer)
Sweeteners	sugar cane, sugar beet, sugar maple, corn
Spices	pepper, cinnamon, vanilla, paprika, cloves, saffron, nutmeg, ginger, allspice
Herbs	rosemary, thyme, sage, dill, basil, oregano, mint
Flavorings	cacao (chocolate), coconut, carob, licorice, quinine
Colorings	red beet, annatto, turmeric, saffron, carrot
Additives	guar, locust bean, citrus (pectin), gum arabic, chicle tree
Garnishes	sesame, caraway, and poppy seeds; parsley; pimento
Snacks	popcorn, sunflower seeds, pumpkin seeds

Root Crops

Root crops are roots or underground stems that are rich in carbohydrates. In many parts of the world, root crops substitute for cereals in providing the major part of the diet. However, diets of root crops or cereals alone are usually low in some important amino acids. To correct this deficiency, people must eat other foods, such as legumes or animal protein.

Root crops include beets, carrots, radishes, rutabagas, turnips, and sweet potatoes. Other kinds of potatoes and yams are actually *tubers* (modified underground stems) but are considered root crops because they grow underground. You may have eaten tapioca pudding, which comes from cassava, a root crop grown in the Tropics and shown in Figure 27-2.

Legumes

Legumes are members of the pea family and bear seeds in pods. Soybean, shown in Figure 27-3, is the most important legume crop because it is produced in the largest amount and has many important uses. Soybean is used to make vegetable oil, soy milk, soy sauce, tofu, and margarine. Alfalfa and clover are legumes used mainly as feed for livestock. Legumes are important in agriculture because they improve the nitrogen content of soil. Recall that some bacteria form a symbiotic relationship with many legumes and convert atmospheric nitrogen into a form that plants can use.

Fruits, Vegetables, and Nuts

Many “vegetables” we know, such as tomatoes, green beans, and squash, are actually botanically classified as fruits. A **fruit** is the part of a flowering plant that usually contains seeds. Foods derived from the leaves, stems, seeds, and roots of nonwoody plants are often called **vegetables**. Fruits and vegetables are excellent sources of many important vitamins and minerals, making them essential parts of a healthy diet. A **nut** is a dry, hard fruit that does not split open to release its seed. Nuts include almonds, walnuts, pecans, and hazelnuts. Peanuts are commonly considered to be nuts but are actually classified as legumes. Nuts and legumes are higher in protein than other plant foods.

Spices, Herbs, and Flavorings

Other food crops add variety and pleasure to our diet by flavoring our water, beverages, and food. More than half the population ingests caffeine through drinking coffee, tea, and cola drinks. Both **spices** and **herbs** are used to add taste to food. In general, spices come from plant parts other than the leaf and are tropical. Herbs usually come from leaves and usually can be grown in a home garden. Flavorings, such as chocolate and coconut, are not usually considered spices or herbs and are therefore placed in a separate category. Another flavor, quinine, is used to make tonic water. **Quinine** comes from the bark of the cinchona tree and is used to treat malaria.



FIGURE 27-2

An important root crop in the Tropics is cassava, which has thick roots that are eaten like potatoes. The starch-filled roots of cassava can be 30–120 cm (1–4 ft) long.



FIGURE 27-3

Soybean is an important legume crop grown in the midwestern and southern parts of the United States. The soybean plant is covered with short, fine fibers and is usually 60–120 cm (2–4 ft) tall. It is an inexpensive and useful source of protein.

Eco Connection

Making Your Own Fertilizer—Composting

Many people are making their own fertilizer through a technique called *composting*. Compost is a type of organic fertilizer that is made from decayed plant matter. Compost improves the texture of soil and provides inorganic nutrients that plants need.

It's easy to start your own compost pile. Collect dead plant matter, such as grass clippings, leaves, coffee grounds, or sawdust. Make a pile by alternating layers of plant matter with a thin layer of soil or manure. Sprinkle water on the pile to speed the process of decay. After the compost has been allowed to decay for about six months, it should be ready for use in your garden.

Word Roots and Origins

pesticide

contains the suffix *-cide*, from the Latin *cida*, meaning "cut down" or "kill"

Food Production

For decades, experts have been predicting widespread food shortages due to the continuing increase in the world population. However, massive food shortages have not occurred mainly because of increased use of irrigation, fertilizers, and pesticides. Improvements in cultivars; farm machinery; food preservation techniques; and methods of controlling diseases, weeds, and pests have also helped improve food production. **Fertilizers** supply plants with essential mineral nutrients like nitrogen and phosphorus. **Pesticides** are chemicals that kill undesirable organisms that harm crops, such as some insects.

People have made many trade-offs to support an adequate food supply. The negative consequences include massive soil erosion, depletion of fossil fuel and water supplies, pollution, and destruction of wild populations of plants and animals as more land is cultivated.

NONFOOD USES OF PLANTS

In addition to providing us with food, plants provide us with thousands of other essential products. It is hard to imagine how we could live without plants, given the variety of products that contain substances from plants.

Medicines

The ancient Greeks treated headaches with the bark of the white willow tree, which contains the chemical salicin. This use gave scientists the idea to test the chemical acetylsalicylic (uh-SEET-uhl-SAL-uh-SIL-ik) acid. The willow is in the genus *Salix*, hence the names *salicin* and *salicylic*. Acetylsalicylic acid is **aspirin**, the world's most widely used medicine. Besides pain relief, aspirin is used to thin blood and thereby prevent heart attacks and strokes. Plants were our first medicines, and early plant biologists, like Linnaeus, were often doctors.

TABLE 27-2 *Plants in Medicine*

Plant	Genus name	Drug	Use
Cinchona	<i>Cinchona</i>	quinine	treat malaria and certain disorders of heart rhythm
Foxglove	<i>Digitalis</i>	digitalis	treat heart disease, help regulate heart rate
Yam	<i>Dioscorea</i>	cortisone	treat inflammation and allergies
White willow	<i>Salix</i>	acetylsalicylic acid (aspirin)	relieve pain, prevent heart attacks and strokes
Yew	<i>Taxus</i>	taxol	treat ovarian cancer, breast cancer, and some types of lung cancer



(a)



(b)

Many modern medicines either still come from plants or were originally obtained from plants and are now synthesized in the laboratory. Table 27-2 lists examples of plants that are used in medicine. Two of these plants, yew and foxglove, are shown in Figure 27-4. Scientists are currently evaluating thousands of plant species that may have medicinal properties. One of the reasons scientists are very concerned about the destruction of rain forests is because many rain-forest plant species have yet to be researched. In addition to medicines, plants provide many other products, which are summarized on the next page in Table 27-3.

Your local health-food store carries a wide range of plant products that claim to prevent disease or improve health. These substances are not regulated by the Food and Drug Administration (FDA). Consumers should remember that the effectiveness and safety of herbal remedies have not been confirmed by the rigorous scientific testing that new medicines must undergo before receiving FDA approval. The FDA, pharmaceutical companies, and health-care providers are working together to investigate the claims of those who market these remedies.

Clothing and Fabric Dyes

Figure 27-5 shows cotton, which is used to make most of our clothing. Some clothing is woven with linen, which is made from the flax plant. Artificial fabrics, like rayon, arnel, and cellulose acetate, are made from processed wood fibers. Leather is made from animal hides, but it is usually treated with tannin, a chemical obtained from many tree species. Tannin makes leather stronger and prevents it from rotting.

Prior to the mid-1800s, fabrics were dyed with natural plant dyes. Today most clothing is colored with dyes manufactured from coal, which is formed from the remains of ancient plants.

FIGURE 27-4

(a) Taxol, originally derived from the bark of the Pacific yew, is a recently discovered cancer drug. This evergreen tree or shrub produces seeds that look like berries. (b) Foxglove is the source of digitalis, which is used in the treatment of heart disease. The beautiful flowers grow in a cluster.

FIGURE 27-5

Cotton, the world's most widely used source of clothing, consists of fibers attached to the seed.



TABLE 27-3 Nonfood Uses of Plants

Use	Example plants
Brooms/brushes	broomcorn, palms, coconut
Building materials	trees, bamboo, reeds, palms, grasses
Carpets/mats	jute, coconut (coir), cotton, trees
Clothing	cotton, flax (linen), ramie, pineapple, trees (rayon and arnel)
Cosmetics	corn, avocado, carrot, almond, cacao, soybean, macadamia, aloe
Fabric dyes	indigo (blue), madder (red), onion (yellow), black walnut (brown), peach (green), maple (pink)
Fuels	trees, bamboo, water hyacinth, grain alcohol, vegetable oils, gopher plant
Furniture	redwood, oak, rattan, teak, willow (wicker), rushes
Hair dyes	henna, rhubarb, chamomile, black walnut
Incense	frankincense, myrrh, cinnamon
Inks	soybean, flax (linseed oil), tung-oil tree
Leather	black wattle, quebracho, Spanish chestnut (tannin)
Lipstick	jojoba, castor bean, carnauba palm, soybean, coconut
Medicines and remedies	foxglove (digitalis), cinchona (quinine), yew (taxol), opium poppy (morphine and codeine), yam (cortisone), aloe, ipecac, ginseng, ginkgo, guarana, purple coneflower, kudzu, saw palmetto
Miscellaneous	cork oak (cork), incense cedar (pencil shafts), trees (disposable baby diapers and cellulose acetate plastic), kapok (life preserver stuffing), rosary pea (bead necklaces), water hyacinth (water purification), lignum vitae (submarine engine bearings)
Musical instruments	ebony (black piano keys), maple (violins), reed (woodwind reeds), African blackwood (woodwinds)
Ornamentals	shade trees, shrubs, lawns, cut flowers, Christmas trees, houseplants
Paints	flax (linseed oil), tung-oil tree, soybean, pine (turpentine)
Paper/cardboard	trees, cotton, flax, hemp, bamboo, papyrus
Perfumes	rose, orange, lavender, orchids, sandalwood, lilac, jasmine, lily of the valley, pine
Pesticides/repellents	tobacco (nicotine sulfate), derris (rotenone), chrysanthemum (pyrethrum), citronella, garlic, citrus
Rope	hemp, agave (sisal)
Rubber	rubber tree, guayule
Shampoo	palm oil, coconut, jojoba, aloe, trees, herbs, fruits
Soaps	coconut, palm oil, cacao, lavender, herbs, fruits
Sports equipment	balata (golf balls), persimmon (golf club heads), ash (baseball bats), ebony and ash (pool cues)
Toothpaste	mint, wheat, palm oil, coconut
Tourist attractions	redwoods, giant sequoias, saguaro cactuses, fall foliage, Holland tulips
Waxes	carnauba palm, cauassu, candelilla, bayberry

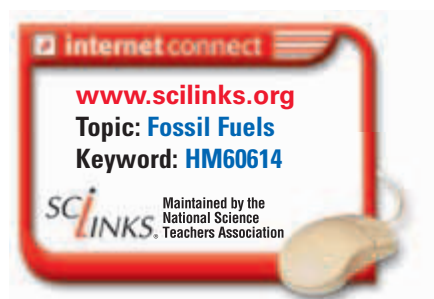


FIGURE 27-6

Coal is a dark-colored, organic rock. Complex chemical and physical changes produced coal from the remains of plants that grew in prehistoric swamps millions of years ago.

Fuels

Most of the energy we use for heat, electricity, and machine fuel comes from fossil fuels—coal, oil, and natural gas. Figure 27-6 shows a coal deposit being uncovered by an earth-moving machine. Fossil fuels are composed of stored photosynthetic energy from millions of years ago. In developing nations, much of the fuel comes from wood or other plant materials. For example, grains can be fermented into alcohol and mixed with gasoline to make gasohol. **Gasohol**, which is made of about 10 percent alcohol, is an alternative fuel for automobiles.



Careers in BIOLOGY

Ethnobotanist

Job Description Ethnobotanists are scientists who study the ways in which people make use of plants, whether for food, medicine, or other purposes. Ethnobotanists are often involved in the collection of plants, the conservation of endangered species, and the research of traditional plant medicines.

Focus On an Ethnobotanist

Ethnobotanist Paul Cox travels to remote places to look for plants that can help cure diseases. He seeks the advice of native healers in his search. For example, Cox traveled to the Pacific island of Samoa in 1984 to meet a 78-year-old healer named Epenesa. She was able to identify more than 200 medicinal plants, and she had an accurate understanding of human anatomy. Epenesa gave Cox

samples of her medicines, which he brought back to the United States for study. American researchers studying her remedies discovered antiviral and anti-inflammatory compounds. Many other plant substances, obtained by ethnobotanists with the assistance of native healers, are being studied in laboratories for their healing properties. Many of the practitioners of traditional medicine are elderly. When they die, generations of medical knowledge often die with them. The need for Cox and other ethnobotanists to record the ancient wisdom of native healers is urgent.

Education and Skills

- **High school**—three years of science courses and four years of math courses.



- **College**—bachelor of science (B.S.) in biology, including course work in botany, chemistry, and anthropology, followed by a doctoral degree (Ph.D.) in botany, chemistry, anthropology, or linguistics, plus field and lab experience.
- **Skills**—patience, ability to learn new languages, self-motivation, respect for other cultures, and field survival skills.



For more about careers, visit go.hrw.com and type in the keyword **HM6 Careers**.



FIGURE 27-7

The California redwood trees are a majestic sight. Redwoods usually grow 60–84 m (200–275 ft) high. The bark is very thick, making the trees resistant to fires.

Other Uses of Plants

Ornamental trees, shrubs, and other plants outside our homes do much more than provide beauty. Besides their decorative function, they improve the environment by preventing soil erosion, reducing noise, providing habitats for wild animals, acting as windbreaks, providing shade, and moderating temperatures, which, in turn, reduces home heating and cooling costs. Scientists have also found that ornamental plants improve our mental well-being. Gardening has long been a popular hobby in the United States, and it is an important form of exercise for millions of people.

Many plants have become major tourist attractions, such as the California redwoods shown in Figure 27-7. Another popular American tourist attraction is the Petrified Forest National Park in Arizona. The park features large areas of fossilized trees. And many yearly festivals, including the Tournament of Roses in California every New Year's Day and the Cherry Festival in Michigan each May, are held around the United States to celebrate plants. In addition, many people visit the forests of the northeastern United States every fall to view the spectacular changing leaf colors.

Plants are essential to our survival because they produce virtually all of our food, and they enhance our lives in many ways. Growing cut flowers is now a multibillion-dollar-a-year industry, and it is only a small part of the huge business of growing and using plants.

Plants can also provide the inspiration to develop innovative products. The cocklebur plant provided the idea for hook and loop fasteners when the hooked fruit was caught in the inventor's clothing. Plants have made our lives better in numerous ways, and they undoubtedly will continue to do so in the future.

SECTION 1 REVIEW

1. Define the term *agriculture*.
2. Distinguish between the food crops known as *cereals*, *root crops*, *legumes*, *fruits*, and *vegetables*.
3. List three activities that humans have conducted to increase production of food crops, thus avoiding predicted food shortages.
4. List four personal grooming products that are manufactured using ingredients from plants.

CRITICAL THINKING

5. **Applying Information** How might transferring specific genes from legumes into rice plants help reduce malnutrition?
6. **Analyzing Information** How might farmers use legumes to reduce their use of fertilizers?
7. **Relating Concepts** Swiss physician Paracelsus (1493–1541) stated that the difference between a poison and a remedy is the dose. Explain how this statement might apply to the investigation of herbal remedies.

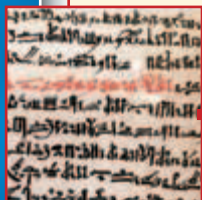
MILESTONES

IN

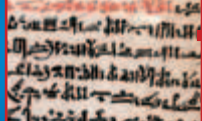
Using Plants for Medicine

Timeline

4000 BC Sumerian clay tablets list plant remedies.



1500 BC Egyptian scroll lists 850 plant remedies.



1000 BC The *Pen Tsao* lists 375 botanicals, including opium and ephedra.

300 BC Theophrastus writes *Historia Plantarum*.



100 AD Dioscorides mentions white willow tree in *de Materia Medica*.

1658 Cinchona bark is advertised for sale in England.

1899 Aspirin, a buffered form of salicin, is introduced.



1957 Extracts from the rosy periwinkle are found to cure childhood leukemia.

1993 Taxol, derived from the Pacific yew, is approved by the FDA as an anticancer drug.



Today, one-quarter of all prescription drugs contain a useful plant ingredient. More than 100 prescription drugs are made from plants. When did the relationship between plants, people, and medicine begin?

Throughout human history, people have used and passed down information about plants that can ease pain, infections, and physical maladies. About 4000 BC, Sumerian healers engraved cuneiform characters for various medicinal plants onto clay tablets. These may have been the first *herbals*, or lists of healing plants. A 70-foot-long Egyptian scroll dated to 1500 BC recorded 850 plant remedies, including garlic, for preventing illness.

An Indian poem from this same era, called the *Rig Veda*, describes many plant remedies, including snakeroot (*Rauvolfia serpentina*) for snakebite. Modern researchers discovered that snakeroot contains the compound reserpine, which lowers blood pressure. A Chinese herbal from 1000 BC, the *Pen Tsao*, lists 375 botanicals, including opium and ephedra, that are still used as painkillers and stimulants.

Around 300 BC, Aristotle described various herbal remedies, and his student Theophrastus wrote the text *Historia Plantarum*. In 100 AD, the Roman scholar Dioscorides wrote *de Materia Medica*, in which he mentions white willow, later found to be a source of aspirin. Both books were still in use in the Middle Ages. At least one herbal written in 1250 AD includes foxglove (*Digitalis purpurea*), the source of a heart stimulant.

The modern era of medicinal plants began with the use of quinine for treating malaria. Peruvian Indians had long recognized the value of the cinchona tree for treating feverish patients. Bark from this tree was first advertised for sale in England in 1658. However, it wasn't until 1820 that quinine was isolated from the bark of the cinchona tree by French chemists.

The development of aspirin presents a similar story. Extracts from the white willow tree had been known from the time of the ancient Greeks to ease pain. However, the pain-relieving substance, salicin, wasn't isolated until 1829. Because salicin causes stomach problems, more research was required for the development of a buffered derivative, aspirin, which was offered to the public in 1899.

During the 1950s, drug companies began to seek medicinal plants all over the world. In 1957, scientists discovered that extracts from the Madagascar rosy periwinkle are effective against childhood leukemia. Deaths from childhood leukemia have since dropped by 80 percent. Taxol, derived from the Pacific yew and approved for medical use in 1993, is an effective treatment against ovarian cancer. So far, however, only a small percentage of the world's plants have been tested for useful medicines such as these.

Review

1. Name two medicinal plants used in ancient times and today.
2. **Critical Thinking** Why is it important to preserve the medicinal knowledge of indigenous cultures?
3. **Critical Thinking** How might the destruction of rain forests impact the future of medicine?



SECTION 2

OBJECTIVES

- **Summarize** the contributions of plants to the environment.
- **Describe** the ways that plants interact with other organisms.
- **Explain** how some plants can cause harm.

VOCABULARY

plant ecology
weed
hay fever

PLANTS AND THE ENVIRONMENT

Based solely on weight, algae and photosynthetic bacteria are dominant organisms in the oceans, and plants are dominant on land. Photosynthetic plants are called producers because they make food for other living things. Organisms that eat other organisms, like animals, are called consumers and depend on plants for a source of organic compounds.

PLANT ECOLOGY

The study of the interactions between plants and the environment is called **plant ecology**. The most important interaction involves the ability of plants to capture solar energy through photosynthesis. In photosynthesis, plants absorb carbon dioxide from the air, produce sugar and starch, and break apart water, releasing oxygen into the air. Consumers, like the one shown in Figure 27-8, use sugar and oxygen in cellular respiration and produce carbon dioxide and water. Organic compounds from plants provide consumers with energy, cellular “building blocks,” and essential compounds such as vitamins and fiber.

Plants also provide organisms with inorganic nutrients. Plant roots are very efficient at mining the soil for inorganic nutrients, such as nitrogen, phosphorus, potassium, iron, and magnesium. Plants use these inorganic nutrients for metabolism and to make organic compounds. Consumers ingest these organic compounds and incorporate the inorganic nutrients into their own bodies.

FIGURE 27-8

About half of the world's species of plants and animals live in tropical rain forests. One of the reasons scientists are very concerned about the destruction of rain forests is because many plant species have yet to be researched.



Eventually, these same inorganic nutrients are returned to the soil when the consumer's waste material or dead body is decomposed by bacteria and fungi. Plants thus play a major role in the continuous cycling of the Earth's water, oxygen, carbon dioxide, and inorganic nutrients.

Plants are also responsible for the formation and maintenance of soil. Roots bind soil particles together, leaves reduce the soil-eroding impact of wind and rain, and dead plant parts add organic matter to the soil.

Plant-Animal Interactions

Plants interact with animals in many fascinating ways. Many flowering plants attract pollinators, animals that carry pollen from one plant to another. Usually the pollinator gets a reward for its efforts in the form of food from nectar. The size, shape, color, and odor of many flowers make them attractive to their pollinators. For example, Figure 27-9 shows that in some orchid species, the flowers have evolved to look and smell like the female of their wasp or bee pollinators. A male wasp or bee lands on a flower believing he has located a mate. The pollen he touches sticks to his body and is transferred to the next orchid he visits. In this case, the flower lures the pollinator with the promise of a mate, but fools the insect into picking up pollen without receiving a reward.

Plant-Microbe Interactions

Two important aspects of plant ecology are plant interactions with fungi and with bacteria. Plant-microbe interactions may be harmful to plants, as in the case of fungal and bacterial diseases. Diseases often cause major crop losses. However, bacteria and fungi also form important beneficial relationships with plants.

The majority of plant species form *mycorrhizae*, which are symbiotic relationships between fungi and the roots of a plant. A mycorrhizal fungus penetrates a root, often changing the root structure. However, the fungus does not harm the root. Instead, it greatly increases the root's ability to absorb water and other inorganic nutrients, such as phosphorus and potassium. In return, the root supplies the fungus with energy.

The roots of many plant species also form beneficial associations with bacteria. Some bacteria can take nitrogen gas from the air and "fix" it, or convert it to a form that plants can use. Plants of the legume family, such as peas, beans, and peanuts, commonly host bacteria that fix nitrogen.

Plant-Human Interactions

We protect and care for many plants that provide us with food, clothing, shelter, medicine, and many other products. However, humans have drastically changed natural plant populations by introducing foreign plant species, diseases, and animals. Introduced plants, such as the water hyacinth shown in Figure 27-10, kudzu, crabgrass, and dandelion have become widespread weeds.

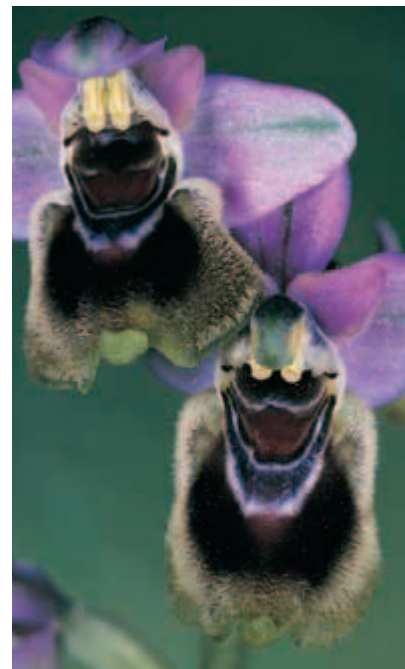


FIGURE 27-9

Some orchid species have evolved to resemble their wasp or bee pollinators.

FIGURE 27-10

The water hyacinth has become a weed that clogs waterways in the southeastern United States.



Word Roots and Origins

pollen

Latin word meaning "fine flour"

Weeds are undesirable plants that often crowd out crop plants or native plant species. For example, water hyacinths float on lakes and rivers, growing so fast and dense that they impede boats and shade underwater plants. The introduction of a fungal disease, chestnut blight, in 1904 virtually wiped out the American chestnut as a dominant forest tree in the eastern United States. Government inspectors now carefully screen plant materials entering the country to prevent the introduction of plant pests and diseases that native plants have no resistance against.

FIGURE 27-11

Giant ragweed, which can grow to more than 4 m (13 ft.) tall, produces massive amounts of pollen that is a major cause of hay fever. The small, dull flowers indicate that it is wind-pollinated.



HARMFUL PLANTS

Despite the many benefits plants provide, some plants can also cause harm. Many deaths are caused by addictive plant products, such as tobacco, cocaine, opium, and alcohol. Some plant species are harmful when eaten or touched. Poison ivy and poison oak give an itchy rash to millions of Americans each year. Children are often poisoned, though usually not fatally, when they eat the leaves or colorful berries of house or garden plants. Despite widespread reports to the contrary, the popular Christmas plant poinsettia is not deadly, but its sap may cause skin irritation. However, holly berries and parts of mistletoe are poisonous.

Tens of millions of people suffer from pollen allergies, one cause of hay fever. **Hay fever** is an allergic reaction that results in sneezing, a runny nose, and watering eyes. Pollen allergies occur at three seasons. In early spring, deciduous trees, such as oak, ash, birch, and sycamore release pollen. In late spring or early summer, it is mainly wild and pasture grasses that cause allergy problems. Of the cereal crops, only rye pollen seems to be an important cause of allergies. In late summer and fall, the highly allergenic pollen of ragweeds, shown in Figure 27-11, affects many people. Contrary to popular belief, large, colorful flowers do not cause hay fever. Pollen that causes allergies comes from small, drab flowers that are wind-pollinated.

SECTION 2 REVIEW

1. What is the name for the study of the interactions that occur between plants and the environment?
2. How do plants play a role in the continuous recycling of many of the Earth's inorganic nutrients?
3. How do plants benefit from their interactions with animals?
4. List three ways that some plants can cause harm to people.

CRITICAL THINKING

5. **Recognizing Differences** Purple loosestrife, a weed that can dominate American wetlands, does not cause problems in its native Europe and Asia. Explain a possible reason for this.
6. **Relating Concepts** Explain the adaptive benefits for plants that have evolved to look like insects.
7. **Recognizing Relationships** Considering how pollination occurs, why do you think hay fever is caused by pollen from grasses and trees?

CHAPTER HIGHLIGHTS

SECTION 1

Plants and People

- The study of plants is called botany. The practical applications of botany are evident in agriculture, which is the raising of crops and livestock for food or other uses.
- Humans have cultivated plants for approximately 11,000 years and have changed, by selection, many plant species so much that these plants can no longer survive in the wild.
- Food crops can be classified in many ways, including by their use and by their taxonomic classification.
- The major part of the human diet is provided by a few cereal crops in the grass family, especially corn, wheat, and rice.
- Everyday definitions of fruits and vegetables are different from botanical definitions. Many common vegetables, such as green beans, tomatoes, squash, and pumpkins, are actually fruits. Botanically speaking, a fruit is the part of a flowering plant that usually contains seeds.
- Plants provide many important medicines, such as digitalis, quinine, morphine, and anti-cancer drugs.
- Several factors have increased food production, including the use of fertilizers and pesticides. As land is cultivated to produce an adequate food supply, the health of the environment is compromised by soil erosion, depleted water supplies, and pollution.
- Plants provide thousands of nonfood products, including clothing, fabric dye, lumber, paper, cosmetics, fuel, cork, rubber, turpentine, and pesticides.
- Ornamental plants improve the human environment in many important ways: they provide shade, minimize soil erosion, reduce noise, and lower home energy costs.

Vocabulary

botany (p. 545)
agriculture (p. 545)
cultivar (p. 545)
cereal (p. 546)

root crop (p. 547)
legume (p. 547)
fruit (p. 547)
vegetable (p. 547)

nut (p. 547)
spice (p. 547)
herb (p. 547)
quinine (p. 547)

fertilizer (p. 548)
pesticide (p. 548)
aspirin (p. 548)
gasohol (p. 551)

SECTION 2

Plants and the Environment

- Plant ecology is the study of the interactions between plants and the environment.
- Plants play a major role in recycling the Earth's water, oxygen, carbon dioxide, and inorganic nutrients.
- Plants provide animals with inorganic nutrients as well as organic nutrients.
- Most plant roots are penetrated by beneficial mycorrhizal fungi, which greatly increase the roots' ability to absorb inorganic nutrients.
- Most nitrogen in living organisms must first be fixed by bacteria, which may live in association with plant roots, especially the roots of legumes.
- Plants associate with animals in many mutually beneficial ways. For example, plants provide food to animals that protect them or carry their pollen.
- People have affected wild plant populations negatively by introducing foreign species of plants, animals, and disease organisms.
- Plants can cause harm in several ways. Many deaths are caused by addictive plant products. Some plant species are poisonous when eaten or touched. And millions of people suffer from allergies to pollen.

Vocabulary

plant ecology (p. 554)

weed (p. 556)

hay fever (p. 556)

CHAPTER REVIEW

USING VOCABULARY

- For each pair of terms, explain the relationship between the terms.
 - botany* and *agriculture*
 - fertilizer* and *pesticide*
 - fruit* and *vegetable*
 - aspirin* and *quinine*
- Choose the term that does not belong in the following group, and explain why it does not belong: *legume*, *nut*, and *root crop*.
- Use the following terms in the same sentence: *gasohol*, *plant ecology*, and *cereal*.
- Word Roots and Origins** The word *herb* is derived from the Latin *herba*, which means “grass.” Using this information, explain why the term *herb* is a good name for the plants that the term describes.

UNDERSTANDING KEY CONCEPTS

- Describe** how people have acted as selecting agents in the evolution of food plants.
- Distinguish** between a fruit and a vegetable.
- State** two negative impacts of modern agricultural methods.
- List** three nonfood, nonmedicinal uses of plants.
- Name** three types of plants that provide fiber used in clothing.
- List** five medicines that are derived from plants.
- Describe** how ornamental trees improve the environment.
- Identify** one inorganic plant nutrient, and describe the role of plants in recycling this nutrient.
- Explain** how plants and microbes interact in mutually beneficial ways.

- Explain** why plant materials entering the country are screened by government inspectors.
- Name** three addictive plant products.
- CONCEPT MAPPING** Use the following  terms to create a concept map that illustrates plant ecology: *humans*, *pollination*, *plants*, *animals*, *hay fever*, *mycorrhizae*, *bees* and *microbes*.

CRITICAL THINKING

- Predicting Results** If all animals disappeared from the Earth, what would the positive and negative effects be on plants?
- Analyzing Information** During the rainy season in the Brazilian rain forest, the rivers flood the land. Many fish from these rivers then swim among the land plants and eat their fruit. How might this intermingling of fish and plants help the plants?
- Analyzing Concepts** Suppose a friend asks you why corn, which he or she considers to be a vegetable, is listed as a cereal crop in the encyclopedia. To answer this question, write a paragraph that explains why corn is a cereal crop, agriculturally, and why corn is a fruit, botanically. Identify other examples of foods that are classified as vegetables or grains but also are fruits.
- Relating Concepts** How did artificial selection by humans play a role in the origin of agricultural crops? How is artificial selection similar to natural selection?
- Interpreting Graphics** Many athletes consume carbohydrates before a competition to increase their endurance. The table below shows a variety of legumes and grains that can be purchased at a grocery store. Compare their prices and nutritional composition. Determine which food is the least expensive source of carbohydrates.

Common Legumes and Grains

Food	Package size	Price	Serving (g)	Per serving			
				Calories	Total fat (g)	Total carbohydrate (g)	Total protein (g)
Navy beans	454 g	.69	45	80	0	23	8
Rice	907 g	.79	45	150	0	35	3
Barley	454 g	.49	45	100	0	24	3
Soybeans	454 g	.89	45	170	8	14	15
Spaghetti	907 g	.79	45	200	1	34	6

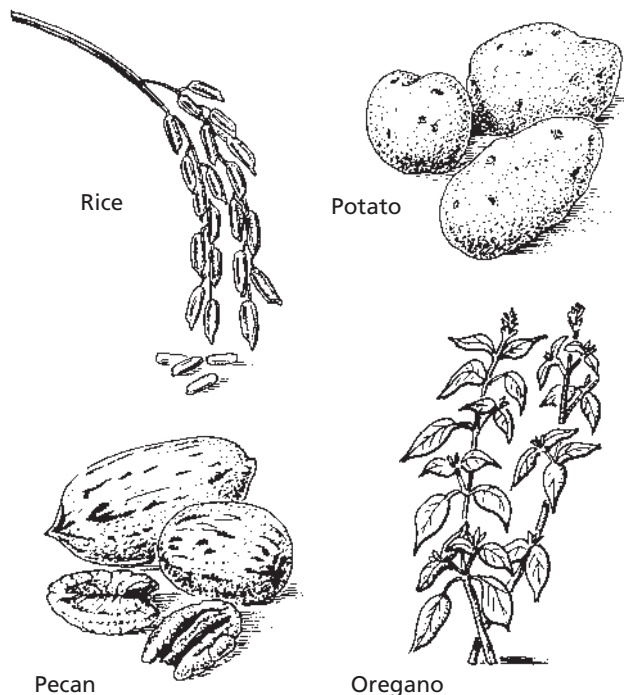


Standardized Test Preparation

DIRECTIONS: Choose the letter of the answer choice that best answers the question.

1. Early farmers selected wheat plants that had which of the following characteristics?
A. fewest grains
B. largest seed pods
C. easily dispersed seeds
D. stalks not easily broken in the wind
2. Which plants are the major source of food for the world today?
F. spices
G. cereals
H. legumes
J. root crops
3. Which of the following is an inorganic nutrient recycled by plants?
A. sugar
B. starch
C. cellulose
D. phosphorus

INTERPRETING GRAPHICS: The illustration below shows four types of food plants. Use the illustration to answer the question that follows.

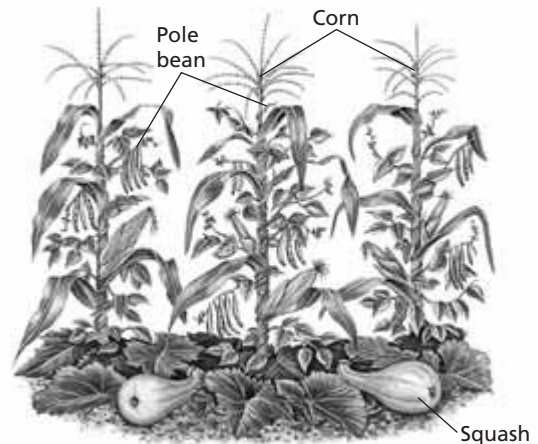


4. Plants classified as root crops have edible underground structures. Which of the food plants shown above is a root crop?
F. rice
G. pecan
H. potato
J. oregano

DIRECTIONS: Complete the following analogy.

5. vegetable : root crop :: grain :
- A. herb
 - B. spice
 - C. cereal
 - D. quinine

INTERPRETING GRAPHICS: The illustration below shows how plants might be grown together. Use the illustration to answer the question that follows.



6. How does this arrangement help these plants?
F. The squash can use the corn as a support.
G. The beans grow apart from the squash and corn.
H. The corn covers the ground and reduces erosion.
J. Beans, which are legumes, will provide nitrogen fertilizer for the corn and squash.

SHORT RESPONSE

Look again at the illustration above.

Explain the advantage to the human diet of growing and eating these food crops together.

EXTENDED RESPONSE

Plants have been cultivated for at least 11,000 years.

Part A Identify 5 plants people use for food and 5 plants people use for other purposes.

Part B How has human cultivation changed some plants?

Test TIP

For a question that uses a word you don't know, try to determine the meaning of the word by breaking it down into smaller parts and inferring the meaning of these parts.

Comparing Soil-Grown Plants with Hydroponic Plants

OBJECTIVES

- Compare hydroponic plant-cultivation techniques with conventional plant-cultivation techniques.
- Observe the germination of wheat seeds over a two-week period.

PROCESS SKILLS

- analyzing data
- measuring
- comparing and contrasting

MATERIALS

- 2 clear plastic cups
- plastic-foam floater with 6 holes in it
- 50 mL of potting soil
- cheesecloth (must be large enough to cover the plastic-foam floater)
- 12 wheat seeds
- 50 mL of complete nutrient solution
- plastic dropper
- labeling tape
- marking pen
- 50 mL graduated cylinder
- metric ruler

Background

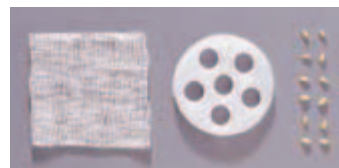
1. Hydroponic cultivation is a technique for growing plants in a solution that contains all of the inorganic nutrients the plant needs. Plants that are grown hydroponically do not require soil.
2. The beginning of growth in a seed is called *germination*.

PART A Day 1

1. Using the marking pen and the labeling tape, label one clear plastic cup "Soil Cultivated," and label the other plastic cup "Hydroponically Cultivated."
2. Fill the cup labeled "Soil Cultivated" halfway with moist potting soil. Place six wheat seeds on the surface of the soil; use the distance between the holes in the foam

floater as a guide to determine the spacing of the wheat seeds. (Do not place the floater on the soil.)

3. Press the seeds into the soil until they are approximately 0.5 cm below the surface. Cover the seeds with soil, and press down firmly.
4. Water the seeds with 10 mL of distilled water.
5. Add 50 mL of complete nutrient solution to the cup labeled "Hydroponically Cultivated," and place the plastic-foam floater on the surface of the solution, as shown in the figure below.
6. Place the cheese-cloth on top of the floater, as illustrated in the figure. Press lightly at the location of the holes in the floater to moisten the cheesecloth.
7. Place the remaining six wheat seeds on top of the cheese-cloth in the cup labeled "Hydroponically Cultivated."



- Position the seeds so that each one lies in an indentation formed by the cheesecloth in a hole in the floater. Press each seed lightly into the hole until the seed coat is moistened.
8. Place both cups in a warm, dry location. Water the soil-cultivated seeds as needed, monitoring the amount of water added. Aerate the roots of the hydroponic plants every day by using a clean plastic dropper to blow air into the nutrient solution.
 9. In your lab report, prepare data tables similar to Table A and Table B. Write your observations of the seeds in your data tables.
 10.  Clean up your lab materials and wash your hands before leaving the lab.

**TABLE A OBSERVATIONS OF
SOIL-GROWN PLANTS**

Day	Appearance of seedlings	Average height (mm)
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		
11		
12		
13		
14		

**TABLE B OBSERVATIONS OF
HYDROPONICALLY GROWN SEEDS**

Day	Appearance of seedlings	Average height (mm)
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		
11		
12		
13		
14		

PART B Days 2–14

11. Compare the contents of each cup every day for two weeks, and record the appearance of the wheat seedlings in your data tables. If you are unable to observe your seedlings over the weekend, be sure to note in your data table that no observations were made on those days.
12. Each time that you observe the seedlings after they have begun to grow, measure their height and record the average height of the seedlings in each cup in your data tables. To find the average height for one cup, add the heights of each seedling in the cup together and divide by the number of seedlings (6).
13. After the seeds in the cup containing nutrient solution have germinated and formed roots, allow an air pocket to form between the floater and the surface of the nutrient solution. A portion of the roots should still be submerged in the nutrient solution. The air pocket allows the roots of the seeds to absorb the oxygen necessary for metabolic processes while continuing to absorb nutrients from the nutrient solution. Continue to observe and record the progress of the seedlings in each cup on a daily basis.
14.  Clean up your lab materials and wash your hands before leaving the lab.

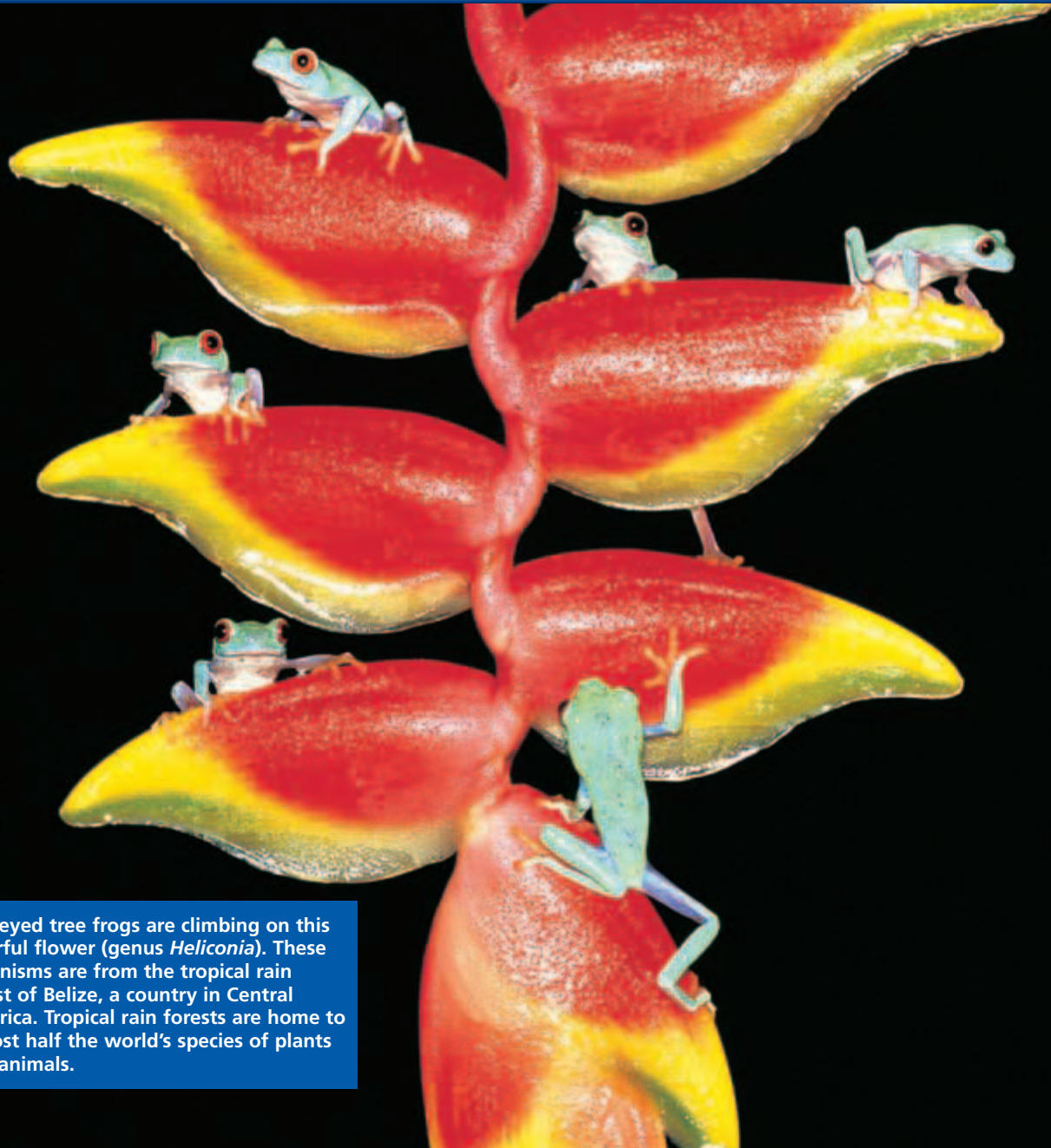
Analysis and Conclusions

1. Based on the data you recorded, which seeds germinated more quickly? Which seeds attained the greatest height?
2. Compare your results with those of your classmates. Were the results the same for each group of students?
3. You planted six seeds in each cup. Why do you think the lab had you do this instead of having you plant a single seed in each cup? Why is the use of more than one sample important?
4. How could hydroponic growing techniques be useful to countries that have either a growing season that is too short to grow a variety of crops or soil that does not support most agricultural crops?

Further Inquiry

The nutrient solution you used in this investigation should have provided all of the inorganic nutrients that the wheat seeds needed for proper growth. How could you determine exactly which inorganic nutrients a plant requires by using hydroponic cultivation?

PLANT EVOLUTION AND CLASSIFICATION



Red-eyed tree frogs are climbing on this colorful flower (genus *Heliconia*). These organisms are from the tropical rain forest of Belize, a country in Central America. Tropical rain forests are home to almost half the world's species of plants and animals.

SECTION 1 *Overview of Plants*

SECTION 2 *Nonvascular Plants*

SECTION 3 *Vascular Plants*

OVERVIEW OF PLANTS

Plants dominate the land and many bodies of water. Plants exhibit tremendous diversity. Some plants are less than 1 mm (0.04 in.) in width, and some plants grow to more than 100 m (328 ft) in height. The 12 phyla, or divisions, of kingdom Plantae include more than 270,000 species. Some plants complete their life cycle in a few weeks, but others may live nearly 5,000 years.

ADAPTING TO LAND

Although life had flourished in the oceans for more than 3 billion years, no organisms lived on land until about 475 million years ago, when a layer of ozone formed. The ozone protected organisms from the sun's ultraviolet radiation. Eventually, small club-shaped plants began to grow in the mud at the water's edge. Three adaptations allowed plants to thrive on land: the ability to prevent water loss, the ability to reproduce in the absence of water, and the ability to absorb and transport nutrients.

Preventing Water Loss

The move to land offered plants distinct advantages, including more exposure to sunlight for photosynthesis, increased carbon dioxide levels, and a greater supply of inorganic nutrients. However, the land environment also presented challenges. Plants on land are susceptible to drying out through evaporation.

The **cuticle** (KYOOT-i-kuhl), a waxy protective covering on plant surfaces that prevents water loss, was one early adaptation to life on land. Although it protects a plant by keeping water in the plant, the cuticle also keeps out carbon dioxide. Plants that had small openings in their surfaces, called *stomata*, were able to survive. Stomata allow the exchange of carbon dioxide and oxygen.

Reproducing by Spores and Seeds

Successful land plants also developed structures that helped protect reproductive cells from drying out. A **spore** is a haploid reproductive cell surrounded by a hard outer wall. Spores allowed the widespread dispersal of plant species. Eventually, most plants developed seeds. A **seed** is an embryo surrounded by a protective coat. Some seeds also contain *endosperm*, a tissue that provides nourishment for the developing plant. Figure 28-1 shows the unusual adaptation of the sugar maple tree for seed dispersal. Seeds are more effective at dispersal than spores are.

OBJECTIVES

- **Name** three adaptations plants have made to life on land.
- **Summarize** the classification of plants.
- **Describe** alternation of generations.

VOCABULARY

cuticle
spore
seed
vascular tissue
xylem
phloem
nonvascular plant
vascular plant
seed plant
gymnosperm
angiosperm
sporophyte
gametophyte
alternation of generations



FIGURE 28-1

The seeds of a sugar maple tree are found inside a winged fruit. Wind can carry the winged fruit away from the tree, thus helping the seeds to disperse away from the parent plant.

Word Roots and Origins

vascular

from the Latin *vasculum*, meaning
"small vessel"

Absorbing and Transporting Materials

Aquatic plants take nutrients from the water around them. On land, most plants absorb nutrients from the soil with their roots. Although the first plants had no roots, fossils show that fungi lived on or within the underground parts of many early plants.

Certain species of plants evolved a type of tissue known as **vascular** (VAS-kyuh-luhr) **tissue**, which transports water and dissolved substances from one part of the plant to another. Two types of specialized tissue make up vascular tissue. **Xylem** (ZIE-luhm) carries absorbed water and inorganic nutrients in one direction, from the roots to the stems and leaves. **Phloem** (FLOH-EM) carries organic compounds, such as carbohydrates, and some inorganic nutrients in any direction, depending on the plant's needs. In addition to transporting absorbed materials, vascular tissue also helps support the plant, which is an important function for land plants.

CLASSIFYING PLANTS

Study the classification of plants in Table 28-1. The 12 phyla of plants, formerly referred to as divisions, can be divided into two groups based on the presence of vascular tissue. The three phyla of **nonvascular plants** have neither true vascular tissue nor true roots, stems, or leaves. The nine phyla of **vascular plants** have vascular tissue and true roots, stems, and leaves.

TABLE 28-1 The 12 Phyla of the Plant Kingdom

Type of plant	Phylum	Common name	Approximate number of existing species
Nonvascular	Bryophyta	mosses	10,000
	Hepatophyta	liverworts	8,000
	Anthoceroophyta	hornworts	100
Vascular, seedless	Psilophyta	whisk ferns	10–13
	Lycophyta	club mosses	1,000
	Sphenophyta	horsetails	15
	Pteridophyta	ferns	12,000
Vascular, seed Gymnosperms	Cycadophyta	cycads	100
	Ginkgophyta	ginkgoes	1
	Coniferophyta	conifers	550
	Gnetophyta	gnetophytes	70
Angiosperms	Anthophyta	flowering plants	240,000
	class Monocotyledones	monocots	70,000
	class Dicotyledones	dicots	170,000

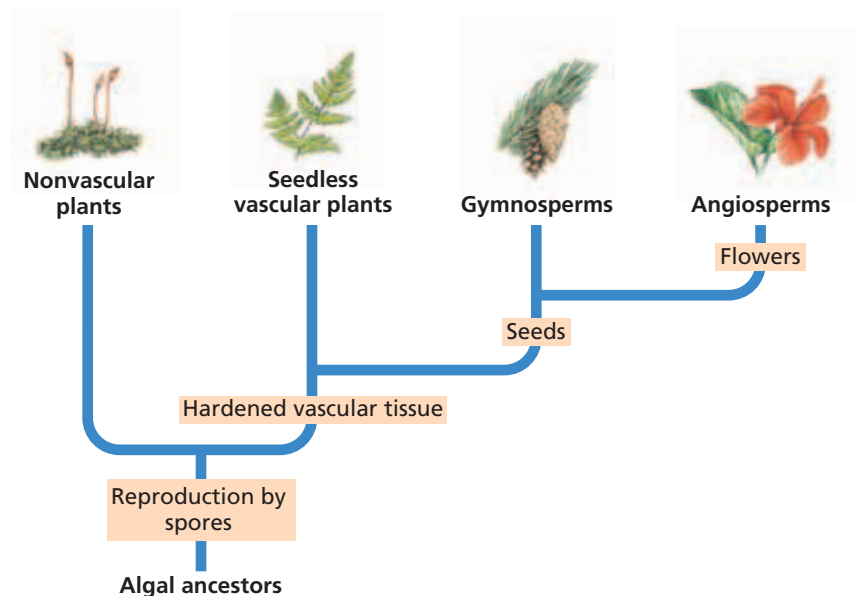


FIGURE 28-2

This phylogenetic diagram represents a hypothesis for the evolutionary relationships between plants and green algae. The earliest plants were nonvascular. These plants evolved into more-complex forms of plants, the vascular plants. For updates on phylogenetic diagrams, visit go.hrw.com. Enter the keyword **HM6 Phylo**.

Notice in Table 28-1 that vascular plants can be further divided into two groups, seedless plants and seed plants. Seedless plants include the phylum of ferns and three phyla made up of plants closely associated with ferns. **Seed plants**—plants that produce seeds for reproduction—include four phyla of gymnosperms (JIM-noh-SPUHRMZ) and one phylum of angiosperms (AN-jee-oh-SPUHRMZ). **Gymnosperms**, which include pine trees, are seed plants that produce seeds that are not enclosed in fruits. **Angiosperms**, also known as flowering plants, are seed plants that produce seeds within a protective fruit. Examples are apple and orange trees.

The Fossil Record of Plants

Figure 28-2 shows the possible origin of major plant groups. Much of what is now known about plant phylogeny comes from the fossil record. The fossil record is incomplete, but scientists hypothesize that plants evolved from algal ancestors. The strongest evidence lies in the similarities between modern green algae and plants. Both have the same photosynthetic pigments—chlorophylls *a* and *b*, both store energy as starch, and both have cell walls made of cellulose.

ALTERNATING LIFE CYCLES

All plants have a life cycle that involves two phases, which are named for the type of reproductive cells they produce. Recall that cells having two sets of chromosomes are referred to as *diploid*, and cells having only one set of chromosomes are referred to as *haploid*. The first phase of a plant's life cycle consists of a diploid ($2n$) **sporophyte** (SPOH-ruh-FIET) plant that produces spores. The second phase consists of a haploid ($1n$) **gametophyte** (guh-MEET-uh-FIET) plant that produces eggs and sperm. A life cycle that alternates between the gametophyte phase and sporophyte phase is called **alternation of generations**.

Eco Connection

Reforestation Efforts

The process of replacing trees that have died or been cut down is called *reforestation*. Natural reforestation occurs when seeds grow into new seedlings. Throughout the world, many governments and private landowners do not replace trees after land has been cleared to produce timber, build roads, and construct buildings.

Such *deforestation* is occurring at a rate of several hundred thousand square miles per year. Although this rate seems overwhelming, people can help reforestation efforts in once naturally forested areas by planting seeds or seedlings. It's best to choose trees that are well adapted to your area. For more information, call your local or state parks department, the U.S. Forest Service, a county extension agent, or a nearby college forestry department.

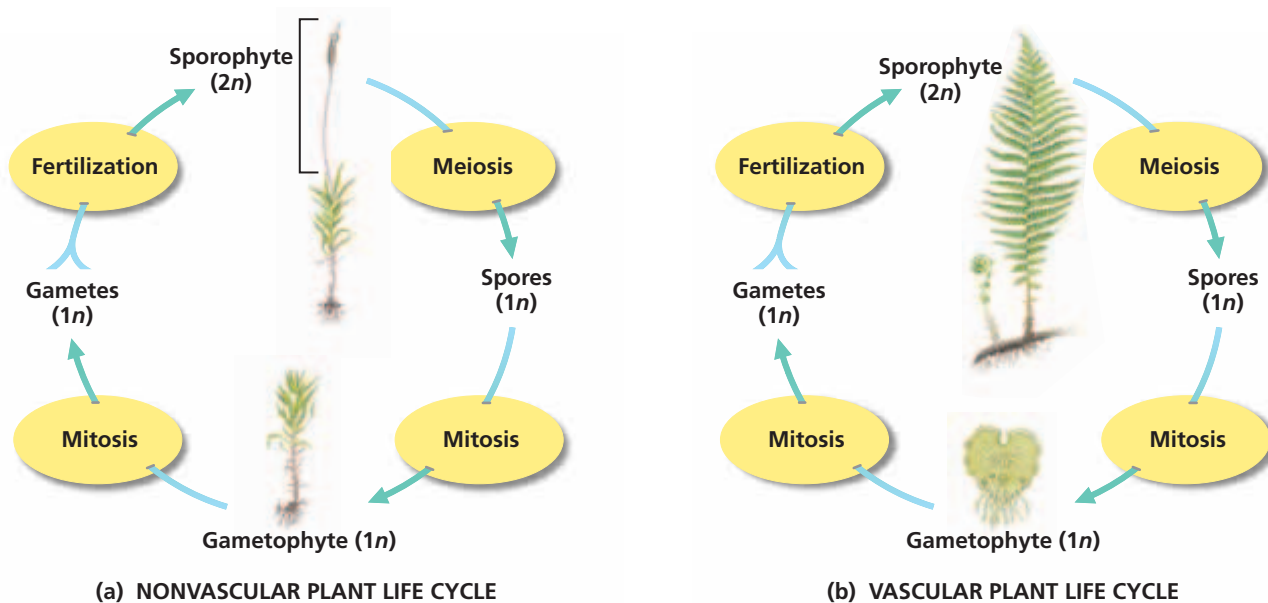


FIGURE 28-3

In the life cycle of a plant, there is an alternation of the haploid gametophyte generation and the diploid sporophyte generation. (a) The life cycle of a nonvascular plant, such as a moss, is characterized by a long, thin sporophyte growing up from the top of the more prominent gametophyte. (b) The life cycle of a vascular plant, such as a fern, is characterized by a large sporophyte and a very small gametophyte.

Figure 28-3 shows the life cycles of a nonvascular plant and a vascular plant. In alternation of generations, the gametophyte ($1n$) undergoes mitosis to form gametes—eggs and sperm. Once an egg is fertilized by a sperm and produces a zygote, the plant begins the diploid phase of its life cycle. The zygote divides by mitosis to form a sporophyte plant. The sporophyte ($2n$) produces cells that undergo meiosis to form haploid spores. These spores are released by most seedless plants but are retained by seed plants. The life cycle begins again when spores divide by mitosis to form new gametophytes.

In nonvascular plants, the gametophyte is the dominant phase. In contrast, the sporophyte is the dominant phase of vascular plants. Oak trees are large sporophytes that dominate some landscapes. In seedless vascular plants, the gametophyte is usually a separate small organism quite different from the sporophyte. In seed plants, the gametophyte is a very small parasite of the sporophyte. For example, gametophytes of flowering plants are microscopic parts of their flowers that are not photosynthetic.

SECTION 1 REVIEW

1. How does the cuticle represent an adaptive advantage for early land plants?
2. Compare the structure of a spore with that of a seed.
3. What is the main difference between the nine phyla of vascular plants and the three phyla of nonvascular plants?
4. Explain how alternation of generations between the gametophyte phase and the sporophyte phase differs in vascular plants and nonvascular plants.

CRITICAL THINKING

5. **Justifying Conclusions** Why are vascular plants more successful than nonvascular plants as land plants?
6. **Inferring Relationships** How have a vascular system and a dominant sporophyte contributed to the success of plants on land?
7. **Organizing Information** For each of the following pairs, identify the most recently evolved characteristic: spore/seed; vascular tissue/no vascular tissue; and cuticle/no cuticle.

NONVASCULAR PLANTS

*The three phyla of nonvascular plants are collectively called **bryophytes**. Botanists have identified about 16,600 species of bryophytes. They lack vascular tissue and do not form true roots, stems, and leaves. These plants usually grow on land near streams and rivers.*

CHARACTERISTICS OF BRYOPHYTES

Bryophytes are the most primitive type of plants. Overall, their characteristics are more like those of plants than of algae. Bryophytes are mostly terrestrial and have an alternation-of-generations life cycle. Bryophytes are seedless, and they produce spores. Because they do not have vascular tissue, they are very small, usually 1–2 cm (0.4–0.8 in.) in height.

Bryophytes need water to reproduce sexually because the sperm must swim through water to an egg. In dry areas, bryophytes can reproduce sexually only when adequate moisture is available. The asexual production of haploid spores does not require water.

PHYLUM BRYOPHYTA

Almost every land environment is home to at least one species of moss in the phylum Bryophyta (brie-AHF-uh-tuh). The thick green carpets of moss on shady forest floors actually consist of thousands of moss gametophytes. Each gametophyte is attached to the soil by rootlike structures called *rhizoids* (RIE-ZOYDZ). Unlike roots, rhizoids do not have vascular tissue. But rhizoids do function like roots by anchoring the moss and by absorbing water and inorganic nutrients.

Moss gametophytes are usually less than 3 cm (1.2 in.) tall. The moss sporophyte grows up from the top of the gametophyte, as shown in Figure 28-4. Gametophytes may be male, may be female, or may contain both male and female reproductive parts.

Mosses are called *pioneer plants* because they are often the first species to inhabit a barren area. Mosses gradually accumulate inorganic and organic matter on rock surfaces, creating a layer of soil in which other plants can grow. In areas devastated by fire, volcanic action, or human activity, pioneering mosses can help trigger the development of new biological communities. They also help prevent soil erosion by covering the soil surface and absorbing water.

OBJECTIVES

- **Identify** the characteristics of bryophytes.
- **Describe** plants in the phylum Bryophyta.
- **Describe** plants in the phylum Hepatophyta.
- **Describe** plants in the phylum Anthocerotophyta.

VOCABULARY

bryophyte
liverwort
hornwort



FIGURE 28-4

Moss sporophytes (thin, brown stalks) depend on the gametophytes (green, “mossy” parts) to which they are attached, because sporophytes do not undergo photosynthesis.

FIGURE 28-5

A peat farmer cuts blocks of peat, produced from moss of the genus *Sphagnum*, to heat his farmhouse in County Clare, Ireland.



Peat moss (of the genus *Sphagnum*) is a major component of bogs in northern parts of the world. Peat moss produces an acid that slows down decomposition in the swamplike bogs. Compression and chemical breakdown in peat bogs produces *peat*, which consists of partially decomposed plant matter. In many northern European and Asian countries, peat is mined and dried for use as fuel, as shown in Figure 28-5. Dried peat moss is widely used to enhance the water-retaining ability of potting and gardening soils. Dried peat moss is also used by florists to pack bulbs and flowers for shipping.

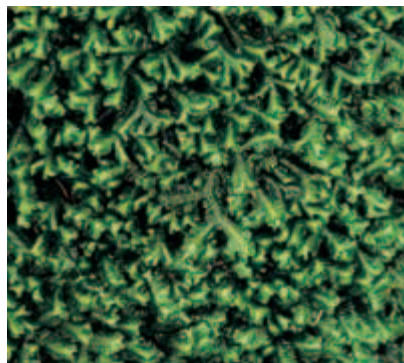
Word Roots and Origins

liverwort

from the familiar *liver* (body organ) and the Old English *wort*, meaning “herb”

PHYLUM HEPATOPHYTA

Phylum Hepatophyta (HEP-uh-TAHF-uh-tuh) includes the **liverworts**, unusual-looking plants that grow in moist, shady areas. Most liverworts have thin, transparent leaflike structures arranged along a stemlike axis, as shown on the left side of Figure 28-6. Some liverworts have a *thalloid* (THAL-oyd) form—that is, a flat body with distinguishable upper and lower surfaces, as shown on the right side of Figure 28-6. All liverworts lie close to the ground. This adaptation allows them to absorb water readily. In some species, the gametophyte is topped by an umbrella-shaped structure that holds the reproductive cells.



(a)



(b)

FIGURE 28-6

A leafy liverwort (a) has two rows of leaflike structures growing on a stem. A thalloid liverwort (b) has very flat, thin lobes.



FIGURE 28-7

Hornworts grow in warm, moist habitats, such as along roads or near streams.

PHYLUM ANTHOCEROPHYTA

Phylum Anthocerophyta (AN-thoh-suh-RAHF-uh-tuh) includes the **hornworts**, which also grow in moist, shaded areas. The name “hornwort” refers to the long, thin, hornlike sporophytes that grow out from the top of the plant, as shown in Figure 28-7. When the sporophytes are not present, hornworts look very similar to thalloid liverworts. Hornworts do not have a stem or leaves; the gametophyte of a hornwort is anchored to the ground by rhizoids. Hornworts share an unusual characteristic with algae: Each cell usually has a single large chloroplast rather than numerous small chloroplasts.

The sporophytes of hornworts are different from the sporophytes of mosses and liverworts in that they are green and carry out photosynthesis. The sporophytes continue to grow throughout the plant’s life. They also are covered with a cuticle and have stomata. Tubelike cells at the center of the sporophytes resemble cells of vascular tissue and may transport materials in these plants. These characteristics indicate that the hornworts are closely related to vascular plants.

SECTION 2 REVIEW

1. List two characteristics shared by all nonvascular plants.
2. Describe the role that mosses play in the early development of biological communities.
3. Explain why liverworts lie close to the ground.
4. Describe how the sporophytes of hornworts differ from the sporophytes of mosses or liverworts.

CRITICAL THINKING

5. **Making Comparisons** What advantage do gametophytes that are either male or female have over gametophytes with both male and female structures?
6. **Evaluating Information** Why can’t mosses grow as large as maple or oak trees?
7. **Applying Information** How is a moss’s ability to absorb water advantageous?

SECTION 3

OBJECTIVES

- **Describe** the adaptive advantages that vascular plants have over nonvascular plants.
- **Summarize** the characteristics of the four phyla of seedless vascular plants.
- **State** the major differences between gymnosperms and angiosperms.
- **Determine** why angiosperms have been so successful.
- **Compare** monocots and dicots.

VOCABULARY

strobilus
rhizome
fiddlehead
frond
germinate
seedling
cone
deciduous
ovary
cotyledon
monocot
dicot
vein
parallel venation
net venation

VASCULAR PLANTS

Vascular plants contain specialized conducting tissues (xylem and phloem) that transport water and dissolved substances from one part of the plant to another. Vascular plants can grow larger and live in more environments than nonvascular plants. The strong stems of vascular plants allow the plants to grow tall, enabling them to rise above other plants and receive more sunlight than shorter plants do.

SEEDLESS VASCULAR PLANTS

Seedless vascular plants dominated the Earth until about 200 million years ago. Characteristics of the four phyla of seedless vascular plants are summarized in Table 28-2. The first three phyla are called *fern allies*, while members of the last phylum are *ferns*. Spores are the mobile sexual reproductive parts of all seedless plants.

Phylum Psilophyta

The phylum Psilophyta (sie-LAHF-uh-tuh) is represented by whisk ferns, illustrated in Figure 28-8. Despite their name, whisk ferns are not ferns at all. They have no roots or leaves and produce spores on the ends of short branches. These features suggest that whisk ferns resemble early land plants. Some species of the phylum Psilophyta are *epiphytes*, which means they grow on other plants. But they are not parasites because they do not harm their host plant.



FIGURE 28-8

The whisk fern produces spores at the ends of branched stem tips. These plants are found in tropical and subtropical regions.

TABLE 28-2 Seedless Vascular Plants

Example plant	Phylum	Features	Size	Location
Whisk ferns	Psilophyta	• produce reproductive structures on the ends of forked branches • no roots or leaves	• about 30 cm (1 ft) tall	• tropical and temperate regions, as far north as South Carolina
Club mosses	Lycophyta	• evergreens that produce spores in cones • have roots	• about 5 cm (2 in.) tall	• tropical and temperate regions, on forest floors, in swamps, or as epiphytes
Horsetails	Sphenophyta	• jointed stems • outer cells of stems contain silica, the major component of sand	• about 60–90 cm (2–3 ft) tall	• tropical and temperate regions, usually in moist soil
Ferns	Pteridophyta	• leaves • most have an underground stem • most produce spores on the underside of their leaves	• range from less than 1 cm (0.4 in.) to 25 m (82 ft) tall	• all climates, on forest floors, as epiphytes, some in full sun, some aquatic

Phylum Lycophyta

The phylum Lycophyta (lie-KAHF-uh-tuh) contains the club mosses, an example of which is shown in Figure 28-9. Because they look like miniature pine trees, club mosses are also called ground pines. They produce a **strobilus** (stroh-BIE-luhs), or cone, which is a cluster of sporangia-bearing modified leaves. Club mosses were once widely collected as Christmas decorations.

Another member of phylum Lycophyta is a spike moss called *Selaginella lepidophylla*, native to the American Southwest. *Selaginella* turns brown and curls up in a ball during drought. However, when moistened, the plant uncurls and turns green again after a few hours.

**FIGURE 28-9**

The club mosses, sometimes known as ground pines, are members of the phylum Lycophyta. The tips of the aerial stems contain conelike structures.



FIGURE 28-10

This horsetail, of the genus *Equisetum*, has hollow, jointed stems that contain silica. About 300 million years ago, some species of the genus *Equisetum* were large trees, growing with large club moss trees and ferns in steaming swamps. Over millions of years, the trees and other plants died, became buried, and turned into coal.



FIGURE 28-11

Tree ferns, such as the *Dicksonia antarctica* shown here, look like palm trees but are actually the largest living ferns. Sometimes, epiphytic ferns grow on the trunk of tree ferns.

Phylum Sphenophyta

The phylum Sphenophyta (sfee-NAHF-uh-tuh) includes horsetails of the genus *Equisetum*. Horsetails have jointed photosynthetic stems that contain silica, with scalelike leaves at each joint. The vertical stems of horsetails, which grow from a rhizome, are hollow and have joints. Spores form in cones located at the tips of stems.

American pioneers used horsetails to scrub pots and pans; hence, they are frequently called *scouring rushes*. As you can see in Figure 28-10, the shoots are often highly branched and remind some people of a horsetail.

Phylum Pteridophyta

Ferns probably originated over 350 million years ago. Ferns belong to the phylum Pteridophyta (tuhr-uh-DAHF-uh-tuh) and represent a diverse group. Some are floating plants that are less than 1 cm (0.4 in.) across. Ferns also grow above the Arctic Circle and in desert regions. The largest living ferns are tree ferns, shown in Figure 28-11. These ferns can reach 25 m (82 ft) in height, and some have leaves 5 m (16 ft) long. Tree ferns live in tropical and subtropical areas.

Most ferns have an underground stem called a **rhizome** (RIE-ZOHM). The fibrous rhizomes of a few species of ferns are used as a growing medium for orchids. The tightly coiled new leaves of ferns are called **fiddleheads**. The young fiddleheads of some species are eaten by humans as a vegetable. Fiddleheads uncoil and develop into mature leaves called **fronds**.

VASCULAR SEED PLANTS

The mobile sexual reproductive part of seed plants is the multicellular seed. Seeds are an evolutionary success story. Plants with seeds have a greater chance of reproductive success than seedless plants. Inside the tough, protective outer coat of a seed is an embryo and a nutrient supply. When conditions are too hot or too cold, or too wet or too dry, the seed remains inactive. When conditions favor growth, the seed sprouts, or **germinates**—that is, the embryo begins to grow into a young plant, called a **seedling**.

There are two main groups of seed-bearing vascular plants, gymnosperms and angiosperms. The four phyla of gymnosperms produce naked seeds, which means the seeds are not enclosed and protected in fruits. Most gymnosperms are evergreen and bear their seeds in cones. A **cone** is a reproductive structure composed of hard scales. The seeds lie open on the surface of the scales. The one phylum of angiosperms produces seeds that are enclosed and protected in fruits. Angiosperms are commonly referred to as flowering plants. Cones serve some of the same functions for gymnosperms that flowers serve for angiosperms.



FIGURE 28-12

The cycad is a gymnosperm that looks like a palm or fern. Cycads can sometimes grow to 18 m (60 ft) in height.

Phylum Cycadophyta

Cycads (SIE-KADZ), such as the one shown in Figure 28-12, are gymnosperms of the phylum Cycadophyta (sie-kad-AHF-uh-tuh). Although cycads flourished during the age of the dinosaurs, only about 100 species survive today. Most are native to the Tropics and grow slowly. Some cycads live for almost a thousand years. Many are endangered because of habitat loss, overcollection, and their slow growth. Most cycads have fernlike, leathery leaves at the top of a short, thick trunk. Cycad plants are either male or female, and they bear large cones. Cycads are mostly used as ornamental plants.

Phylum Ginkgophyta

Like cycads, ginkgoes (GING-kohz) flourished during the time of the dinosaurs. The only species existing today is *Ginkgo biloba*, which is native to China. It is called a living fossil because it closely resembles fossil ginkgoes that are 125 million years old. The ginkgo tree has fan-shaped leaves that fall from the tree at the end of each growing season—an unusual characteristic for a gymnosperm. Trees that lose their leaves at the end of the growing season, like the ginkgo, are called **deciduous**. Most gymnosperms are evergreens and retain their leaves year-round.

Ginkgoes are tolerant of air pollution, making them good plants for urban settings. Ginkgo seeds are considered a delicacy in China and Japan. Notice the plum-shaped, fleshy seeds on the ginkgo shown in Figure 28-13. They are often mistakenly called berries or fruits.

Phylum Coniferophyta

The conifers (KAHN-uh-fuhrz), which are gymnosperms of the phylum Coniferophyta (kahn-uh-fuhr-AHF-uh-tuh), include pine, cedar, redwood, fir, spruce, juniper, cypress, and bald cypress trees. They are important sources of wood, paper, turpentine, ornamental plants, and Christmas trees. Juniper seeds can be used to flavor food. Amber is yellow or brownish yellow fossilized resin that once flowed from ancient conifers. Prehistoric insects are often preserved in amber.



Quick Lab

Examining Ferns

Materials disposable gloves, lab apron, potted fern, hand lens, water

Procedure



1. Put on your disposable gloves and lab apron.
2. Choose a frond of the fern, and examine its underside for the structures that contain spores.
3. Wash the soil from the underground structures. Examine the fern's horizontal stems and rhizome.

Analysis How do ferns differ from nonvascular plants? What enables ferns to surpass nonvascular plants in height and size? From what part of the fern do the fronds grow?



FIGURE 28-13

The ginkgo, *Ginkgo biloba*, has large seeds and unusual fan-shaped leaves. This gymnosperm tree can reach heights of 24 m (80 ft).



(a) Fir needles and cones



(b) Pine needles and cones



(c) Yew needles and seeds

FIGURE 28-14

The needles and cones of conifers come in many shapes and sizes. (a) The fir tree displays its female cones. Its needle-shaped leaves grow evenly all around the branch. (b) The pine tree shows its small male and larger female cones. Some pines reach heights of 60 m (200 ft). (c) The seed of the yew tree is surrounded by a red covering that looks like a berry. Its leaves are flat, pointed needles that are dark green on top and pale green underneath.

Conifers are woody plants, and most have needle or scalelike leaves, as shown in Figure 28-14. A conifer usually bears both male and female cones. Small male cones typically grow in clusters. Male cones release clouds of dustlike pollen, and then the cones fall from the branches. The pollen falls or blows into the larger female cones, where the egg cells are attached to the scales of the cone. After pollination, the female cone closes up tightly. This protects the developing seeds, which mature after one or two years. The mature seeds are released when the female cone opens.

Redwoods and giant sequoia trees provide a majestic forest setting along the West Coast of the United States. These conifers are the Earth's tallest and most massive living organisms. The tallest living coastal redwood, *Sequoia sempervirens*, is about 110 m (360 ft) tall, the height of a 30-story building. The most massive tree is a giant sequoia, *Sequoiadendron giganteum*, estimated to weigh 5,600 megagrams (6,200 tons).

Phylum Gnetophyta

Gnetophytes (NEE-tuh-FIETS), an odd group of cone-bearing gymnosperms, have vascular systems that more closely resemble those of angiosperms. As Figure 28-15 shows, *Ephedra* (ih-FED-ruh) is a genus of desert shrubs with jointed stems that look like horsetails. It is the source of the drug *ephedrine*, which can be used as a decongestant.



FIGURE 28-15

Ephedra viridis, called Mormon tea, grows on the rim of the Grand Canyon. This highly branched shrub has small, scalelike leaves. It is the source of the drug ephedrine and can be brewed to make a tea.

Figure 28-16 shows the unique *Welwitschia mirabilis* plant. The plant's stem is only a few centimeters tall but can grow to 1 m (3.3 ft) in diameter. Two leaves elongate from their base on the stem and then become tattered and split lengthwise by the wind. A mature leaf may be nearly 1 m (3.3 ft) wide and 3 m (10 ft) long. *Welwitschia* grows in the Namib Desert of southwestern Africa. The Namib Desert lies near the Atlantic Ocean, so a thick night fog often rolls in over the desert. *Welwitschia* apparently gets most of its water from the dew that condenses from the fog.

Phylum Anthophyta

Anthophyta (an-THAHF-uh-tuh), the largest phylum of plants, includes over 240,000 species of flowering plants. Angiosperms, or the flowering plants, are seed plants characterized by the presence of a flower and fruit. Botanists define a *fruit* as a ripened ovary that surrounds the seeds of angiosperms. The **ovary** is the female part of the flower that encloses the egg(s).

Angiosperms grow in many forms and occupy diverse habitats. Some are herbaceous plants with showy flowers, such as violets and impatiens. Others, such as rose bushes, are shrubs. Some angiosperms are vines, such as grape plants. Oak, aspen, and birch trees are all flowering plants that have woody stems, although you may never have noticed their small flowers. Grasses are also angiosperms, but you must look closely to see their small, highly modified flowers. The world's largest flower, which can grow to 1 m (3.3 ft) in diameter, is shown in Figure 28-17.

THE EVOLUTION OF ANGIOSPERMS

Angiosperms first appeared in the fossil record about 135 million years ago. By about 90 million years ago, angiosperms had probably begun to outnumber gymnosperms. Several factors probably led to the success of this new kind of plant. In many angiosperms, seeds germinate and produce mature plants, which in turn produce new seeds, all in one growing season. This is a tremendous advantage over gymnosperms, which often take 10 or more years to reach maturity and produce seeds. Also, the fruits of flowering plants protect seeds and aid in their dispersal. Angiosperms also have a more efficient vascular system and are more likely to be associated with mycorrhizae than gymnosperms are. Angiosperms also may gain an advantage by using animal pollination rather than the less-efficient wind pollination method used by gymnosperms. However, wind pollination is used by many successful angiosperms, including many deciduous trees. Finally, angiosperms are more diverse than gymnosperms, so they occupy more niches, such as in aquatic, epiphytic, and parasitic environments.



FIGURE 28-16

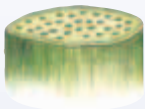
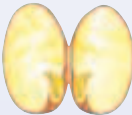
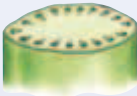
Welwitschia mirabilis has a short, wide stem and twisting leaves. The female plants of this unusual gymnosperm bear large seed cones that are reddish in color.

FIGURE 28-17

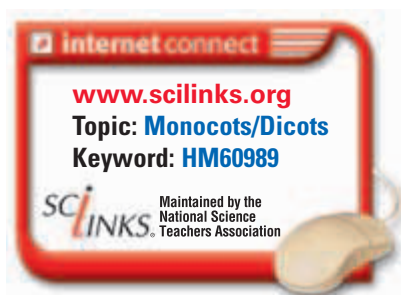
The stinking-corpse lily, *Rafflesia arnoldii*, has the world's largest flowers but no leaves or stems. The flowers can be male or female, and they are pollinated by flies. The plant lacks chlorophyll and is parasitic on a woody vine native to Southeast Asia.



TABLE 28-3 Comparing Monocots and Dicots

Plant type	Embryos	Leaves	Stems	Flower parts	Examples
Monocots	One cotyledon 	Parallel venation 	Scattered vascular bundles 	Usually occur in threes 	lilies, irises, orchids, palms, tulips, bananas, pineapples, onions, bamboo, coconut, grasses (including wheat, corn, rice, and oats)
Dicots	Two cotyledons 	Net venation 	Radially arranged vascular bundles 	Usually occur in fours or fives 	beans, lettuce, oaks, maples, elms, roses, carnations, cactuses, most broad-leaved forest trees

MONOCOTS AND DICOTS



The flowering plants are divided into Monocotyledones (monocots) and Dicotyledones (dicots). The primary feature that distinguishes these two classes is the number of **cotyledons** (KAHT-uh-LEED'nz), or seed leaves, in a plant embryo. **Monocots** (MAHN-oh-KAHTS) usually have one cotyledon, while **dicots** (DIE-KAHTS) typically have two. By comparison, gymnosperms usually have two or more cotyledons.

Several characteristics can be used to identify monocots and dicots, as shown in Table 28-3. For example, most mature monocot leaves have several main **veins**, or bundles of vascular tissue, running parallel to each other. This vein arrangement is called **parallel venation**. Most dicots have one or more nonparallel veins that branch repeatedly, forming a network. This vein arrangement is called **net venation**. More than one characteristic should be used to determine whether a species is a monocot or a dicot.

SECTION 3 REVIEW

1. Identify the reproductive advantages seed plants have over seedless vascular plants and nonvascular plants.
2. Name the reproductive characteristic that all seedless vascular plants share.
3. Describe the main differences between gymnosperms and angiosperms.
4. Explain why angiosperms have been more successful than gymnosperms.
5. List three characteristics that distinguish monocots from dicots.

CRITICAL THINKING

6. **Making Comparisons** Compare the vascular tissue of vascular plants with the circulatory system of vertebrate animals.
7. **Analyzing Information** Explain how the evolutionary advancements present in both seed plants and seedless vascular plants benefit land plants.
8. **Inferring Relationships** Conifers, which typically have needles or scalelike leaves, are the dominant plants in areas with cold winter climates. Identify the adaptive advantage of having narrow or small leaves.

CHAPTER HIGHLIGHTS

SECTION 1

Overview of Plants

- Three adaptations have allowed plants to be successful on land: a cuticle to prevent water loss, spores and seeds to protect reproductive cells, and special tissues for absorbing and transporting materials within the plant.
- The 12 phyla of plants are divided into two groups based on the presence of vascular tissue. The three phyla of nonvascular plants have neither true vascular tissue nor roots, stems, or leaves. Most members of the nine phyla of vascular plants have vascular tissue and true roots, stems, and leaves.
- Vascular plants can be further divided into two groups, seedless plants and seed plants. Seed plants include four phyla of gymnosperms and one phylum of angiosperms.
- All plants have a life cycle known as alternation of generations. In alternation of generations, a haploid gametophyte produces gametes. Gametes unite and give rise to a diploid sporophyte. Through meiosis, the sporophyte produces haploid spores, which develop into gametophytes.

Vocabulary

cuticle (p. 563)

spore (p. 563)

seed (p. 563)

vascular tissue (p. 564)

xylem (p. 564)

phloem (p. 564)

nonvascular plant (p. 564)

vascular plant (p. 564)

seed plant (p. 565)

gymnosperm (p. 565)

angiosperm (p. 565)

sporophyte (p. 565)

gametophyte (p. 565)

alternation of generations
(p. 565)

SECTION 2

Nonvascular Plants

- The three phyla of nonvascular plants are collectively called bryophytes. These plants do not have true roots, stems, or leaves. They are very small and are usually found in moist areas.
- Bryophytes in the phylum Bryophyta are mosses. Mosses are attached to the soil by structures called rhizoids. Peat moss is a moss that has many uses.
- Bryophytes in the phylum Hepatophyta are liverworts. Liverworts lie close to the ground, which allows them to absorb water readily.
- Bryophytes in the phylum Anthocerophyta are hornworts. Hornworts do not have a stem or leaves. Hornworts have long, thin, hornlike sporophytes that grow out of the top of the plant.

Vocabulary

bryophyte (p. 567)

liverwort (p. 568)

hornwort (p. 569)

SECTION 3

Vascular Plants

- Vascular plants have several adaptive advantages over nonvascular plants, including specialized conducting tissues, the ability to grow large and live in many environments, and strong stems that allow them to grow tall and receive more sunlight.
- Seedless vascular plants include the four phyla Psilophyta, Lycophyta, Sphenophyta, and Pteridophyta. Seedless vascular plants include ferns and fernlike plants. Ferns are the dominant phylum of seedless plants. Most ferns have a rhizome, an underground stem.
- There are two main groups of seed-bearing vascular plants, gymnosperms and angiosperms. Gymnosperms are characterized by naked seeds and no flowers. Angiosperms have flowers and seeds enclosed by a fruit.
- Angiosperms have been successful for many reasons, including the production of fruit that protects seeds, quick germination, and an efficient vascular system.
- Dicots are distinguished from monocots on the basis of several characteristics: cotyledon number, leaf venation, arrangement of stem vascular tissue, and number of flower parts.

Vocabulary

strobilus (p. 571)

rhizome (p. 572)

fiddlehead (p. 572)

frond (p. 572)

germinate (p. 572)

seedling (p. 572)

cone (p. 572)

deciduous (p. 573)

ovary (p. 575)

cotyledon (p. 576)

monocot (p. 576)

dicot (p. 576)

vein (p. 576)

parallel venation (p. 576)

net venation (p. 576)

CHAPTER REVIEW

USING VOCABULARY

- For each pair of terms, explain the relationship between the terms.
 - seed* and *spore*
 - vascular plant* and *nonvascular plant*
 - phloem* and *xylem*
- Choose the term that does not belong in the following group, and explain why it does not belong: *cuticle*, *spore*, *seed*, and *strobilus*.
- For each pair of terms, explain how the meanings of the terms differ.
 - gymnosperm* and *angiosperm*
 - gametophyte* and *sporophyte*
 - monocot* and *dicot*
- Use the following terms in the same sentence: *cotyledon*, *germinate*, and *seedling*.
- Word Roots and Origins** The word *deciduous* is derived from the Latin *de*, which means “down,” and *cadere*, which means “to fall.” Using this information, explain why the term *deciduous* is a good name for the plants that this term describes.

UNDERSTANDING KEY CONCEPTS

- Summarize** how plants are adapted to living successfully on land.
- Name** two basic differences between nonvascular plants and vascular plants.
- Describe** how alternation of generations in nonvascular plants differs from that in vascular plants.
- Explain** why bryophytes are the most primitive type of plant.
- Describe** why plants in the phylum Bryophyta are sometimes called pioneer plants.
- Differentiate** between leafy liverworts and thaloid liverworts.
- State** the characteristic that hornworts do not share with other bryophytes.
- Explain** how specialized conducting tissues give vascular plants an adaptive advantage over nonvascular plants.
- Relate** how ferns and fern allies are able to reproduce without seeds.
- Differentiate** between a cone and a fruit.
- Explain** how the diversity of angiosperms has helped them be more successful than gymnosperms.
- State** the primary basis used for classifying angiosperms.

- 
CONCEPT MAPPING Use the following terms to create a concept map that describes the possible evolutionary relationships of plants: *angiosperm*, *cone*, *flower*, *dicot*, *gymnosperm*, *monocot*, *ovary*, and *seed plant*.

CRITICAL THINKING

- Analyzing Information** Fossil trees are easier to find than fossils of small plants. Give two possible explanations.
- Interpreting Graphics** Look at the plants in the photo below. Do you think the cuticles of these plants are thicker on the upper surface of the leaf or on the lower surface? Explain.



- Organizing Information** Copy the chart below into your notebook. Indicate with a check mark (✓) which groups of plants have vascular tissue. Also, indicate which groups of plants have a dominant sporophyte as part of their life cycle.

Plant Evolution

Phylum	Has vascular tissue	Has dominant sporophyte
Anthophyta		
Bryophyta		
Coniferophyta		
Hepatophyta		
Pteridophyta		

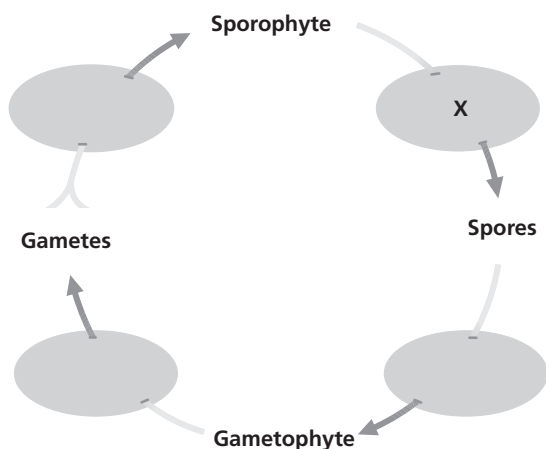


Standardized Test Preparation

DIRECTIONS: Choose the letter of the answer choice that best answers the question.

- Which of the following plants are bryophytes?
 - ferns and cycads
 - conifers and ginkgoes
 - hornworts and liverworts
 - horsetails and club mosses
- In which way do mosses help start new biological communities?
 - forming new soil
 - producing spores
 - detecting air pollution
 - slowing decomposition
- True roots, stems, and leaves are characteristics of which types of plants?
 - all plants
 - only seed plants
 - only angiosperms
 - all vascular plants
- Which of the following is a vascular seed plant?
 - ferns
 - cycads
 - horsetails
 - club mosses

INTERPRETING GRAPHICS: The diagram below shows a plant life cycle. Use the diagram to answer the question that follows.

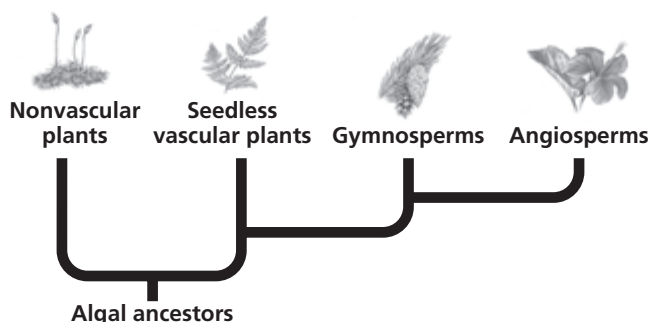


- Which process occurs at X in this life cycle?
 - mitosis
 - meiosis
 - alternation
 - fertilization
- Which of the following phrases describes monocots?
 - bear seeds in cones
 - have parallel venation
 - do not produce flowers
 - have vascular bundles arranged in a circle

DIRECTIONS: Complete the following analogy.

7. bryophyte : spore :: angiosperm :
- seed
 - cone
 - ovary
 - cuticle

INTERPRETING GRAPHICS: The phylogenetic diagram below shows a possible evolutionary relationship between plants and algae. Use the diagram to answer the question that follows.



8. According to this diagram, which plants have evolved most recently?
- angiosperms
 - gymnosperms
 - nonvascular plants
 - seedless vascular plants

SHORT RESPONSE

The plant kingdom is very diverse, from small non-vascular plants to large flowering vascular plants.

Explain why some plants have been more successful on land than other plants have been.

EXTENDED RESPONSE

All plants have a life cycle that alternates between haploid and diploid phases.

Part A Name the two phases of a plant life cycle and describe how they differ from each other.

Part B Describe the life cycle of one nonvascular plant and one vascular plant, including the relative sizes of the different forms and other characteristics.

Test TIP

For questions involving life cycles, draw as much of the life cycle as you can remember. Looking at such a model may help you understand the question better and determine the answer.

Observing Plant Diversity

OBJECTIVES

- Compare similarities and differences among phyla of living plants.
- Relate structural adaptations to the evolution of plants.

PROCESS SKILLS

- observing
- comparing and contrasting
- classifying
- relating structure and function

MATERIALS

- live and preserved specimens representing four plant phyla
- stereomicroscope or hand lens
- compound light microscope
- prepared slides of male and female gametophytes of mosses and ferns

Background

1. How do plants you commonly see compare with their ancestors, the green algae?
2. What are the differences between nonvascular plants and vascular plants? How do these differences relate to the size of a plant?
3. What is alternation of generations? Is it found in all plants?
4. Do all plants produce spores? Do all plants produce seeds? What are the advantages of producing seeds?
5. What do you think was the evolutionary pressure that resulted in colorful flowers?

Procedure

1.   **CAUTION** Put on protective gloves. Keep your hands away from your face while handling plants. You will travel to four stations to observe plants from four phyla. Record the answers to the questions in your lab report.



STATION 1 Mosses

2. Use a stereomicroscope or a hand lens to examine the samples of mosses, which are bryophytes. Which part of the moss is the gametophyte? Which part of the moss is the sporophyte? Make a sketch of your observations in your lab report. In your drawing, label the gametophyte and sporophyte portions of the moss plant and indicate whether each is haploid or diploid.
3. Use a compound microscope to look at the prepared slides of male and female gametophytes. What kinds of reproductive cells are produced in each of these structures? Draw the cells in your lab report.
4. Do mosses have roots? How do mosses obtain water and nutrients from the soil?

STATION 2 Ferns

5. Look at the examples of ferns at this station. The fern leaf is called a frond. Use the hand lens to examine the fronds.
 - a. How does water travel throughout a fern? List observations supporting your answer.
 - b. Make a drawing of the fern plant in your lab report.



- c. Indicate whether the leafy green frond in your drawing is haploid or diploid.
- d. Search the underside of the fern fronds for evidence of reproductive structures. Make a drawing of your findings in your lab report. What kind of reproductive cells are produced by these structures?
- 6. Examine the examples of fern gametophytes.
 - a. Locate and identify the reproductive organs found on the gametophytes. In your lab report, sketch and label these organs and identify the reproductive cells produced by each.
 - b. Are the gametophytes haploid or diploid?
- 7. In what ways are ferns like bryophytes? In what ways are they different?

STATION 3 Conifers

- 8. The gymnosperms most familiar to us are conifers. Look at the samples of conifers at this station.
 - a. When you look at the limb of a pine tree, which portion (gametophyte or sporophyte) of the plant life cycle are you seeing?
 - b. In what part of the conifer would you find reproductive structures?
- 9. Name an evolutionary advancement found in gymnosperms but lacking in ferns.



STATION 4 Angiosperms

- 10. Draw one of the representative angiosperms at this station in your lab report. Label the representative angiosperm as a monocot or a dicot, and list at least two characteristics you used to identify it.
- 11. Name an evolutionary development that is present in both gymnosperms and angiosperms but absent in bryophytes and ferns.



- 12. How do the seeds of angiosperms differ from those of gymnosperms?
- 13. Examine the fruits found at this station. How have fruits benefited angiosperms?
- 14.   Clean up your materials and wash your hands before leaving the lab.

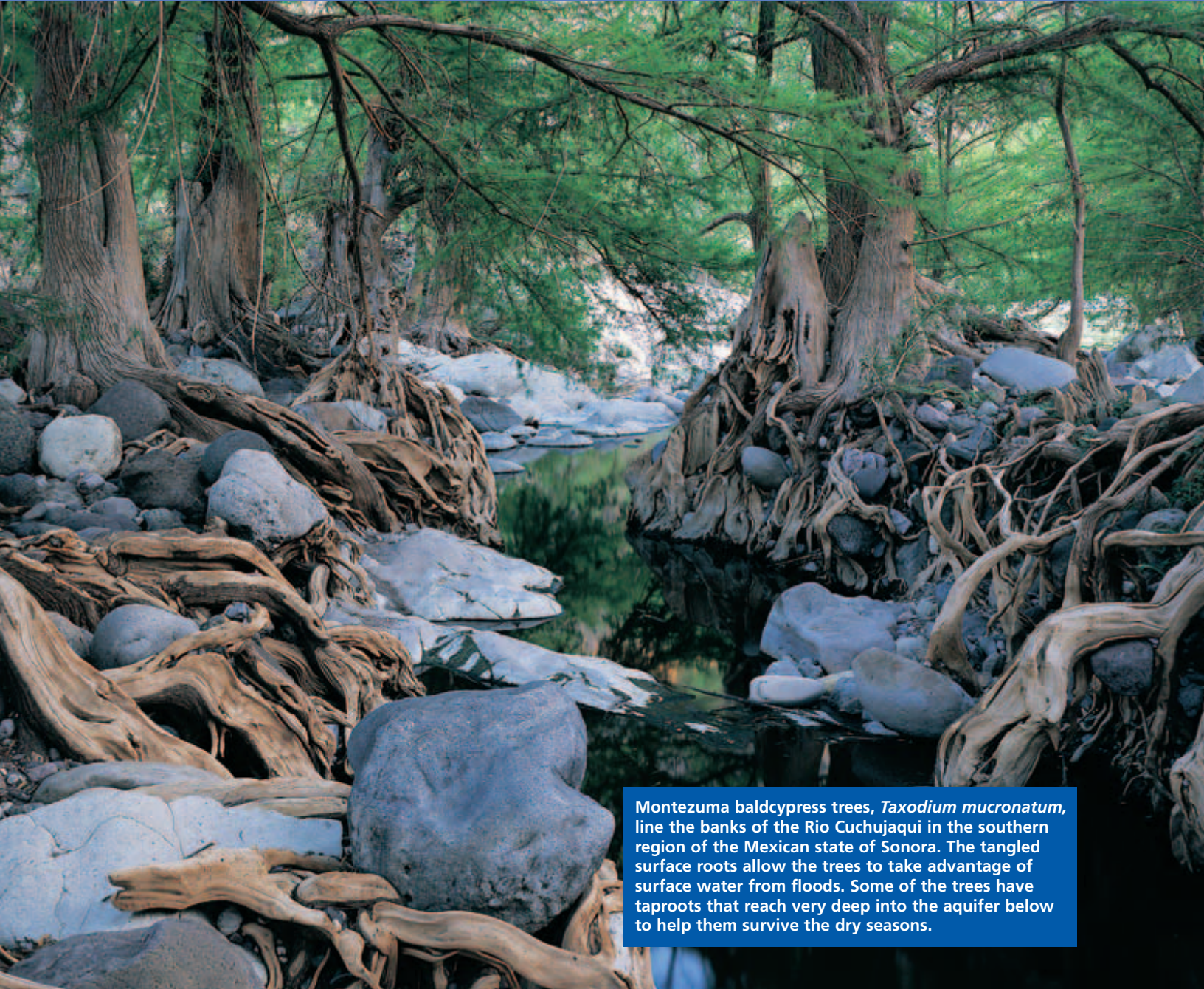
Analysis and Conclusions

- 1. In bryophytes, how do the sperm travel from the male gametophyte to the female gametophyte?
- 2. In angiosperms, how do the sperm get to the part of the flower containing the egg?
- 3. Which portion of the plant life cycle is dominant in bryophytes? Which portion is dominant in ferns, gymnosperms, and angiosperms?
- 4. What is a seed? Why is the seed a helpful adaptation for terrestrial plants?
- 5. Why are gymnosperms referred to as *naked* seed plants?
- 6. Which group of plants is the most successful and diverse today? What are some adaptations found among members of this group?

Further Inquiry

- 1. Find out how the geographic distribution of the phyla of living plants relates to their structures.
- 2. Research the deforestation of tropical rain forests. How are the different groups of plants affected by deforestation?

PLANT STRUCTURE AND FUNCTION



Montezuma baldcypress trees, *Taxodium mucronatum*, line the banks of the Rio Cuchujaqui in the southern region of the Mexican state of Sonora. The tangled surface roots allow the trees to take advantage of surface water from floods. Some of the trees have taproots that reach very deep into the aquifer below to help them survive the dry seasons.

SECTION 1 *Plant Cells and Tissues*

SECTION 2 *Roots*

SECTION 3 *Stems*

SECTION 4 *Leaves*



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PLANT CELLS AND TISSUES

Plants have adapted to a range of environments over the course of their evolution. As plants grow, their cells become specialized for particular functions. The tissue patterns vary in each plant's roots, stems, and leaves. They also vary depending on the plant's stage of growth and taxonomic group. This chapter examines the structure and function of roots, stems, and leaves.

PLANT CELLS

All organisms are composed of cells. Recall that plant cells have unique structures, including a central vacuole, plastids, and a cell wall that surrounds the cell membrane. These common features are found in three basic types of plant cells—parenchyma, collenchyma, and sclerenchyma—which are shown in Figure 29-1. Small changes in the structure of these plant cells help make different functions possible. The three types of plant cells are arranged differently in roots, stems, and leaves.

Parenchyma (puh-REN-kuh-muh) cells are usually loosely packed cube-shaped or elongated cells with a large central vacuole and thin, flexible cell walls. Parenchyma cells are involved in metabolic functions, including photosynthesis, storage of water and nutrients, and healing. These cells usually form the bulk of nonwoody plants (plants with flexible, green stems). The fleshy parts of most fruit are made up mostly of parenchyma cells.

The cell walls of **collenchyma** (koh-LEN-kuh-muh) cells are thicker than those of parenchyma cells. Collenchyma cell walls are also irregular in shape. The thicker walls provide support for the plant.

OBJECTIVES

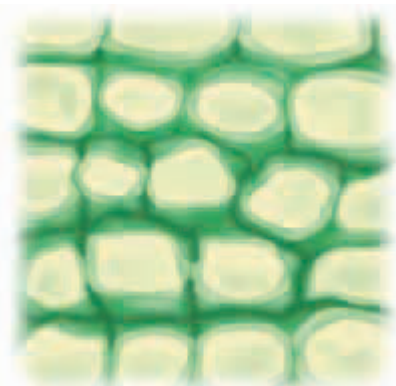
- **Describe** the three basic types of plant cells.
- **Compare** the three plant tissue systems.
- **Describe** the type of growth that occurs in each of the three main types of meristems.
- **Differentiate** between primary and secondary growth.

VOCABULARY

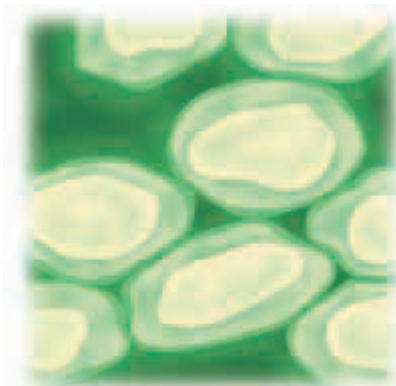
parenchyma
collenchyma
sclerenchyma
epidermis
cuticle
tracheid
pit
vessel element
vessel
sieve tube member
sieve tube
sieve plate
companion cell
meristem
apical meristem
lateral meristem
vascular cambium
cork cambium

FIGURE 29-1

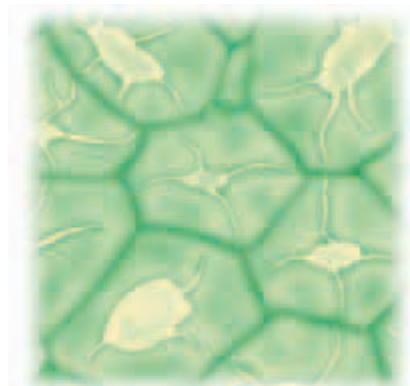
Plants are composed of three basic types of cells: (a) parenchyma, (b) collenchyma, and (c) sclerenchyma.



(a) PARENCHYMA



(b) COLLENCYMA



(c) SCLERENCHYMA

Collenchyma cells are usually grouped in strands. They are specialized for supporting regions of the plant that are still lengthening. Celery stalks contain a great amount of collenchyma cells.

Sclerenchyma (skluh-REN-kuh-muh) cells have thick, even, rigid cell walls. They support and strengthen the plant in areas where growth is no longer occurring. This type of cell is usually dead at maturity, providing a frame to support the plant. The hardness of the shells around nuts is due to the presence of sclerenchyma cells.

PLANT TISSUE SYSTEMS

Cells that work together to perform a specific function form a *tissue*. Tissues are arranged into systems in plants, including the *dermal system*, *ground system*, and *vascular system*, which are summarized in Table 29-1. These systems are further organized into the three major plant organs—the roots, stems, and leaves. The organization of each organ reflects adaptations to the environment.

Dermal Tissue System

The *dermal tissue system* forms the outside covering of plants. In young plants, it consists of the **epidermis** (EP-uh-DUHR-muhs), the outer layer made of parenchyma cells. The outer epidermal wall is often covered by a waxy layer called the **cuticle**, which prevents water loss. Some epidermal cells of the roots develop hairlike extensions that increase water absorption. Openings in the leaf and stem epidermis are called *stomata*. Stomata regulate the passage of gases and moisture into and out of the plant. In woody stems and roots, the epidermis is replaced by dead cork cells.

TABLE 29-1 Characteristics of Plant Tissue Systems

Tissue system	Type of cells	Location	Function in roots	Function in stems	Function in leaves
Dermal tissue system	flat, living parenchyma (epidermal cells) in nonwoody parts; flat, dead parenchyma (cork cells) in woody parts	outermost layer(s) of cells	absorption, protection	gas exchange, protection	gas exchange, protection
Ground tissue system	mostly parenchyma, usually with some collenchyma and fewer sclerenchyma	between dermal and vascular in nonwoody plant parts	support, storage	support, storage	photosynthesis
Vascular tissue system	elongated sclerenchyma cells, with some parenchyma	tubes throughout plant	transport, support	transport, support	transport, support

Ground Tissue System

Dermal tissue surrounds the *ground tissue system*, which consists of all three types of plant cells. Ground tissue functions in storage, metabolism, and support. Parenchyma cells are the most common type of cell found in ground tissue. Nonwoody roots, stems, and leaves are made up primarily of ground tissue. Cactus stems have large amounts of parenchyma cells for storing water in dry environments. Plants that grow in waterlogged soil often have parenchyma with large air spaces that allow air to reach the roots. Nonwoody plants that must be flexible to withstand wind have large amounts of collenchyma cells. Sclerenchyma cells are found where hardness is an advantage, such as in the seed coats of hard seeds and in the spines of cactuses.

Vascular Tissue System

Ground tissue surrounds the *vascular tissue system*, which functions in transport and support. Recall that the term *vascular tissue* refers to both xylem and phloem. Xylem tissue conducts water and mineral nutrients primarily from roots upward in the plant. Xylem tissue also provides structural support for the plant. Phloem tissue conducts organic compounds and some mineral nutrients throughout the plant. Unlike xylem, phloem is alive at maturity.

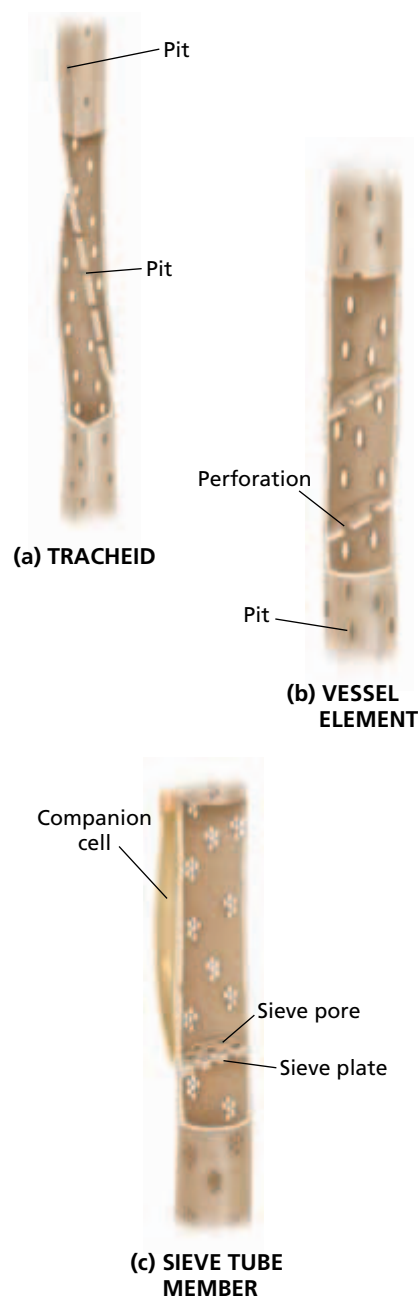
In angiosperms, xylem has two major components—tracheids and vessel elements. Both are dead cells at maturity. Look at Figure 29-2a. A **tracheid** (TRAY-kee-id) is a long, thick-walled sclerenchyma cell with tapering ends. Water moves from one tracheid to another through **pits**, which are thin, porous areas of the cell wall. A **vessel element**, shown in Figure 29-2b, is a sclerenchyma cell that has either large holes in the top and bottom walls or no end walls at all. Vessel elements are stacked to form long tubes called **vessels**. Water moves more easily in vessels than in tracheids. The xylem of most seedless vascular plants and most gymnosperms contains only tracheids, which are considered a primitive type of xylem cell. The vessel elements in angiosperms probably evolved from tracheids. Xylem also contains parenchyma cells.

The conducting parenchyma cell of angiosperm phloem is called a **sieve tube member**. Look at Figure 29-2c. Sieve tube members are stacked to form long **sieve tubes**. Compounds move from cell to cell through end walls called **sieve plates**. Each sieve tube member lies next to a **companion cell**, a specialized parenchyma cell that assists in transport. Phloem also usually contains sclerenchyma cells called *fibers*. Commercially important hemp, flax, and jute fibers are phloem fibers.

Vascular tissue systems are adapted to different environmental conditions. For example, xylem forms the wood of trees, providing the plants with strength while conducting water and mineral nutrients. In aquatic plants, such as duckweed, xylem is not necessary for support or water transport and may be nearly absent from the mature plant.

FIGURE 29-2

(a) Tracheids are long and thin, and they contain pits in their cell walls. (b) Vessel elements are shorter and wider than tracheids. Both tracheids and vessel elements transport water. (c) Sieve tube members are long and tubular, and they contain pores in their cell walls. Sugar is transported through sieve tube members.



Word Roots and Origins

meristem

from the Greek *meristos*,
meaning “divided”

TABLE 29-2 *Types of Meristems*

Type	Location	Growth function
Apical meristem	tips of stems and roots	increase length at tips
Intercalary meristem	between the tip and base of stems and leaves	increase length between nodes
Lateral meristem	sides of stems and roots	increase diameter

PLANT GROWTH

Plant growth originates mainly in **meristems** (MER-i-STEMZ), regions where cells continuously divide. Look at Table 29-2. Most plants grow in length through **apical** (AP-i-kuhl) **meristems** located at the tips of stems and roots. Some monocots grow in length through *intercalary* (in-TUHR-kah-ler-ee) *meristems* located above the bases of leaves and stems. Intercalary meristems allow grass leaves to quickly regrow after being grazed or mowed.

Gymnosperms and most dicots also have **lateral meristems**, which allow stems and roots to increase in diameter. Lateral meristems are located near the outside of stems and roots. There are two types of lateral meristems. The **vascular cambium**, located between the xylem and phloem, produces additional vascular tissues. The **cork cambium**, located outside the phloem, produces cork. *Cork* cells replace the epidermis in woody stems and roots. Cork cells are dead cells that provide protection and prevent water loss.

You have probably noticed that trees grow taller and wider over time. Growth in length is called *primary growth* and is produced by apical and intercalary meristems. Growth in diameter is called *secondary growth* and is produced by the lateral meristems.

SECTION 1 REVIEW

1. How do the structures of parenchyma cells, collenchyma cells, and sclerenchyma cells differ?
2. Describe the structures of the three major plant tissue systems.
3. What kind of meristems do monocots and dicots have in common?
4. Distinguish between primary growth and secondary growth in a tree.

CRITICAL THINKING

5. **Making Comparisons** What vertebrate animal structures have cells that carry out functions similar to those of sclerenchyma?
6. **Applying Information** Describe some factors that may influence the transport of water through xylem tissue.
7. **Analyzing Information** Describe how the plant kingdom would be different if plant growth only occurred in lateral meristems.

ROOTS

Plants have three kinds of organs—roots, stems, and leaves. A plant's root system includes the structures that typically grow underground. Roots are important because they anchor the plant in the soil. They also absorb and transport water and mineral nutrients. The storage of water and organic compounds is provided by roots.

TYPES OF ROOTS

When a seed sprouts, it produces a primary root. If this first root becomes the largest root, it is called a **taproot**, as illustrated in Figure 29-3a. Many plants, like carrots and certain trees, have taproots. Contrary to what you might think, even taproots rarely penetrate the ground more than a meter or two. A few species, such as cottonwoods, do have some roots that grow 50 m (164 ft) deep to tap into underground water supplies.

In some plants, the primary root does not become large. Instead, numerous small roots develop and branch to produce a **fibrous root system**, like that shown in Figure 29-3b. Many monocots, such as grasses, have fibrous root systems. Fibrous roots of monocots often develop from the base of the stem rather than from other roots.

OBJECTIVES

- **Explain** the difference between a taproot, a fibrous root system, and an adventitious root.
- **Describe** the structure of roots.
- **Distinguish** between primary growth and secondary growth in roots.
- **List** the three major functions of roots.

VOCABULARY

taproot
fibrous root system
adventitious root
root cap
root hair
cortex
endodermis
pericycle
macronutrient
micronutrient



(a) TAPROOT



(b) FIBROUS ROOT SYSTEM

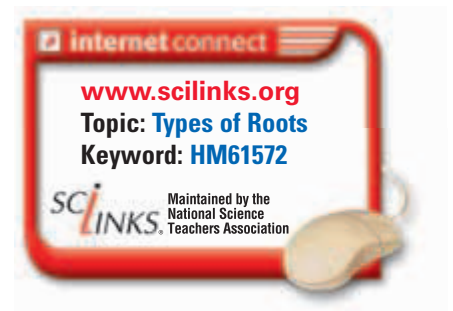


FIGURE 29-3

Plants can have either a taproot or a fibrous root system. (a) Many dicots, including the radish, have a large central taproot with small lateral roots. (b) Most monocots, including grasses, have a highly branched fibrous root system.

FIGURE 29-4

Some plants grow adventitious roots from above-ground parts, including stems and leaves. (a) Corn plants grow prop roots at their base to provide additional stability. (b) This orchid grows *aerial roots*, which absorb water and mineral nutrients from the surface of the tree and from the air.



(a)



(b)

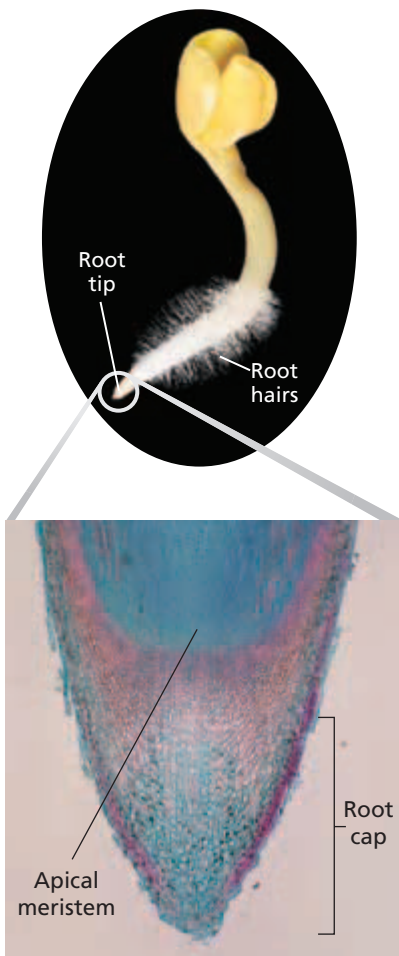


FIGURE 29-5

The root tip of this seedling has many root hairs, which increase surface area and help the plant absorb water and mineral nutrients from the soil. The root cap protects the apical meristem of a root tip.

Specialized roots that grow from uncommon places, such as stems and leaves, are called **adventitious roots**. Figure 29-4a shows the prop roots of corn, which help keep the plant's stems upright. The aerial roots of an epiphytic orchid, shown in Figure 29-4b, obtain water and mineral nutrients from the air. Aerial roots on the stems of ivy and other vines enable them to climb walls and trees.

ROOT STRUCTURES

Root structures are adapted for several functions. Study Figure 29-5. Notice that the root tip is covered by a protective **root cap**, which covers the apical meristem. The root cap produces a slimy substance that functions like lubricating oil, allowing the root to move more easily through the soil as it grows. Cells that are crushed or knocked off the root cap as the root moves through the soil are replaced by new cells produced in the apical meristem, where cells are continuously dividing.

Root hairs, which are extensions of epidermal cells, increase the surface area of the root and thus increase the plant's ability to absorb water and minerals from the soil. Root hairs are shown in Figure 29-5. Most roots form symbiotic relationships with fungi to form a *mycorrhiza*. The threadlike hyphae of a mycorrhiza also increase the surface area for absorption. The spreading, usually highly branched root system increases the amount of soil that the plant can “mine” for water and mineral nutrients and aids in anchoring the plant. The large amount of root parenchyma usually functions in storage and general metabolism. Roots are dependent on stems and leaves for their energy, so they must store starch to use as an energy source during periods of little or no photosynthesis, such as winter.

Primary Growth in Roots

Roots increase in length through cell division, elongation, and maturation in the apical meristem in the root tip (shown in Figure 29-5 on the previous page). Dermal tissue matures to form the epidermis, which is the outermost boundary of the root. Ground tissue in roots matures into two different, specialized regions: the cortex and the endodermis. The **cortex** is located just inside the epidermis, as you can see below in Figure 29-6. This largest region of the primary root is made of loosely packed parenchyma cells.

The innermost boundary of the cortex is the **endodermis** (EN-doh-DUHR-mis), also shown in Figure 29-6. Endodermal cell walls contain a narrow band of a waterproof substance that stops the movement of water beyond the endodermal cells. To enter farther into the root than the endodermis, water and dissolved substances must pass through a selectively permeable membrane. Once past this membrane, dissolved substances can move from cell to cell via small channels in cell walls that interconnect the cytoplasm of adjoining cells, including those of the endodermis. The endodermis thus controls the flow of dissolved substances into the vascular tissue of the root. The endodermis also prevents dissolved substances from backing into the cortex.

Vascular tissue in roots matures into the innermost core of the root. In most dicots and gymnosperms, xylem makes up the central core of the root, as shown in Figure 29-6a. Dicot root xylem usually forms an X-shaped structure with pockets of phloem between the xylem lobes. In monocots, the center of the root usually contains a pith of parenchyma cells, as Figure 29-6b shows. Monocot root xylem occurs in many patches that circle the pith. Small areas of phloem occur between the xylem patches.

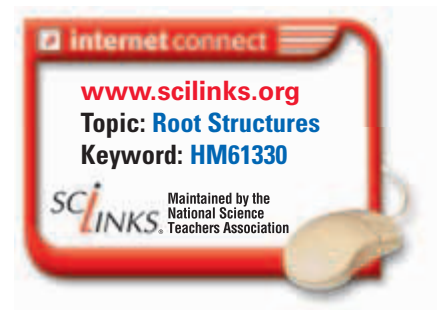
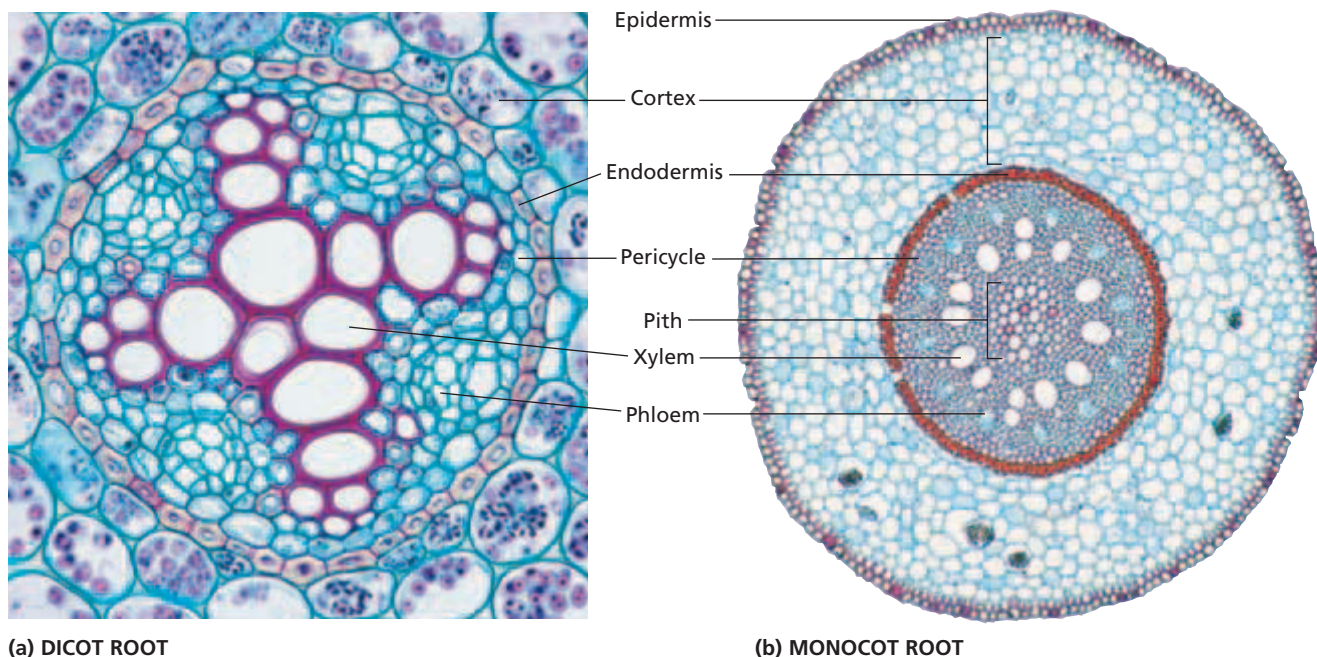


FIGURE 29-6

(a) This cross section of a dicot root shows the arrangement of vascular tissue and ground tissue. Note how the xylem tissue forms an "X." The cortex and the endodermis, which are composed of ground tissue, surround the vascular tissue. (b) This cross section of a monocot root shows a prominent endodermis, the innermost boundary of the cortex. The center of the root, called the *pith*, is made up of parenchyma cells.

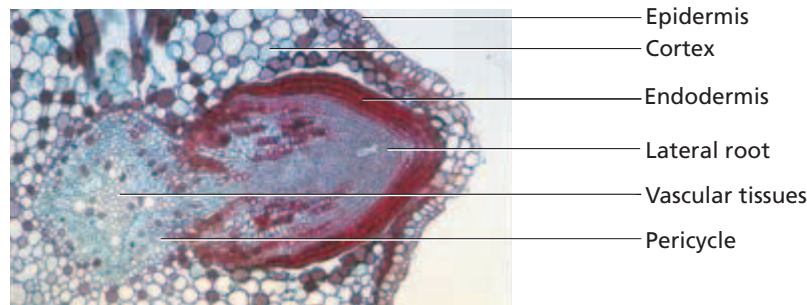


(a) DICOT ROOT

(b) MONOCOT ROOT

FIGURE 29-7

The vascular tissues of a primary root are surrounded by the pericycle, a tissue that produces lateral roots. The pericycle is also shown in Figure 29-6 on the previous page.



The outermost layer or layers of the central vascular tissues is termed the **pericycle** (PER-i-SIE-kuhl). Lateral roots are formed by the division of pericycle cells. The developing lateral root connects its vascular tissues and endodermis to those of the parent root. Figure 29-7 shows how a lateral root grows out through the parent root's endodermis and cortex, finally emerging from the epidermis.

Secondary Growth in Roots

Dicot and gymnosperm roots often experience secondary growth. Secondary growth begins when a pericycle and other cells form a vascular cambium between primary xylem and primary phloem. The vascular cambium produces secondary xylem toward the inside of the root and secondary phloem toward the outside. The expansion of the vascular tissues in the center of the root crushes all the tissues external to the phloem, including the endodermis, cortex, and epidermis. A cork cambium develops in the pericycle, replacing the crushed cells with cork.



Quick Lab

Observing Roots

Materials wilted radish seedlings, hand lens, Petri dish, water, pipet

Procedure

1. Place the wilted radish seedlings in a Petri dish. Observe them with the hand lens. Record your observations.
2. Using a pipet, cover only the roots with water. Observe the seedlings with the hand lens every 5 minutes for 15 minutes. Record each of your observations.
3. Use the hand lens to observe the roots. Draw and label what you see through the lens.

Analysis What happened to the wilted seedlings when you put them in water? How can you explain what happened? Describe two functions of a root.

ROOT FUNCTIONS

Besides anchoring a plant in the soil, roots serve two other primary functions. First, they absorb water and a variety of minerals or mineral nutrients that are dissolved in water in the soil. Roots are selective about which minerals they absorb. Roots absorb some minerals and exclude others. Table 29-3 on the next page lists the 13 minerals that are essential for all plants. They are absorbed mainly as ions. Carbon, hydrogen, and oxygen are not listed because they are absorbed as water and carbon dioxide.

Plant cells use some minerals, such as nitrogen and potassium, in large amounts. These elements are called **macronutrients** and usually are required in amounts greater than 1,000 mg/kg of dry matter. Plant cells use other minerals, such as manganese, in smaller amounts. These are called **micronutrients** and usually are required in amounts less than 100 mg/kg of dry matter.

Adequate amounts of all 13 mineral nutrients in Table 29-3 are required for normal growth. Plants with deficiencies show characteristic symptoms and reduced growth. Severe mineral deficiencies can kill a plant. Excess amounts of some essential mineral nutrients also can be toxic to a plant.

TABLE 29-3 *Essential Mineral Nutrients in Plants***Macronutrients**

Element	Absorbed as	Role in plants
Nitrogen	NO_3^- , NH_4^+	part of proteins, nucleic acids, chlorophyll, ATP
Phosphorus	H_2PO_4^-	part of nucleic acids, ATP, phospholipids, coenzymes
Potassium	K^+	required for stomatal opening and closing, enzyme cofactor
Calcium	Ca^{2+}	part of cell walls and cell membranes
Magnesium	Mg^{2+}	part of chlorophyll
Sulfur	SO_4^{2-}	part of proteins

Micronutrients

Element	Absorbed as	Role in plants
Iron	Fe^{2+}	part of molecules in electron transport
Manganese	Mn^{2+}	required by many enzymes
Boron	$\text{B}(\text{OH})_3$	thought to be involved in carbohydrate transport
Chlorine	Cl^-	required to split water in photosynthesis
Zinc	Zn^{2+}	essential part of many enzymes
Copper	Cu^{2+}	essential part of many enzymes
Molybdenum	MoO_4^{2-}	required for nitrogen metabolism

Some roots also store carbohydrates or water. Phloem tissue carries sugars made in leaves to roots. Sugars that roots do not immediately use for energy or building blocks are stored. In roots, these carbohydrates are usually converted to starch and stored in parenchyma cells. You are probably familiar with the storage roots of carrots, turnips, and sweet potatoes. The roots of some species in the pumpkin family store large amounts of water, which helps the plants survive during dry periods.

SECTION 2 REVIEW

1. What are the differences between a taproot, a fibrous root system, and an adventitious root?
2. Explain how root hairs increase the ability of a plant to absorb water from the soil the plant grows in.
3. State one function of a root cap.
4. What is the difference between primary growth and secondary growth in roots?
5. What are the major functions performed by the root systems of plants?

CRITICAL THINKING

6. **Analyzing Information** Why might a taproot be an advantage to some plants, whereas a fibrous root system might be an advantage to others?
7. **Applying Information** How are the roots of most plants adapted to perform the major root functions?
8. **Relating Concepts** Suggest an environmental condition in which the regulation of water movement by endodermal cells might help ensure a plant's survival.

SECTION 3

OBJECTIVES

- **Explain** why stems differ in shape and growth.
- **Describe** the structure of stems.
- **Distinguish** between primary growth and secondary growth in stems.
- **Discuss** the three main functions of stems.

VOCABULARY

node
internode
bud
bud scale
pith
wood
heartwood
sapwood
bark
springwood
summerwood
annual ring
source
sink
translocation
pressure-flow hypothesis
transpiration
cohesion-tension theory

STEMS

In contrast to roots, which are mainly adapted for absorption and anchoring, stems are usually adapted to support leaves. Whatever their sizes and shapes, stems also function in transporting materials and providing storage.

TYPES OF STEMS

A typical stem grows upright and is either woody or nonwoody. The many different forms of stems seen in nature, including those shown in Figure 29-8, represent adaptations to the environment. Strawberry *stolons* grow along the soil surface and produce new plants. Stems such as the edible white potato tuber are modified for storing energy as starch. Cactuses have green fleshy stems that both store water and carry on all the plant's photosynthesis. Stems of the black locust and the honeylocust develop sharp thorns that can protect the plant from animals.



(a) STRAWBERRY (STOLONS)



(b) POTATO (TUBERS)



(c) CACTUS (FLESHY STEMS)

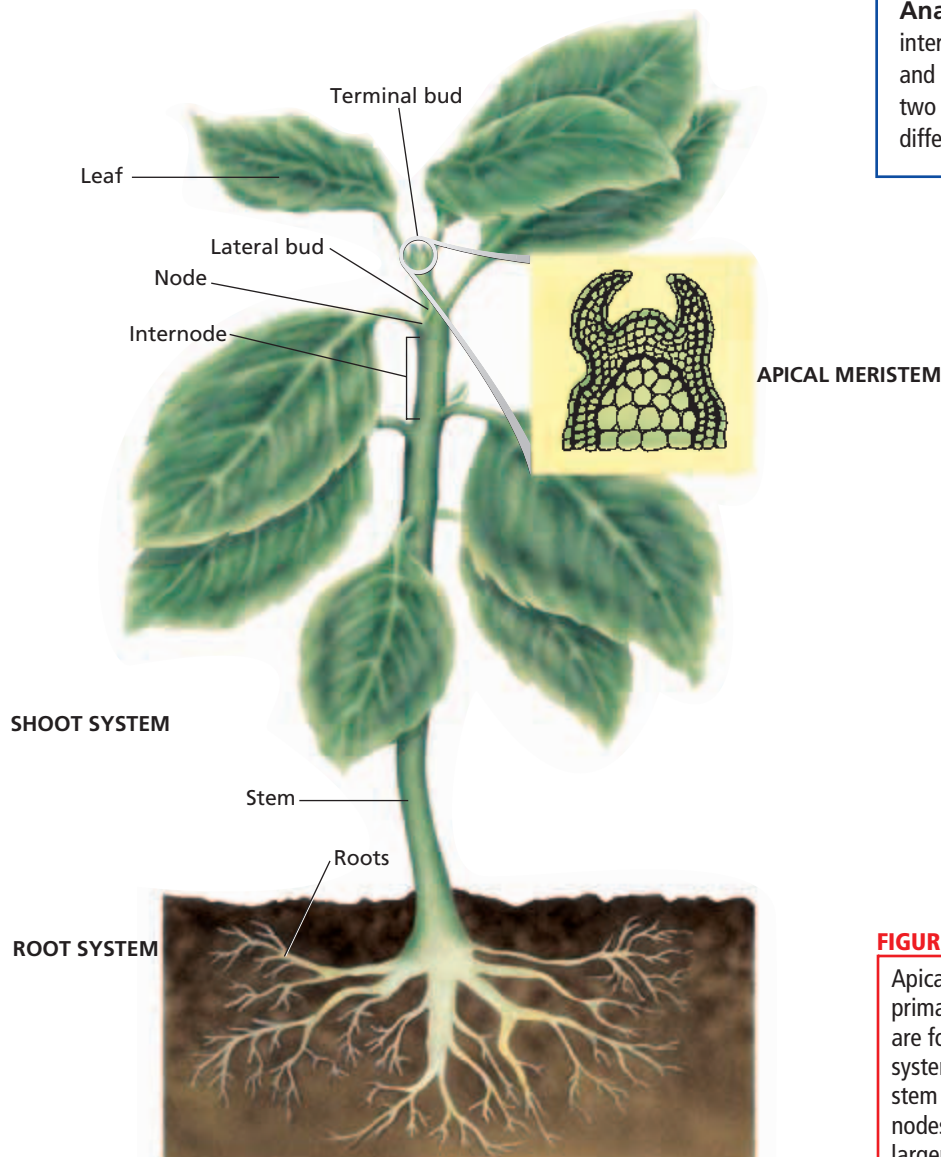
FIGURE 29-8

Stems provide a supporting framework for leaves. Some plants produce stems that are modified for other functions. (a) A strawberry plant has stolons, which are horizontal, above-ground stems that form new plants. (b) The potato plant has tubers, which are enlarged, short, underground stems used for storing starch. (c) A cactus is called a *succulent* because of its fleshy, water-storing stems.

STEM STRUCTURES

Stems have a more complex structure than roots, yet they are similar in many ways. Did you know that a sign nailed 2 m (about 7 ft) high on a tree will remain 2 m high, even though the tree may grow much taller? That is because most stems, like roots, grow in length only at their tips, where apical meristems produce new primary tissues. Stems, like roots, grow in circumference through lateral meristems.

Stems have several features that roots lack, as Figure 29-9 shows. Each leaf is attached to the stem at a location called a **node**. The spaces between nodes are called **internodes**. At the point of attachment of each leaf, the stem bears a lateral bud. A **bud** is capable of developing into a new *shoot system* (the above-ground part of a plant, consisting of stems and leaves). A bud contains an apical meristem and is enclosed by specialized leaves called **bud scales**. The tip of each stem usually has a terminal bud.



Quick Lab

Observing Stems

Materials winter twig, fresh stem with leaves, hand lens or dissecting microscope

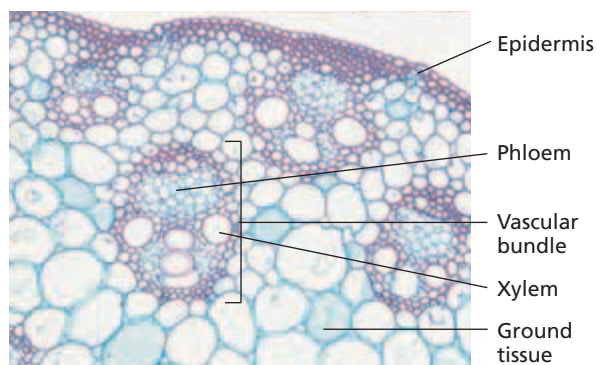
Procedure

1. Observe the twig with the hand lens. Locate several buds, and identify the bud scales. Locate leaf scars. Identify a node and an internode. Draw and label the twig.
2. Observe the fresh stem with the hand lens. Locate and identify the stem, nodes, internodes, buds, and a leaf. Draw and label the fresh stem.

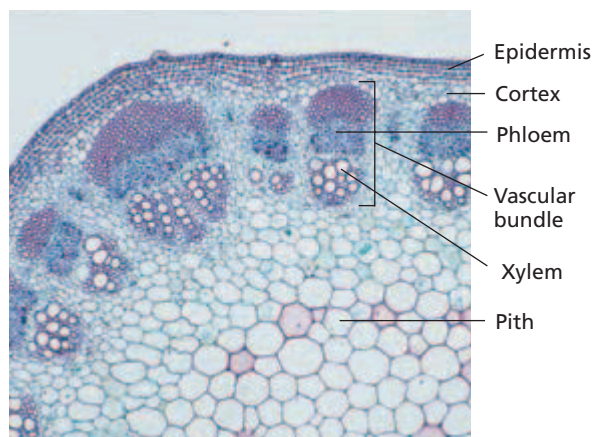
Analysis How are a node and an internode related? How are a bud and a node related? How are the two stems alike? How are they different?

FIGURE 29-9

Apical meristems, responsible for the primary elongation of the plant body, are found in the shoot system and root system. Each leaf is attached to the stem at a node. The space between nodes, called the *internode*, is much larger in the older part of the stem.



(a) MONOCOT STEM



(b) DICOT STEM

FIGURE 29-10

Primary growth in stems produces dermal, ground, and vascular tissues. (a) The cross section of a herbaceous stem of corn, a monocot, shows the vascular bundles scattered throughout the ground tissue. (b) In the sunflower, a dicot, the vascular tissues appear as a single ring of bundles between the cortex and the pith. Notice that the cortex lies just inside the epidermis, as it does in the root.

In some plants that live in cold winter environments, the terminal bud opens and the bud scales fall off when growth resumes in the spring. The bud scales leave scars on the stem surface. Trees and shrubs often are identified in winter by stem characteristics.

Root tips usually have a protective layer, the root cap. The stem apical meristem is protected by bud scales only when the stem is not growing. A surface bud forms very close to the stem tip with one or more buds at each node. In contrast, lateral roots originate farther back from the root tip and form deep inside the root at no particular location along the root axis.

Primary Growth in Stems

As in roots, apical meristems of stems give rise to the dermal, ground, and vascular tissues. Locate each of these tissues in Figure 29-10. The dermal tissue is represented by the epidermis, or the outer layer of the stem. Its main functions are to protect the plant and to reduce the loss of water to the atmosphere while still allowing gas exchange through stomata.

In gymnosperm and dicot stems, ground tissue usually forms a cortex and a pith. The cortex frequently contains flexible collenchyma cells. The **pith** is located in the center of the stem. The ground tissue of monocot stems is usually not clearly separated into pith and cortex.

Vascular tissue formed near the apical meristem occurs in bundles—long strands that are embedded in the cortex. Look at the vascular bundles in Figure 29-10. Each bundle contains xylem tissue and phloem tissue. Xylem is usually located toward the inside of the stem, while phloem is usually located toward the outside.

Compare the arrangement of vascular bundles in monocots and dicots. Monocot stem vascular bundles are usually scattered throughout the ground tissue. Stem vascular bundles of dicots and gymnosperms usually occur in a single ring. Most monocots have little or no secondary growth, and they retain the primary growth pattern their entire lives. However, in many dicots and gymnosperms, the primary tissues are eventually replaced by secondary tissues.

Secondary Growth in Stems

Stems increase in thickness due to the division of cells in the vascular cambium. The vascular cambium in dicot and gymnosperm stems first arises between the xylem and phloem in a vascular bundle. Eventually, the vascular cambium forms a boundary. The vascular cambium produces secondary xylem to the inside and secondary phloem to the outside. It usually produces much more secondary xylem than it does secondary phloem. Secondary xylem is called **wood**. Occasionally, the vascular cambium produces new cambium cells, which increase the stem's diameter.

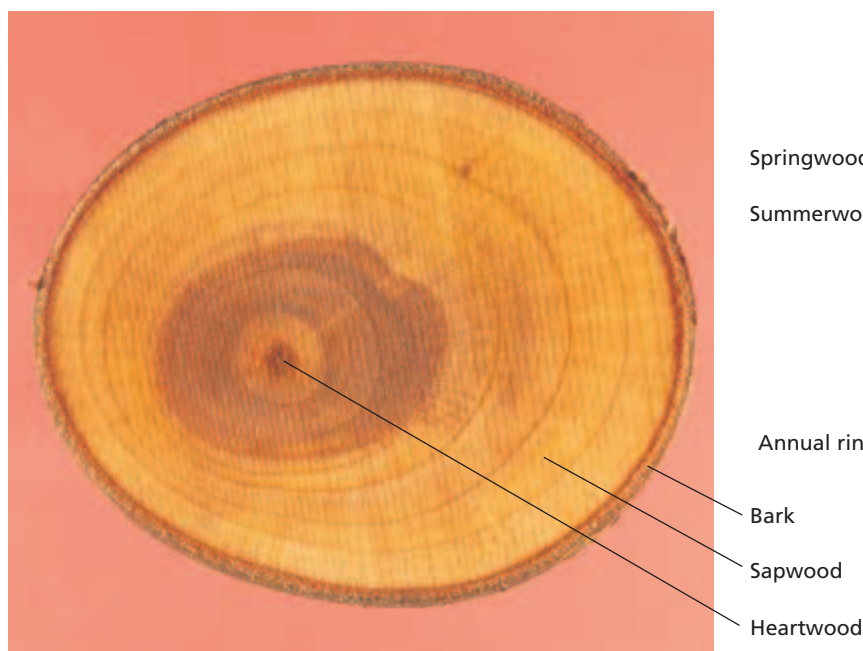
Older portions of the xylem eventually stop transporting water. They often become darker than the newer xylem due to the accumulation of resins and other organic compounds produced by the few live cells remaining in the xylem. This darker wood in the center of a tree is called **heartwood**, as shown in Figure 29-11a. The functional, often lighter-colored wood nearer the outside of the trunk is **sapwood**. In a large-diameter tree, the heartwood keeps getting wider while the sapwood remains about the same thickness.

The phloem produced near the outside of the stem is part of the bark. **Bark** is the protective outside covering of woody plants. It consists of cork, cork cambium, and phloem. The cork cambium produces cork near the outside. However, cork cells are dead at maturity and cannot elongate, so the cork ruptures as the stem continues to expand in diameter. This results in the bark pattern of some trees, such as oaks and maples, appearing rough or irregular in texture.

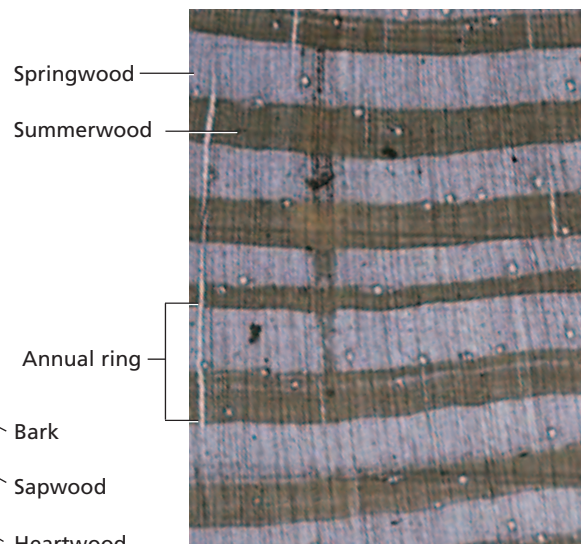
During spring, if water is plentiful, the vascular cambium can form new xylem with cells that are wide and thin walled. This wood is called **springwood**, as shown in Figure 29-11b. In summer, when water is limited, the vascular cambium produces **summerwood**, which has smaller cells with thicker walls. In a stem cross section, the abrupt change between small summerwood cells and the following year's large springwood cells produces an **annual ring**. The circles that look like rings on a target in Figure 29-11a are the annual rings of the stem. Because one ring is usually formed each year, you can estimate the age of the stem by counting its annual rings. Annual rings also form in dicot and gymnosperm roots, but they are often less pronounced because the root environment is more uniform. Annual rings often do not occur in tropical trees because of their uniform year-round environment.

FIGURE 29-11

(a) Wood consists of secondary xylem. The dark-colored wood is called *heartwood*, and the light-colored wood is called *sapwood*. Bark is the outside protective covering of woody plants. (b) You can see the annual rings produced by alternating springwood cells and summerwood cells. The abundant water of the spring season helps expand the cell walls of springwood cells.



(a) HEARTWOOD AND SAPWOOD



(b) SPRINGWOOD AND SUMMERWOOD

STEM FUNCTIONS

Stems function in the transportation of nutrients and water, the storage of nutrients, and the support of leaves. Sugars, some plant hormones, and other organic compounds are transported in the phloem. The movement of sugars occurs from where the sugars are made or have been stored, called a **source**. The place where they will be stored or used is called a **sink**. Botanists use the term **translocation** to refer to the movement of sugars through the plant. For example, most of the time, sugars move from the leaves to the roots, to the shoot apical meristems, and to the developing flowers or fruits. Sugars may be newly made in photosynthetic cells or may have been stored as starch in roots or other stems.

Movement of sugars in the phloem is explained by the **pressure-flow hypothesis**, which states that sugars are actively transported into sieve tubes. Look at Figure 29-12. As sugars enter the sieve tubes, water is also transported in by osmosis. Thus, a positive pressure builds up at the source end of the sieve tube. This is the pressure part of the pressure-flow hypothesis.

At the sink end of the sieve tube, this process is reversed. Sugars are actively transported out, water leaves the sieve tube by osmosis, and pressure is reduced at the sink. The difference in pressure causes water to flow from source to sink. This water is also carrying dissolved substances with it. Transport in the phloem can occur in different directions at different times.

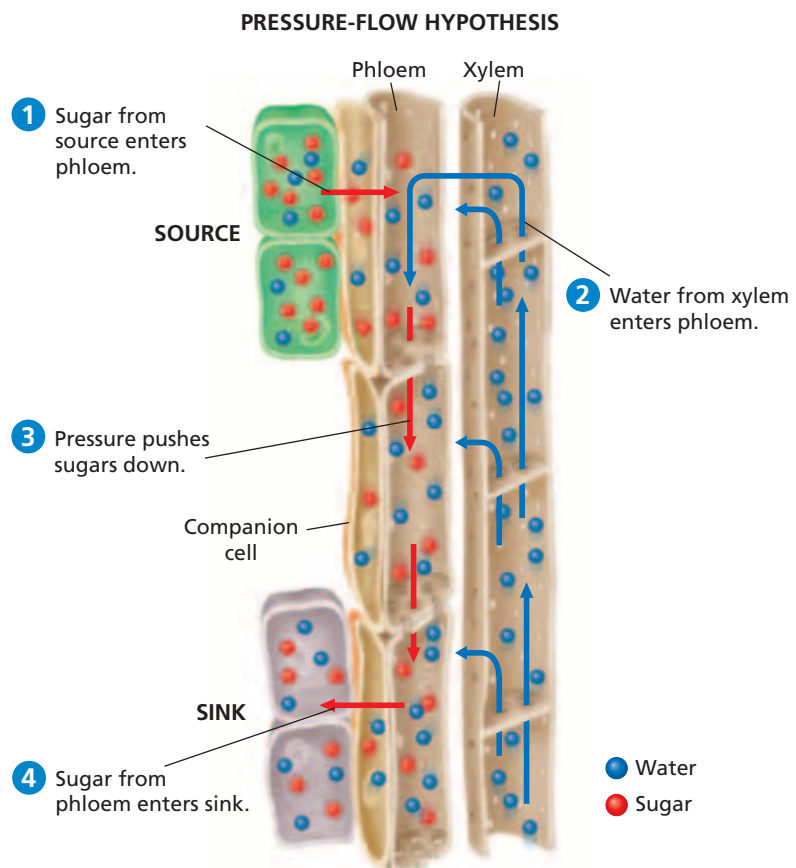


FIGURE 29-12

1 Sugars from a source enter the sieve tubes of the phloem. 2 Water moves from the xylem to the phloem by osmosis. 3 The water creates pressure that moves the sugars down the phloem. 4 The sugars exit the phloem and enter the sink, where they will be used or stored.

Transport of Water

The transport of water and mineral nutrients occurs in the xylem of all plant organs, but it occurs over the greatest distances in the stems of tall trees. During the day, water is constantly evaporating from the plant, mainly through leaf stomata. This water loss is called **transpiration** (TRAN-spuh-RAY-shuhn). The large amount of water lost from the plant is a result of the plant's need to obtain carbon dioxide from the air through the stomata. Exactly how do huge trees, like redwoods, move water and mineral nutrients up their 100 m tall trunks?

According to the **cohesion-tension theory**, water is pulled up the stem xylem by the strong attraction of water molecules to each other, a property of water called *cohesion*. The movement also depends on the rigid xylem walls and the strong attraction of water molecules to the xylem wall, which is called *adhesion*. The thin, continuous columns of water extend from the leaves through the stems and into the roots. As water evaporates from the leaf, the water column is subject to great tension. However, the water column does not break because of cohesion, and it does not pull away from the xylem walls because of adhesion. The only other possibility is for it to be pulled upward. The pull at the top of the tree extends all the way to the bottom of the column. As water is pulled up the xylem, more water enters the roots from the soil.

Storing Water and Nutrients

With abundant parenchyma cells in the cortex, plant stems are adapted for storage in most species. In some species, storage is a major function. Cactus stems are specialized for storing water. The roots of a cactus are found close to the soil surface, enabling them to absorb rainwater quickly and transport it to the cactus's fleshy stem. The edible white potato is an underground stem that is specialized for storing starch. The "eyes" of white potatoes are buds that have the ability to develop into new shoots.

Word Roots and Origins

transpiration

from the Latin *trans*, meaning "across," and *spirare*, meaning "to breathe"



SECTION 3 REVIEW

1. Explain why a plant species might develop thorny stems in response to its environment.
2. Describe the similarities of the apical meristems of roots to the apical meristems of stems.
3. Compare primary growth in monocots with primary growth in dicots.
4. Describe how summerwood and springwood form annual rings.
5. Describe how water is transported through xylem tissue.

CRITICAL THINKING

6. **Analyzing Information** Some squirrels damage trees by stripping off portions of the bark. Why might a squirrel eat bark?
7. **Evaluating Differences** What adaptive advantages might the dead cells of the xylem tissue provide over transporting cells that are alive?
8. **Inferring Relationships** The transport cells of the phloem tissue are connected end to end, and sugars flow through them. What inferences can you make about the contents of these cells?

Science in Action

Do Wild Animals Self-Medicate?

Many plants have chemical compounds that defend the plants against herbivores. Such compounds are the active ingredients in many familiar medicines. With so many plants in existence, how do scientists know which ones to test for possible medical benefit? Michael Huffman studies animal feeding behavior for clues.



Dr. Michael Huffman

HYPOTHESIS: Chimpanzees Eat Some Plants for Medicinal Purposes

Researchers studying chimpanzees in Africa have noted that chimps have an unusual feeding habit when they are ill. Before eating the shoots of the plant *Vernonia amygdalina*, chimps carefully remove the outer bark and leaves to chew on the exposed pith, the spongy material found in some vascular plants. The chimps extract juice from the pith, which the researchers found odd because the juice is extremely bitter. Even though the plant is available year-round, chimps rarely eat it. Based on his observations and those of other scientists, Michael A. Huffman, a researcher at the Primate Research Institute in Japan, wanted to find out whether chimps are self-medicating by eating these plants.

METHODS: Observe Behavior, Collect Samples, and Test Plants

Huffman began by collecting chimpanzee fecal samples at Mahale, Tanzania. He also made detailed observations of as many individual chimps as possible. The fecal samples were from chimps seen eating *V. amygdalina* during times of apparent illness. Huffman collaborated with other scientists already studying chemicals found in *V. amygdalina*.

RESULTS: Chimps Recover; Fecal Samples Contain Parasites; Plant Analysis Reveals Antiparasitic Compounds

Huffman and his colleagues found that within 24 hours after eating *V. amygdalina*, chimps regain their appetites, have reduced numbers of parasites, and recover from constipation or diarrhea. Scientists working with Huffman showed that the pith of *V. amygdalina* contains several bioactive chemicals. Tests revealed that these chemicals are effective against many different parasites. For example, vernonioside B1 and vernoniol B1 were shown to suppress movement and egg laying in *Schistosoma japonicum*, a parasitic worm. Scientists also found that chemicals from the pith are effective against drug-resistant malarial parasites.

CONCLUSIONS: Chimpanzees Self-Medicate by Eating Plants with Beneficial Chemical Compounds

Interestingly, a more toxic compound, vernodalin, was found in the leaves. The pith of the plant instead contained large amounts of vernonioside B1. This same pattern was later verified in analyses of other *V. amygdalina* specimens collected at various locations in Mahale during different seasons. Huffman and his colleagues think that because chimps discard the bark and leaves and just eat the pith, the chimps have learned that certain parts of the plant are harmful and certain parts are beneficial.

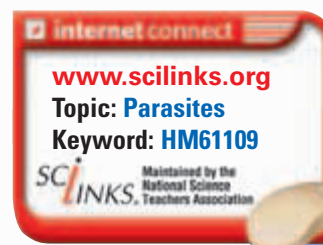
Additional research has revealed that *V. amygdalina* has been part of Tanzanian folk medicine for years. The WaTongwe people of this area use *V. amygdalina* for stomachaches and parasitic infections. Other African tribes use *V. amygdalina* to treat ailments in their livestock, which suggests agricultural applications for other countries.

An adult male chimpanzee with a nematode infection chews on the pith of Vernonia amygdalina.



REVIEW

1. What observations led Huffman to propose his hypothesis?
2. Can chimpanzees learn from experience? Explain your reasoning.
3. **Critical Thinking** Why was it important for Huffman to work with other scientists?



LEAVES

Most leaves are thin and flat, an adaptation that helps them capture sunlight for photosynthesis. Although this structure may be typical, it is certainly not universal. Like roots and stems, leaves are extremely variable. This variability represents adaptations to environmental conditions.

TYPES OF LEAVES

Look at the leaves in Figure 29-13a. The coiled structure is a **tendrill**, a modified leaf found in many vines, such as peas and pumpkins. It wraps around objects and supports the climbing vine. In some species, like grape, tendrils are specialized stems.

An unusual leaf modification occurs in carnivorous plants such as the pitcher plant, shown in Figure 29-13b. In carnivorous plants, leaves function as food traps. These plants grow in soil that is poor in several mineral nutrients, especially nitrogen. The plant receives substantial amounts of mineral nutrients when it traps and digests insects and other small animals.

Leaves, or parts of leaves, are often modified into spines that protect the plant from being eaten by animals, as shown in Figure 29-13c. Because spines are small and nonphotosynthetic, they greatly reduce transpiration in desert species such as cactuses.

OBJECTIVES

- **Describe** adaptations of leaves.
- **Identify** the difference between a simple leaf, a compound leaf, and a doubly compound leaf.
- **Describe** the tissues that make up the internal structure of a leaf.
- **Describe** the major functions of leaves.

VOCABULARY

tendrill
blade
petiole
simple leaf
compound leaf
leaflet
mesophyll
palisade mesophyll
spongy mesophyll
vein
venation
parallel venation
net venation
guard cell



(a) TENDRIL



(b) TUBULAR LEAF



(c) SPINES

FIGURE 29-13

Many plant species have developed leaf adaptations. (a) The pea plant has tendrils that climb. (b) The pitcher plant has tubular leaves that trap insects. (c) The barberry has spines that may protect against herbivores.

LEAF STRUCTURES



(a) SIMPLE LEAF



(b) COMPOUND LEAF



(c) DOUBLY COMPOUND LEAF

FIGURE 29-14

(a) A sugar-maple leaf is called a simple leaf because it has only one blade. (b) A white clover leaf is called a compound leaf because the leaf blade is divided into distinct leaflets. (c) The honeylocust has a doubly compound leaf because each leaflet is subdivided into smaller leaflets.

Leaves come in a wide variety of shapes and sizes and are an important feature used for plant identification. Leaves can be round, straplike, needlelike, or heart-shaped. The broad, flat portion of a leaf, called the **blade**, is the site of most photosynthesis. The blade is usually attached to the stem by a stalklike **petiole**. The maple leaf shown in Figure 29-14a is a **simple leaf**; it has a single blade. In **compound leaves**, such as the white clover in Figure 29-14b, the blade is divided into **leaflets**. In some species, the leaflets themselves are divided. The result is a doubly compound leaf, such as that of the honeylocust shown in Figure 29-14c.

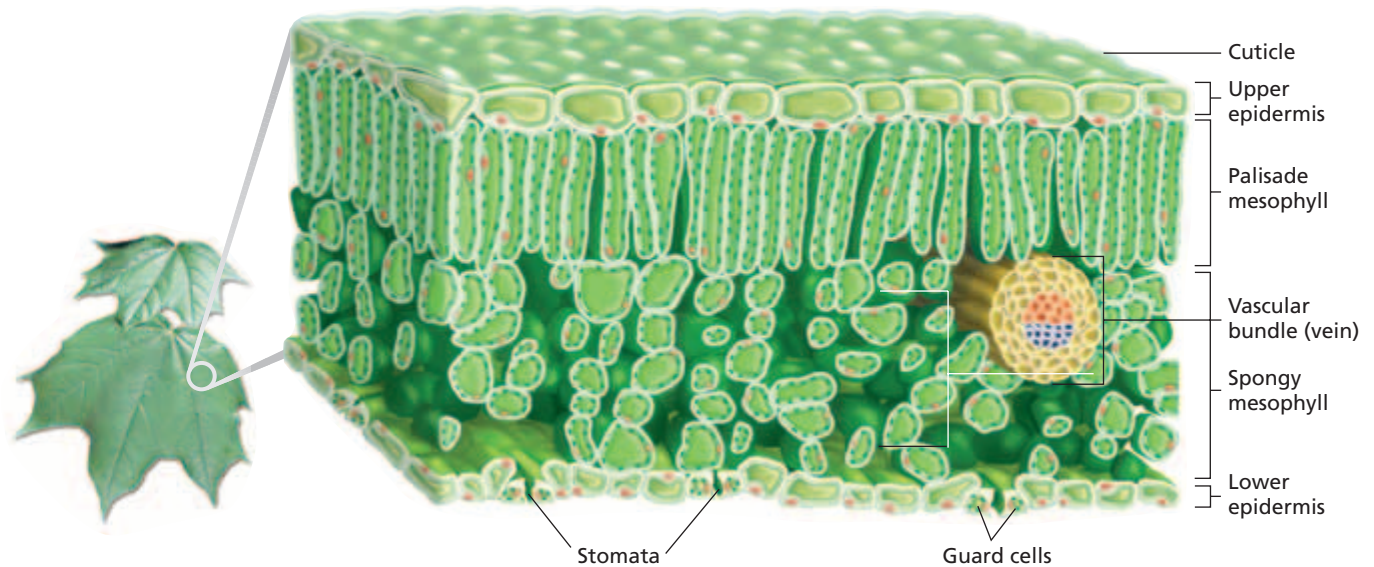
Leaves consist of three tissue systems. The dermal tissue system is represented by the epidermis. In most leaves the epidermis is a single layer of cells coated with a nearly impermeable cuticle. Water, oxygen, and carbon dioxide enter and exit the leaf through stomata in the epidermis. Epidermal hairs are often present and usually function to protect the leaf from insects and intense light.

The number of stomata per unit area of leaf varies by species. For example, submerged leaves of aquatic plants have few or no stomata. Corn leaves have up to 10,000 stomata per square centimeter on both upper and lower surfaces. Scarlet oak has over 100,000 stomata per square centimeter on the lower leaf surface and none on the upper surface. Regardless of their exact distribution, stomata are needed to regulate gas exchange.

In most plants, photosynthesis occurs in the leaf **mesophyll** (MEZ-oh-FIL), a ground tissue composed of chloroplast-rich parenchyma cells. In most plants, the mesophyll is organized into two layers, which are shown in Figure 29-15 on the next page. The **palisade mesophyll** layer occurs directly beneath the upper epidermis and is the site of most photosynthesis. Palisade cells are columnar and appear to be packed tightly together in one or two layers. However, there are air spaces between the long side walls of palisade cells. Beneath the palisade layer is the **spongy mesophyll**. It usually consists of irregularly shaped cells surrounded by large air spaces, which allow oxygen, carbon dioxide, and water to diffuse into and out of the leaf.

The vascular tissue system of leaves consists of vascular bundles called **veins**. Veins are continuous with the vascular tissue of the stem and the petiole, and they lie embedded in the mesophyll. Veins branch repeatedly so that each cell is usually less than 1 mm (0.04 in.) from a vein.

Venation is the arrangement of veins in a leaf. Leaves of most monocots, such as grasses, have **parallel venation**, meaning that several main veins are roughly parallel to each other. The main veins are connected by small, inconspicuous veins. Leaves of most dicots, such as sycamores, have **net venation**, meaning that the main vein or veins repeatedly branch to form a conspicuous network of smaller veins.



LEAF FUNCTIONS

Leaves are the primary site of photosynthesis in most plants. Mesophyll cells in leaves use light energy, carbon dioxide, and water to make sugars. Light energy also is used by mesophyll cells to synthesize amino acids and a variety of other organic molecules. Sugars made in a leaf can be used by the leaf as an energy source or as building blocks for cells. They also may be transported to other parts of the plant, where they are either stored or used for energy or building blocks.

A major limitation to plant photosynthesis is insufficient water due to transpiration. For example, up to 98 percent of the water that is absorbed by a corn plant's roots is lost through transpiration. However, transpiration may benefit the plant by cooling it and by speeding the transport of mineral nutrients through the xylem.

Modifications for Capturing Light

Leaves absorb light, which, in turn, provides the energy for photosynthesis. The leaves of some plant species have adapted to their environment to maximize light interception. On the same tree, leaves that develop in full sun are thicker, have a smaller area per leaf, and have more chloroplasts per unit area. Shade-leaf chloroplasts are arranged so that shading of one chloroplast by another is minimized, while sun-leaf chloroplasts are not.

In dry environments, plants often receive more light than they can use. Structures have evolved in these plants that reduce the amount of light absorbed. For example, many desert plants have evolved dense coatings of hairs that reduce light absorption. The cut window plant shown in Figure 29-16 protects itself from its dry environment by growing underground. Only its leaf tips protrude above the soil to gather light for photosynthesis.

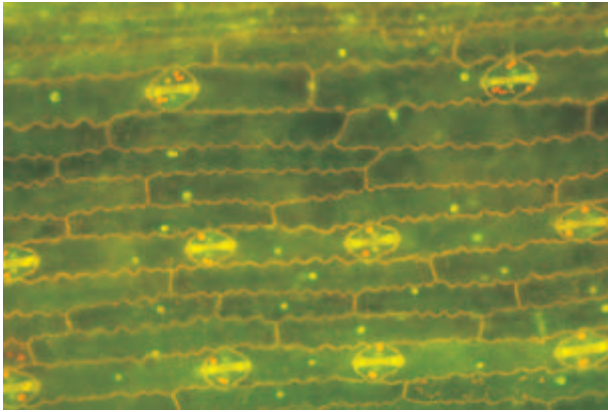
FIGURE 29-15

Cells from all three tissue systems are represented in a leaf. The epidermis is part of the dermal system, and the vascular bundle (vein) is part of the vascular system. The mesophyll is ground tissue made of parenchyma cells, and it generally contains chloroplasts.

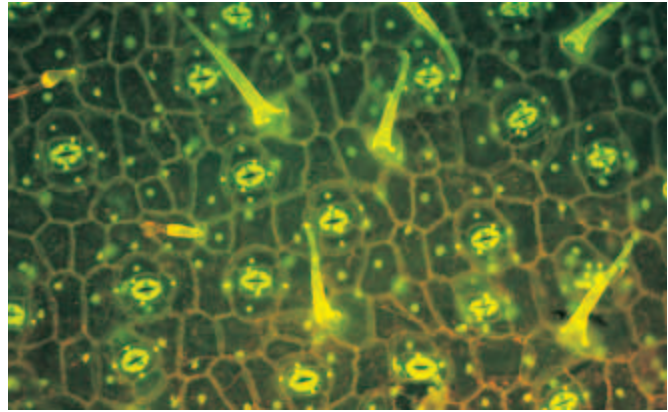
FIGURE 29-16

The cut window plant (*Haworthia truncata*) has flat-topped leaves that look as if they have been cut. The transparent leaf tips are sacs that diffuse sunlight before channeling it to the plant's buried leaves.





(a) MONOCOT LEAF



(b) DICOT LEAF

FIGURE 29-17

These micrographs of monocot and dicot leaves show how the arrangement of stomata in each is different. (a) In the corn leaf, the stomata are in a parallel arrangement, which is typical of monocots. (b) In the potato leaf, the stomata are in a random arrangement, which is typical of dicots.

Gas Exchange

Plants must balance their need to open their stomata to receive carbon dioxide and release oxygen with their need to close their stomata to prevent water loss through transpiration. A stoma is bordered by two kidney-shaped guard cells. **Guard cells**, shown in Figure 29-15 on the previous page, are modified cells on the leaf epidermis that regulate gas and water exchange. Figure 29-17 shows how the stomata are arranged differently in monocots and dicots.

The stomata of most plants open during the day and close at night. The opening and closing of a stoma is regulated by the amount of water in its guard cells. When epidermal cells of leaves pump potassium ions (K^+) into guard cells, water moves into the guard cells by osmosis. This influx of water makes the guard cells swell, which causes them to bow apart and form a pore. During darkness, potassium ions are pumped out of the guard cells. Water then leaves the guard cells by osmosis. This causes the guard cells to shrink slightly and the pore to close.

Stomata also close if water is scarce. The closing of stomata greatly reduces further water loss and may help the plant survive until the next rain. However, stomata closure virtually shuts down photosynthesis by cutting off the supply of carbon dioxide.

SECTION 4 REVIEW

1. Describe three adaptations found in different types of leaves.
2. What is the difference between a simple leaf, a compound leaf, and a doubly compound leaf?
3. Describe the basic function of each of the three leaf tissues.
4. Explain the function of the guard cells in regulating the opening and closing of stomata on a leaf's surface.

CRITICAL THINKING

5. **Evaluating Information** Why might it be an advantage for a plant to have most of its stomata on the underside of a horizontal leaf?
6. **Inferring Relationships** Why do plants grown in greenhouses in winter rarely grow as fast as the same type of plant grown outside in the summer, even if the temperatures are the same?
7. **Predicting Results** Suppose a plant had a mutation that produced defective potassium ion pumps. What might happen to this plant?

CHAPTER HIGHLIGHTS

SECTION 1

Plant Cells and Tissues

- Parenchyma, collenchyma, and sclerenchyma are basic plant cell types.
- The three types of plant tissue systems are dermal, ground, and vascular.
- Plant growth occurs in the apical, intercalary, and lateral meristems.
- Primary growth occurs mainly in the apical meristems. Secondary growth occurs in the lateral meristems.

Vocabulary

parenchyma (p. 583)
collenchyma (p. 583)
sclerenchyma (p. 584)
epidermis (p. 584)
cuticle (p. 584)

tracheid (p. 585)
pit (p. 585)
vessel element (p. 585)
vessel (p. 585)
sieve tube member (p. 585)

sieve tube (p. 585)
sieve plate (p. 585)
companion cell (p. 585)
meristem (p. 586)
apical meristem (p. 586)

lateral meristem (p. 586)
vascular cambium (p. 586)
cork cambium (p. 586)

SECTION 2

Roots

- A taproot is a large primary root. A fibrous root system has many small branching roots. Adventitious roots are specialized roots that grow from uncommon places.
- Roots are made up of root caps, apical meristems, and root hairs.
- Primary growth in roots occurs at the root tip. Secondary growth occurs when a vascular cambium forms between primary xylem and primary phloem.
- Roots anchor the plant and store and absorb water and mineral nutrients from the soil.

Vocabulary

taproot (p. 587)
fibrous root system (p. 587)

adventitious root (p. 588)
root cap (p. 588)
root hair (p. 588)

cortex (p. 589)
endodermis (p. 589)
pericycle (p. 590)

macronutrient (p. 590)
micronutrient (p. 590)

SECTION 3

Stems

- Stem shape and growth are adaptations to the environment.
- Both nonwoody and woody stems contain xylem and phloem.
- Most monocot stems have only primary growth. Dicot stems often produce abundant secondary growth.
- Stems support leaves and transport and store nutrients and water.

Vocabulary

node (p. 593)
internode (p. 593)
bud (p. 593)
bud scale (p. 593)
pith (p. 594)

wood (p. 594)
heartwood (p. 595)
sapwood (p. 595)
bark (p. 595)
springwood (p. 595)

summerwood (p. 595)
annual ring (p. 595)
source (p. 596)
sink (p. 596)
translocation (p. 596)

pressure-flow hypothesis (p. 596)
transpiration (p. 597)
cohesion-tension theory (p. 597)

SECTION 4

Leaves

- The variability in leaf structures represents adaptations to environmental conditions. Leaves may be simple, compound, or doubly compound.
- Photosynthesis occurs mostly in the mesophyll.
- Some leaves are modified to maximize light interception. Gas exchange is controlled by stomata.

Vocabulary

tendrils (p. 600)
blade (p. 600)
petiole (p. 600)
simple leaf (p. 600)

compound leaf (p. 600)
leaflet (p. 600)
mesophyll (p. 600)
palisade mesophyll (p. 600)

spongy mesophyll (p. 600)
vein (p. 600)
venation (p. 600)
parallel venation (p. 600)

net venation (p. 600)
guard cell (p. 602)

CHAPTER REVIEW

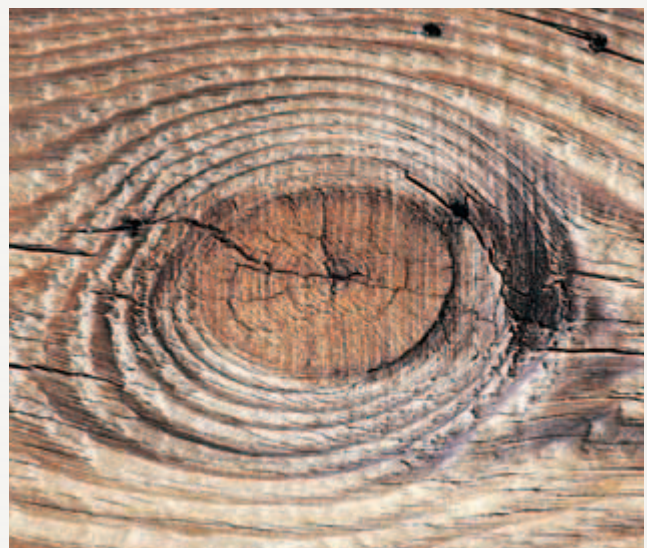
USING VOCABULARY

- For each pair of terms, explain the relationship between the terms.
 - taproot* and *fibrous root*
 - sieve tube member* and *vessel element*
 - transpiration* and *translocation*
- Use the following terms in the same sentence: *parenchyma*, *collenchyma*, and *sclerenchyma*.
- For each pair of terms, explain how the meanings of the terms differ.
 - apical meristem* and *lateral meristem*
 - simple leaf* and *compound leaf*
 - root hair* and *root cap*
- Word Roots and Origins** The word *mesophyll* is derived from the Latin *meso*, which means “middle,” and the Greek *phyllon*, which means “leaf.” Using this information, explain why the term *mesophyll* is a good name for the part of the leaf that the term describes.

UNDERSTANDING KEY CONCEPTS

- Compare** the structure and function of parenchyma, collenchyma, and sclerenchyma cells.
- Compare** the functions of the three plant tissue systems.
- Differentiate** the type of growth that occurs in each of the three kinds of meristems.
- Explain** the difference between primary growth and secondary growth.
- Name** two types of adventitious roots.
- Sequence** the structures of a root that a water molecule would pass through as it enters and then moves through a plant.
- Relate** what causes a plant stem or root to grow in diameter.
- Name** two familiar examples of carbohydrate storage in roots.
- Explain** why stems vary in shape and growth patterns.
- Distinguish** the structure of a root from the structure of a stem.
- Explain** how annual rings form in a woody plant.
- Differentiate** between the mechanisms of water and sugar transport in a plant.
- Explain** why all leaves do not look alike.
- Describe** the structure of a doubly compound leaf.
- Name** the type of plant tissue that makes up the mesophyll.
- Identify** a leaf adaptation for maximizing light interception.
- CONCEPT MAPPING** Use the following terms to create a concept map that illustrates the structures and functions of one type of plant tissue: *phloem*, *vessel*, *sieve tube member*, *tracheid*, *vascular tissue*, *vessel element*, *carbohydrates*, and *xylem*.
- Applying Information** When transplanting a plant, it is important to not remove any more soil than necessary from around the root system. From your knowledge of the function of roots and root hairs, why do you think this is so important?
- Analyzing Data** Suppose you examine a tree stump and notice that the annual rings are thinner and closer together 50 rings in from the edge. What would you conclude about the climate in the area 50 years ago?
- Applying Information** Why would an agricultural practice that eliminated transpirational water loss be disadvantageous for plants?
- Evaluating Differences** Would you expect water absorption in a plant to be greater in parts of roots that have undergone secondary growth or in parts that have not? Explain your reasoning.
- Interpreting Graphics** What causes a knot in a board, like the one shown in the photo below?

CRITICAL THINKING



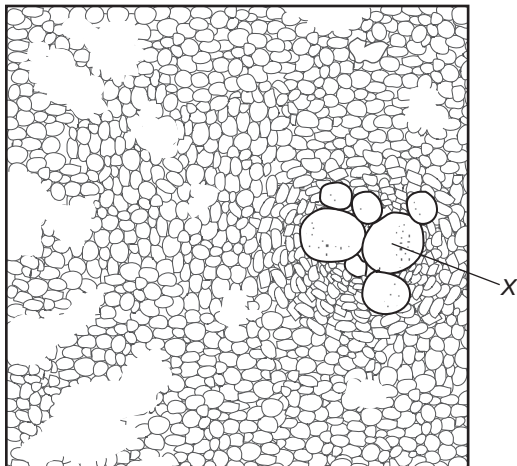


Standardized Test Preparation

DIRECTIONS: Choose the letter of the answer choice that best answers the question.

- Which of the following are the main function of collenchyma and sclerenchyma?
A. storage
B. support
C. transport
D. photosynthesis
- Most monocots do not have which of the following?
F. xylem
G. phloem
H. primary growth
J. secondary growth
- Which of the following structures is found in stems but not in roots?
A. node
B. cortex
C. epidermis
D. vascular tissue
- The movement of water through a plant is driven by the loss of water vapor from which structure?
F. buds
G. nodes
H. leaves
J. root hairs

INTERPRETING GRAPHICS: The illustration below shows a cross section of a monocot root. Use the illustration to answer the question that follows.



- What is the structure marked with an X?
A. xylem
B. cortex
C. phloem
D. endodermis

DIRECTIONS: Complete the following analogy.

6. Mesophyll : photosynthesis :: sapwood :
F. storage
G. sugar transport
H. water transport
J. production of new cells

INTERPRETING GRAPHICS: The illustration below shows a leaf. Use the illustration to answer the question that follows.



7. What type of leaf is shown?
A. tendril
B. simple
C. compound
D. doubly compound

SHORT RESPONSE

The primary function of leaves is to carry out photosynthesis.

Explain how a leaf's structure is an adaptation that allows intake of carbon dioxide with minimal water loss.

EXTENDED RESPONSE

A stem contains tissues that transport water along with dissolved minerals and sugars.

Part A Identify the cells that transport water, and describe how water moves through a plant.

Part B Identify the cells that transport sugar, and describe how sugars move through a plant.

Test TIP

For the short- or extended-response questions, be sure to write in complete sentences. When you have finished writing your answer, be sure to proofread for spelling, grammar, and punctuation errors.

Observing Roots, Stems, and Leaves

OBJECTIVES

- Observe the tissues and structures that make up roots, stems, and leaves.
- Explain how roots, stems, and leaves are adapted for the functions they perform.

PROCESS SKILLS

- observing
- identifying
- relating structure and function

MATERIALS

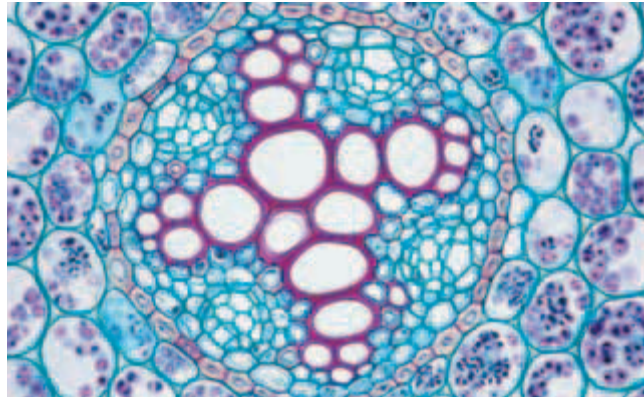
- prepared slides of the following tissues:
 - *Allium* root tip longitudinal section
 - *Syringa* leaf cross section
 - *Ranunculus* root cross section
 - *Ranunculus* stem cross section
 - *Zea mays* stem cross section
- compound light microscope

Background

1. Which plant tissues are responsible for the absorption of water and mineral nutrients?
2. How is sugar, which is produced in the leaf, moved to other parts of the plant?
3. How are woody and nonwoody stems different, and how are they alike?
4. What tissues are continuous in the root, stem, and leaf?
5. How does the leaf conserve water?

PART A Roots

1.  **CAUTION** Glass slides may break and cut you. Observe the prepared slide of the *Allium* root tip under low power. Locate the root cap and the root tip meristematic cells. You may look at the photograph in the chapter for help. Note the long root hairs in the area above the root tip.
2. In your lab report, draw the root tip that you observe, and label the root cap, meristem, and root hairs in your drawings.

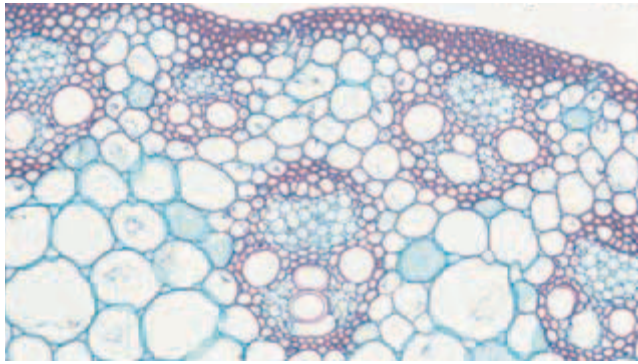


Ranunculus root (275 $\times$)—a dicot

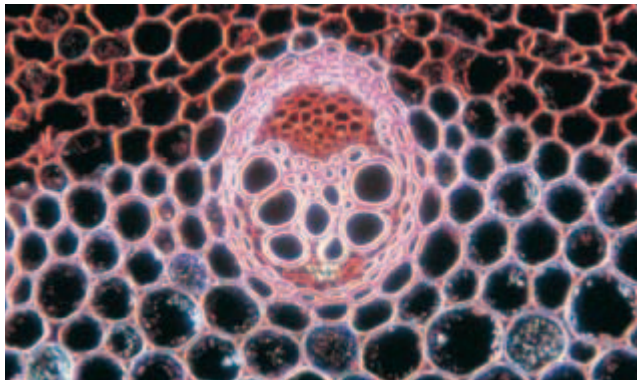
3. Change slides to the root cross section of the dicot *Ranunculus*. This slide should look similar to the photograph above. The inner core is the vascular tissue, which is surrounded by the endodermis. This area is involved in the transport of water, mineral nutrients, and organic compounds. Look for the X-shaped xylem and the smaller phloem cells surrounding the xylem.
4. In your lab report, draw what you see, and identify the xylem tissue and the phloem tissue in your drawing.
5. Locate the cortex, where starch is stored, which surrounds the vascular cylinder. Outside the cortex, find the epidermal cells and their root hairs. Draw a one-quarter section of the root tissues. Label all the tissues in your lab report.

PART B Stems

6. Observe a prepared slide of the stem of *Zea mays*, a monocot, and find the epidermis and the photosynthetic layer. It should look like the top photograph on the next page. In the center, look for the vascular bundles made up of xylem and phloem. Draw a diagram showing the location of the vascular bundles and the epidermis layer as they appear when viewed under low power.
7. Switch to high power, and observe a vascular bundle. Draw the vascular bundle, and label the tissues.
8. Observe a cross section of *Ranunculus*, a nonwoody dicot stem. Compare your slide with the photograph



Zea mays stem (130 $\times$)—a monocot



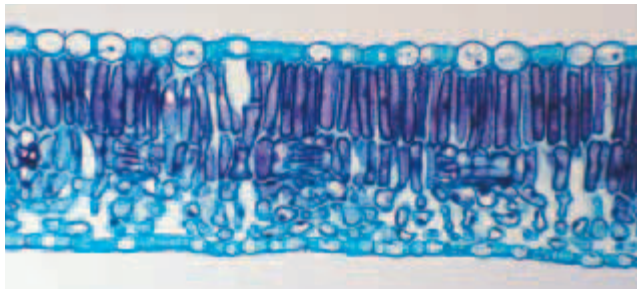
Ranunculus stem (151 $\times$)—a dicot

of *Ranunculus* shown above. Look for the epidermis and cortex layers. Notice that in a dicot, the stem is more complex than the root. Note the arrangement of the vascular bundles. In your lab report, draw what you observe. Label the epidermis, cortex, and individual vascular bundles.

9. Focus on a vascular bundle under high power. Draw and label a diagram of a vascular bundle in your lab report.

PART C Leaves

10. Observe a prepared slide of a lilac leaf, *Syringa*, shown below, under low power, and find the lower epidermis.



Syringa leaf (530 $\times$)

11. Identify the stomata on the lower epidermis of the lilac leaf. Find the guard cells that open and close a particular stoma. Locate an open stoma and a closed stoma. Draw and label diagrams of the stomata and the guard cells in your lab report.
12. Look at the center part of the cross section. Note the spongy texture of the mesophyll layer. Locate a vein containing xylem and phloem. Now identify the palisade layer, the upper epidermis, and finally the clear, continuous, noncellular layer on top. This layer is called the *cuticle*. Draw and label a diagram of your observations in your lab report.
13.  Clean up your materials before leaving the lab.

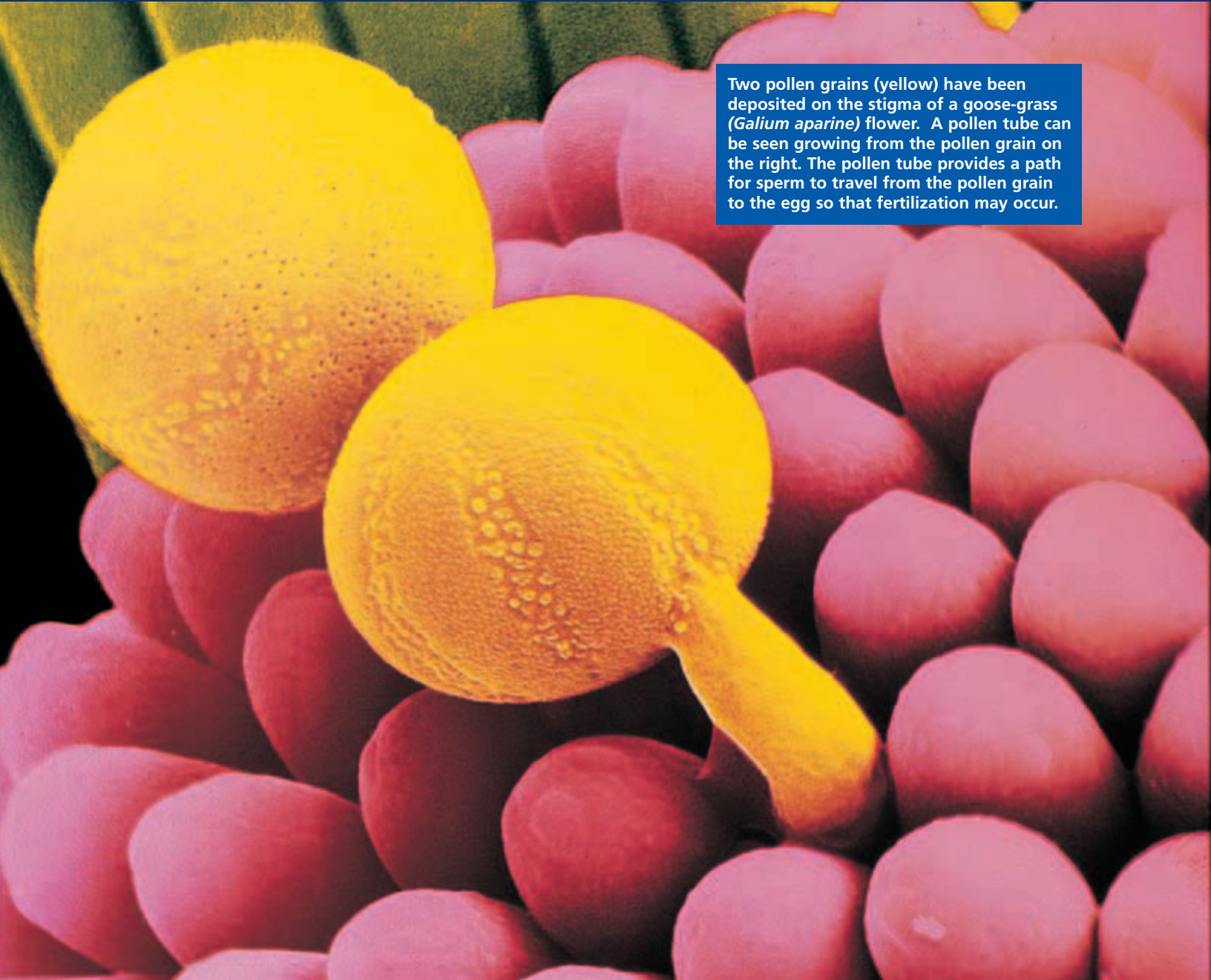
Analysis and Conclusions

1. In the dicot root that you observed, where are phloem and xylem located?
2. Where are the xylem and phloem found in the non-woody stem that you observed?
3. How are the vascular bundles different in the monocot and dicot stems that you observed?
4. How are the root cap cells different from the root tip meristematic cells?
5. What is the function of the root hairs?
6. How do the arrangements of xylem and phloem differ in roots, stems, and leaves?
7. What is the function of a stoma?
8. What is the function of the air space in the mesophyll of the leaf?
9. Which leaf structures help to conserve water?
10. Which tissues of the leaf are continuous with the stem and root tissues? How is this functional?
11. Look at the various tissues found in your drawings of roots, stems, and leaves. Classify each tissue as either dermal tissue, ground tissue, or vascular tissue.

Further Inquiry

The parts of a flower are actually modified stems and leaves. Design—but do not carry out—a procedure for dissecting a flower. Include a diagram of the parts of the flower to be viewed. Use references from the library to determine which kinds of flowers are best for dissection.

PLANT REPRODUCTION



Two pollen grains (yellow) have been deposited on the stigma of a goose-grass (*Galium aparine*) flower. A pollen tube can be seen growing from the pollen grain on the right. The pollen tube provides a path for sperm to travel from the pollen grain to the egg so that fertilization may occur.

SECTION 1 *Plant Life Cycles*

SECTION 2 *Sexual Reproduction in Flowering Plants*

SECTION 3 *Dispersal and Propagation*



For project ideas from *Scientific American*, visit go.hrw.com and type in the keyword HM6SAG.

PLANT LIFE CYCLES

A life cycle includes all of the stages of an organism's growth and development. Recall that a plant's life cycle involves two alternating multicellular stages—a diploid ($2n$) sporophyte stage and a haploid ($1n$) gametophyte stage. This type of life cycle is called alternation of generations. Also recall that the size of gametophytes and sporophytes varies among the plant groups.

THE LIFE CYCLE OF MOSSES

The dominant form of a moss is a clump of green gametophytes. Look at the moss life cycle illustrated in Figure 30-1. Moss gametophytes produce gametes in two types of reproductive structures—**antheridia** and **archegonia**. An **antheridium** (AN-thuhr-ID-ee-uhm) is a male reproductive structure that produces hundreds of flagellated sperm by mitosis. An **archegonium** (AWR-kuh-GOH-nee-uhm) is a female reproductive structure that produces a single egg by mitosis. During moist periods, sperm break out of the antheridia and swim to the archegonia. One sperm fertilizes the egg at the base of an archegonium, forming a diploid zygote. Through repeated mitotic divisions, the zygote forms an embryo and develops into a sporophyte.

OBJECTIVES

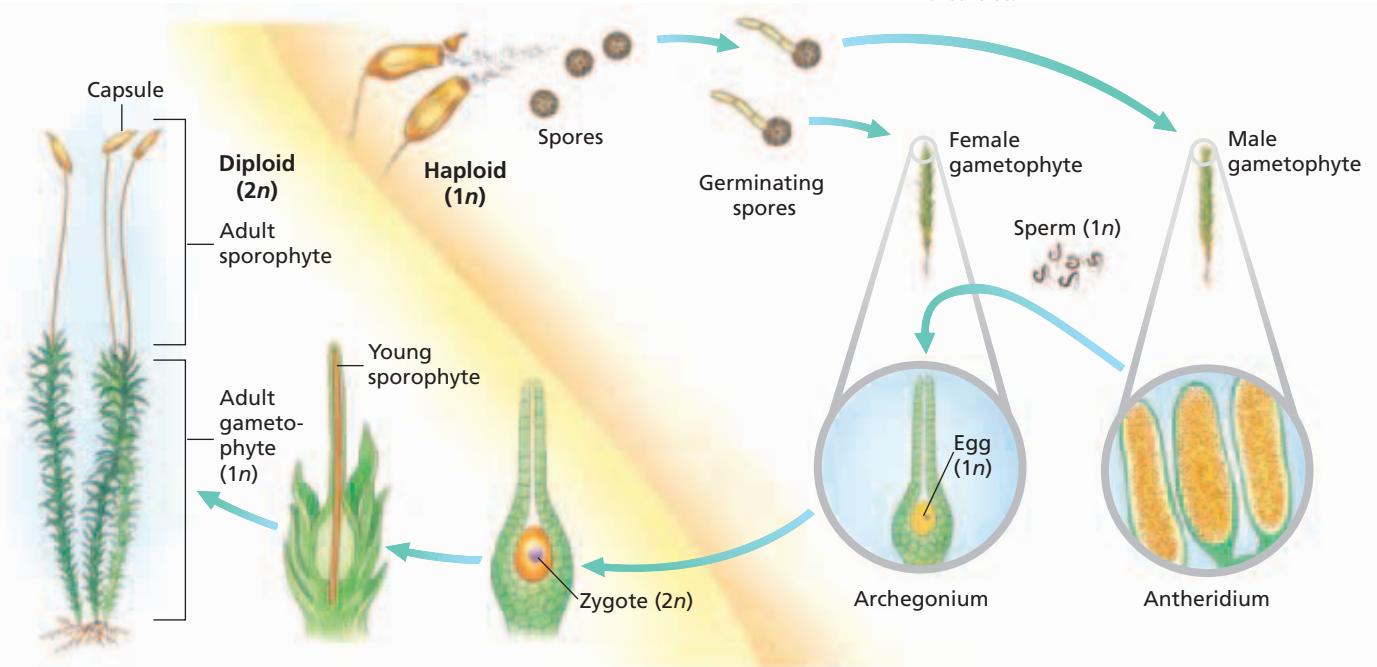
- Describe the life cycle of a moss.
- Describe the life cycle of a fern.
- Describe the life cycle of a gymnosperm.
- Compare homospory and heterospory.

VOCABULARY

antheridium
archegonium
homospory
sorus
microspore
megaspore
heterospory
integument
micropyle
ovule
pollen grain
pollination
pollen tube

FIGURE 30-1

The life cycle of mosses alternates between clumps of gametophytes (the dominant generation) and a sporophyte that consists of a spore capsule on a bare stalk.



A moss sporophyte begins as a thin stalk that grows from the tip of a gametophyte. The sporophyte remains attached to the gametophyte and depends on it for nourishment. Soon, cells at the tip of a stalk form a sporangium, called the *capsule*. Cells in the capsule undergo meiosis to form haploid spores, which are all the same. The production of one type of spore is called **homospory** (hoh-MAHS-puh-ree). Therefore, the life cycle of mosses is called *homosporous* (hoh-MAHS-puh-ruhs) *alternation of generations*. When the spores are mature, the capsule splits open, and the spores are carried away by the wind. Spores that land in favorable environments may germinate and grow into new gametophytes.

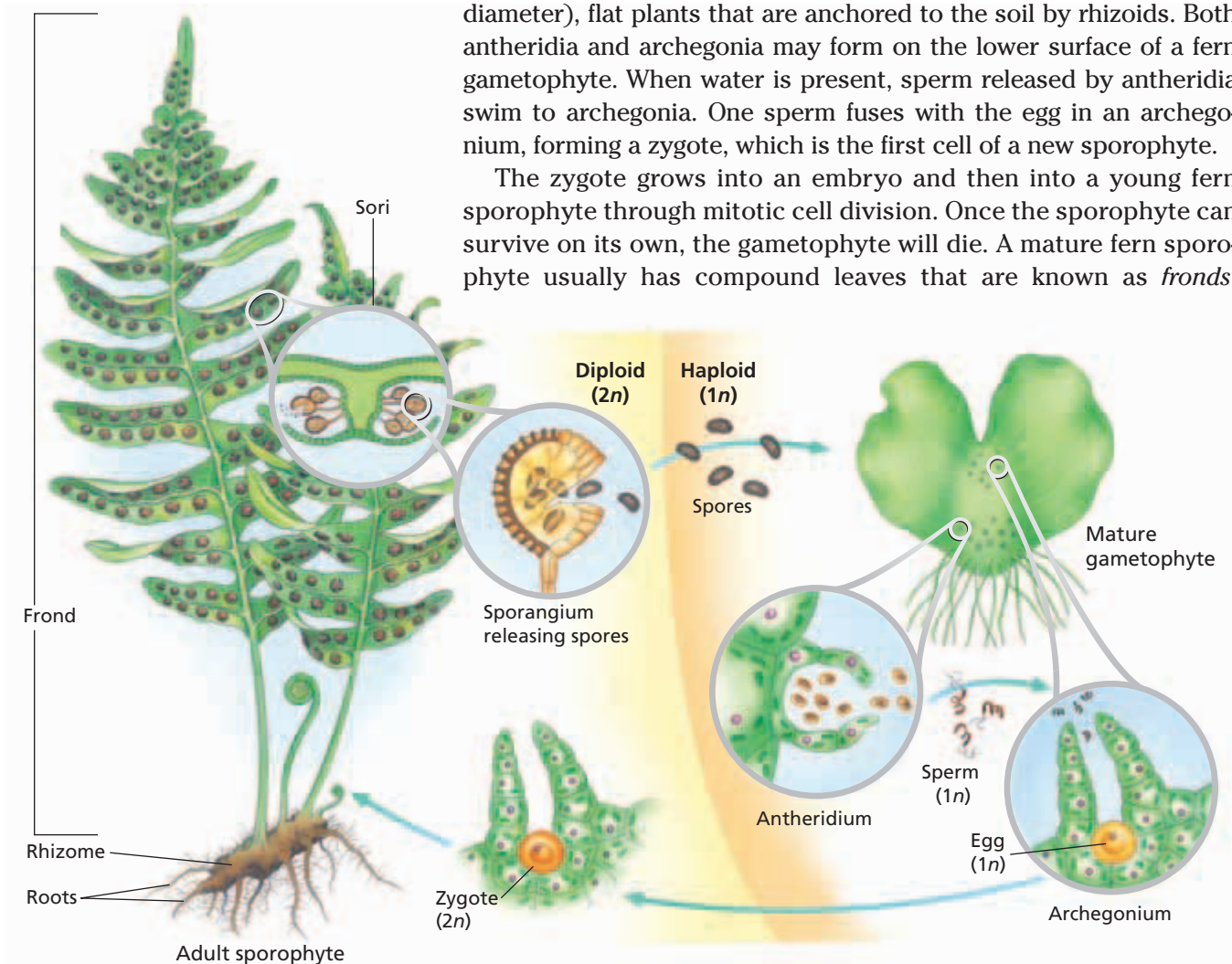
FIGURE 30-2

The life cycle of most ferns alternates between a large sporophyte (the dominant generation) and a small gametophyte. Both the egg and sperm may be produced on the same gametophyte.

THE LIFE CYCLE OF FERNS

The life cycle of a typical fern, shown in Figure 30-2, is similar to the moss life cycle. Like mosses, most ferns are homosporous. And as in mosses, the fern sporophyte grows from the gametophyte. But in the fern life cycle, the sporophyte, not the gametophyte, is the dominant generation. Fern gametophytes are tiny (about 10 mm, or 0.5 in., in diameter), flat plants that are anchored to the soil by rhizoids. Both antheridia and archegonia may form on the lower surface of a fern gametophyte. When water is present, sperm released by antheridia swim to archegonia. One sperm fuses with the egg in an archegonium, forming a zygote, which is the first cell of a new sporophyte.

The zygote grows into an embryo and then into a young fern sporophyte through mitotic cell division. Once the sporophyte can survive on its own, the gametophyte will die. A mature fern sporophyte usually has compound leaves that are known as *fronds*.



Fronds grow from an underground stem, or rhizome. In most ferns, certain cells on the underside of the fronds develop into sporangia. In many ferns, the sporangia are clustered together. A cluster of sporangia is called a **sorus** (SOH-ruhs), plural, *sori* (SOH-ree). Cells inside a sporangium undergo meiosis, forming haploid spores. At maturity, the sporangium opens and the spores are catapulted 1 cm (0.4 in.) or more and are then carried away by air currents. When the spores land, they may grow into new gametophytes.

Word Roots and Origins

sorus

from the Greek *soros*, meaning "heap"

THE LIFE CYCLE OF GYMNOSPERMS

Unlike mosses and most ferns, gymnosperms produce two types of spores—male **microspores** and female **megaspores**. Microspores grow into male gametophytes, while megaspores grow into female gametophytes. The production of different types of spores is called **heterospory** (HET-uh-AHS-puh-ree). Thus, the gymnosperm life cycle is called *heterosporous* (HET-uh-AHS-puh-ruhs) *alternation of generations*. All seed plants, spike mosses, quillworts, and a few fern species have heterospory. The microspores of heterosporous plants produce male gametophytes that stay attached to the much larger sporophyte and develop into pollen. Pollen can be transported through the air to female gametophytes. Sexual reproduction in seed plants therefore can take place independent of seasonal rains or other periods of moisture. Figure 30-3 shows the life cycle of a conifer, the most common kind of gymnosperm.

FIGURE 30-3

The life cycle of a gymnosperm alternates between a large sporophyte (the dominant generation), which produces two types of spores in cones, and microscopic gametophytes, which produce gametes. Female gametophytes produce eggs, and male gametophytes produce sperm.

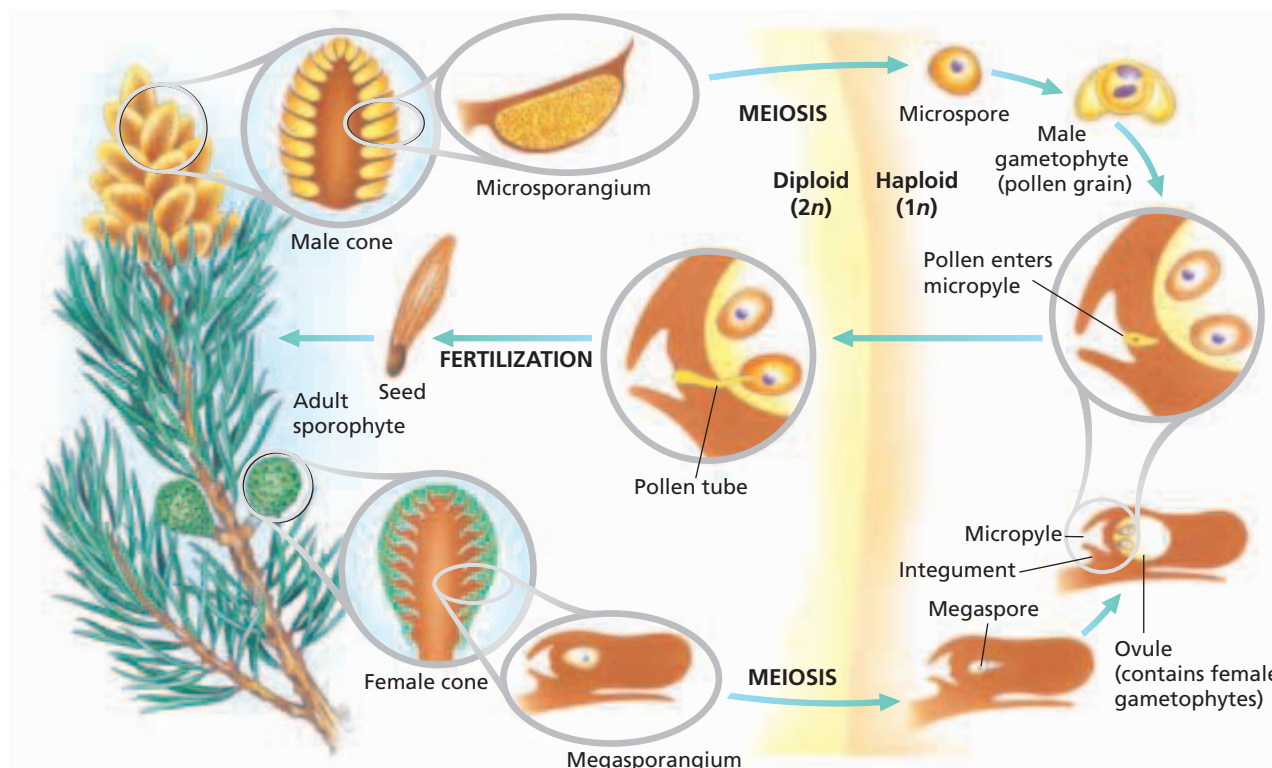


FIGURE 30-4

The male cones (pollen cones) of pines produce millions of pollen grains. The male cones then die. Wind-pollinated species typically produce large amounts of pollen. Large numbers of pollen grains increase the odds that female cones (seed cones) will be pollinated.



In the pine (a conifer), sexual reproduction takes more than two years. During the first summer, a mature pine tree produces separate female and male cones. The female cones produce *megasporangia*, while the male cones produce *microsporangia*. The following spring, cells in all sporangia undergo meiosis and divide to produce haploid spores. These spores never leave the parent to develop independently. *Megasporangia* produce megaspores, which develop into *megagametophytes*, or female gametophytes. A thick layer of cells called an **integument** (in-TEG-yoo-muhnt) surrounds each megasporangium. The integument has a small opening called the **micropyle** (MIE-kroh-PIEL). Together, a megasporangium and its integument form a structure called an **ovule** (AHV-yool). Two ovules develop on each scale of a female cone. *Microsporangia* produce microspores, which develop into *microgametophytes*, or male gametophytes. A **pollen grain** is a microgametophyte of a seed plant.

The male cones of a pine release huge numbers of pollen grains, as seen in Figure 30-4. Pine pollen travels on the wind, and only a few grains may land on a female cone. The pollen grains drift between the cone scales until they reach the ovules. The transfer of pollen to ovules is called **pollination**. A drop of fluid at the micropyle captures the pollen grain. As the fluid dries, the pollen grain is drawn into the micropyle. After pollination, the female gametophyte within the ovule produces archegonia and eggs.

After pollination, the pollen grain begins to grow a **pollen tube**, a slender extension of the pollen grain that enables sperm to reach an egg. Unlike the sperm of seedless plants, pine sperm do not have flagella and they do not swim to an egg. The pollen tube takes about a year to reach an egg only a few millimeters away. During this time, two sperm develop in the pollen tube. When the pollen tube reaches an archegonium, one sperm unites with an egg to form a zygote. The other sperm and the pollen tube die. Over the next few months, the zygote develops into an embryo as the ovule matures into a seed.

SECTION 1 REVIEW

1. Distinguish between an antheridium and an archegonium in mosses, including their structures and functions.
2. List three differences between the life cycle of a typical fern and the life cycle of a pine tree (a gymnosperm).
3. Explain why a pine tree (a gymnosperm) has two different types of cones.
4. What is the difference between homosporous alternation of generations and heterosporous alternation of generations?

CRITICAL THINKING

5. **Evaluating Differences** How do the differences in the environments in which mosses and gymnosperms live relate to the differences in sexual reproduction in these plants?
6. **Forming Reasoned Opinions** How might the presence of a haploid generation (the gametophyte) in plants serve as a “filter” for natural selection?
7. **Making Comparisons** Compare male and female gametophytes of mosses with those of gymnosperms in terms of their resistance to harsh environmental conditions.

SEXUAL REPRODUCTION IN FLOWERING PLANTS

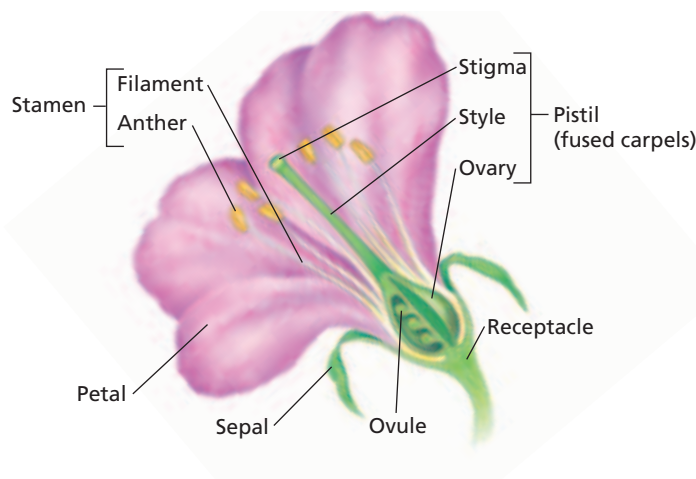
Many flowers have bright colors, attractive shapes, and pleasing aromas. These adaptations help ensure successful sexual reproduction by attracting animals that will transfer pollen. But some flowers are not colorful, large, or fragrant. Such flowers rely on wind or water for the transfer of pollen.

PARTS OF A FLOWER

Recall that early land plants lacked leaves and roots and consisted of only stems. Leaves evolved from branches of stems. Botanists consider flowers to be highly specialized branches and the parts of a flower to be specialized leaves. These specialized leaves form on the swollen tip of a floral “branch” or **receptacle**.

Flower parts are usually found in four concentric whorls, or rings, as shown in Figure 30-5. **Sepals** (SEE-puhlz) make up the outer whorl. They protect the other parts of a developing flower before it opens. **Petals** make up the next whorl. Most animal-pollinated flowers have brightly colored petals. The petals and sepals of wind-pollinated plants are usually small or absent.

The two innermost whorls of flower parts contain the reproductive structures. The male reproductive structures are **stamens** (STAY-muhnz), each of which consists of an anther and a filament. An **anther** contains microsporangia, which produce microspores that develop into pollen grains. A stalklike **filament** supports an anther. The innermost whorl contains the female reproductive structures, which are called **carpels** (KAHR-puhlz).



OBJECTIVES

- **Identify** the four main flower parts, and state the function of each.
- **Describe** gametophyte formation in flowering plants.
- **Relate** flower structure to methods of pollination.
- **Describe** fertilization in flowering plants.

VOCABULARY

receptacle
sepal
petal
stamen
anther
filament
carpel
pistil
ovary
style
stigma
embryo sac
polar nuclei
tube cell
generative cell
nectar
double fertilization

FIGURE 30-5

This diagram shows a complete flower that has all four whorls of flower parts—sepals, petals, stamens, and carpels. Most species of flowering plants have flowers that have both stamens and carpels.

FORMATION OF A FEMALE GAMETOPHYTE

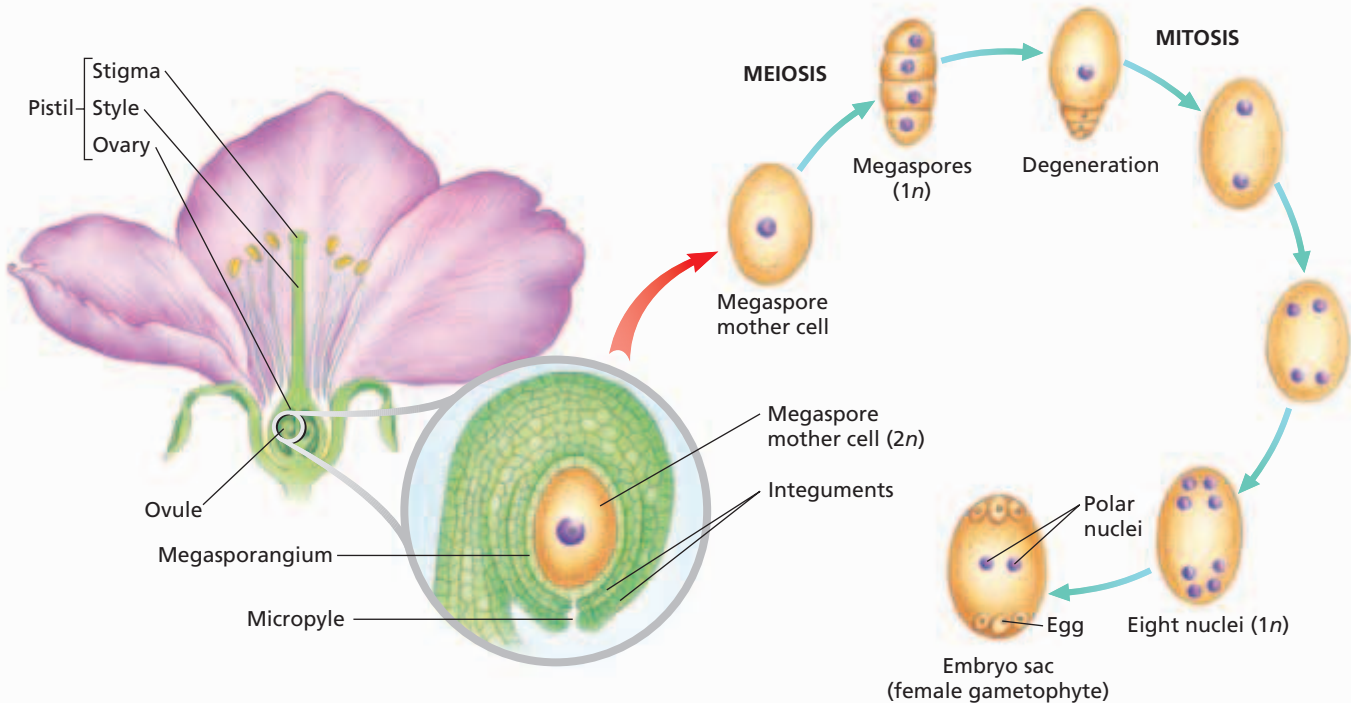


FIGURE 30-6

A cross section of an immature ovule from a flower reveals a single large cell, a megaspore mother cell. This cell undergoes meiosis and produces four megaspores. One of the megaspores undergoes a series of mitotic divisions that results in the formation of an embryo sac (a female gametophyte).

One or more carpels fused together make up the structure called a **pistil** (also shown in Figure 30-6). The enlarged base of a pistil is called the **ovary**. A **style**, which is usually stalklike, rises from the ovary. The tip of the style is called the **stigma**. Generally, a stigma is sticky or has hairs, enabling it to trap pollen grains.

GAMETOPHYTE FORMATION

In angiosperms, gametophytes develop within the reproductive structures of flowers. **Embryo sacs**, which are the female gametophytes in angiosperms, form within the ovary of the pistil. Pollen grains, the male gametophytes, form within the anthers of the stamens.

Embryo Sac Formation

In flowering plants, ovules form in the ovary of a pistil. As Figure 30-6 shows, an angiosperm ovule consists of a megasporangium surrounded by two integuments. These two integuments do not completely enclose the megasporangium. At one end of the ovule is the micropyle, through which a pollen tube can enter.

An ovule contains a large diploid cell called a *megaspore mother cell*. A megaspore mother cell undergoes meiosis and produces four haploid megaspores. In many species of flowering plants, only one megaspore enlarges. The other three degenerate. The remaining megaspore undergoes three mitotic divisions, which produce a cell that has eight haploid nuclei. These nuclei migrate to certain locations within the cell. The nuclei are initially arranged in two groups of four, with one group of nuclei at each end of the cell.

FORMATION OF A MALE GAMETOPHYTE

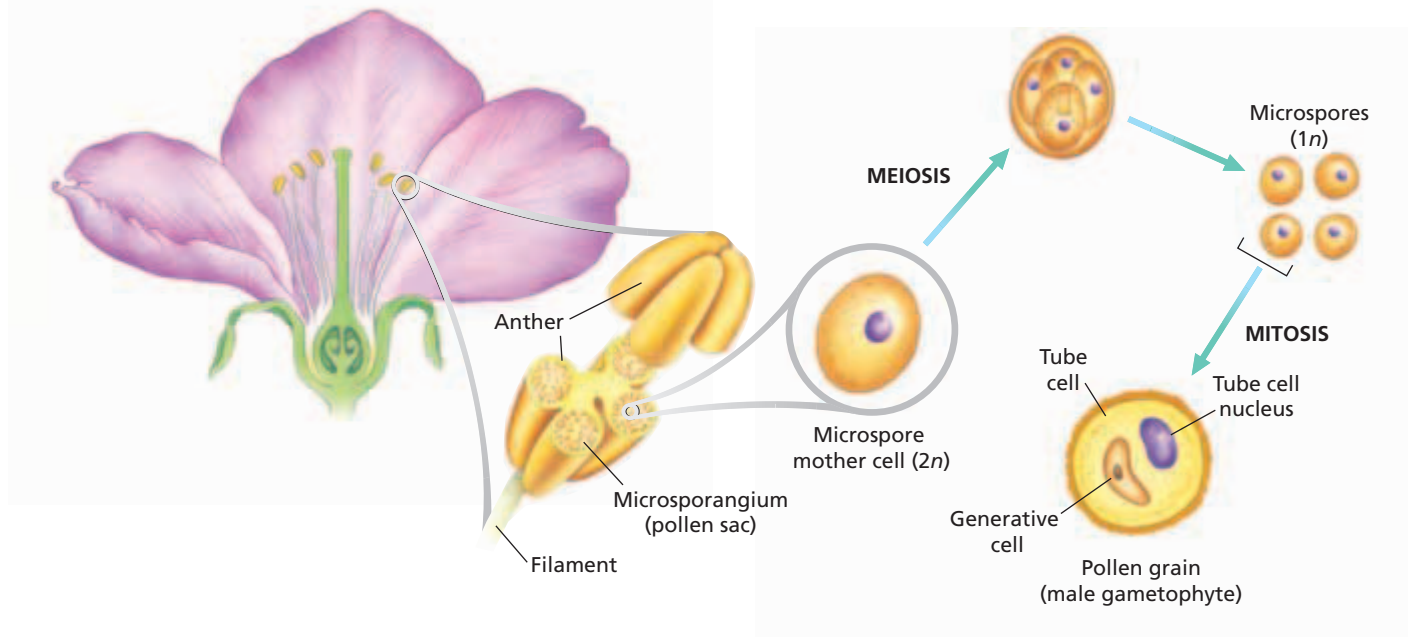


FIGURE 30-7

A cross section of an anther of a flower reveals four pollen sacs (microsporangia). Microspore mother cells within the microsporangia undergo meiosis and produce microspores. Each microspore then develops into a two-celled pollen grain (a male gametophyte).

One nucleus from each group migrates to the center of the cell. These two nuclei are called **polar nuclei** because they came from the ends, or poles, of the cell. Cell walls then form around each of the remaining six nuclei. One of the three cells that are nearest to the micropyle enlarges and becomes the egg. The two cells on either side of the egg will later help attract the pollen tube toward the egg. The function of the three cells at the other end of the cell is unclear. These five non-egg cells will eventually die after fertilization occurs.

The resulting structure, which usually contains eight nuclei and seven cells (the egg, the five non-egg cells, and the large central cell enclosing these cells), is the embryo sac. The embryo sac is the mature female gametophyte, or megagametophyte. The embryo sac is another feature that is seen in angiosperms but is not seen in gymnosperms. The surrounding integuments and the embryo sac now form a mature ovule, which may eventually develop into a seed.

Pollen Grain Formation

An anther contains four microsporangia, or *pollen sacs*, as shown in Figure 30-7. Initially, the pollen sacs contain many diploid cells. These diploid cells are called *microspore mother cells*. Each of these microspore mother cells undergoes meiosis and produces four haploid microspores. A microspore undergoes mitosis and produces two haploid cells, but these cells do not separate. A thick wall then develops around the microspore. The resulting two-celled structure is a pollen grain, which is the male gametophyte, or microgametophyte. The larger of the two cells is the **tube cell**, from which the pollen tube will form. The **generative cell**, which is enclosed in the tube cell, will divide by mitosis to form two sperm.



FIGURE 30-8

These long, tubular flowers are adapted for pollination by hummingbirds, which have long beaks adapted for reaching the nectar located deep in the flowers.

POLLINATION

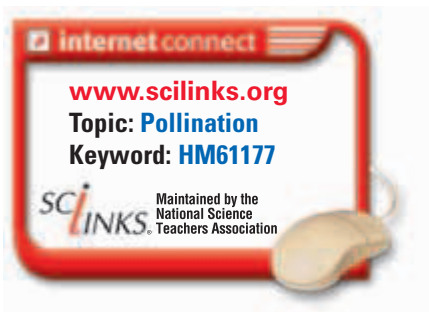
Before a sperm can fertilize the egg contained in an embryo sac, pollination must occur. In flowering plants, pollination occurs when pollen grains are transferred from an anther to a stigma. Pollination that involves just one flower, flowers on the same plant, or flowers from two genetically identical plants is called *self-pollination*. In contrast, pollination that involves two genetically different plants is called *cross-pollination*.

The pollination of flowering plants occurs in several ways. Flower structure promotes self-pollination in plants, such as peas and beans, that have flowers with petals that completely enclose both the male and female flower parts. Some aquatic plants, such as sea grasses, have pollen that is dispersed by water. Many plants, such as oak trees and grasses, release their pollen into the air. The flowers of such wind-pollinated angiosperms are small and lack showy petals and sepals. Successful wind pollination depends on four conditions: the release of large amounts of pollen, the ample circulation of air to carry pollen, the relative proximity of other plants for the pollen to be transferred to, and dry weather to ensure that pollen is not washed from the air by rain.

Most plants that have colorful or fragrant flowers are pollinated by animals. Bright petals and distinctive odors attract animals that feed on pollen and **nectar**, a nourishing solution of sugars. Animal pollinators include bats, bees, beetles, moths, butterflies, mosquitoes, monkeys, and hummingbirds. When these animals gather nectar, pollen sticks to their bodies. As they collect more nectar, the animals deposit some of the pollen on other flowers. For example, as the hummingbird in Figure 30-8 collects nectar from a flower, pollen from the anthers is deposited on the hummingbird's beak and head. When the hummingbird moves on to another flower, the pollen may be transferred to the stigma of the second flower.

FERTILIZATION

Fertilization, the union of haploid gametes resulting in a diploid zygote, may follow pollination, as shown in Figure 30-9. For fertilization to occur in angiosperms, a pollen grain must land on a stigma and then absorb moisture. The pollen grain will then germinate. Germination occurs when the nucleus of the tube cell causes the tube cell to form a pollen tube that grows through the stigma and style toward the ovary. The pollen tube grows to an ovule within the ovary and enters the ovule through the micropyle. After the pollen tube penetrates the embryo sac, two sperm (produced from the mitotic division of the generative cell in the pollen grain) can travel through the pollen tube and reach the egg. Unlike a gymnosperm's pollen tubes, an angiosperm's pollen tubes usually reach an egg a day or two after pollination.



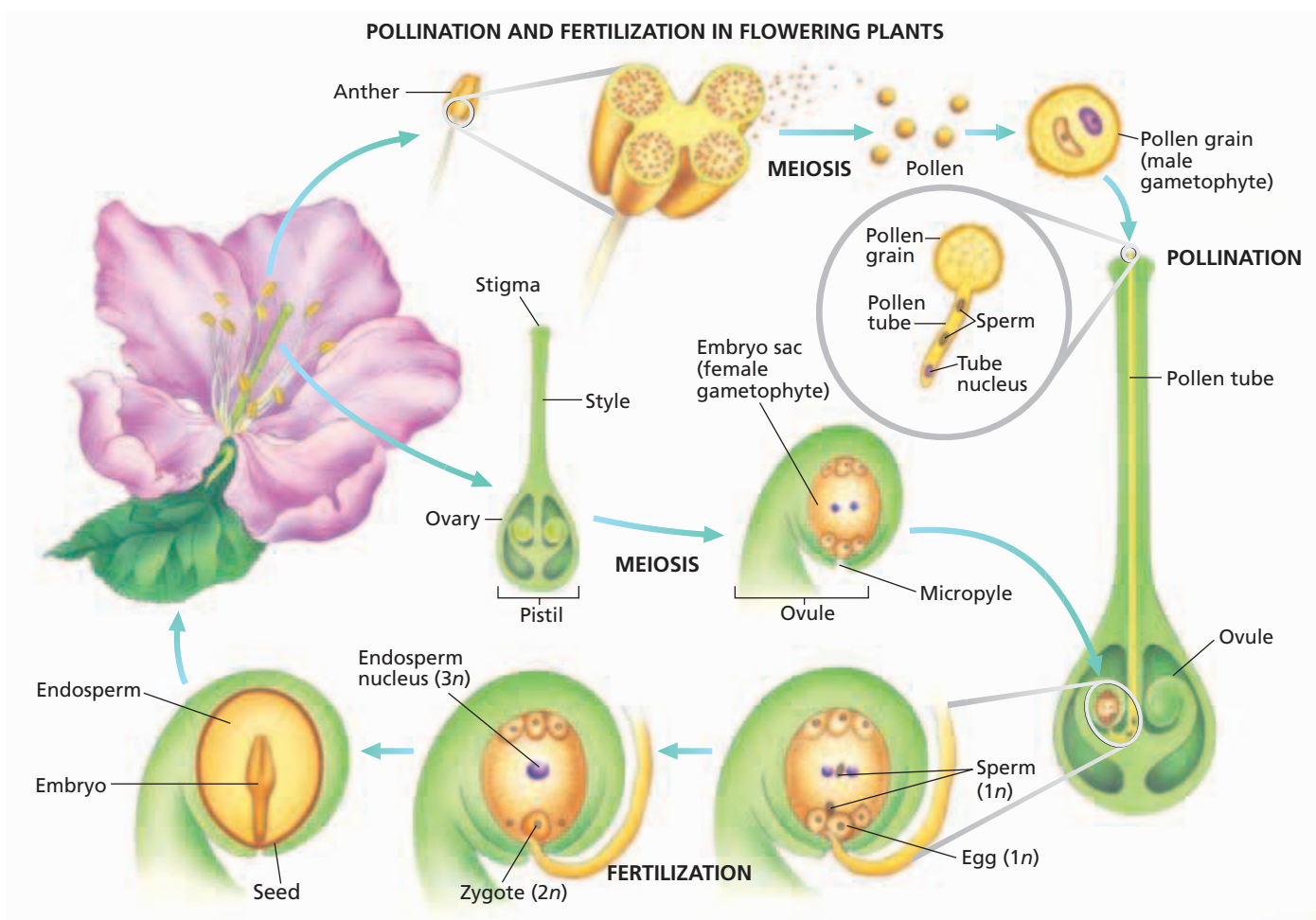


FIGURE 30-9

In a process called *double fertilization*, one sperm fuses with an egg, forming a zygote. Another sperm fuses with two polar nuclei, forming an endosperm.

One of the two sperm fuses with the egg, forming a diploid zygote. The zygote eventually develops into an embryo. The second sperm fuses with the two polar nuclei, producing a triploid ($3n$) nucleus. This nucleus then develops into tissue called *endosperm*. The endosperm provides nourishment for the embryo. Kernels of corn are mostly endosperm. In many plants, however, the endosperm is absorbed by the embryos as the seeds mature. This process of two cell fusions, which is called **double fertilization**, is unique to angiosperms.

SECTION 2 REVIEW

1. Draw a generalized flower and show the four types of flower parts in relation to each other. Be sure to label each structure.
2. Name the male and female gametophytes in flowering plants.
3. How do the flowers of wind-pollinated plants differ from the flowers of animal-pollinated plants?
4. Why is fertilization in flowering plants called *double fertilization*?

CRITICAL THINKING

5. **Interpreting Graphics** What type of pollination is shown in the illustration above?
6. **Distinguishing Relevant Information** The vast majority of the known species of plants are angiosperms. What are some characteristics of this group of plants that help explain their evolutionary success?
7. **Making Comparisons** How is meiosis in flowering plants like meiosis in humans?

SECTION 3

OBJECTIVES

- **Describe** adaptations for fruit and seed dispersal.
- **Name** the three major categories of fruits.
- **Compare** the structure and germination of different types of seeds.
- **Recognize** the advantages and disadvantages of asexual reproduction.
- **Describe** human methods of plant propagation.

VOCABULARY

propagation
seed coat
plumule
epicotyl
hypocotyl
radicle
hilum
dormancy
clone
vegetative reproduction
cutting
layering
grafting
tissue culture

FIGURE 30-10

Milkweed seeds have “parachutes” that help them drift with the wind.



DISPERSAL AND PROPAGATION

*Fruits and seeds normally result from sexual reproduction in flowering plants. Fruits are adapted for dispersing, or distributing, seeds. Seeds are adapted for the dispersal and **propagation** (production of new individuals) of plants. Many plants also propagate through asexual reproduction.*

DISPERSAL OF FRUITS AND SEEDS

Recall that one property of populations is dispersion, the spatial distribution of individuals. If individual plants are too close together, they must compete with each other for available water, nutrients, and sunlight. One reason for the success of the seed plants is the development of structures that are adapted for dispersing offspring—fruits and seeds.

Fruits and seeds are dispersed by animals, wind, water, forcible discharge, and gravity. You may have walked through a field and unwittingly collected burrs, or stickers, on your shoes or socks. These burrs are fruits, and you helped disperse them. The smell, bright color, or flavor of many fruits attract animals. When animals eat such fruits, the seeds are transported to new locations and often pass unharmed through the animals’ digestive systems.

Fruits and seeds dispersed by wind or water are adapted for those methods of dispersal. Orchids have tiny, dustlike seeds that can easily be carried by a slight breeze. Figure 30-10 shows an example of seeds adapted for wind dispersal. Many plants that grow near water produce fruits and seeds that contain air chambers, which allow them to float. Coconuts, for example, may float thousands of kilometers on ocean currents.

The most dramatic method of seed dispersal occurs in plants that forcibly discharge their seeds from their fruits. The tropical sandbox tree, which has fruits that hurl seeds up to 100 m (328 ft), appears to hold the distance record.

Although gymnosperms do not produce fruits, their cones may help protect seeds and aid in seed dispersal. Female pine cones close after pollination and open again only when the seeds are mature. Pine seeds may be dispersed when gravity causes cones to drop and roll away from the parent tree. Pine seeds have “wings” that aid in wind dispersal.

TYPES OF FRUITS

Botanists define a *fruit* as a mature ovary. Many different types of fruits have evolved among the flowering plants. Figure 30-11 shows examples of some of these fruit types. Fertilization usually initiates the development of fruits. Fruits protect seeds, aid in their dispersal, and often delay their sprouting. Fruits are classified mainly on the basis of two characteristics: how many pistils or flowers form the fruit, and whether the fruit is dry or fleshy. Table 30-1 presents a classification system for fruits. Notice that fruits with common names that include “nut” or “berry” may not be actual nuts or berries. For example, a peanut is actually a legume, not a nut. You may have heard the fleshy seeds of ginkgo, juniper, and yew trees referred to as berries. These names are misleading because ginkgo, juniper, and yew trees are gymnosperms, which do not form fruits.



FIGURE 30-11

A pea pod is a simple fruit. A raspberry is an aggregate fruit. A pineapple is a multiple fruit.

TABLE 30-1 Fruit Classification

Major categories and types of fruits	Examples
Simple fruit —formed from one pistil of a single flower A. Dry at maturity 1. Usually splits open a. Legume—splits along two sides to form two halves b. Follicle—splits along one side c. Capsule—splits in a variety of other ways 2. Usually does not split open a. Grain—thin ovary wall fused to seed coat b. Nut—thick, woody ovary wall not fused to single seed c. Achene—thin ovary wall not fused to single seed d. Samara—like an achene but with a thin, flat wing B. Fleshy at maturity and usually not opening 1. Usually contains only one seed a. Drupe—stony inner layer around the seed 2. Usually contains many seeds a. Pome—core with seeds surrounded by papery ovary walls; outer part formed from sepals b. Typical berry—thin skin c. Pepo—berry with a thick, hard rind d. Hesperidium—berry with leathery, easily removed skin	pea, peanut, black locust milkweed, columbine poppy, tulip corn, wheat oak, chestnut sunflower, dandelion ash, elm, maple cherry, coconut, pecan apple, pear grape, tomato, banana watermelon, cucumber orange, grapefruit, lemon
Aggregate fruit —formed from several pistils of a single flower A. Dry at maturity B. Fleshy at maturity	tulip tree, magnolia raspberry, strawberry
Multiple fruit —formed from several flowers growing together A. Dry at maturity B. Fleshy at maturity	sweetgum, sycamore pineapple, fig

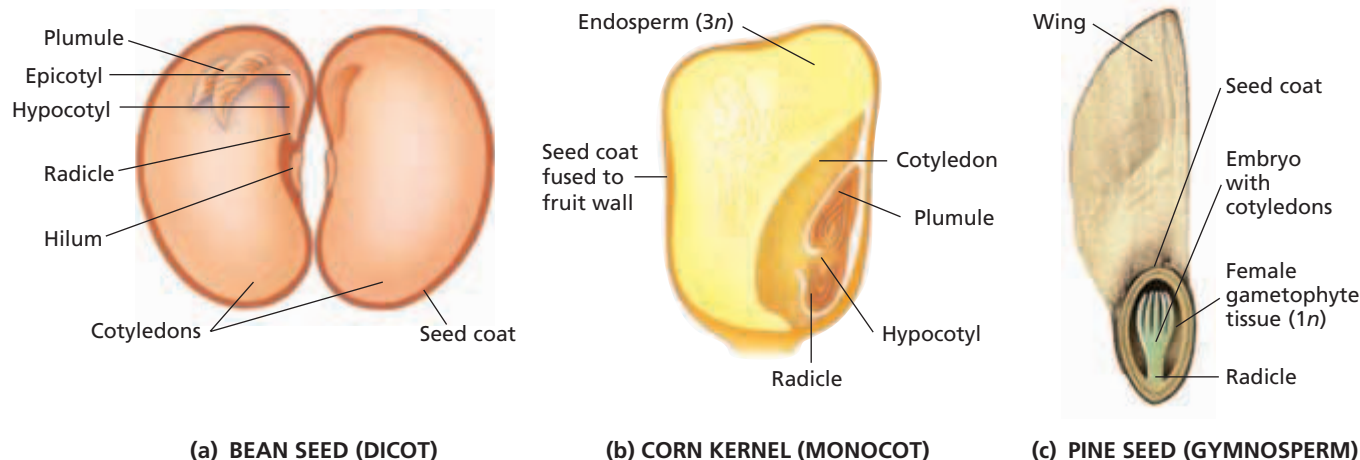


FIGURE 30-12

(a) A bean seed has two cotyledons and no endosperm. (b) A corn kernel contains a single seed, which has one cotyledon and endosperm. (c) A pine seed has cotyledons and tissue from the female gametophyte.

STRUCTURE OF SEEDS

A **seed** is a plant embryo surrounded by a protective coat called the **seed coat**. The structure of seeds differs among the major groups of seed plants—angiosperms, which include monocots and dicots, and gymnosperms. To understand some of the differences, examine the seeds shown in Figure 30-12.

Look at the bean seed in Figure 30-12a, which has been opened to reveal the structures inside. Most of the interior of a bean seed is filled by two large, fleshy cotyledons (seed leaves), which are part of the embryo. Recall that dicots have two cotyledons in their embryos. Therefore, beans are dicots. A mature bean seed has no endosperm. The endosperm was absorbed by the fleshy cotyledons.

Between the two cotyledons of a bean seed are the parts that make up the rest of the embryo. The shoot tip, along with any embryonic leaves, is called the **plumule** (PLOO-MYOOl). The **epicotyl** (EP-I-KAHT-uhl) extends from the plumule to the attachment point of the cotyledons. The **hypocotyl** (HIE-poh-KAHT-uhl) extends from the attachment point of the cotyledons to the radicle. The **radicle** is the embryonic root. Along the concave edge of the seed, beneath the radicle, is the **hilum** (HIE-luhm), which is a scar that marks where the seed was attached to the ovary wall.

Now examine the corn kernel in Figure 30-12b. Technically, a corn kernel is a fruit, but the seed occupies almost the entire kernel. The wall of the fruit is very thin and is fused to the seed coat. A single umbrella-shaped cotyledon is pressed close to the endosperm. The cotyledon of a monocot seed does not store nutrients, as bean cotyledons do. Instead, it absorbs nutrients from the endosperm and transfers them to the embryo.

Finally, look at the pine seed in Figure 30-12c. A pine seed contains a sporophyte embryo that has needle-like cotyledons. The embryo is surrounded by tissue of the female gametophyte. Like the triploid endosperm of angiosperm seeds, the haploid tissue of the female gametophyte functions as a source of nourishment for the embryo.



Quick Lab

Predicting Seed Dispersal

Materials five different fruits, balance or scale

Procedure

1. Create a data table that has at least five rows. Your table should have six columns with the following headings: "Fruit name," "Fruit type" (from Table 30-1), "Dry/fleshy," "Seed mass in grams," "Whole fruit mass in grams," and "Dispersal method."
2. Examine your fruits, and fill in your data table. Discuss with your group how characteristics of fruits and seeds might relate to dispersal methods.

Analysis Form a hypothesis about a dispersal method for one of the fruits you have examined. Describe how you might test your hypothesis.

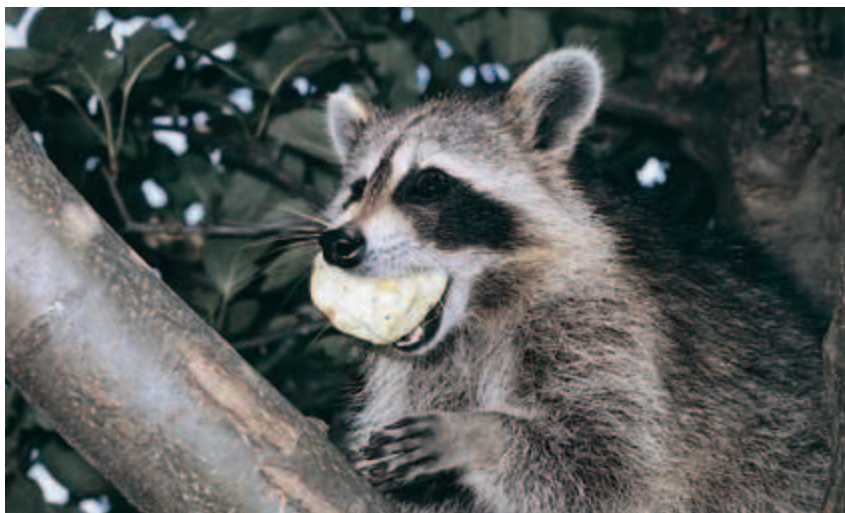
SEED GERMINATION

Many plants are easily grown from seeds. Although its embryo is alive, a seed will not germinate, or sprout, until it is exposed to certain environmental conditions. The delay of germination often assures the survival of a plant. For example, if seeds that mature in the fall were to sprout immediately, the young plants could be killed by cold weather. Similarly, if a plant's seeds were to sprout all at once and all of the new plants died before producing seeds, the species could become extinct. Many seeds will not germinate even when exposed to conditions ideal for germination. Such seeds exhibit **dormancy**, which is a state of reduced metabolism in which growth and development do not occur. The longevity of dormant seeds is often remarkable. A botanist once germinated lotus seeds that were almost 1,000 years old.

Conditions Needed for Germination

Environmental factors, such as water, oxygen, and temperature, can trigger seed germination. Most mature seeds are very dry and must absorb water to germinate. Water softens the seed coat and activates enzymes that convert starch in the cotyledons or endosperm into simple sugars, which provide energy for the embryo to grow. As the embryo begins to grow, the softened seed coat cracks open. This enables the oxygen needed for cellular respiration to reach the embryo. Many small seeds need light for germination. This adaptation prevents the seeds from sprouting if they are buried too deeply in the soil. In addition, some seeds germinate only if exposed to temperatures within a certain range.

Some seeds germinate only after being exposed to extreme conditions. For example, animals often swallow the seeds of fruits they eat, as shown in Figure 30-13. Acids in the digestive system wear away the hard seed coat. The seeds may germinate after passing through the digestive systems of these animals. As an added bonus, the seed is deposited with a bit of natural fertilizer.



Word Roots and Origins

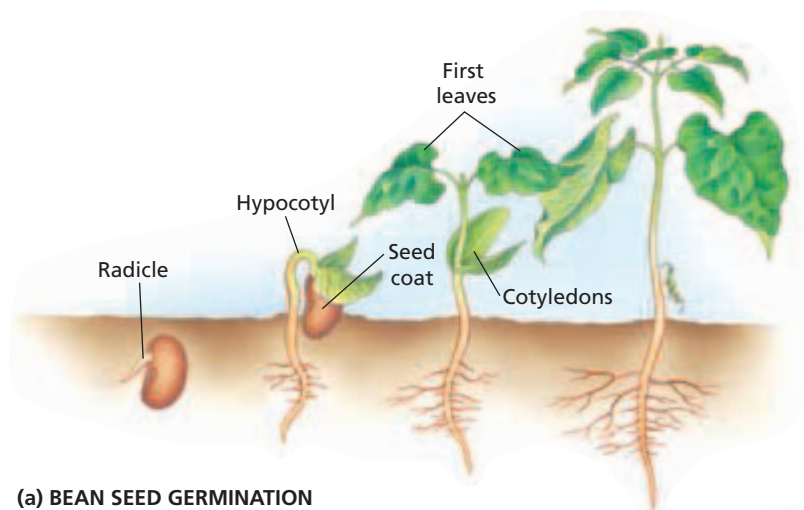
dormancy

from the Latin *dormire*,
meaning "to sleep"



FIGURE 30-13

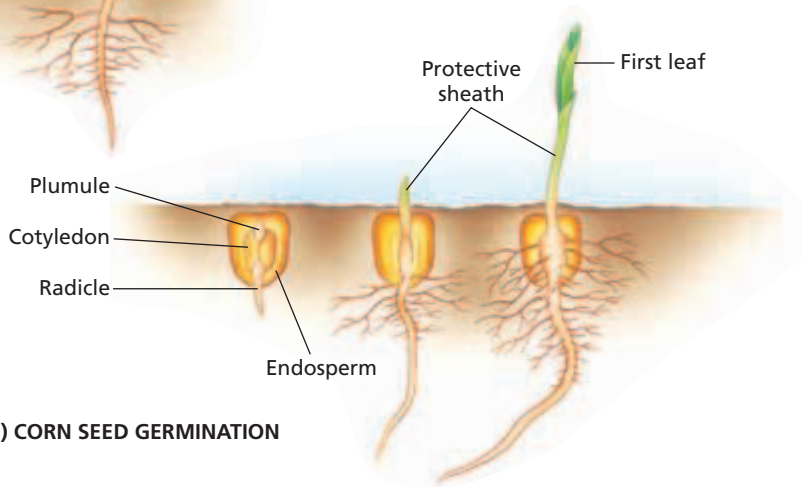
Many animals eat apples or other fruits. The seeds are swallowed and are exposed to acids as they pass through the animal's digestive system. The acids wear away the seed coat. Once the seed is on the ground, water and oxygen can enter the seed and enable the growing embryo to break out.



(a) BEAN SEED GERMINATION

FIGURE 30-14

(a) During the germination of a bean seed, the cotyledons and seed coat emerge, and the plumule is protected by a hook in the hypocotyl. (b) During the germination of a corn seed, the cotyledon and seed coat remain below ground and the plumule is protected by a sheath.



(b) CORN SEED GERMINATION

Other seeds, including apple seeds, must be exposed to near-freezing temperatures for several weeks before they sprout. This temperature requirement prevents the seeds from germinating in the fall and thus ensures that the seedlings will not be killed by the cold temperatures of winter. The cold temperatures cause chemical changes within the seed. These changes enable the embryo to grow.

Process of Germination

Figure 30-14 compares germination in corn and bean seeds. The first visible sign of seed germination is the emergence of the radicle. In beans, the entire root system develops from the radicle. In corn, most of the root system develops from the lower part of the radicle. Soon after the radicle breaks the seed coat, the shoot begins to grow.

In some seeds, such as bean seeds (Figure 30-14a), the hypocotyl curves and becomes hook-shaped. Once the hook breaks through the soil, the hypocotyl straightens. This straightening pulls the cotyledons and the embryonic leaves into the air. The embryonic leaves unfold, synthesize chlorophyll, and begin photosynthesis. After their stored nutrients are used up, the shrunken bean cotyledons fall off.

In contrast, the cotyledon of the corn seed (Figure 30-14b) remains underground and transfers nutrients from the endosperm to the growing embryo. Unlike the bean shoot, the corn cotyledon remains below ground. The corn plumule is protected by a sheath that pushes through the soil. Once the sheath has broken through the soil surface, the plumule grows up through the sheath and the first leaf unfolds.



ASEXUAL REPRODUCTION

Recall that asexual reproduction is the production of an individual without the union of gametes. Elaborate technology has recently yielded some success in the asexual reproduction of mammals to create **clones**, or exact duplicates of organisms. But people who create clones of themselves are still the subjects of science fiction movies and books. However, asexual reproduction to create clones is common in the plant kingdom.

Asexual reproduction can be an advantage to individuals that are well-adapted to their environment. Many new individuals can be produced in a short time, which enables the clones to spread rapidly and fill available space. A disadvantage of asexual reproduction is the lack of genetic variation among the offspring. All of the offspring are genetically identical. Thus, the clones have the same tolerance to an environment and are attacked by the same diseases and pests. Many of our plant cultivars are clones.

In nature, plants reproduce asexually in many ways, including the development of spores and vegetative reproduction. **Vegetative reproduction** is asexual reproduction involving vegetative (nonreproductive) parts of a plant, such as leaves, stems, or roots. Many structures specialized for vegetative reproduction have evolved in plants. Table 30-2 lists some of these structures. One example of vegetative reproduction is shown in Figure 30-15.



FIGURE 30-15

New plants are vegetatively reproduced from the stolons (runners) of an airplane plant, also called a *spider plant*.

PROPAGATION OF PLANTS BY HUMANS

People often use vegetative structures to propagate plants. Many species of plants are vegetatively propagated by humans from the structures described in Table 30-2. People have also developed several methods of propagating plants from other vegetative parts, such as roots and even tissue samples. These methods include layering, grafting, and using cuttings and tissue cultures.

TABLE 30-2 *Plant Structures Adapted for Vegetative Reproduction*

Name	Description	Examples
Stolon (runner)	horizontal, aboveground stem that produces leaves and roots at its nodes; a new plant can grow from each node	strawberry, airplane (spider) plant, Boston fern
Rhizome	horizontal, belowground stem that produces leaves and roots at its nodes; a new plant can grow from each node	ferns, horsetails, iris, ginger, sugar cane
Bulb	very short, underground monocot stem with thick, fleshy leaves adapted for storage; bulbs divide naturally to produce new plants	tulip, daffodil, onion, garlic, hyacinth
Tuber	underground, swollen, fleshy stem specialized for storage; the buds on a tuber can grow into new plants	potato, caladium, Jerusalem artichoke

Cuttings

In some plants, roots will form on a cut piece of a stem, or shoots will form on a piece of a root. Pieces of stems and roots that are cut from a plant and used to grow new plants are called **cuttings**. Plants such as African violets can be grown from leaf cuttings, which will form both roots and shoots. Cuttings are widely used to propagate houseplants, ornamental trees and shrubs, and some fruit crops.

Layering

In some species, such as raspberries, roots form on stems where they make contact with the soil. People often stake branch tips to the soil or cover the bases of stems with soil to propagate such plants. The process of causing roots to form on a stem is called **layering**. Air layering, wounding a stem and placing moist peat moss around the wound, is another common form of layering.

Grafting

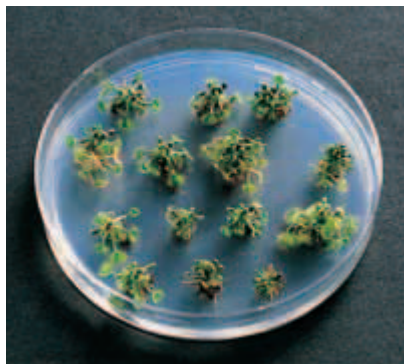
Grafting is the joining of two or more plant parts to form a single plant. In grafting, a bud or small stem of one plant is attached to the roots or stems of a second plant. The vascular cambium of both parts must be aligned for a graft to be successful. Grafting enables the desirable characteristics of two cultivars to be combined. Grafting is used to propagate virtually all commercial fruit and nut trees and many ornamental trees and shrubs.

Tissue Culture

Figure 30-16 shows plants grown by **tissue culture**, the production of new plants from pieces of tissue placed on a sterile nutrient medium. Unlike most animal cells, plant cells contain functional copies of all the genes needed to produce a new plant. Thus, it is possible for a whole plant to regrow from a single cell. Millions of identical plants can be grown from a small amount of tissue. Tissue culture is used in the commercial production of orchids, houseplants, cut flowers, fruit plants, and ornamental trees, shrubs, and nonwoody plants.

FIGURE 30-16

Tissue culture can be used to grow round-leaved sundews, *Drosera rotundifolia*. The jelly in the Petri dish is a sterile soil substitute that contains needed nutrients.



SECTION 3 REVIEW

1. Name three common methods of fruit and seed dispersal.
2. Name one fruit from each of the three major categories of fruit.
3. How do the structure and germination of a bean seed and a corn seed differ from each other?
4. Compare asexual reproduction with sexual reproduction.
5. Make a table that compares structures and methods for propagation of plants by humans.

CRITICAL THINKING

6. **Justifying Conclusions** How might seed dormancy be an evolutionary advantage?
7. **Applying Information** Many fruits are green until they are mature. How might this be an advantage in terms of seed dispersal by animals?
8. **Evaluating Information** Instructions on packets of large seeds state that they should be planted deeply. Instructions on packets of small seeds state that they should be planted shallowly. Why are these instructions different?

CHAPTER HIGHLIGHTS

SECTION 1

Plant Life Cycles

- Plants have a life cycle called *alternation of generations*, in which a multicellular haploid gametophyte stage alternates with a multicellular diploid sporophyte stage.
- In the moss life cycle, a spore develops into a leafy green gametophyte that produces eggs in archegonia and flagellated sperm in antheridia. A moss sporophyte grows from a fertilized egg.
- In the fern life cycle, a spore develops into a small, flat gametophyte that produces eggs in archegonia and flagellated sperm in antheridia. A fern sporophyte grows from a fertilized egg.
- In the gymnosperm life cycle, a large sporophyte produces two types of spores, and microscopic gametophytes produce gametes. Nonflagellated sperm reach an ovule through a pollen tube.
- Mosses and most ferns are homosporous (produce only one type of spore). All seed plants as well as a few other species are heterosporous (produce male microspores and female megaspores).

Vocabulary

antheridium (p. 609)
archegonium (p. 609)
homospory (p. 610)
sorus (p. 611)

microspore (p. 611)
megaspore (p. 611)
heterospory (p. 611)

integument (p. 612)
micropyle (p. 612)
ovule (p. 612)

pollen grain (p. 612)
pollination (p. 612)
pollen tube (p. 612)

SECTION 2

Sexual Reproduction in Flowering Plants

- Flowers are reproductive structures. Most flowers consist of four parts—protective sepals, colorful petals, pollen-producing stamens, and egg-containing carpels.
- An immature ovule contains a megaspore mother cell that undergoes meiosis and produces four megaspores. One of the megaspores divides by mitosis to form an embryo sac.
- Pollen grains form in stamens. Microspore mother cells in pollen sacs undergo meiosis and produce microspores, each of which develops into a pollen grain.
- Many flowering plants have flowers adapted for animal pollination or for wind pollination.
- In double fertilization, one sperm combines with the egg to form a zygote. A second sperm combines with two polar nuclei to form the endosperm.

Vocabulary

receptacle (p. 613)
sepal (p. 613)
petal (p. 613)
stamen (p. 613)
anther (p. 613)

filament (p. 613)
carpel (p. 613)
pistil (p. 613)
ovary (p. 613)

style (p. 613)
stigma (p. 613)
embryo sac (p. 614)
polar nuclei (p. 615)

tube cell (p. 615)
generative cell (p. 615)
nectar (p. 616)
double fertilization (p. 617)

SECTION 3

Dispersal and Propagation

- Fruits and seeds are dispersed by animals, wind, water, forcible discharge, and gravity. Many fruits and seeds are adapted for a particular type of dispersal.
- The three major categories of fruits are simple, aggregate, and multiple.
- The structure of seeds differs among the major groups of seed plants. Seeds need water, oxygen, suitable temperatures, and sometimes light to germinate. Delayed germination often assures the survival of a plant.
- Asexual reproduction enables plants to spread rapidly in a favorable environment but can result in a lack of genetic variation among offspring.
- Human methods of plant propagation include using cuttings, layering, grafting, and tissue culture.

Vocabulary

propagation (p. 618)
seed coat (p. 620)
plumule (p. 620)
epicotyl (p. 620)

hypocotyl (p. 620)
radicle (p. 620)
hilum (p. 620)
dormancy (p. 621)

clone (p. 623)
vegetative reproduction (p. 623)
cutting (p. 624)

layering (p. 624)
grafting (p. 624)
tissue culture (p. 624)

CHAPTER REVIEW

USING VOCABULARY

- For each pair of terms, explain how the meanings of the terms differ.
 - microspore* and *megaspore*
 - pollination* and *double fertilization*
 - sepal* and *petal*
- For each pair of terms, explain the relationship between the terms.
 - homospory* and *heterospory*
 - polar nuclei* and *embryo sac*
 - integument* and *micropyle*
- Use the following terms in the same sentence: *dormancy*, *plumule*, *radicle*, and *seed coat*.
- Word Roots and Origins** The word *carpel* is derived from the Greek word *karpos*, which means “fruit.” Using this information, explain why the term *carpel* is a good name for the structure that the term describes.

UNDERSTANDING KEY CONCEPTS

- Name** the sexual reproductive structures in a moss life cycle, and describe the function of each.
- Identify** the dominant stage in the life cycle of a fern.
- Summarize** the adaptation that allows gymnosperms to sexually reproduce in the absence of environmental moisture.
- Explain** how heterospory can be an advantage over homospory.
- Describe** the roles of the parts of a flower that do not participate directly in reproduction.
- Describe** the roles of the parts of a flower that do participate directly in reproduction.
- Distinguish** between embryo sac formation and pollen grain formation.
- Draw** a diagram showing the events and plant structures involved in angiosperm pollination.
- Distinguish** the process of fertilization in gymnosperms from the process of fertilization in flowering plants.
- Identify** one example of a plant structure that is an adaptation for fruit or seed dispersal.
- Describe** the two characteristics used for classifying fruits.
- Explain** the advantage of the hypocotyl hook that the embryos of some seeds form as the plant emerges after germination.
- Explain** why asexual reproduction can be an advantage for plants that are well-adapted to their environment.
- Compare** two methods of plant propagation used by humans.
- CONCEPT MAPPING** Use the following terms to create a concept map that describes male reproductive structures in flowering plants: *anther*, *generative cell*, *microspore mother cell*, *stamen*, *microspore*, *pollen grain*, and *tube cell*.

CRITICAL THINKING

- Recognizing Relationships** In many flowers with both stamens and pistils, the stigma is located well above the anthers. What might be the value of such an arrangement?
- Analyzing Information** Following the self-pollination of some plants, the pollen tubes die before reaching ovules. What is the significance of this event?
- Justifying Conclusions** If you were to discover a new type of rose, would you use vegetative parts or seeds to propagate the rose and produce large numbers of identical plants? Justify your answer.
- Interpreting Graphics** In the photograph below, the houseplant on the left shows a *Sansevieria trifasciata*, which has yellow-edged leaves. A section of the leaf, or a leaf cutting, can be used to produce new plants. However, as shown on the right side of the photograph, only shoots with all-green leaves and no yellow edges will form. How do you explain this?



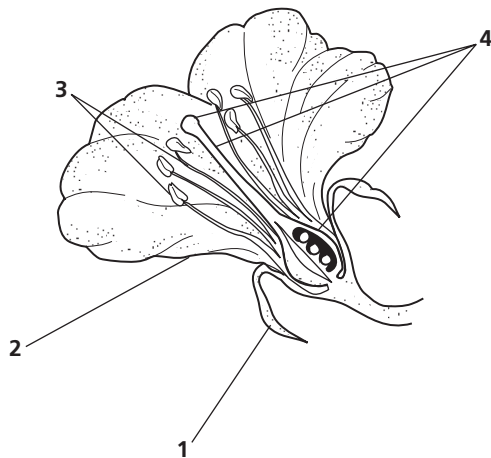


Standardized Test Preparation

DIRECTIONS: Choose the letter of the answer choice that best answers the question.

- The production of a single type of spore is a characteristic of the life cycles of which kinds of plants?
 - mosses and most ferns
 - most ferns and gymnosperms
 - mosses and most gymnosperms
 - mosses, most ferns, and gymnosperms
- Why is sexual reproduction in gymnosperms and other seed plants independent of seasonal rains?
 - These plants grow only near streams and rivers.
 - Pollinators or wind carry the sperm to the eggs.
 - Fertilization occurs inside structures within the spores.
 - Fertilization always involves eggs and sperm of the same plant.

INTERPRETING GRAPHICS: The illustration below shows the four concentric whorls of a typical flower. Use this illustration to answer the question that follows.

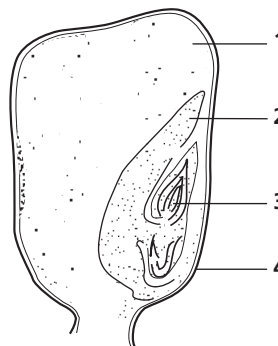


- In which whorl are the male reproductive structures found?
 - 1
 - 2
 - 3
 - 4
- Pollination in flowering plants is said to take place when which of the following occurs?
 - when insects ingest nectar
 - when pollen lands on a stigma
 - when a sperm fuses with an egg
 - when a spore leaves a sporangium

DIRECTIONS: Complete the following analogy.

- Generative cell : pollen grain :: egg cell :
 - ovary
 - carpel
 - receptacle
 - embryo sac

INTERPRETING GRAPHICS: The illustration below shows the structure of a corn kernel, which is a monocot seed. Use this illustration to answer the question that follows.



- Which structure of the corn kernel is the endosperm, the source of nutrients?
 - 1
 - 2
 - 3
 - 4

SHORT RESPONSE

A seed will not germinate until it is exposed to certain environmental conditions.

List two environmental factors that influence seed germination, and describe the mechanisms by which they trigger the germination process.

EXTENDED RESPONSE

Double fertilization is unique to angiosperms.

Part A What cells participate in double fertilization, and what products do they form?

Part B What are the evolutionary advancements seen only in angiosperms, and what advantages do they confer?

Test TIP

If you are not sure about the spelling of certain words when answering the short- or extended-response questions, look at the question to see if the same word appears in the question.

Comparing Seed Structure and Seedling Development

OBJECTIVES

- Observe the structures of dicot and monocot seeds.
- Compare the structure of dicot and monocot embryos.
- Compare the development of dicot and monocot seedlings.

PROCESS SKILLS

- relating structure to function
- comparing and contrasting
- identifying
- collecting data

MATERIALS

- 1 pea seed soaked overnight
- 6 bean seeds soaked overnight
- 6 corn kernels soaked overnight
- stereomicroscope
- scalpel
- Lugol's iodine solution in dropper bottle
- paper towels
- 2 rubber bands
- 2 150 mL beakers
- glass-marking pen
- metric ruler
- microscope slide and cover slip
- medicine dropper
- compound light microscope

Background

1. What are the parts of a seed?
2. In what ways are seeds like their parent plant?
3. How do monocotyledons and dicotyledons differ?
4. What changes occur as a seed germinates?

PART A Seed Structure

1. Obtain one each of the following seeds—pea, bean, and corn.
2. Remove the seed coats of the pea and bean seeds. Open the seeds, and locate the two cotyledons in each seed.

3. Using the stereomicroscope, examine the embryos of the pea seed and the bean seed.

4. In your lab report, draw the pea and bean embryos and label all of the parts that you can identify.



5. **CAUTION** Put on a lab apron, safety goggles, and

gloves. In this lab, you will be working with a chemical that may irritate your skin or stain your skin and clothing. If you get a chemical on your skin or clothing, wash it off at the sink while calling to your teacher. If you get a chemical in your eyes, immediately flush it out at the eyewash station while calling to your teacher. Use the scalpel carefully to avoid injury. Examine a corn kernel, and find a small, oval, light-colored area that shows through the seed coat. Use the scalpel to cut the kernel in half along the length of this area. Place a drop of iodine solution on the cut surface.

6. Use the stereomicroscope to examine the corn embryo. In your lab report, sketch the embryo and label all the parts that you can identify.

PART B Seedling Development

7. Set five corn kernels on a folded paper towel. Roll up the paper towel, and put a rubber band around the roll. Stand the roll in a beaker with 1 cm of water in the bottom. The paper towel will soak up water and moisten the corn. Keep water at the bottom of the beaker, but do not allow the corn kernels to be covered by the water.



8. Repeat step 7 with five bean seeds.
9. After three days, unroll the paper towels and examine the corn and bean seedlings. Use a glass-marking pen to mark the roots and shoots of the developing seedlings. Starting at the seed of the corn and at the cotyledon node of the bean, make a mark every 0.5 cm along the root of each seedling. And again starting at the seed of the corn and at the cotyledon node of the bean, make a mark every 0.5 cm along the stem of each seedling. Measure the distance from the last mark on the root to the root tip of each seedling. Also measure the distance from the last mark on the stem to the shoot tip of each seedling. Record these data in your lab report.
10. Draw a corn seedling and a bean seedling in your lab report. Using a fresh paper towel, roll up the seeds, place the rolls in the beakers, and add fresh water to the beakers.
11. Make a data table similar to the one shown below in your lab notebook. Expand your table by adding columns under each "Roots" and "Stems" head to account for every section of the roots and stems of your seedlings.
12. After two days, reexamine the seedlings. Measure the distance between the marks, and record the data in your data table.
13.  **CAUTION** Use the scalpel carefully to avoid injury. Using the scalpel, make a cut about 2 cm from the tip of the root of a bean seedling. Place the root tip on a microscope slide and add a drop of water. Place a cover slip over the root tip. Using a compound light microscope on low power, observe the root tip. In your lab report, draw the root tip.

14.   Clean up your materials and wash your hands before leaving the lab.

Analysis and Conclusions

1. What protects the tips of corn shoots as they push through the soil? What protects the bean shoots?
2. What types of leaves first appear on the bean seedling?
3. What substance does the black color in the corn kernel indicate? Why might you expect to find this substance in the seed?
4. Examine the data you recorded for steps 9 and 12. Has the distance between the marks changed? If so, where has it changed?
5. What parts of the embryo were observed in all seeds on the third day?
6. How does the structure and development of the corn kernel differ from the structure and development of the pea and bean seeds?
7. What was the source of nutrients for each seed embryo? What is your evidence?
8. Describe the growth in the seedlings you observed.
9. Corn and beans are often cited as representative examples of monocots and dicots, respectively. Relate the seed structure of each to the terms *monocotyledon* and *dicotyledon*.

Further Inquiry

Design an experiment to find out how monocots and dicots compare in general plant growth and in the structure of their leaves and flowers.

CORN AND BEAN SEEDLING GROWTH AFTER TWO DAYS

Seedling	Corn seedlings				Bean seedlings			
	Roots		Stems		Roots		Stems	
	Seed to first mark	Root tip to last mark	Seed to first mark	Shoot tip to last mark	Seed to first mark	Root tip to last mark	Seed to first mark	Shoot tip to last mark
1								
2								
3								
4								
5								

PLANT RESPONSES



Venus' flytrap (*Dionaea muscipula*), a carnivorous plant, has leaves that are hinged in the middle. In response to an insect landing on an open leaf, the leaf can snap shut in less than a second. Glands on the leaf's surface then secrete enzymes that digest the insect, providing the plant with nutrients it does not get from the poor soil in which it grows.

SECTION 1 *Plant Hormones*

SECTION 2 *Plant Movements*

SECTION 3 *Seasonal Responses*



For project ideas from *Scientific American*, visit go.hrw.com and type in the keyword **HM6SAH**.

PLANT HORMONES

The growth and development of a plant are influenced by genetic factors, external environmental factors, and chemicals. Plants respond to chemicals that occur naturally inside plants and to synthetic chemicals.

GROUPS OF HORMONES

Plant hormones are chemical messengers that affect a plant's ability to respond to its environment. Hormones are organic compounds that are effective at very low concentrations; they may be made in one part of the plant and transported to another part. They interact with specific target tissues to cause physiological responses, such as growth or fruit ripening. Each response may be the result of two or more hormones acting together.

Because hormones stimulate or inhibit plant growth, many botanists also refer to them as plant **growth regulators**. Many hormones can be synthesized in the laboratory, which increases the quantity of hormones available for commercial applications. Botanists recognize at least five major groups of hormones: auxins, gibberellins, ethylene, cytokinins, and abscisic acid. These groups of hormones are examined in Table 31-1 on the next page.

AUXINS

Auxins (AWK-suhnz) are hormones involved in plant-cell elongation, shoot and bud growth, and rooting. A well-known natural auxin is **indoleacetic** (IN-DOHL-uh-SEET-ik) **acid**, or IAA. Developing seeds produce IAA, which stimulates the development of fleshy structures such as fruit. An experiment with strawberries, shown in Figure 31-1, provides one form of experimental evidence about the role of IAA in plants. The removal of the seed-containing parts from a strawberry prevents the fleshy structure from enlarging. However, if IAA is then applied, the strawberry enlarges normally.



(a)

(b)

(c)

OBJECTIVES

- **List** the actions of the five major types of plant hormones.
- **Describe** agricultural or gardening applications for each of the five major types of plant hormones.
- **Discuss** how growth retardants are used commercially.

VOCABULARY

plant hormone
growth regulator
auxin
indoleacetic acid
naphthalene acetic acid
apical dominance
2,4-D
Agent Orange
gibberellin
ethylene
ethephon
abscission
cytokinin
abscisic acid
growth retardant

FIGURE 31-1

(a) Strawberry plants normally produce a fleshy, heart-shaped "berry." The "berry" is not truly a fruit—it is a swollen flower receptacle. The yellowish "seeds" covering the "berry" are truly complete fruits, each containing a seed. (b) If the "seeds" are removed from a young "berry," it does not enlarge. (c) But if the "berry" is then treated with IAA, it grows normally.

TABLE 31-1 *Five Groups of Plant Hormones*

Plant hormone	Actions	Features	Examples
Auxins	• promote cell growth • promote root formation on stem and leaf cuttings • promote shoot and bud growth • promote setting of fruits • prevent dropping of fruit	• produced in growing regions of plant (shoot tips, young leaves, developing seeds) • important role in tropisms	• indoleacetic acid, IAA (natural) • indolebutyric acid, IBA (natural and synthetic) • naphthalene acetic acid, NAA (synthetic) • herbicides 2,4-D and Agent Orange (synthetic)
Gibberellins (GA)	• promote elongation growth • promote germination and seedling growth • increase size of fruit • overcome bud dormancy • substitute for long-day or vernalization requirements for flowering	• produced in young shoots and immature seeds • more than 100 naturally occurring types	• gibberellic acid, GA₃ (natural)
Ethylene	• promotes ripening of fruit • promotes flowering in mangoes and pineapples • promotes abscission	• produced in fruits, flowers, leaves, and roots • colorless gas	• ethephon (synthetic) breaks down and releases ethylene (natural)
Cytokinins	• promote cell division • promote lateral bud growth in dicots	• produced in developing roots, fruits, and seeds • auxin-to-cytokinin ratio is important	• zeatin (natural) • kinetin (synthetic) • benzyladenine (synthetic)
Abscisic acid (ABA)	• promotes stomatal closure • promotes dormancy • inhibits other hormones • blocks growth	• produced in leaves • expensive to synthesize	• ABA (natural or synthetic)

Indoleacetic acid is produced in actively growing shoot tips and developing seeds, and it is involved in elongation. Before a cell can elongate, the cell wall must become less rigid so that it can expand. Indoleacetic acid triggers an increase in the plasticity, or stretchability, of cell walls, which allows elongation to occur.

Synthetic Auxins

Auxins have a variety of possible effects, so they are used for several purposes in gardening, commercial agriculture, and scientific research. Chemists have synthesized several inexpensive compounds similar in structure to IAA. Synthetic auxins, such as **naphthalene** (NAF-thuh-LEEN) **acetic** (uh-SEET-ik) **acid**, or NAA, are used to promote root formation on stem and leaf cuttings, as shown in Figure 31-2 on the next page. Such vegetative propagation of genetically identical plants is useful in agriculture and in laboratory research.

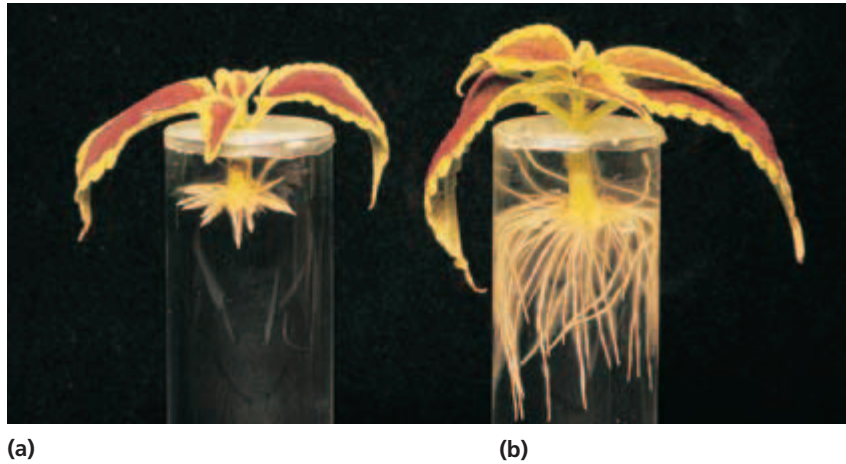


FIGURE 31-2

(a) This stem cutting from a coleus plant, *Coleus hybridus*, is growing roots in pure water. (b) This coleus stem cutting has been treated with a synthetic auxin—naphthalene acetic acid (NAA). Roots form more rapidly when the stem is treated with NAA. Both cuttings were taken at the same time.

When NAA is sprayed on young fruits of apple and olive trees, some of the fruits drop off so that the remaining fruits grow larger. When NAA is sprayed directly on some other fruits—such as pears and citrus fruits—several weeks before they are ready to be picked, NAA prevents the fruits from dropping off the trees before they are mature. The fact that auxins can have opposite effects—they cause fruit to drop or prevent fruit from dropping—illustrates an important point: The effects of a hormone on a plant often depend on the stage of the plant's development.

Naphthalene acetic acid is used to prevent the undesirable sprouting of stems from the base of ornamental trees. Recall that stems contain a lateral bud at the base of each leaf. In many stems, these buds fail to sprout as long as the plant's shoot tip is intact. The inhibition of lateral buds by the presence of a shoot tip is called **apical dominance**. If the shoot tip is removed, the lateral buds begin to grow. If IAA or NAA is applied to the cut tip of the stem, the lateral buds remain dormant. NAA is used commercially to prevent buds from sprouting on potato tubers during storage.

Another important synthetic auxin is **2,4-D**, which is a weed-killer. At certain concentrations, it kills dicots, such as dandelions, without injuring monocots, such as lawn grasses and cereal crops. Given our dependence on cereals for food, 2,4-D has been important to agriculture. **Agent Orange**, a mixture of 2,4-D and another auxin, was used to defoliate jungles in the Vietnam War. A nonauxin contaminant in Agent Orange is thought to have caused health problems in many people who were exposed to it.

GIBBERELLINS

In the 1920s, scientists in Japan discovered that a substance produced by fungi of the genus *Gibberella* caused fungus-infected rice plants to grow abnormally tall. This type of hormone, named **gibberellin** (JIB-uh-REL-uhn), was later found to be produced in small quantities by plants themselves.

FIGURE 31-3

These bird's nest ferns, *Asplenium nidus*, are the same age. The fern in (b), however, has been treated with gibberellin, which stimulates the leaves to grow larger.



(a)

(b)

Gibberellins have many effects on a plant, but they primarily stimulate elongation growth. Spraying some plants with gibberellins may cause them to grow larger than normal, as shown in Figure 31-3.

Like auxins, gibberellins are hormones that have important commercial applications. Many seedless grapes, such as those shown in Figure 31-4, are sprayed with gibberellins to increase the size of the fruit. Beer makers use gibberellins to increase the alcohol content of beer by increasing the amount of starch converted to sugar in the brewing process. Gibberellins are also used to treat seeds, because they break seed dormancy and promote uniform germination.

FIGURE 31-4

Almost all of the raisins produced in California are made from Thompson Seedless grapes. Normal Thompson Seedless grapes are smaller than those treated with a gibberellin hormone. The grapes on the left were treated with a gibberellin. As you can see, the addition of a gibberellin to grapes stimulates the grapes to grow larger.



ETHYLENE

The hormone **ethylene** (ETH-uh-LEEN) plays a role in the ripening of fruits. Unlike other plant hormones, ethylene is a gas at room temperature. Ethylene gas diffuses easily through the air from one plant to another. The saying “One bad apple spoils the barrel” has its basis in ethylene gas. One rotting apple will produce ethylene gas, which stimulates nearby apples to ripen and then spoil.

In commercial use, ethylene is usually applied in a solution of **ethephon** (ETH-uh-fohn), a synthetic chemical that breaks down to release ethylene gas. It is used to ripen bananas, honeydew melons, and tomatoes. Oranges, lemons, and grapefruits often remain green when they are ripe. Although the green fruit may taste good, consumers will not usually buy them. The application of ethylene to green citrus fruits causes desirable citrus colors, such as yellow and orange, to develop.

In some plant species, ethylene promotes **abscission**, the detachment of leaves, flowers, or fruits. Growers can use mechanical tree shakers to harvest cherries and other crops if the fruit is sprayed with ethylene before harvest. Leaf abscission can be an adaptive advantage. Dead, damaged, or infected leaves drop off rather than shading healthy leaves or spreading disease. The plant can minimize water loss in winter, when water in the environment is often frozen.